

CloudEngine 16800 Series Switches

Hardware Description (V200)

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Intended Audience

This document is intended for network engineers responsible for network design and deployment. You should understand your network well, including the network topology and service requirements.

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

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1 Using the Info-Finder

Info-Finder ([enterprise users](#) / [carrier users](#)) is a tool platform. It allows you to search for key product information by product series and model. The key product information includes basic information such as the software specifications, life cycles, and hardware information, and operation and maintenance information such as the licenses, alarms, logs, commands, and MIBs. The hardware-related tools are as follows:

- **Product Image Gallery:** provides product photos and network element icons for you to produce design drawings and networking diagrams.
- **Hardware Configuration:** automatically generates hardware configuration diagrams after you select components are required and calculates the weight, power consumption, and heat consumption.
- **Hardware Center:** provides the technical specifications of devices and components, as well as the mapping between devices, components, and versions.
- **3D Model:** Using this function, you can query product images, product overview, and component insertion/removal videos, enabling you to quickly obtain product information in one-stop mode.

2 Version Requirements for Components

NOTE

- This document describes all the device models and modules supported in a version. To obtain accurate subscription information, visit <http://e.huawei.com> or contact Huawei local sales offices. You can also pay attention to the product change notices (PCNs) and lifecycle management bulletins on this website.
- The figures in this document are for reference only.

3 Cabinet

- 3.1 Introduction to the A610-22 Cabinet
- 3.2 Introduction to the A612-22 Cabinet
- 3.3 Introduction to the A812-20 Cabinet

3.1 Introduction to the A610-22 Cabinet

Overview

This document details the A610-22 cabinet. If a third-party cabinet needs to be evaluated, see **Checking the Cabinet or Rack**.

Naming Conventions

The cabinet name A610-22 is divided into four fields, as shown in **Figure 3-1**. **Table 3-1** explains the meaning of each field.

Figure 3-1 Cabinet name

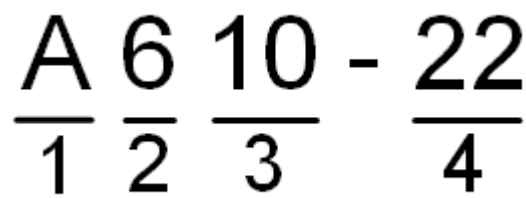


Table 3-1 Cabinet naming conventions

Field	Description
1	Cabinet model

Field	Description
2	Cabinet width: 600 mm
3	Cabinet depth: 1000 mm
4	Cabinet height: 2200 mm

Appearance

 NOTE

Figures in the document are for reference only, and the actual appearance of the cabinet may vary depending on the exact model.

[Figure 3-2](#) shows the A610-22 cabinet appearance.

Figure 3-2 Cabinet appearance



Functions

The cabinet complies with International Electrotechnical Commission (IEC) 60297-1. The distance between front and rear mounting rails is adjustable in 25 mm increments. Its assembled architecture facilitates capacity expansion.

- The cabinet has the following features and functions:
- Static load bearing capability: 1 ton
 - Sufficient space to place and connect components.
 - Ability in protecting internal components from contamination.
 - Ability in preventing damage to internal components.

Physical Structure

Figure 3-3 shows components of the A610 cabinet.

Figure 3-3 Cabinet components



1. Front door	2. Top cover	3. Rear door
4. Side panel	5. Mounting rail	6. Rack

Cabinet Material

The cabinet is made of class-A high-strength carbon cold rolled steel and galvanized sheet, which complies with the Restriction of the use of certain Hazardous Substances (RoHS). The fire endurance of interior materials complies with Underwriter Laboratories (UL) standards.

The cabinet complies with International Electrotechnical Commission (IEC) 60297-1. The distance between front and rear mounting rails is adjustable in 25 mm increments. Its assembled architecture facilitates capacity expansion.

The cabinet has the following features and functions:

- Sufficient space to place and connect components.
- Ability in protecting internal components from contamination.
- Ability in preventing damage to internal components.

Cabling

The cabinet supports overhead or underfloor cabling.

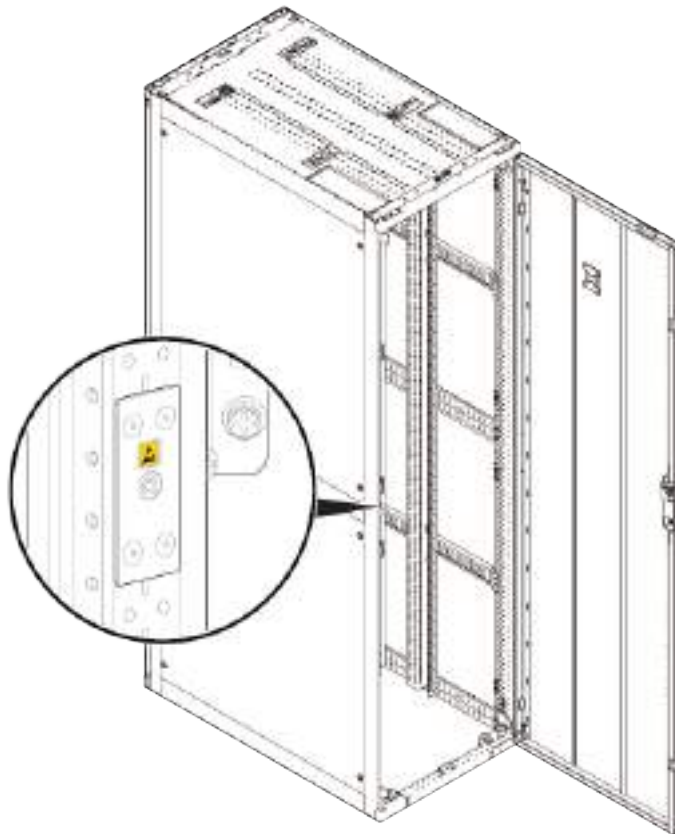
ESD Jack

The cabinet has an ESD jack, as shown in [Figure 3-4](#). When installing or maintaining devices in the cabinet, wear an ESD wrist strap and insert the other end of the ESD wrist strap in the ESD jack.

NOTE

A cabinet provides multiple grounding points. You can choose the nearest one.

Figure 3-4 ESD jack on the cabinet

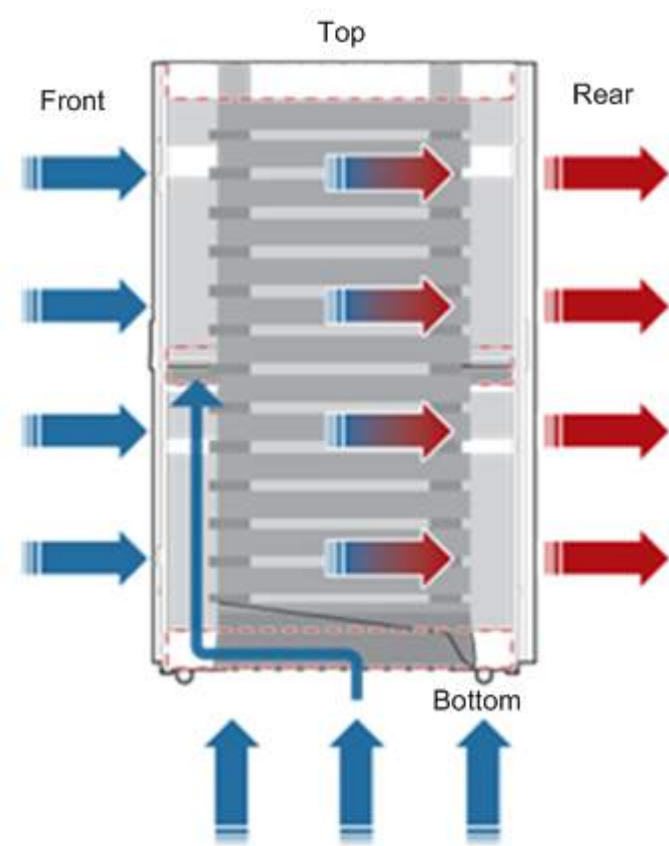


Heat Dissipation

The front and rear doors of the cabinet are of mesh construction. Air enters the chassis from the front door and base and is exhausted through the rear door.

Figure 3-5 shows the airflow.

Figure 3-5 Airflow in the cabinet



Technical Specifications

Table 3-2 lists technical specifications of the A610-22 cabinet.

Table 3-2 Technical specifications of the A610-22 cabinet

Item	Details
Part number	02116250
Dimensions without packaging (H x W x D)	2200 mm x 600 mm x 1050 mm (86.61 in. x 23.62 in. x 41.34 in.)
Dimensions with packaging (H x W x D)	2320 mm x 720 mm x 1290 mm (91.34 in. x 28.35 in. x 50.79 in.)
Weight (empty cabinet)	135 kg (297.62 lb)
Weight (assembly cabinet)	215 kg (473.99 lb)
Distance between the front and rear mounting rails	650 mm
Available space	47 U

Item	Details
Environment parameters	- Long-term temperature: 0°C to 40°C - Short-term temperature: -5°C to +55°C - Long-term relative humidity: 5% RH to 85% RH - Short-term relative humidity: 0% RH to 95% RH
Installation mode	Installed on an ESD floor or a concrete floor
Static load bearing capacity	1000 kg (2204.62 lb)

 NOTE

- 1U equals 1.75 in. (approximately 44.45 mm). U is a unit of height defined in IEC 60297.
- The net weight is the weight of a cabinet with the front door and rear door.
- The weight of an assembly cabinet is the total weight of the empty cabinet, cabinet accessories, and the package.
- The environment parameter values are measured at 0.4 m in front of the equipment and 1.5 m above the ground.
- Short-term means the continuous working time within 96 hours and the accumulated working time within 15 days per year.

3.2 Introduction to the A612-22 Cabinet

Naming Conventions

The cabinet name A612-22 is divided into four fields, as shown in [Figure 3-6](#). [Table 3-3](#) explains the meaning of each field.

Figure 3-6 Cabinet name

A	6	12	-	22
<u>1</u>	<u>2</u>	<u>3</u>		<u>4</u>

Table 3-3 Cabinet naming conventions

Field	Description
1	Cabinet model
2	Cabinet width: 600 mm
3	Cabinet depth: 1200 mm
4	Cabinet height: 2200 mm

Appearance

 NOTE

Figures in the document are for reference only, and the actual appearance of the cabinet may vary depending on the exact model.

Figure 3-7 shows the A612-22 cabinet appearance.

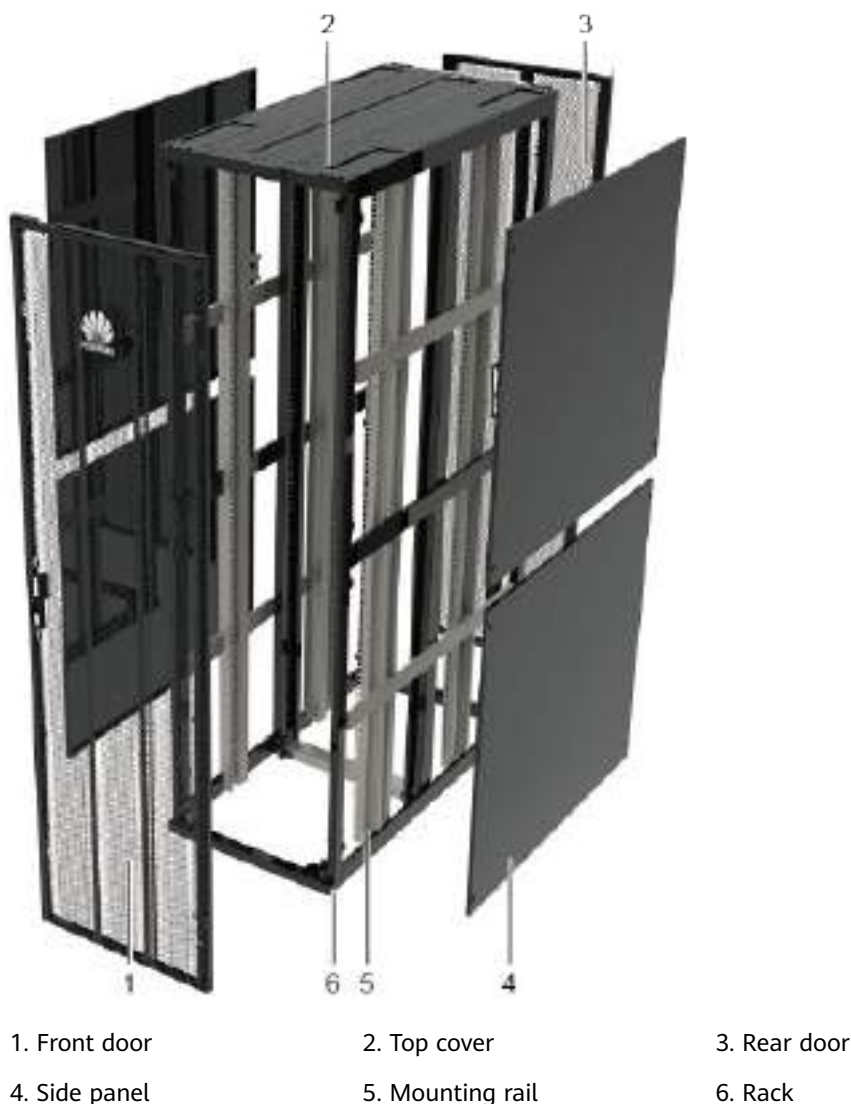
Figure 3-7 Cabinet appearance



Physical Structure

Figure 3-8 shows components of the A612-22 cabinet.

Figure 3-8 Cabinet components



Functions and Features

The cabinet is made of class-A high-strength carbon cold rolled steel and galvanized sheet, which complies with the Restriction of the use of certain Hazardous Substances (RoHS). The fire endurance of interior materials complies with Underwriter Laboratories (UL) standards.

The cabinet supports overhead or underfloor cabling.

The cabinet complies with International Electrotechnical Commission (IEC) 60297-1. The distance between front and rear mounting rails is adjustable in 25 mm increments. Its assembled architecture facilitates capacity expansion.

The cabinet has the following features and functions:

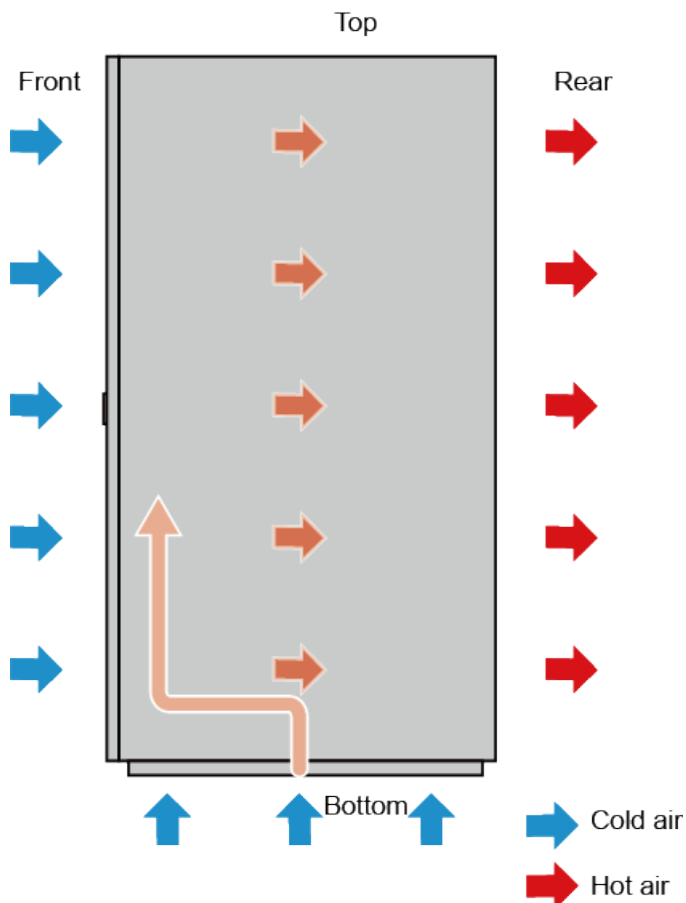
- Space to place and connect components.
- Ability in protecting internal components from contamination.
- Ability in preventing damage to internal components.

Heat Dissipation

The front and rear doors of the cabinet are of mesh construction. Air enters the chassis from the front door and base and is exhausted through the rear door.

[Figure 3-9](#) shows the airflow.

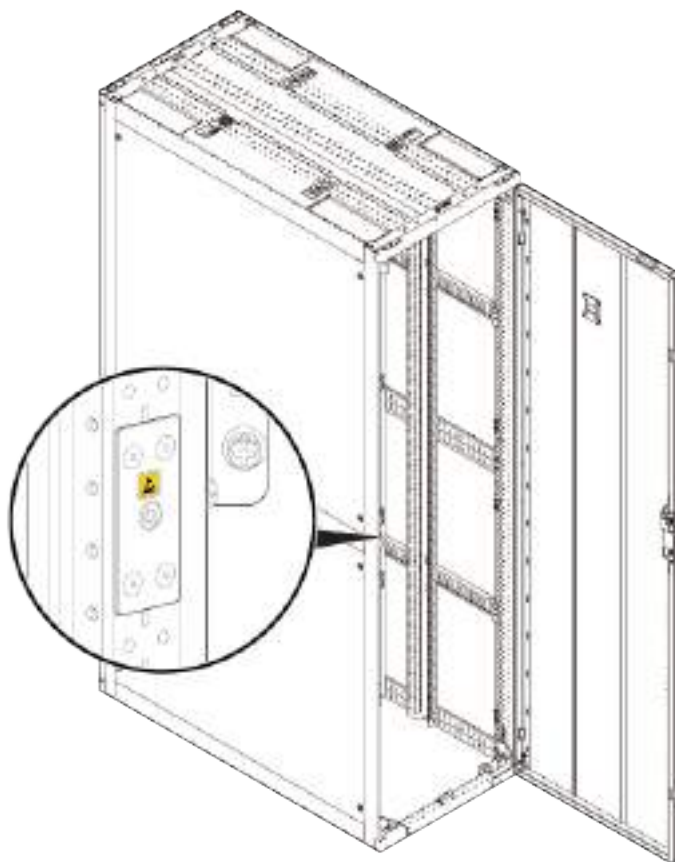
Figure 3-9 Airflow in the cabinet



ESD Jack

The cabinet has an ESD jack, as shown in [Figure 3-10](#). When installing or maintaining devices in the cabinet, wear an ESD wrist strap and insert the other end of the ESD wrist strap in the ESD jack.

Figure 3-10 ESD jack on the cabinet



Technical Specifications

Table 3-4 lists technical specifications of the cabinet.

Table 3-4 Technical specifications of the A612-22 cabinet

Item	Details
Part number	02116249
Dimensions without packaging (H x W x D)	2200 mm x 600 mm x 1200 mm (86.61 in. x 23.62 in. x 47.24 in.)
Dimensions with packaging (H x W x D)	2410 mm x 980 mm x 1455 mm (94.88 in. x 38.58 in. x 57.28 in.)
Weight (empty, without packaging)	160 kg

Item	Details
Weight (assembly cabinet, with packaging)	192 kg
Standards compliance	IEC 60297-1
Distance between mounting rails	725 mm
Available space	47 U
Installation scenario	• Long-term temperature: 0°C to 40°C • Short-term temperature: -5°C to +55°C • Long-term relative humidity: 5% RH to 85% RH • Short-term relative humidity: 0% RH to 95% RH
Installation mode	Installed on a concrete floor or an ESD floor
Static load bearing capacity	1200 kg

 NOTE

- 1U equals 1.75 in. (approximately 44.45 mm). U is a unit of height defined in IEC 60297.
- The net weight is the weight of a cabinet with the front door and rear door.
- The weight of an assembly cabinet is the total weight of the empty cabinet, cabinet accessories, and the package.
- The environment parameter values are measured at 0.4 m in front of the equipment and 1.5 m above the ground.
- Short-term means the continuous working time within 96 hours and the accumulated working time within 15 days per year.

3.3 Introduction to the A812-20 Cabinet

Naming Conventions

The cabinet name A812-20 is divided into four fields, as shown in [Figure 3-11](#). [Table 3-5](#) explains the meaning of each field.

Figure 3-11 Cabinet name

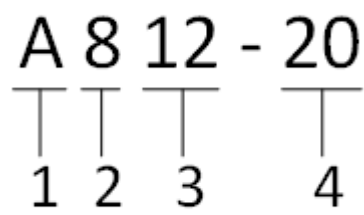


Table 3-5 Cabinet naming conventions

Field	Description
1	Cabinet model
2	Cabinet width: 800 mm
3	Cabinet depth: 1200 mm
4	Cabinet height: 2000 mm

Appearance

NOTE

Figures in the document are for reference only, and the actual appearance of the cabinet may vary depending on the exact model.

Figure 3-12 shows the A812-20 cabinet appearance.

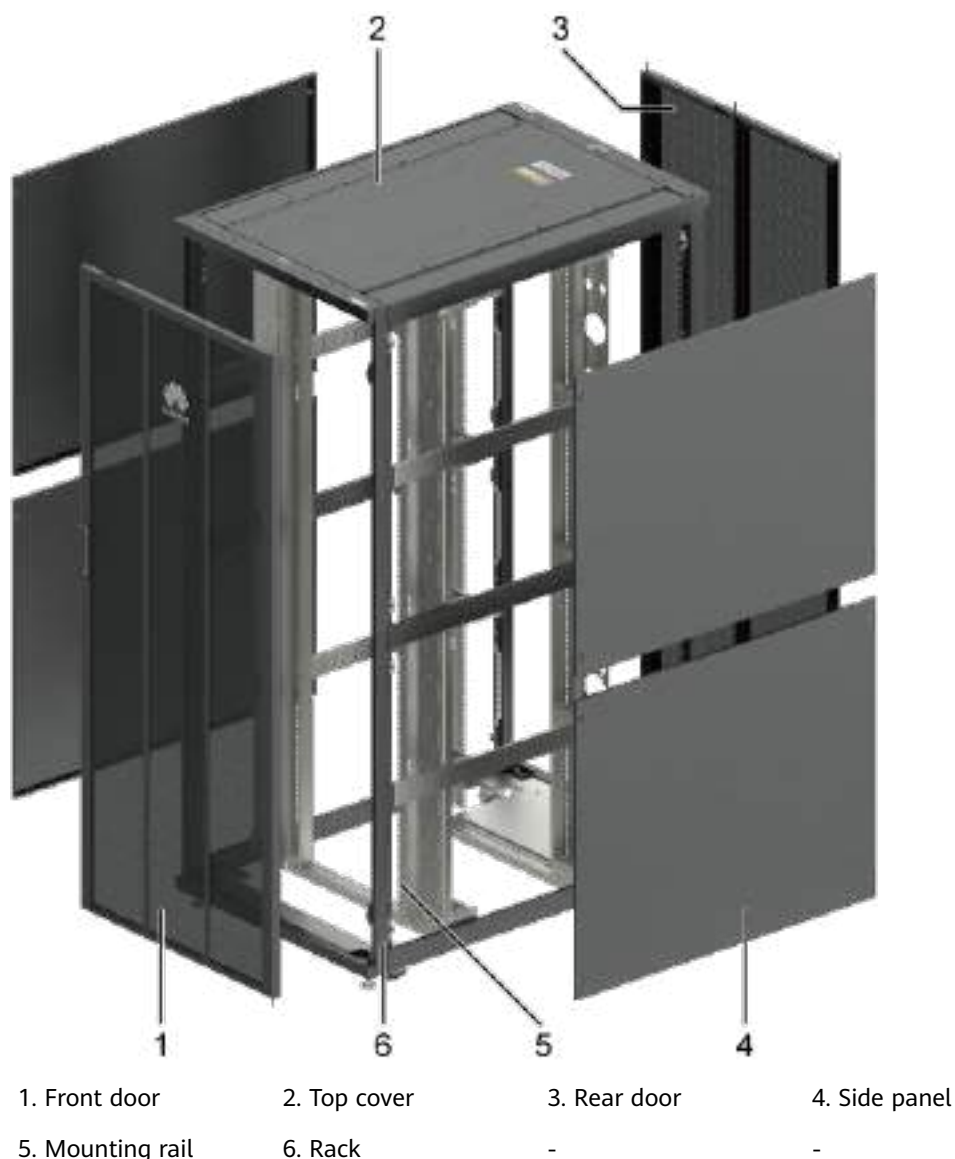
Figure 3-12 Cabinet appearance



Physical Structure

Figure 3-13 shows components of the A812-20 cabinet.

Figure 3-13 Cabinet components



Functions and Features

The cabinet is made of class-A high-strength carbon cold rolled steel and galvanized sheet, which complies with the Restriction of the use of certain Hazardous Substances (RoHS). The fire endurance of interior materials complies with Underwriter Laboratories (UL) standards.

The cabinet supports overhead or underfloor cabling.

The cabinet complies with International Electrotechnical Commission (IEC) 60297-1. The distance between front and rear mounting rails is adjustable in 25 mm increments. Its assembled architecture makes it convenient to expand the cabinet space.

The cabinet has the following features and functions:

- Casters at the bottom to help move the cabinet easily on an inclined plane with a slope less than or equal to 10 degrees.

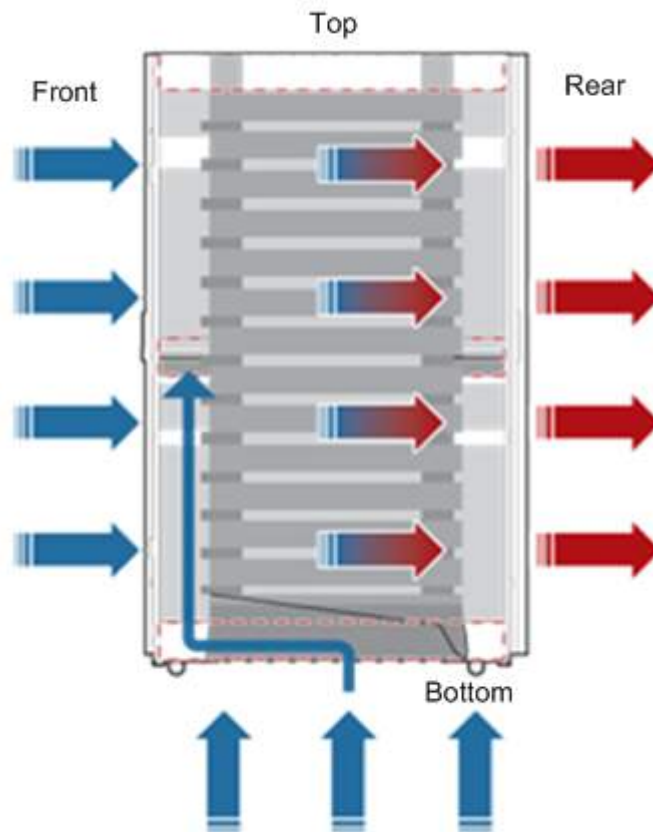
- Sufficient space to place and connect components.
- Ability in protecting internal components from contamination.
- Ability in preventing damage to internal components.

Heat Dissipation

The front and rear doors of the cabinet are of mesh construction. Air enters the chassis from the front door and base and is exhausted through the rear door.

[Figure 3-14](#) shows the airflow.

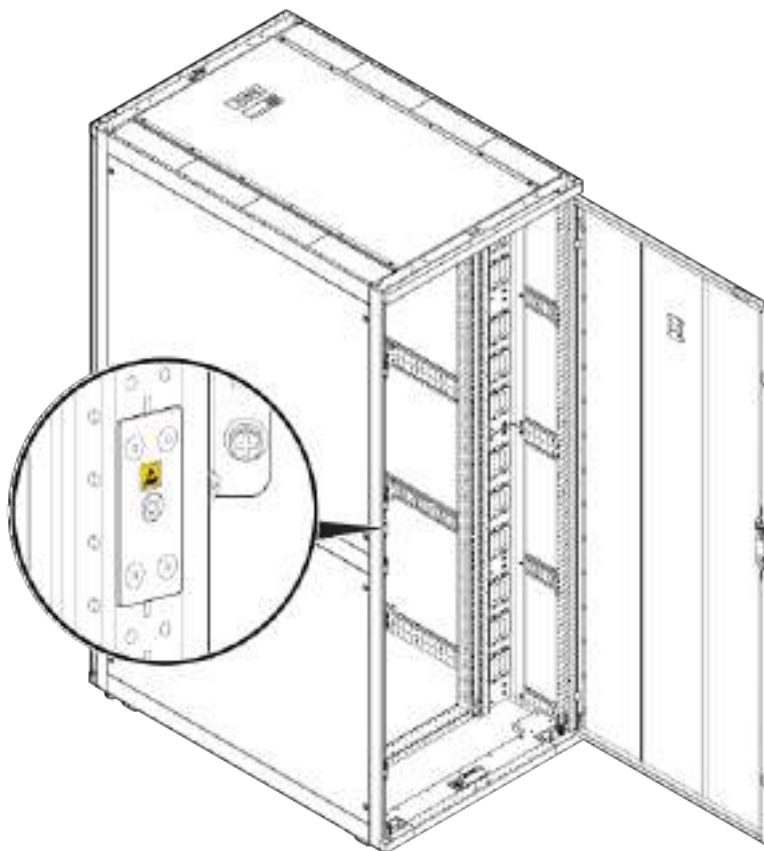
Figure 3-14 Airflow in the cabinet



ESD Jack

The cabinet has an ESD jack, as shown in [Figure 3-15](#). When installing or maintaining devices in the cabinet, wear an ESD wrist strap and insert the other end of the ESD wrist strap in the ESD jack.

Figure 3-15 ESD jack on the cabinet



Technical Specifications

Table 3-6 lists technical specifications of the cabinet.

Table 3-6 Technical specifications of the A812-20 cabinet

Item	Details
Part number	02116085
Dimensions without packaging (H x W x D)	2000 mm x 800 mm x 1200 mm (78.74 in. x 31.50 in. x 47.24 in.)
Dimensions with packaging (H x W x D)	2145 mm x 970 mm x 1375 mm (84.45 in. x 38.19 in. x 54.13 in.)
Weight (empty, without packaging)	190 kg

Item	Details
Weight (assembly cabinet, with packaging)	220 kg
Standards compliance	IEC 60297-1
Distance between mounting rails	725 mm
Available space	42U
Installation scenario	- Long-term temperature: 0°C to 40°C - Short-term temperature: -5°C to +55°C - Long-term relative humidity: 5% RH to 85% RH - Short-term relative humidity: 0% RH to 95% RH
Installation mode	Installed on a concrete floor or an ESD floor.
Dynamic load bearing capacity	1000 kg
Static load bearing capacity	1200 kg

 NOTE

- 1U equals 1.75 in. (approximately 44.45 mm). U is a unit of height defined in IEC 60297.
- The net weight is the weight of a cabinet with a front door and a rear door.
- The weight of an assembly cabinet is the total weight of the empty cabinet, cabinet accessories, and the package.
- The environment parameter values are measured at 0.4 m in front of the equipment and 1.5 m above the ground.

4 Chassis

- 4.1 Chassis Naming Conventions
- 4.2 CloudEngine 16804
- 4.3 CloudEngine 16808
- 4.4 CloudEngine 16816

4.1 Chassis Naming Conventions

Figure 4-1 Chassis naming conventions

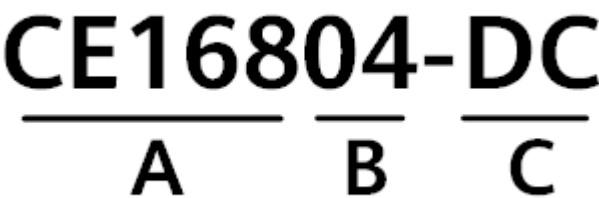


Table 4-1 Chassis naming conventions

Identifier	Meaning	Description
A	CloudEngine series data center core switches	CloudEngine 16800 series data center core switches.
B	Number of LPU slots	04: The chassis has four LPU slots. 08: The chassis has eight LPU slots. 16: The chassis has 16 LPU slots.

Identifier	Meaning	Description
C	Chassis	- AH: The chassis can be configured with AC & high-voltage DC power modules. - DC: The chassis can be configured with DC power modules.

4.2 CloudEngine 16804

Overview

Table 4-2 Basic information about the CloudEngine 16804

Description	Part Number	Model	First supported version
CE16804 AC&high-voltage DC assembly chassis	02352RCU	CE16804-AH	V200: V200R005C20 V300: V300R020C00
CE16804 DC assembly chassis	02352RCR	CE16804-DC	V200: V200R005C20 V300: V300R020C00

 CAUTION

Some cards are attached with labels of different colors near the ejector levers to identify their types. The color of labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the card will be damaged.

Appearance

Figure 4-2 Appearance of the CloudEngine 16804 (front view)



Figure 4-3 Appearance of the CloudEngine 16804 (rear view)



 NOTE

- The DC chassis and AC & high-voltage DC chassis have similar appearances and structures except that they use DC and AC & high-voltage DC power modules (PMs) and power entry modules (PEMs) respectively. The figures show the appearance of an AC & high-voltage DC chassis as an example.
- Figures in the document are for reference only, and the actual appearance of the devices may vary depending on the exact device model.

Components

In the CloudEngine 16804 chassis, MPUs and LPUs are installed horizontally in the front of the chassis, and SFUs are installed vertically at the rear of the chassis, as shown in [Figure 4-4](#).

Figure 4-4 Components of the CloudEngine 16804

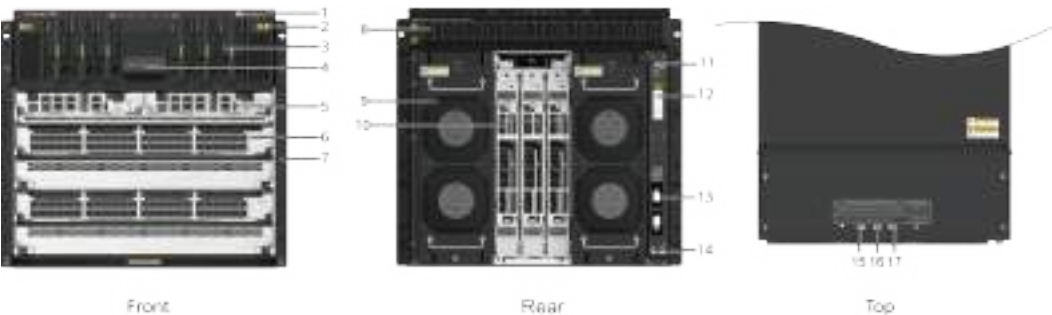


Table 4-3 Components

1. Chassis header	2. Front ESD jack	3. PM	4. Power switch	5. MPU
6. LPU	7. Cable management frame	8. PEM	9. Fan module	10. SFU SFUs are located behind fan modules and are invisible after all fan modules are installed.
11. Location of fan modules and SFUs	12. Two ground points for a two-hole OT terminal There is a yellow grounding label attached.	13. SFU wrench	14. Rear ESD jack	15. ESN label

16. Assembly chassis label	17. MAC address label	-	-	-
----------------------------	-----------------------	---	---	---

 NOTE

- The chassis header shows the following:
- Product series name: CloudEngine 16800
 - Huawei Logo
 - Indicators: For details, see the indicator description table.
- The indicator plate on the chassis header includes indicators and a manufacturing information loading interface. The manufacturing information loading interface is disabled before delivery and cannot be used.

Slot Layout

Figure 4-5 Slot Layout of the CloudEngine 16804

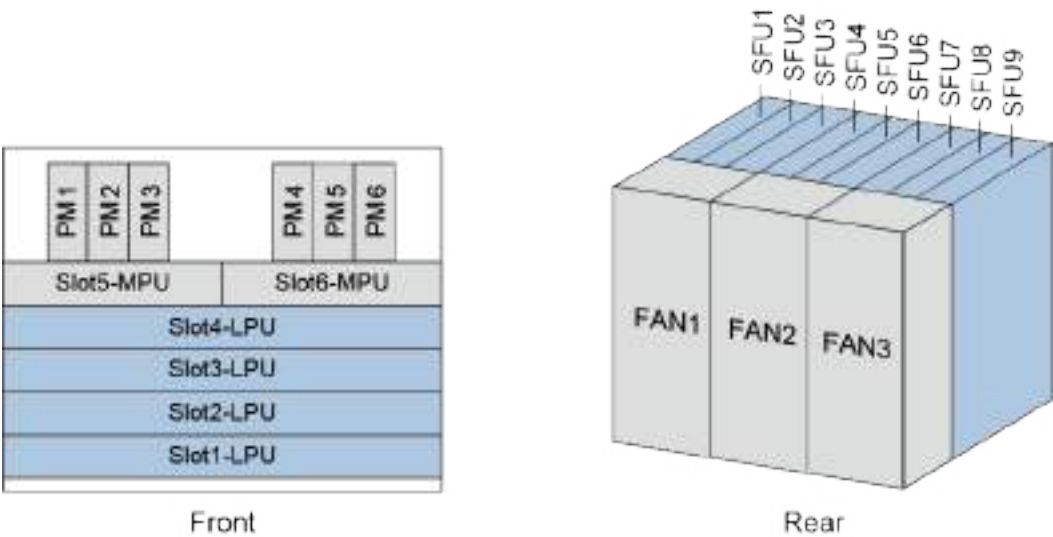


Table 4-4 Slots on the CloudEngine 16804

Slot Type	Slot ID	Slot Direction	Remarks
LPU slot	Slots 1 to 4	The top of the board faces upwards	Different types of LPUs can be installed in the same chassis.

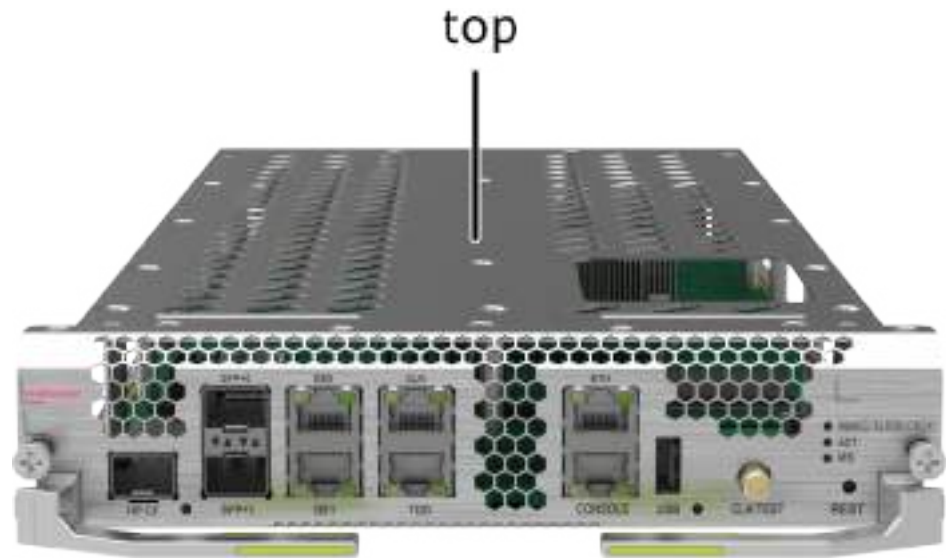
Slot Type	Slot ID	Slot Direction	Remarks
MPU slot	Slots 5 and 6	The top of the board faces upwards	Two MPUs work in active/standby mode for 1:1 hot standby. By default, the MPU with a smaller slot ID is the active MPU, and the MPU with a larger slot ID is the standby MPU. The active and standby MPUs in the same chassis must be of the same model.
SFU slot	1 to 9 Note: The slots in which SFUs can be installed vary according to the SFU model. For details, see the "IDs of Available Slots" column in the "Version Mapping" part of each SFU.	The right side of the board faces upwards	SFUs work in load balancing and redundancy mode. SFUs in the same chassis must be the same model. That is, SFUs of different models cannot be installed in the same chassis.
PM slot	Slots 1 to 6	The indicator light end upwards	PMs are plug-and-play.
Fan module slot	Slots 1 to 3	The indicator light end upwards	At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.

 NOTE

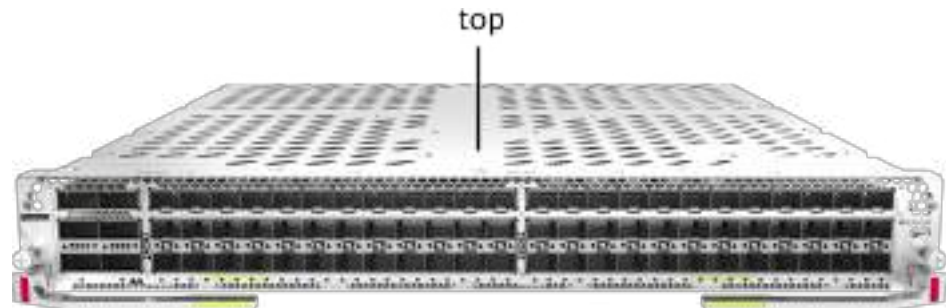
- SFU slots 1, 2, and 9 are unavailable. In V200R005C20 to V200R021C00, after SFUs are installed in slots 1, 2, and 9, the SFUs can be registered but are unavailable. In V200R021C10 and later versions, after SFUs are installed in slots 1, 2, and 9, an alarm will be reported to the NMS and then the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)
- SFU slots 4 and 7:
 - By default, SFU slots 4 and 7 are available.
 - After the **set sfu reduced-mode** command is run to enable the SFU reduction mode, SFUs in slots 4 and 7 are unavailable. In V200R005C20, after SFUs are installed in slots 4 and 7, the SFUs can be registered but are unavailable. In V200R019C10 and later versions, after SFUs are installed in slots 4 and 7, the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)

 NOTE

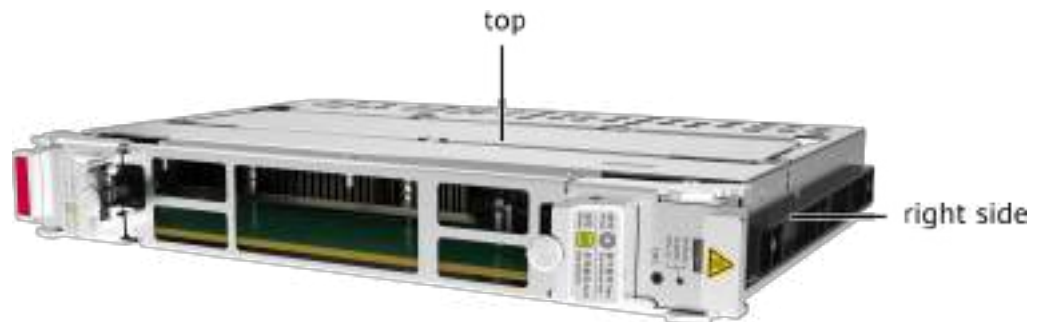
- Top of the MPU:



- Top of the LPU:

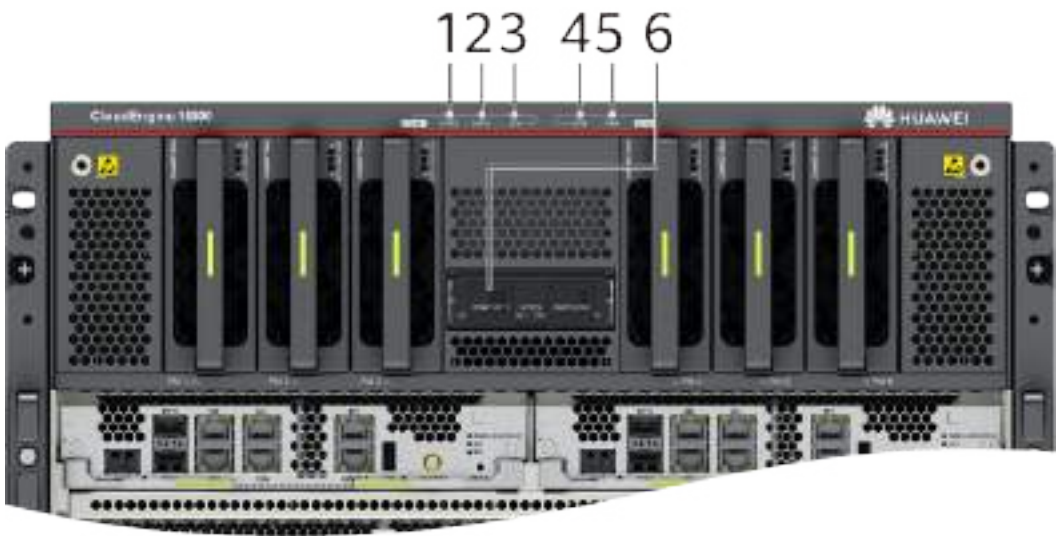


- Top and right side of the SFU:



Panel

Figure 4-6 Panel on the CloudEngine 16804



1. PM indicator	2. MPU indicator	3. LPU indicator	4. SFU indicator	5. Fan module indicator
6. Power switch	-	-	-	-

Table 4-5 Indicators on the CloudEngine 16804

Silkscreen	Name	Color	Status	Description
PWR	PM indicator	Green	Steady on	All PMs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more PMs are not working properly. For example, a PM is present but does not have output power. - One or more PMs are powered off.
MPU	MPU indicator	Green	Steady on	All installed MPUs are working properly.
		Red	Steady on	At least one MPU has failed.
LPU	LPU indicator	Green	Steady on	All installed LPUs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One LPU is not working properly and a hardware alarm has been generated. - No LPU is properly installed.
SFU	SFU indicator	Green	Steady on	All installed SFUs are working properly.
		Red	Steady on	The possible causes are as follows: - One SFU is not working properly and a hardware alarm has been generated. - No SFU is properly installed.
FAN	Fan module indicator	Green	Steady on	All fan modules are installed and working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more fan modules are not working properly. - One or more fan modules are not properly installed or have been removed.

Table 4-6 Buttons on the CloudEngine 16804

Silkscreen	Name	Description
SWITCH 1/SWITCH 2	Power switch	Controls device power-on and power-off. NOTE The device is powered off only when switch1 and switch2 are both OFF. The device is powered on as long as either switch is ON.

Technical Specifications

Table 4-7 Technical specifications of the CE16804-AH

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	730 mm x 770 mm x 1150 mm (28.74 in. x 30.31 in. x 45.28 in.)

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	438 mm x 442 mm x 874 mm (17.24 in. x 17.40 in. x 34.41 in.), dimensions of the chassis body 438 mm x 483 mm x 985 mm (17.24 in. x 19.02 in. x 38.78 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	94.1 kg (207.49 lb) (including two MPUs and filler panels)
Weight without packaging (full configuration) [kg (lb)]	- V200: 176.4 kg/388.89 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 176.4 kg/388.89 lb (fully configured with CEL36CQFD-E cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	138 kg (304.23 lb)
Cabinet installation standards	- A812 - A612 - A610
Chassis height [U]	9.8 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	AC/HVDC pluggable

Item	Specification
Typical power consumption [W]	- V200: 2852 W (fully configured with CEL36CQFD-G cards) - V300: 2892 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Typical heat dissipation [BTU/hour]	- V200: 9731 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 9868 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 4562 W (fully configured with CEL36CQFD-G cards) - V300: 4664 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 15566 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 15914 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	6
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.

Item	Specification
Rated input voltage [V]	- AC: 220 V; 50 Hz/60 Hz - High-voltage DC (HVDC): 240 V/380 V
Input voltage range [V]	- AC: 176–290 V; 45–65 Hz - High-voltage DC (HVDC): 188 V to 288 V or 260 V to 400 V
Maximum input current [A]	- AC: 16 A @ 200 V; 18.5 A @ 176 V - High-voltage DC (HVDC): 18 A @ 188 V; 13 A @ 260 V
Maximum output power [W]	- In 5+1 backup mode: 3000 W x 5 = 15000 W - In 6+0 backup mode: 3000 W x 6 = 18000 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999971695
MTBF [year]	34.93 years
Number of LPU slots	4
Number of MPU slots	2
Number of SFU slots	9
Number of slots	15
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).

Item	Specification
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE - The product has a valid storage period of one year. - The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

Table 4-8 Technical specifications of the CE16804-DC

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	730 mm x 770 mm x 1150 mm (28.74 in. x 30.31 in. x 45.28 in.)
Dimensions without packaging (H x W x D) [mm (in.)]	438 mm x 442 mm x 874 mm (17.24 in. x 17.40 in. x 34.41 in.), dimensions of the chassis body 438 mm x 483 mm x 985 mm (17.24 in. x 19.02 in. x 38.78 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	98.1 kg (216.31 lb) (including two MPUs and filler panels)

Item	Specification
Weight without packaging (full configuration) [kg (lb)]	- V200: 178 kg/392.42 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 178 kg/392.42 lb (fully configured with CEL36CQFD-E cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	142 kg (313.05 lb)
Cabinet installation standards	- A812 - A612 - A610
Chassis height [U]	9.8 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	DC pluggable
Typical power consumption [W]	- V200: 2852 W (fully configured with CEL36CQFD-G cards) - V300: 2892 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.

Item	Specification
Typical heat dissipation [BTU/hour]	- V200: 9731 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 9868 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 4562 W (fully configured with CEL36CQFD-G cards) - V300: 4664 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 15566 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 15914 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	6
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.
Rated input voltage [V]	-48 V DC/-60 V DC
Input voltage range [V]	-40 V DC to -72 V DC
Maximum input current [A]	58.5 A for the voltage of -40 V

Item	Specification
Maximum output power [W]	- In 5+1 backup mode: 2200 W x 5 = 11000 W - In 6+0 backup mode: 2200 W x 6 = 13200 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999971695
MTBF [year]	34.93 years
Number of LPU slots	4
Number of MPU slots	2
Number of SFU slots	9
Number of slots	15
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE - The product has a valid storage period of one year. - The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

 NOTE

- Temperature and humidity are measured at 1.5 m (4.92 ft.) above the floor and 0.4 m (1.31 ft.) in front of the rack. There should be no protection board in the front or rear of the rack.
- The short-term operating conditions mean that the continuous operating period does not exceed 96 hours and the accumulative total period within a year does not exceed 15 days. Otherwise, it is considered as long-term operation.
- If the PMs use a 110 V AC power source, follow these suggestions:
 - Use double live wires. In this case, the output power of a PM is 3000 W.
 - If only one live wire is used, the output power of a PM is 1500 W. In this case, it is recommended that you use more PMs or change the PM backup mode to increase the output power.
- Use the [Hardware Configuration Tool](#) to calculate the weight of the chassis with other card configurations and the power consumption of the chassis with different configurations.

4.3 CloudEngine 16808

Overview

Table 4-9 Basic information about the CloudEngine 16808

Description	Part Number	Model	First supported version
CE16808 AC&high-voltage DC assembly chassis	02352RCV	CE16808-AH	V200: V200R005C20 V300: V300R020C00

Description	Part Number	Model	First supported version
CE16808 DC assembly chassis	02352RCS	CE16808-DC	V200: V200R005C20 V300: V300R020C00

 CAUTION

Some cards are attached with labels of different colors near the ejector levers to identify their types. The color of labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the card will be damaged.

Appearance

Figure 4-7 Appearance of the CloudEngine 16808 (front view)



Figure 4-8 Appearance of the CloudEngine 16808 (rear view)



NOTE

- The DC chassis and AC & high-voltage DC chassis have similar appearances and structures except that they use DC and AC & high-voltage DC power modules (PMs) and power entry modules (PEMs) respectively. The figures show the appearance of an AC & high-voltage DC chassis as an example.
- Figures in the document are for reference only, and the actual appearance of the devices may vary depending on the exact device model.

Components

In the CloudEngine 16808 chassis, MPUs and LPUs are installed horizontally in the front of the chassis, and SFUs are installed vertically at the rear of the chassis, as shown in [Figure 4-9](#).

Figure 4-9 Components of the CloudEngine 16808

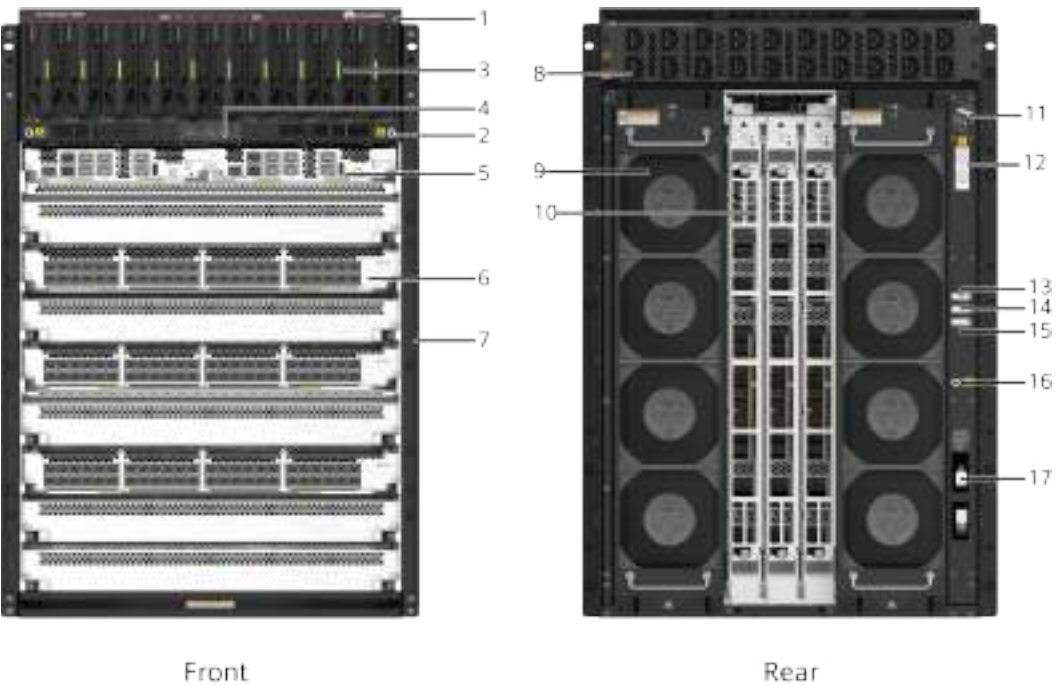


Table 4-10 Components

1. Chassis header	2. Front ESD jack	3. PM	4. Power switch	5. MPU
6. LPU	7. Cable management frame	8. PEM	9. Fan module	10. SFU SFUs are located behind fan modules and are invisible after all fan modules are installed.
11. Location of fan modules and SFUs	12. Two ground points for a two-hole OT terminal There is a yellow grounding label attached.	13. ESN label	14. Assembly chassis label	15. MAC address label
16. Rear ESD jack	17. SFU wrench	-	-	-

 NOTE

- The chassis header shows the following:
- Product series name: CloudEngine 16800
 - Huawei Logo
 - Indicators: For details, see the indicator description table.
- The indicator plate on the chassis header includes indicators and a manufacturing information loading interface. The manufacturing information loading interface is disabled before delivery and cannot be used.

Slot Layout

Figure 4-10 Slot Layout of the CloudEngine 16808

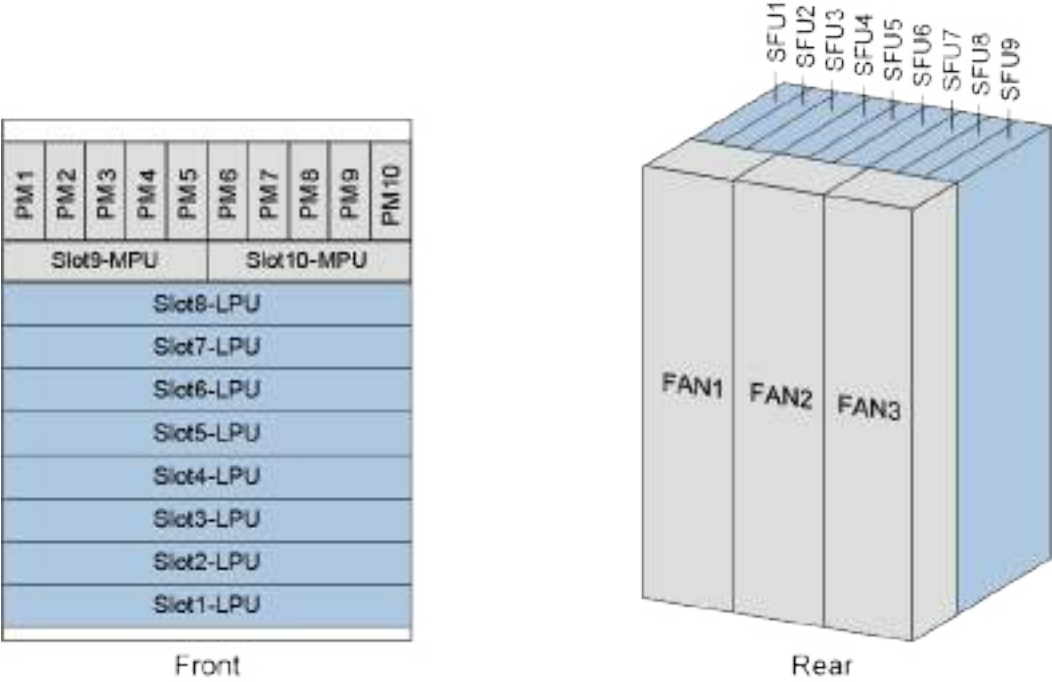


Table 4-11 Slots on the CloudEngine 16808

Slot Type	Slot ID	Slot Direction	Remarks
LPU slot	Slots 1 to 8	The top of the board faces upwards	Different types of LPUs can be installed in the same chassis.

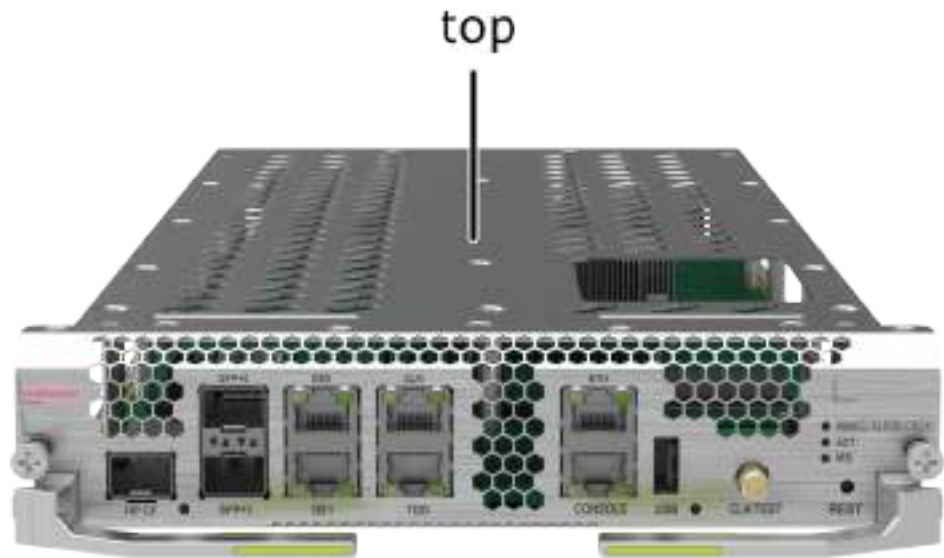
Slot Type	Slot ID	Slot Direction	Remarks
MPU slot	Slots 9 and 10	The top of the board faces upwards	Two MPUs work in active/standby mode for 1:1 hot standby. By default, the MPU with a smaller slot ID is the active MPU, and the MPU with a larger slot ID is the standby MPU. The active and standby MPUs in the same chassis must be of the same model.
SFU slot	1 to 9 Note: The slots in which SFUs can be installed vary according to the SFU model. For details, see the "IDs of Available Slots" column in the "Version Mapping" part of each SFU.	The right side of the board faces upwards	SFUs work in load balancing and redundancy mode. SFUs in the same chassis must be the same model. That is, SFUs of different models cannot be installed in the same chassis.
PM slot	Slots 1 to 10	The indicator light end upwards	PMs are plug-and-play.
Fan module slot	Slots 1 to 3	The indicator light end upwards	At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.

 NOTE

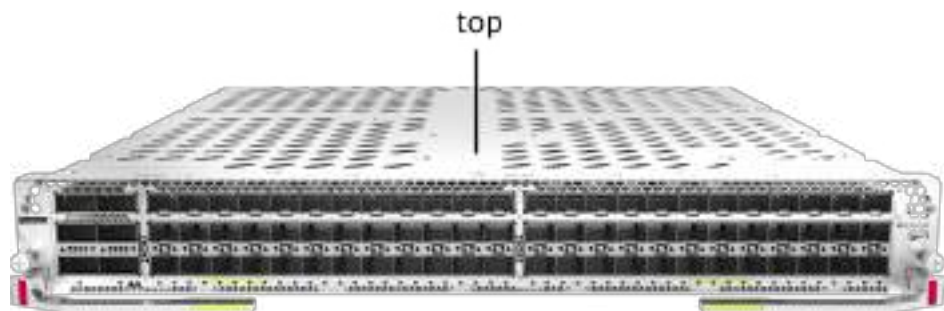
- SFU slots 1, 2, and 9 are unavailable. In V200R005C20 to V200R021C00, after SFUs are installed in slots 1, 2, and 9, the SFUs can be registered but are unavailable. In V200R021C10 and later versions, after SFUs are installed in slots 1, 2, and 9, an alarm will be reported to the NMS and then the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)
- SFU slots 4 and 7:
 - By default, SFU slots 4 and 7 are available.
 - After the **set sfu reduced-mode** command is run to enable the SFU reduction mode, SFUs in slots 4 and 7 are unavailable. In V200R005C20, after SFUs are installed in slots 4 and 7, the SFUs can be registered but are unavailable. In V200R019C10 and later versions, after SFUs are installed in slots 4 and 7, the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)

 NOTE

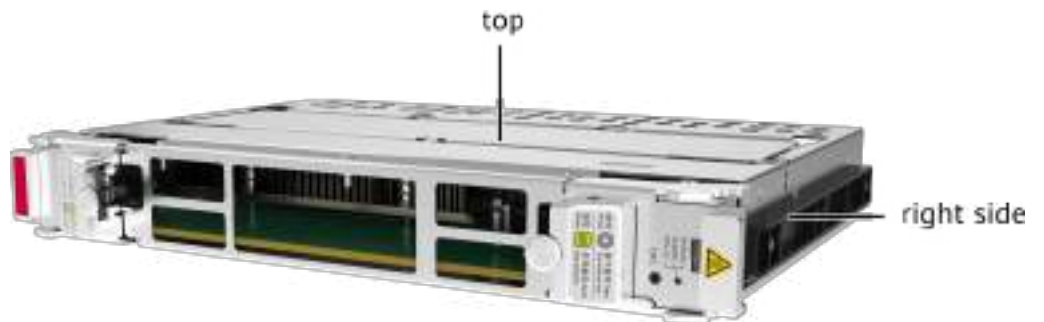
- Top of the MPU:



- Top of the LPU:

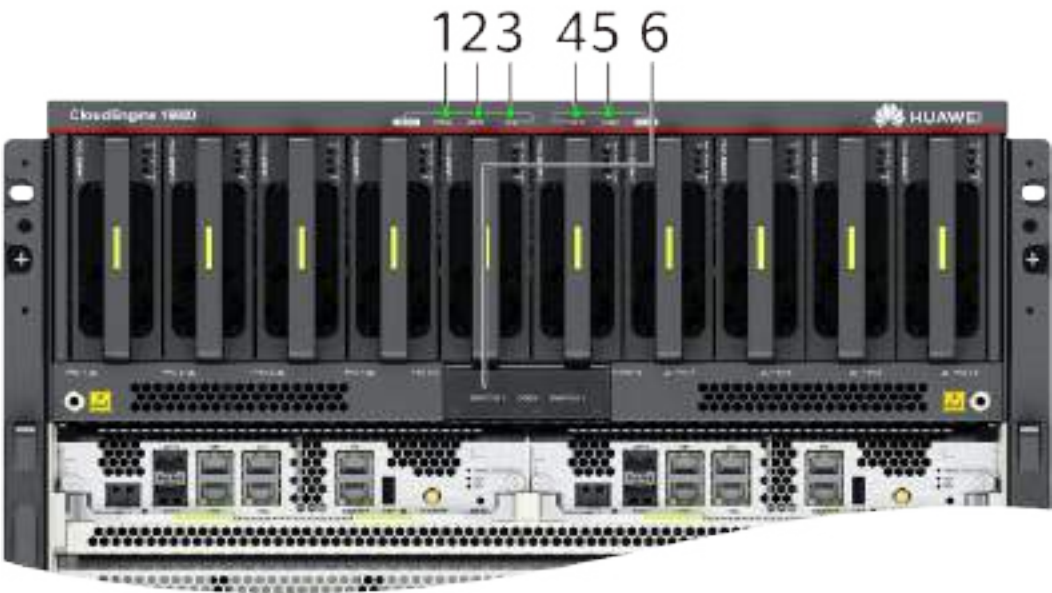


- Top and right side of the SFU:



Panel

Figure 4-11 Panel on the CloudEngine 16808



1. PM indicator	2. MPU indicator	3. LPU indicator	4. SFU indicator	5. Fan module indicator
6. Power switch	-	-	-	-

Table 4-12 Indicators on the CloudEngine 16808

Silkscreen	Name	Color	Status	Description
PWR	PM indicator	Green	Steady on	All PMs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more PMs are not working properly. For example, a PM is present but does not have output power. - One or more PMs are powered off.
MPU	MPU indicator	Green	Steady on	All installed MPUs are working properly.
		Red	Steady on	At least one MPU has failed.
LPU	LPU indicator	Green	Steady on	All installed LPUs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One LPU is not working properly and a hardware alarm has been generated. - No LPU is properly installed.
SFU	SFU indicator	Green	Steady on	All installed SFUs are working properly.
		Red	Steady on	The possible causes are as follows: - One SFU is not working properly and a hardware alarm has been generated. - No SFU is properly installed.
FAN	Fan module indicator	Green	Steady on	All fan modules are installed and working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more fan modules are not working properly. - One or more fan modules are not properly installed or have been removed.

Table 4-13 Buttons on the CloudEngine 16808

Silkscreen	Name	Description
SWITCH 1/SWITCH 2	Power switch	Controls device power-on and power-off. NOTE The device is powered off only when switch1 and switch2 are both OFF. The device is powered on as long as either switch is ON.

Technical Specifications

Table 4-14 Technical specifications of the CE16808-AH

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	990 mm x 770 mm x 1150 mm (38.98 in. x 30.31 in. x 45.28 in.)

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	703 mm x 442 mm x 874 mm (27.68 in. x 17.40 in. x 34.41 in.), dimensions of the chassis body 703 mm x 483 mm x 985 mm (27.68 in. x 19.02 in. x 38.78 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	129.8 kg (286.21 lb) (including two MPUs and filler panels)
Weight without packaging (full configuration) [kg (lb)]	- V200: 281.9 kg/621.48 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 377 kg/831.13 lb (fully configured with CEL36DQHG-X cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	177 kg (390.21 lb)
Cabinet installation standards	- A812 - A612 - A610
Chassis height [U]	15.8 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	AC/HVDC pluggable

Item	Specification
Typical power consumption [W]	- V200: 5570 W (fully configured with CEL36CQFD-G cards) - V300: 12576 W (fully configured with CEL36DQHG-X cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Typical heat dissipation [BTU/hour]	- V200: 19005 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 42909 BTU/hour (fully configured with CEL36DQHG-X cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 8956 W (fully configured with CEL36CQFD-G cards) - V300: 23690 W (fully configured with CEL36DQHG-X cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 30557 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 80830 BTU/hour (fully configured with CEL36DQHG-X cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	10
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.

Item	Specification
Rated input voltage [V]	- AC: 220 V; 50 Hz/60 Hz - High-voltage DC (HVDC): 240 V/380 V
Input voltage range [V]	- AC: 176–290 V; 45–65 Hz - High-voltage DC (HVDC): 188 V to 288 V or 260 V to 400 V
Maximum input current [A]	- AC: 16 A @ 200 V; 18.5 A @ 176 V - High-voltage DC (HVDC): 18 A @ 188 V; 13 A @ 260 V
Maximum output power [W]	- In 9+1 backup mode: 3000 W x 9 = 27000 W - In 10+0 backup mode: 3000 W x 10 = 30000 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999971582
MTBF [year]	34.55 years
Number of LPU slots	8
Number of MPU slots	2
Number of SFU slots	9
Number of slots	19
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).

Item	Specification
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE - The product has a valid storage period of one year. - The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

Table 4-15 Technical specifications of the CE16808-DC

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	990 mm x 770 mm x 1150 mm (38.98 in. x 30.31 in. x 45.28 in.)
Dimensions without packaging (H x W x D) [mm (in.)]	703 mm x 442 mm x 874 mm (27.68 in. x 17.40 in. x 34.41 in.), dimensions of the chassis body 703 mm x 483 mm x 985 mm (27.68 in. x 19.02 in. x 38.78 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	135.8 kg (299.44 lb) (including two MPUs and filler panels)

Item	Specification
Weight without packaging (full configuration) [kg (lb)]	- V200: 283.9 kg/625.89 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 283.9 kg/625.89 lb (fully configured with CEL36CQFD-E cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	183 kg (403.44 lb)
Cabinet installation standards	- A812 - A612 - A610
Chassis height [U]	15.8 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	DC pluggable
Typical power consumption [W]	- V200: 5570 W (fully configured with CEL36CQFD-G cards) - V300: 5610 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.

Item	Specification
Typical heat dissipation [BTU/hour]	- V200: 19005 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 19141 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 8956 W (fully configured with CEL36CQFD-G cards) - V300: 9058 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 30557 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 30906 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	10
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.
Rated input voltage [V]	-48 V DC/-60 V DC
Input voltage range [V]	-40 V DC to -72 V DC
Maximum input current [A]	58.5 A for the voltage of -40 V

Item	Specification
Maximum output power [W]	- In 9+1 backup mode: 2200 W x 9 = 19800 W - In 10+0 backup mode: 2200 W x 10 = 22000 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999971582
MTBF [year]	34.55 years
Number of LPU slots	8
Number of MPU slots	2
Number of SFU slots	9
Number of slots	19
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE - The product has a valid storage period of one year. - The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

 NOTE

- Temperature and humidity are measured at 1.5 m (4.92 ft.) above the floor and 0.4 m (1.31 ft.) in front of the rack. There should be no protection board in the front or rear of the rack.
- The short-term operating conditions mean that the continuous operating period does not exceed 96 hours and the accumulative total period within a year does not exceed 15 days. Otherwise, it is considered as long-term operation.
- If the PMs use a 110 V AC power source, follow these suggestions:
 - Use double live wires. In this case, the output power of a PM is 3000 W.
 - If only one live wire is used, the output power of a PM is 1500 W. In this case, it is recommended that you use more PMs or change the PM backup mode to increase the output power.
- Use the [Hardware Configuration Tool](#) to calculate the weight of the chassis with other card configurations and the power consumption of the chassis with different configurations.

4.4 CloudEngine 16816

Overview

Table 4-16 Basic information about the CloudEngine 16816

Description	Part Number	Model	First supported version
CE16816 AC&high-voltage DC assembly chassis	02352RCW	CE16816-AH	V200: V200R005C20 V300: V300R020C00

Description	Part Number	Model	First supported version
CE16816 DC assembly chassis	02353BXE	CE16816-DC	V200: V200R005C20 V300: V300R020C00

 CAUTION

Some cards are attached with labels of different colors near the ejector levers to identify their types. The color of labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the card will be damaged.

Appearance

Figure 4-12 Appearance of the CloudEngine 16816 (front view)

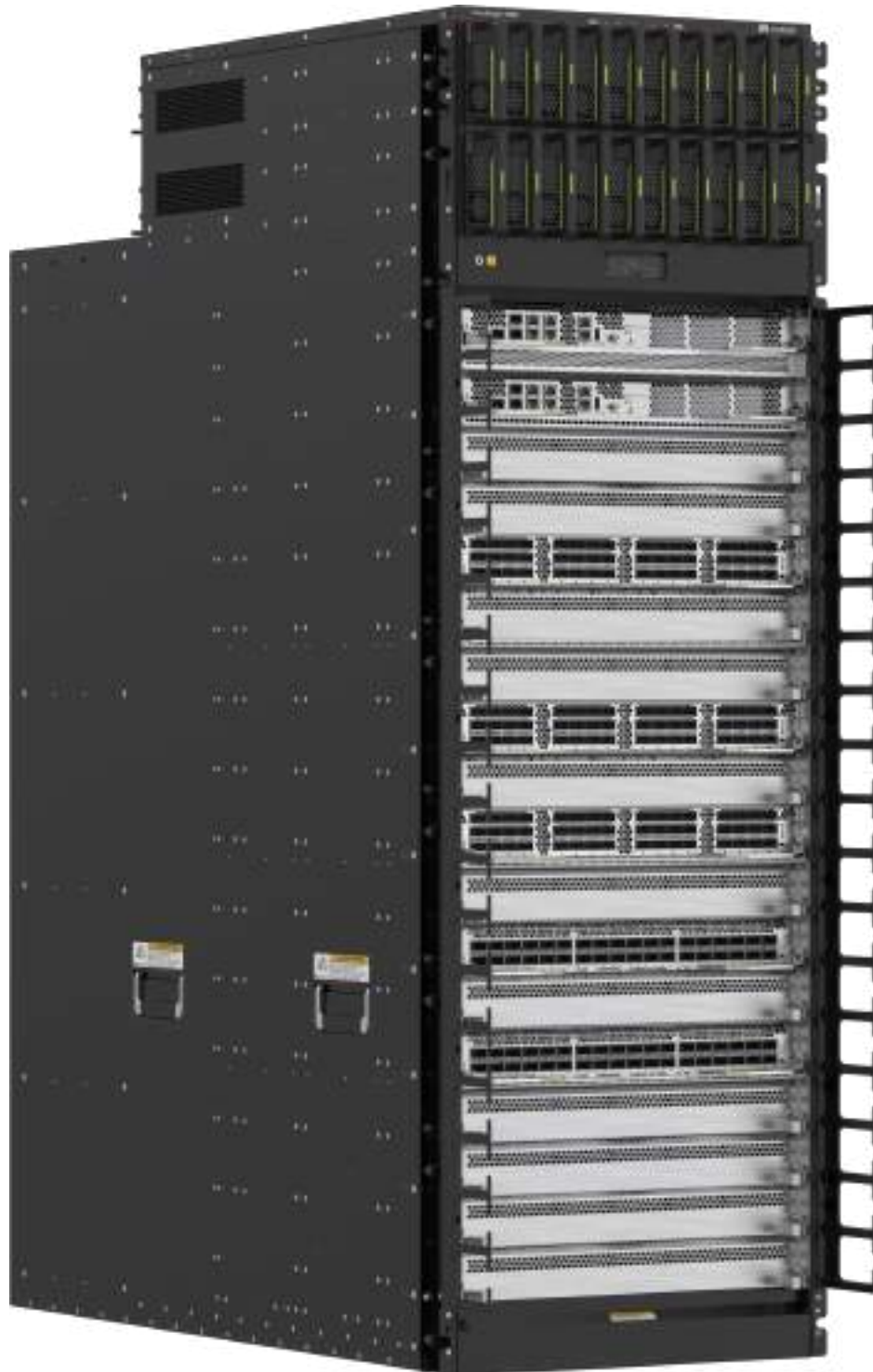


Figure 4-13 Appearance of the CloudEngine 16816 (rear view)



NOTE

- The DC chassis and AC & high-voltage DC chassis have similar appearances and structures except that they use DC and AC & high-voltage DC power modules (PMs) and power entry modules (PEMs) respectively. The figures show the appearance of an AC & high-voltage DC chassis as an example.
- Figures in the document are for reference only, and the actual appearance of the devices may vary depending on the exact device model.

Components

In the CloudEngine 16816 chassis, MPUs and LPUs are installed horizontally in the front of the chassis, and SFUs are installed vertically at the rear of the chassis, as shown in [Figure 4-14](#).

Figure 4-14 Components of the CloudEngine 16816

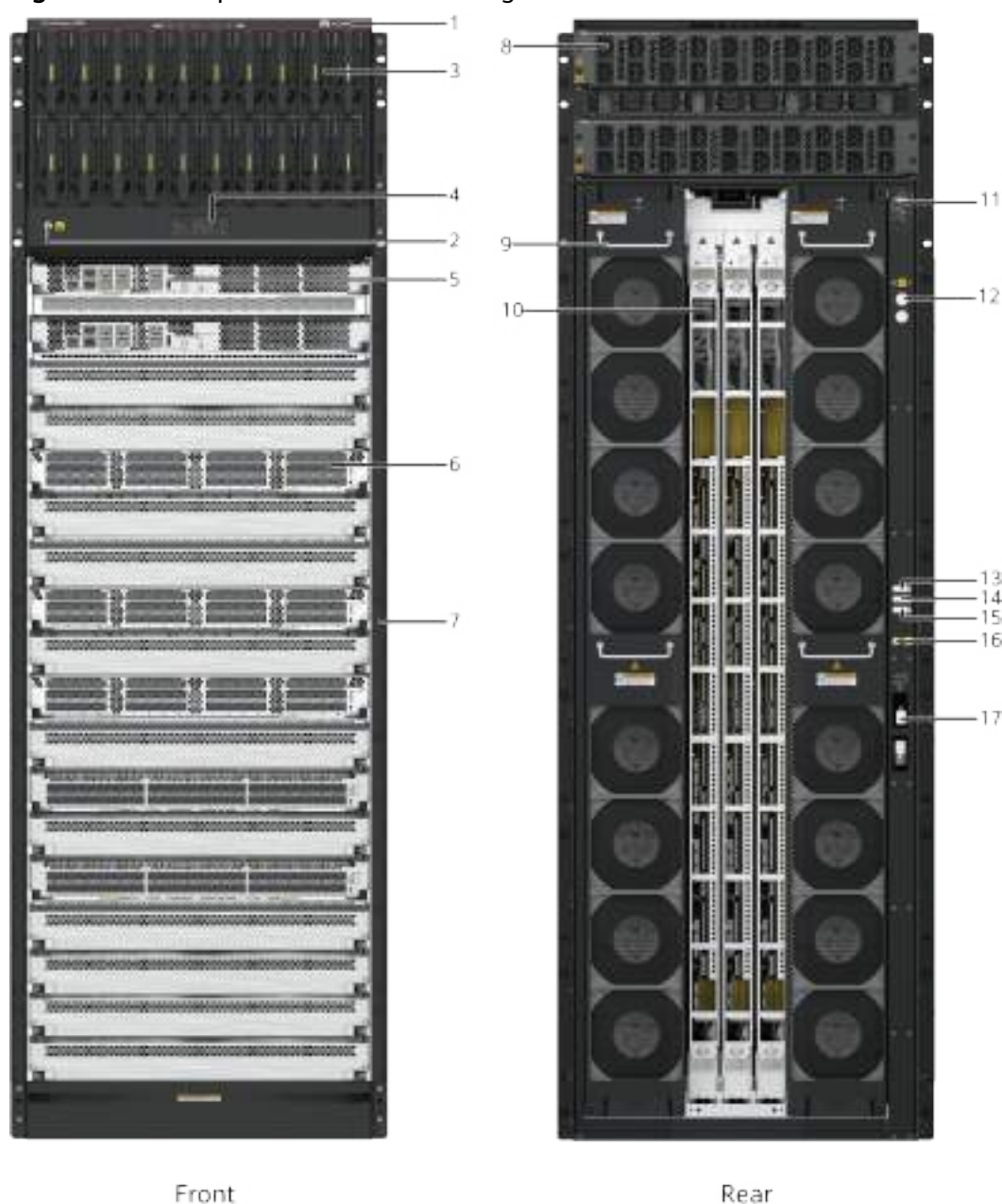


Table 4-17 Components

1. Chassis header	2. Front ESD jack	3. PM	4. Power switch	5. MPU
6. LPU	7. Cable management frame	8. PEM	9. Fan module	10. SFU SFUs are located behind fan modules and are invisible after all fan modules are installed.
11. Location of fan modules and SFUs	12. Two ground points for a two-hole OT terminal There is a yellow grounding label attached.	13. ESN label	14. Assembly chassis label	15. MAC address label
16. Rear ESD jack	17. SFU wrench	-	-	-

 NOTE

- The chassis header shows the following:
- Product series name: CloudEngine 16800
 - Huawei Logo
 - Indicators: For details, see the indicator description table.
- The indicator plate on the chassis header includes indicators and a manufacturing information loading interface. The manufacturing information loading interface is disabled before delivery and cannot be used.

Slot Layout

Figure 4-15 Slot Layout of the CloudEngine 16816

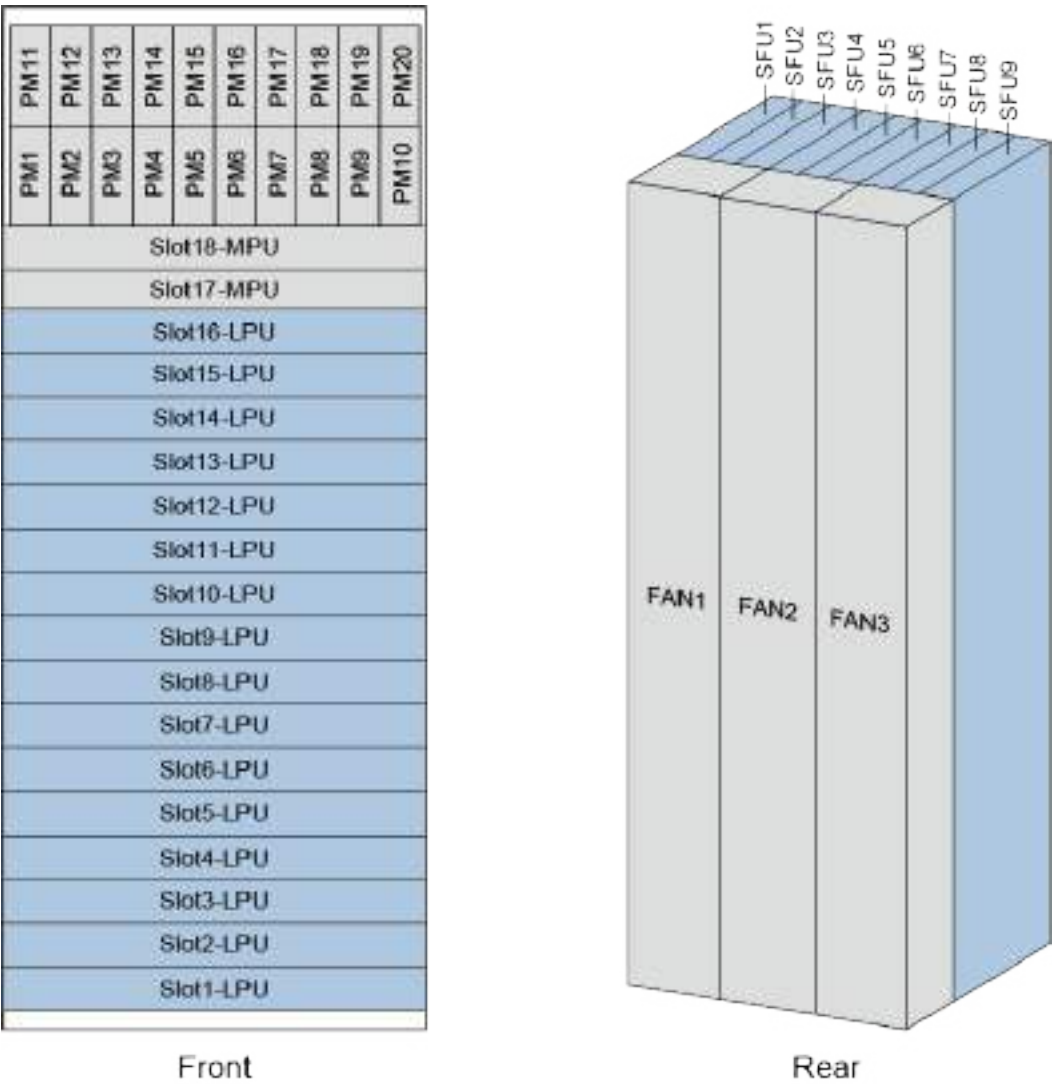


Table 4-18 Slots on the CloudEngine 16816

Slot Type	Slot ID	Slot Direction	Remarks
LPU slot	Slots 1 to 16	The top of the board faces upwards	Different types of LPUs can be installed in the same chassis.

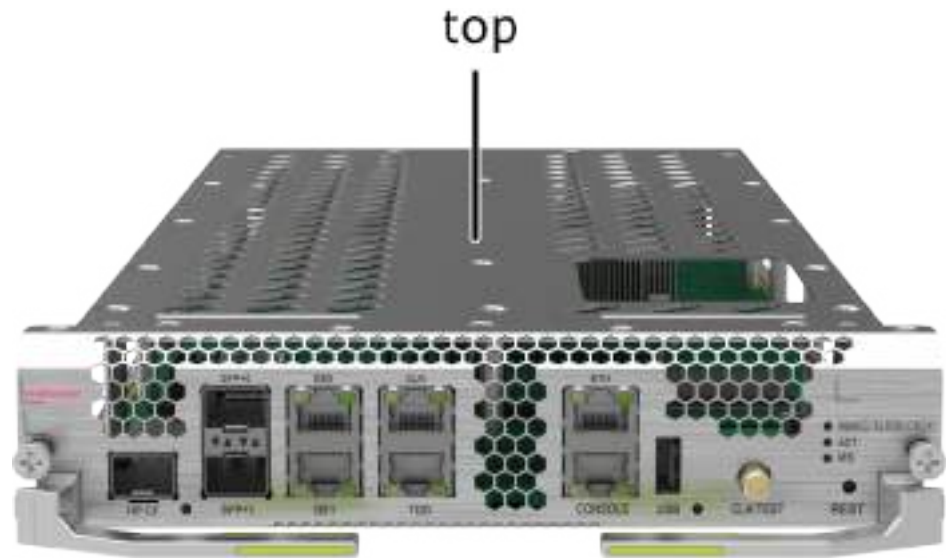
Slot Type	Slot ID	Slot Direction	Remarks
MPU slot	Slots 17 and 18	The top of the board faces upwards	Two MPUs work in active/standby mode for 1:1 hot standby. By default, the MPU with a smaller slot ID is the active MPU, and the MPU with a larger slot ID is the standby MPU. The active and standby MPUs in the same chassis must be of the same model.
SFU slot	1 to 9 Note: The slots in which SFUs can be installed vary according to the SFU model. For details, see the "IDs of Available Slots" column in the "Version Mapping" part of each SFU.	The right side of the board faces upwards	SFUs work in load balancing and redundancy mode. Only the CE-SFU16F-G and CE-SFU16F-G1 SFUs can be used together. For other SFUs, SFUs of different models cannot be installed in the same chassis.
PM slot	Slots 1 to 20	The indicator light end upwards	PMs are plug-and-play.
Fan module slot	Slots 1 to 3	The indicator light end upwards	At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.

 NOTE

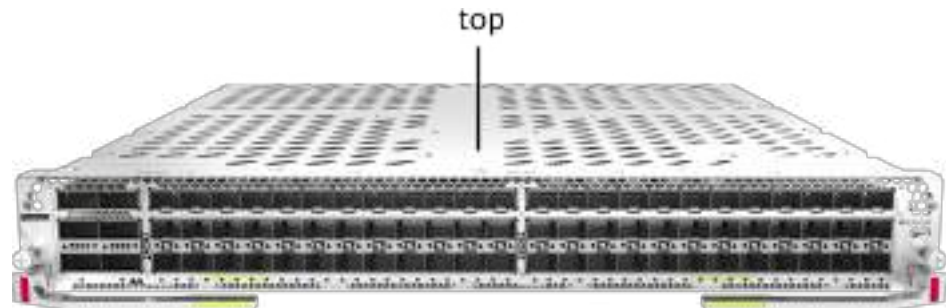
- SFU slots 1, 2, and 9 are unavailable. In V200R005C20 to V200R021C00, after SFUs are installed in slots 1, 2, and 9, the SFUs can be registered but are unavailable. In V200R021C10 and later versions, after SFUs are installed in slots 1, 2, and 9, an alarm will be reported to the NMS and then the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)
- SFU slots 4 and 7:
 - By default, SFU slots 4 and 7 are available.
 - After the **set sfu reduced-mode** command is run to enable the SFU reduction mode, SFUs in slots 4 and 7 are unavailable. In V200R005C20, after SFUs are installed in slots 4 and 7, the SFUs can be registered but are unavailable. In V200R019C10 and later versions, after SFUs are installed in slots 4 and 7, the SFUs will be powered off in the process from startup to the SFU registration phase. (For the specific duration, refer to the specifications of the card startup time.)

 NOTE

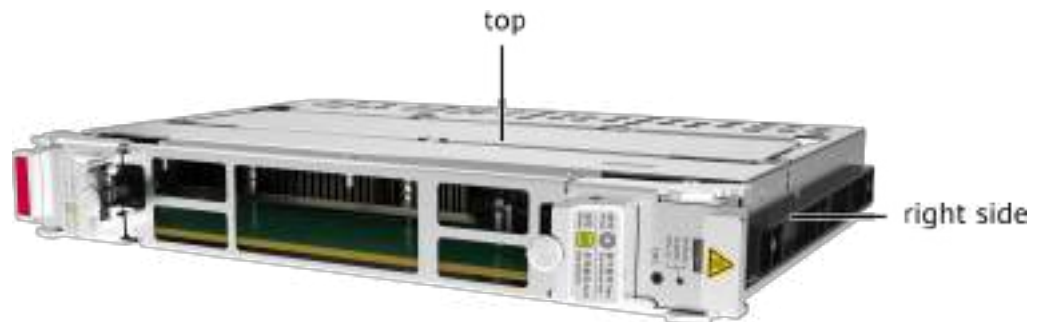
- Top of the MPU:



- Top of the LPU:



- Top and right side of the SFU:



Panel

Figure 4-16 Panel on the CloudEngine 16816



1. PM indicator	2. MPU indicator	3. LPU indicator	4. SFU indicator	5. Fan module indicator
6. Power switch	-	-	-	-

Table 4-19 Indicators on the CloudEngine 16816

Silkscreen	Name	Color	Status	Description
PWR	PM indicator	Green	Steady on	All PMs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more PMs are not working properly. For example, a PM is present but does not have output power. - One or more PMs are powered off.
MPU	MPU indicator	Green	Steady on	All installed MPUs are working properly.
		Red	Steady on	At least one MPU has failed.
LPU	LPU indicator	Green	Steady on	All installed LPUs are working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One LPU is not working properly and a hardware alarm has been generated. - No LPU is properly installed.
SFU	SFU indicator	Green	Steady on	All installed SFUs are working properly.
		Red	Steady on	The possible causes are as follows: - One SFU is not working properly and a hardware alarm has been generated. - No SFU is properly installed.
FAN	Fan module indicator	Green	Steady on	All fan modules are installed and working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The possible causes are as follows: - One or more fan modules are not working properly. - One or more fan modules are not properly installed or have been removed.

Table 4-20 Buttons on the CloudEngine 16816

Silkscreen	Name	Description
SWITCH 1/SWITCH 2	Power switch	Controls device power-on and power-off. NOTE The device is powered off only when switch1 and switch2 are both OFF. The device is powered on as long as either switch is ON.

Technical Specifications

Table 4-21 Technical specifications of the CE16816-AH

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	1695 mm x 770 mm x 1350 mm (66.73 in. x 30.31 in. x 53.15 in.)

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	1436 mm x 442 mm x 1033 mm (56.54 in. x 17.40 in. x 40.67 in.), dimensions of the chassis body 1436 mm x 483 mm x 1144 mm (56.54 in. x 19.02 in. x 45.04 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	262.8 kg (579.47 lb) (including two MPUs and filler panels)
Weight without packaging (full configuration) [kg (lb)]	- V200: 627.2 kg/1382.73 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 627.2 kg/1382.73 lb (fully configured with CEL36CQFD-E cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	335 kg (738.54 lb)
Cabinet installation standards	- A812 - A612
Chassis height [U]	32.3 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	AC/HVDC pluggable

Item	Specification
Typical power consumption [W]	- V200: 11936 W (fully configured with CEL36CQFD-G cards) - V300: 11976 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Typical heat dissipation [BTU/hour]	- V200: 40726 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 40862 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 18548 W (fully configured with CEL36CQFD-G cards) - V300: 18650 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 63286 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 63634 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	20
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.

Item	Specification
Rated input voltage [V]	- AC: 220 V; 50 Hz/60 Hz - High-voltage DC (HVDC): 240 V/380 V
Input voltage range [V]	- AC: 176–290 V; 45–65 Hz - High-voltage DC (HVDC): 188 V to 288 V or 260 V to 400 V
Maximum input current [A]	- AC: 16 A @ 200 V; 18.5 A @ 176 V - High-voltage DC (HVDC): 18 A @ 188 V; 13 A @ 260 V
Maximum output power [W]	- In 19+1 backup mode: 3000 W x 19 = 57000 W - In 20+0 backup mode: 3000 W x 20 = 60000 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999970745
MTBF [year]	32.68 years
Number of LPU slots	16
Number of MPU slots	2
Number of SFU slots	9
Number of slots	27
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).

Item	Specification
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE - The product has a valid storage period of one year. - The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

Table 4-22 Technical specifications of the CE16816-DC

Item	Specification
Dimensions with packaging (H x W x D) [mm (in.)]	1695 mm x 770 mm x 1350 mm (66.73 in. x 30.31 in. x 53.15 in.)
Dimensions without packaging (H x W x D) [mm (in.)]	1436 mm x 442 mm x 1033 mm (56.54 in. x 17.40 in. x 40.67 in.), dimensions of the chassis body 1436 mm x 483 mm x 1144 mm (56.54 in. x 19.02 in. x 45.04 in.), including mounting brackets and cable management frames
Weight without packaging [kg (lb)]	273.8 kg (603.73 lb) (including two MPUs and filler panels)

Item	Specification
Weight without packaging (full configuration) [kg (lb)]	- V200: 630.2 kg/1389.34 lb (fully configured with CEL36CQFD-G cards, including optical modules) - V300: 630.2 kg/1389.34 lb (fully configured with CEL36CQFD-E cards, including optical modules) NOTE You can use Info-Finder's Hardware Configuration tool to determine the weight of a device according to its card configuration.
Weight with packaging [kg (lb)]	346 kg (762.79 lb)
Cabinet installation standards	- A812 - A612
Chassis height [U]	32.3 U
Switching capacity	For more details, see the datasheet/brochure or contact product sales personnel. You can obtain the datasheet/brochure from the following link: https://e.huawei.com/en/products/switches/data-center-switches
Redundant MPUs	1:1
Redundant Switch fabrics	N+M
Power supply mode	DC pluggable
Typical power consumption [W]	- V200: 11936 W (fully configured with CEL36CQFD-G cards) - V300: 11976 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.

Item	Specification
Typical heat dissipation [BTU/hour]	- V200: 40726 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 40862 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Maximum power consumption [W]	- V200: 18548 W (fully configured with CEL36CQFD-G cards) - V300: 18650 W (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the power consumption of a device according to its card configuration.
Maximum heat dissipation [BTU/hour]	- V200: 63286 BTU/hour (fully configured with CEL36CQFD-G cards) - V300: 63634 BTU/hour (fully configured with CEL36CQFD-E cards) NOTE You can use Info-Finder's Hardware Configuration tool to determine the heat dissipation of a device according to its card configuration.
Number of power modules	20
Redundant power supply	- Dual-input power supply system: N +1 backup is recommended. - Single-input power supply system: N+1 backup Dual-input power supply is recommended to ensure reliability.
Rated input voltage [V]	-48 V DC/-60 V DC
Input voltage range [V]	-40 V DC to -72 V DC
Maximum input current [A]	58.5 A for the voltage of -40 V

Item	Specification
Maximum output power [W]	- In 19+1 backup mode: 2200 W x 19 = 41800 W - In 20+0 backup mode: 2200 W x 20 = 44000 W
Certification	- Compliance with safety standards - Compliance with EMC standards - Compliance with environment and environmental protection standards
Redundant fans	Fan modules work in hot standby mode. At the normal temperature range, the system can operate properly for a short time after a single fan module fails. You are advised to replace the faulty fan module immediately.
Heat dissipation mode	Heat dissipation using fans
Airflow direction	Port-side air exhaust
Availability	0.9999970745
MTBF [year]	32.68 years
Number of LPU slots	16
Number of MPU slots	2
Number of SFU slots	9
Number of slots	27
Long-term operating altitude [m (ft.)]	≤ 5000 m (16404 ft.) When the altitude is between 1800 m and 5000 m (5906 ft. and 16404 ft.), the highest operating temperature reduces by 1°C (1.8°F) every time the altitude increases by 220 m (722 ft.).
Long-term operating relative humidity [RH]	5% RH to 85% RH, non-condensing
Long-term operating temperature [°C (°F)]	0°C to 40°C (32°F to 104°F)
Short-term operating relative humidity [RH]	5% RH to 95% RH, non-condensing
Storage altitude [m (ft.)]	≤ 5000 m (16404 ft.)
Storage relative humidity [RH]	5% RH to 95% RH, non-condensing

Item	Specification
Storage temperature [°C (°F)]	-40°C to +70°C (-40°F to +158°F)
Storage environment	ETSI EN 300 019-1-1 Class 1.2 NOTE • The product has a valid storage period of one year. • The valid storage period refers to the period during which the product maintains the required quality when stored with packing materials in an environment that meets the preceding requirements.

 NOTE

- Temperature and humidity are measured at 1.5 m (4.92 ft.) above the floor and 0.4 m (1.31 ft.) in front of the rack. There should be no protection board in the front or rear of the rack.
- The short-term operating conditions mean that the continuous operating period does not exceed 96 hours and the accumulative total period within a year does not exceed 15 days. Otherwise, it is considered as long-term operation.
- If the PMs use a 110 V AC power source, follow these suggestions:
 - Use double live wires. In this case, the output power of a PM is 3000 W.
 - If only one live wire is used, the output power of a PM is 1500 W. In this case, it is recommended that you use more PMs or change the PM backup mode to increase the output power.
- Use the [Hardware Configuration Tool](#) to calculate the weight of the chassis with other card configurations and the power consumption of the chassis with different configurations.

5 Power Supply System

- 5.1 CloudEngine 16804 Power Supply System
- 5.2 CloudEngine 16804 AC&High-Voltage DC PEM
- 5.3 CloudEngine 16804 DC PEM
- 5.4 CloudEngine 16808 Power Supply System
- 5.5 CloudEngine 16808 AC&High-Voltage DC PEM
- 5.6 CloudEngine 16808 DC PEM
- 5.7 CloudEngine 16816 Power Supply System
- 5.8 CloudEngine 16816 AC&High-Voltage DC PEM
- 5.9 CloudEngine 16816 DC PEM
- 5.10 Power Switch
- 5.11 PAH-3000WA (3000W Dual Inputs AC&HVDC Power Module)
- 5.12 PDC-2200WC (PM2200W DC Power Supply)
- 5.13 CR8MP2K3PHC0
- 5.14 CR8MP2KHDCC0
- 5.15 PDU2000-32-11/2-B1-3M
- 5.16 PDU2000-32-3PH-1
- 5.17 PDU2000-32-HVDC-6/4-B1
- 5.18 APD32-6-24
- 5.19 CR5MDPD12500

5.1 CloudEngine 16804 Power Supply System

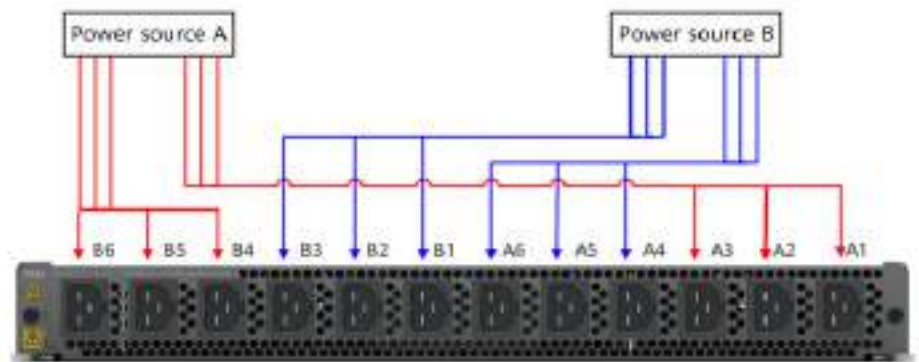
Overview

The CloudEngine 16804 supports DC, AC, and HVDC power supplies. The device connects to an external power distribution system through a power entry module

(PEM) and supports two power inputs: A and B. Each power module (PM) has two inputs and converts the power supplied to the PEM to power the entire device.

NOTE

- If there is only one power supply in the equipment room, connect all A power sockets (A1 to A6) to power source A or all B power sockets (B1 to B6) to power source B.
 - If there are two power supplies in the equipment room:
 - In AC or HVDC scenarios, power source A is used by default. If power source A fails, the system automatically switches to power source B. To prevent power supply fluctuation caused by dual power supplies, the system does not immediately switch back to power source A after the fault is rectified, doing so only after a delay of 4.5 hours. It is advised to connect one power source to all A power sockets (A1 to A6) and the other power source to all B power sockets (B1 to B6).
- If the two power sources need to work in load-balancing mode, connect power sockets as follows:
- Connect power source A to one half of A power sockets (A1 to A3) and one half of B power sockets (B4 to B6).
 - Similarly, connect power source B to the other half of A power sockets (A4 to A6) and the other half of B power sockets (B1 to B3).



- In DC scenarios, you are advised to connect one power source to A power sockets (A1 to A6) and the other power source to B power sockets (B1 to B6). If the two power sources have different voltages, the power source with a higher voltage provides a higher current.
- If AC and HVDC inputs are available to both power sources A and B, the 220 V AC input is used by default. If the 220 V AC input is faulty, the system automatically switches to the HVDC input.
- Requirements for the power supply system of a customer's cabinet: The output power of two power sources must be greater than the maximum input power of the device.

Components

As shown in [Figure 5-1](#), the power supply system of the CloudEngine 16804 is located at the top of the chassis. It consists of PMs and power switches at the front of the chassis, and a PEM at the rear of the chassis. [Table 5-1](#) describes the functions of each module.

Figure 5-1 Key components in the power supply system of the CloudEngine 16804

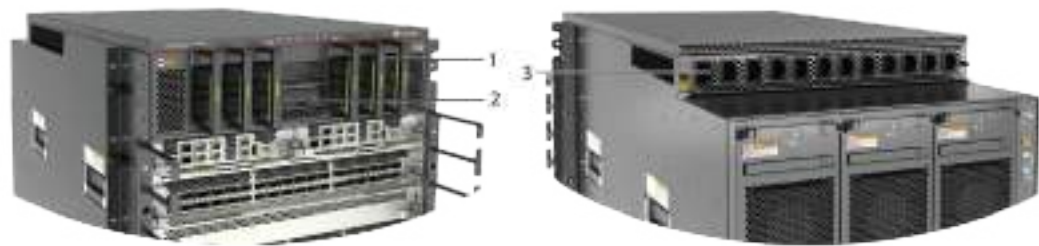


Table 5-1 Functions of key components in the power supply system of the CloudEngine 16804

No.	Component	Description
1	PM	Converts the power supplied to a PEM to power the entire device through the power backplane and busbar.
2	Power switch	Controls the power output of PMs. The two power switches back up each other. When one or both of them are in ON state, the power output of PMs is enabled. NOTICE To ensure reliability, both power switches must be turned on when the device is running.
3	PEM	Connects to an external power distribution system. The PEM can be divided into power supply areas A and B, with each one corresponding to a different power distribution system with a different source. This design implements 1+1 backup of external power distribution systems, ensuring the reliability of the entire power supply system by enabling the device to continue running when a power supply system in the equipment room fails.

PM and PEM Applications

The CloudEngine 16804 supports multiple power supply scenarios, which vary according to the installed PEM and PMs. Table 5-2 describes the power supply scenarios and PEM/PM applications.

Table 5-2 Power supply scenarios and PEM/PM applications on the CloudEngine 16804

Power Supply Scenario	PEM	PM
-48 V DC power supply	CloudEngine 16804 DC PEM	2200 W DC PM
AC power supply	CloudEngine 16804 AC & HVDC PEM	3000 W dual-input AC & HVDC PM

Power Supply Scenario	PEM	PM
240 V HVDC power supply	CloudEngine 16804 AC & HVDC PEM	3000 W dual-input AC & HVDC PM
380 V HVDC power supply	CloudEngine 16804 AC & HVDC PEM	3000 W dual-input AC & HVDC PM

PM and PEM Application Principles

The PEM is integrated into the chassis and does not need to be installed, configured, or removed onsite. PMs can be configured based on the overall power consumption of the device to flexibly supply power.

The DC PMs and AC & HVDC PMs of the CloudEngine 16804 support N+1 backup. N indicates the number of PMs configured based on the estimated power consumption of the chassis. The system's maximum power supply capability is equal to the total maximum output power of N PMs. In N+1 backup mode, N PMs provide power to the chassis, and one PM works as a backup to implement power redundancy.

The following uses a 2200 W DC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to the number of required PMs), the value of $N \times 2200 \text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-3](#).

Table 5-3 DC power distribution example of the CloudEngine 16804

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Backup for PMs
$P < 2200 \text{ W}$	1, 2	A1, B1, A2, B2	4	63 A	1+1
$2200 \text{ W} \leq P < 4400 \text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$4400 \text{ W} \leq P < 6600 \text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
$6600 \text{ W} \leq P < 8800 \text{ W}$	1, 2, 3, 4, 5	A1, B1, A2, B2, A3, B3, A4, B4, A5, B5	10		4+1
$8800 \text{ W} \leq P < 11000 \text{ W}$	1, 2, 3, 4, 5, 6	A1, B1, A2, B2, A3, B3, A4, B4, A5, B5, A6, B6	12		5+1

The following uses a 3000 W AC & HVDC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to

the number of required PMs), the value of $N \times 3000\text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-4](#).

Table 5-4 AC & HVDC PM distribution examples of the CloudEngine 16804

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Backup for PMs
$P < 3000\text{ W}$	1, 2	A1, B1, A2, B2	4	AC: 16 A/20 A HVDC: 16 A	1+1
$3000\text{ W} \leq P < 6000\text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$6000\text{ W} \leq P < 9000\text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
$9000\text{ W} \leq P < 12000\text{ W}$	1, 2, 3, 4, 5	A1, B1, A2, B2, A3, B3, A4, B4, A5, B5	10		4+1
$12000\text{ W} \leq P < 15000\text{ W}$	1, 2, 3, 4, 5, 6	A1, B1, A2, B2, A3, B3, A4, B4, A5, B5, A6, B6	12		5+1

 NOTE

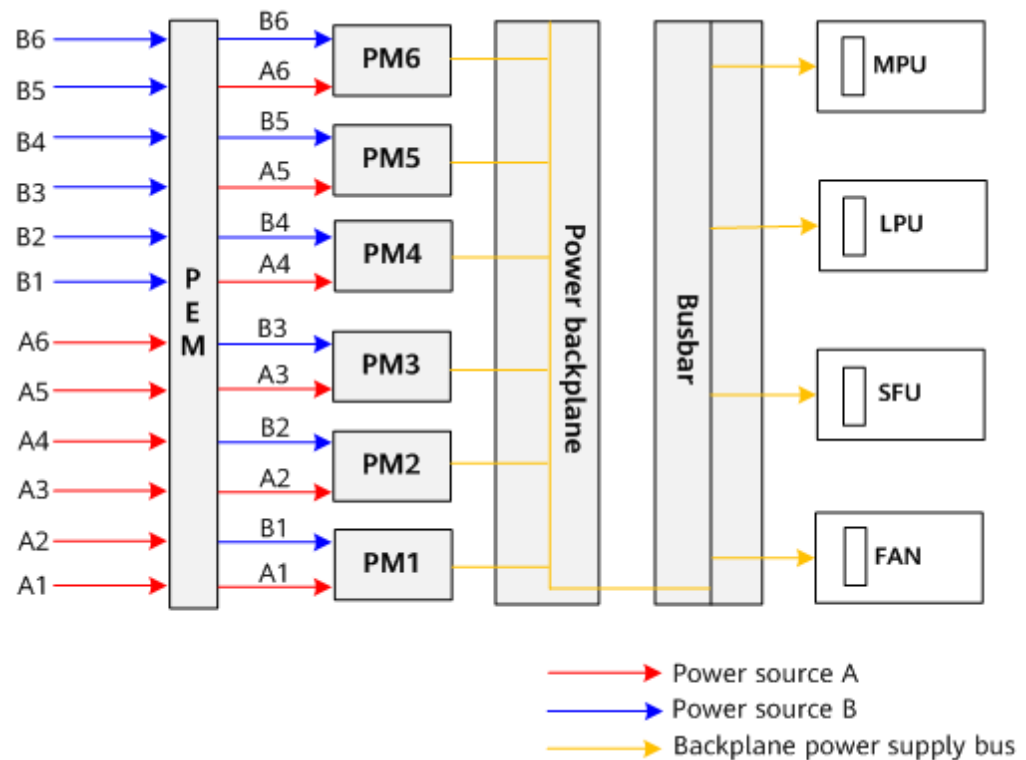
In each vacant PM slot, install a filler panel.

Power Supply Architecture

- DC power supply

The DC power supply system of the CloudEngine 16804 must be equipped with one DC PEM. The PEM has 12 inputs, corresponding to 12 power sockets. Power sources A and B each provide six outputs. Each DC PM has two inputs (from power sources A and B in the equipment room). The DC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other. [Figure 5-2](#) shows the DC power supply architecture of the CloudEngine 16804.

Figure 5-2 DC power supply architecture of the CloudEngine 16804



- AC & HVDC power supply

The AC & HVDC power supply system of the CloudEngine 16804 must be equipped with one AC & HVDC PEM. The PEM has 12 inputs, corresponding to 12 power sockets. Power sources A and B each provide six outputs. Each AC & HVDC PM has two inputs (from power sources A and B in the equipment room). The AC & HVDC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other.

The AC & HVDC power supply system of the CloudEngine 16804 can be directly connected to a power source or power distribution unit (PDU). Connecting the power supply system to a power source is similar to the DC power supply mode, and is not described here. Instead, the following describes the power supply architecture when the power supply system is connected to a PDU.

The CloudEngine 16800 series switches provide the following types of PDUs:

- AC PDU: Each AC PDU provides one input and six outputs.
- HVDC PDU: Each HVDC PDU provides two inputs and eight outputs. Each input corresponds to four outputs.

Power sources A and B can provide AC outputs, HVDC outputs, or both AC and HVDC outputs. [Figure 5-3](#), [Figure 5-4](#), and [Figure 5-5](#) show the power supply architectures of the three modes.

Figure 5-3 AC power supply architecture of the CloudEngine 16804

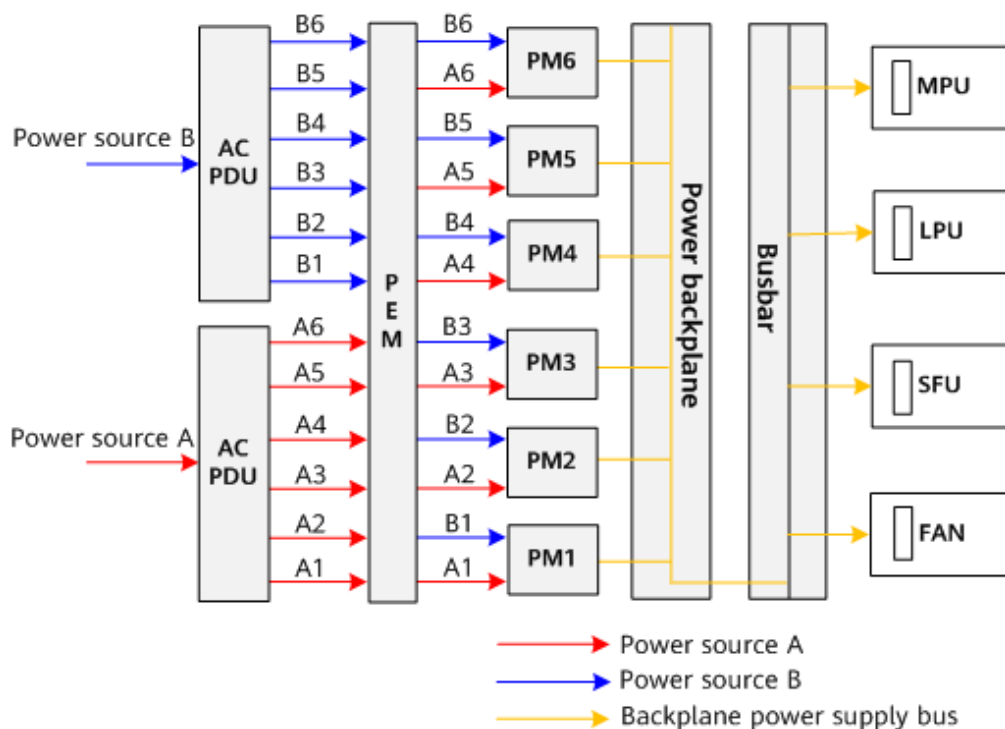


Figure 5-4 HVDC power supply architecture of the CloudEngine 16804

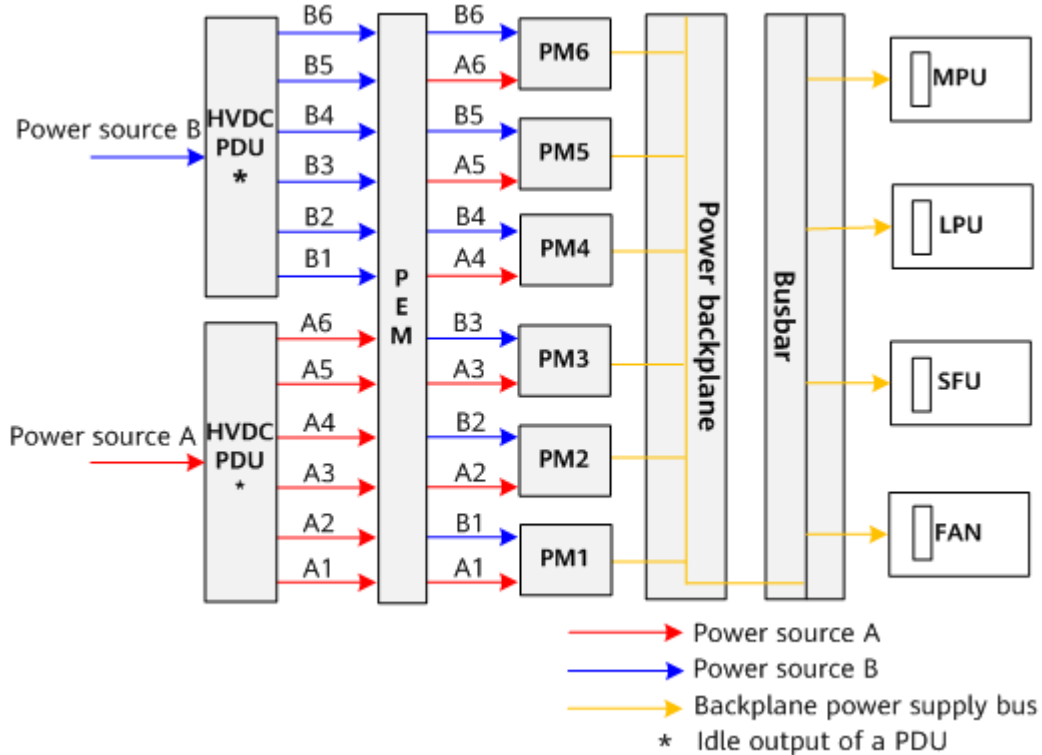
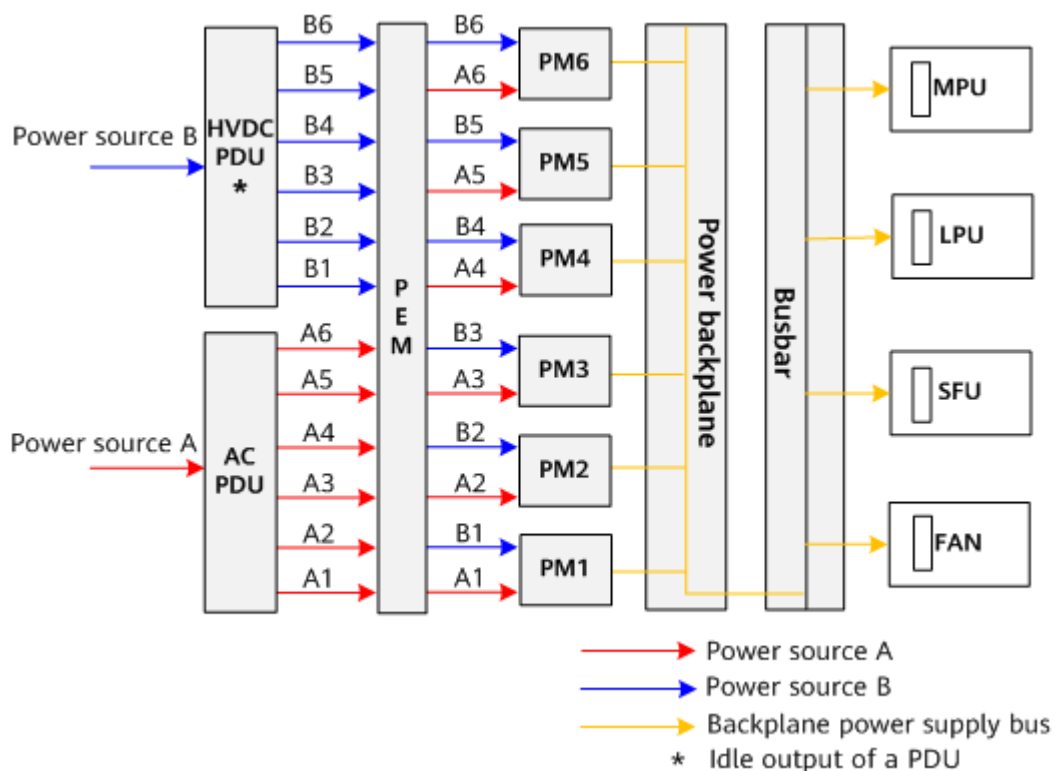


Figure 5-5 AC & HVDC power supply architecture of the CloudEngine 16804



5.2 CloudEngine 16804 AC&High-Voltage DC PEM

Overview

The CloudEngine 16804 AC&high-voltage DC PEM connects the CloudEngine 16804 to AC&high-voltage DC power sources. The PEM is pre-installed by default, and cannot be removed, installed, or replaced onsite.

CAUTION

The PEM is integrated into the chassis and does not need to be configured onsite. Do not remove or install the PEM onsite.

Appearance

Figure 5-6 Appearance of the CloudEngine 16804 AC&high-voltage DC PEM



Panel

Figure 5-7 Power sockets on the CloudEngine 16804 AC&high-voltage DC PEM

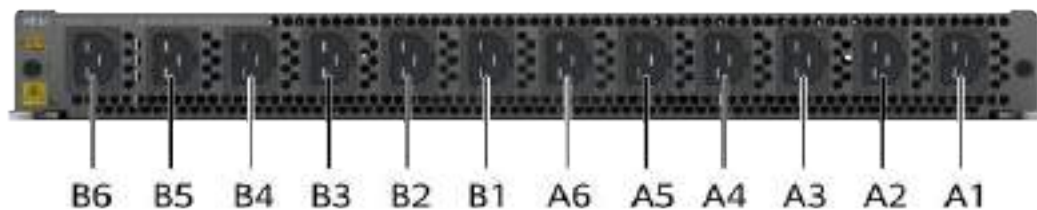


Table 5-5 describes the location mapping between CloudEngine 16804 PEM power sockets and power modules.

Table 5-5 Location mapping between PEM power sockets and power modules

PEM Power Socket	A1/B1	A2/B2	A3/B3	A4/B4	A5/B5	A6/B6
Power Module	1	2	3	4	5	6

Table 5-6 Power socket description

Name	Description	Power Cable Terminal
Power socket	200 V AC to 240 V AC/240 V DC/380 V DC power socket	Dedicated high-voltage DC terminal: Power Connector, 3Pin, 450V, 16A, Straight Female, 1.5mm ² or 14AWG

Input Requirements

- In an AC power distribution scenario, the 220 V AC input is supported.
- 240 V/380 V high-voltage DC input is supported in high-voltage DC power distribution scenarios.
- Protection components must be provided for power distribution frames (PDFs). When the AC and high-voltage DC power is from an uninterruptible power supply (UPS), ensure that the rated current on each line is not less than 16 A. When the AC and high-voltage DC power is from the mains power grid, ensure that the rated current on each line is not less than 20 A.

NOTICE

Note the derating requirements of protection components when using protection components such as fuses and circuit breakers.

NOTE

The input circuit breakers of a power distribution frame (PDF) must match the output circuit breakers.

- If the front-end PDF does not provide enough power inputs, configure a power distribution box (PDB) that meets the following requirements:
 - When the AC power is from a UPS, the output current on each line of the AC PDB is not less than 16 A. When the AC power is from the mains power grid, the output current on each line of the AC PDB is not less than 20 A.
 - The PDB supports double-line backup.
 - The input current of the PDB is not less than the sum of current of its connected output lines.

5.3 CloudEngine 16804 DC PEM

Overview

The CloudEngine 16804 DC PEM connects the CloudEngine 16804 to DC power sources. The PEM is pre-installed by default, and cannot be removed, installed, or replaced onsite.

CAUTION

The PEM is integrated into the chassis and does not need to be configured onsite. Do not remove or install the PEM onsite.

Appearance

Figure 5-8 Appearance of the CloudEngine 16804 DC PEM



Panel

Figure 5-9 Terminal blocks on the CloudEngine 16804 DC PEM

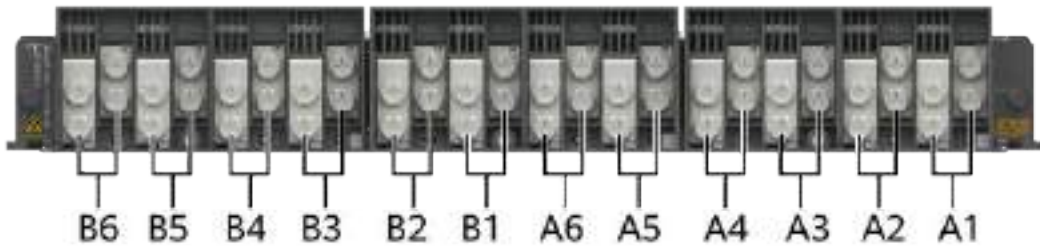


Table 5-7 describes the location mapping between CloudEngine 16804 PEM terminal blocks and PMs.

Table 5-7 Location mapping between PEM terminal blocks and PMs

PEM Terminal Block	A1/B1	A2/B2	A3/B3	A4/B4	A5/B5	A6/B6
PM	1	2	3	4	5	6

Table 5-8 Terminal block description

Name	Description	Power Cable Terminal
NEG (-)	-48 V/-60 V terminal block	- Naked Crimping Connector,JG2,25mm^2,M6,80A,Right angle_Galaxy - Naked Crimping Connector,JG2,35mm^2,M6,160A,Right angle_Galaxy NOTICE Do not bend terminals to 90 degrees onsite.
RTN (+)	BGND terminal block	

5.4 CloudEngine 16808 Power Supply System

Overview

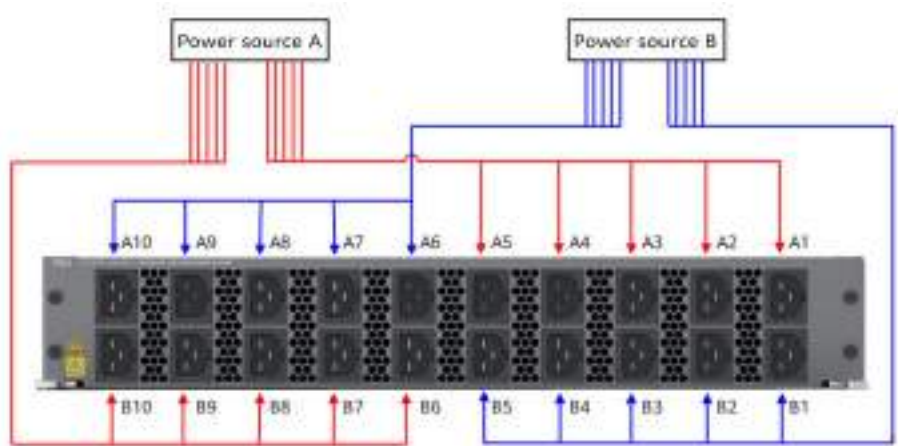
The CloudEngine 16808 supports DC, AC, and HVDC power supplies. The device connects to an external power distribution system through a power entry module (PEM) and supports power sources A and B. Each power module (PM) has two inputs and converts the power supplied to the PEM to power the entire device.

NOTE

- If there is only one power source in the equipment room, connect the power source to all A power sockets (A1 to A10) or all B power sockets (B1 to B10).
- If there are two power sources in the equipment room:
 - In AC or HVDC scenarios, power source A is used by default. If power source A fails, the system automatically switches to power source B. To prevent power supply fluctuation caused by dual power sources, the system does not immediately switch back to power source A after the fault is rectified, doing so only after a delay of 4.5 hours. It is advised to connect one power source to all A power sockets (A1 to A10) and the other power source to all B power sockets (B1 to B10).

If the power sources need to work in load-balancing mode, connect them to power sockets as follows:

- Connect power source A to one half of A power sockets (A1 to A5) and one half of B power sockets (B6 to B10).
- Similarly, connect power source B to the other half of A power sockets (A6 to A10) and the other half of B power sockets (B1 to B5).



- In DC scenarios, you are advised to connect one power source to A power sockets (A1 to A10) and the other power source to B power sockets (B1 to B10). If the two power sources have different voltages, the power source with a higher voltage provides a higher current.
- If AC and HVDC inputs are available to both power sources A and B, the 220 V AC input is used by default. If the 220 V AC input is faulty, the system automatically switches to the HVDC input.
- Requirements for the power supply system of a customer's cabinet: The output power of two power sources must be greater than the maximum input power of the device.

Components

As shown in [Figure 5-10](#), the power supply system of the CloudEngine 16808 is located at the top of the chassis. It consists of PMs and power switches at the front of the chassis, and a PEM at the rear of the chassis. [Table 5-9](#) describes the functions of each module.

Figure 5-10 Key components in the power supply system of the CloudEngine 16808



Table 5-9 Functions of key components in the power supply system of the CloudEngine 16808

No.	Component	Description
1	PM	Converts the power supplied to a PEM to power the entire device through the power backplane and busbar.
2	Power switch	Controls the power output of PMs. The two power switches back up each other. When one or both of them are in ON state, the power output of PMs is enabled. NOTICE To ensure reliability, both power switches must be turned on when the device is running.
3	PEM	Connects to an external power distribution system. The PEM can be divided into power supply areas A and B, with each one corresponding to a different power distribution system with a different source. This design implements 1+1 backup of external power distribution systems, ensuring the reliability of the entire power supply system by enabling the device to continue running when a power supply system in the equipment room fails.

PM and PEM Applications

The CloudEngine 16808 supports multiple power supply scenarios, which vary according to the installed PEMs and PMs. [Table 5-10](#) describes the power supply scenarios and PEM/PM applications.

Table 5-10 Power supply scenarios and PEM/PM applications on the CloudEngine 16808

Power Supply Scenario	PEM	PM
-48 V DC power supply	CloudEngine 16808 DC PEM	2200 W DC PM
AC power supply	CloudEngine 16808 AC & HVDC PEM	3000 W dual-input AC & HVDC PM

Power Supply Scenario	PEM	PM
240 V HVDC power supply	CloudEngine 16808 AC & HVDC PEM	3000 W dual-input AC & HVDC PM
380 V HVDC power supply	CloudEngine 16808 AC & HVDC PEM	3000 W dual-input AC & HVDC PM

PM and PEM Application Principles

The PEM is integrated into the chassis and does not need to be configured onsite. Do not remove or install the PEM onsite. PMs can be configured based on the overall power consumption of the device to flexibly supply power.

DC PMs and AC & HVDC PMs of the CloudEngine 16808 support N+1 backup. N indicates the number of PMs configured based on the estimated power consumption of the chassis. The maximum power supply capability of the system is equal to the sum of the maximum output power of N PMs. In N+1 backup mode, N PMs provide power to the chassis, and one PM works as a backup to implement power redundancy.

The following uses a 2200 W DC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to the number of required PMs), the value of $N \times 2200 \text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-11](#).

Table 5-11 DC PM application examples for the CloudEngine 16808

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Redundancy
$P < 2200 \text{ W}$	1, 2	A1, B1, A2, B2	4	63 A	1+1
$2200 \text{ W} \leq P < 4400 \text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$4400 \text{ W} \leq P < 6600 \text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
...	...	...	...		...
$17600 \text{ W} \leq P < 19800 \text{ W}$	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	A1, B1, A2, B2, A3, B3 ... A10, B10	20		9+1

The following uses a 3000 W AC & HVDC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to

the number of required PMs), the value of $N \times 3000\text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-12](#).

Table 5-12 AC & HVDC PM application examples for the CloudEngine 16808

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Redundancy
$P < 3000\text{ W}$	1, 2	A1, B1, A2, B2	4	AC: 16 A/20 A HVDC: 16 A	1+1
$3000\text{ W} \leq P < 6000\text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$6000\text{ W} \leq P < 9000\text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
...	...	...	...		...
$24000\text{ W} \leq P < 27000\text{ W}$	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	A1, B1, A2, B2, A3, B3 ... A10, B10	20		9+1

 NOTE

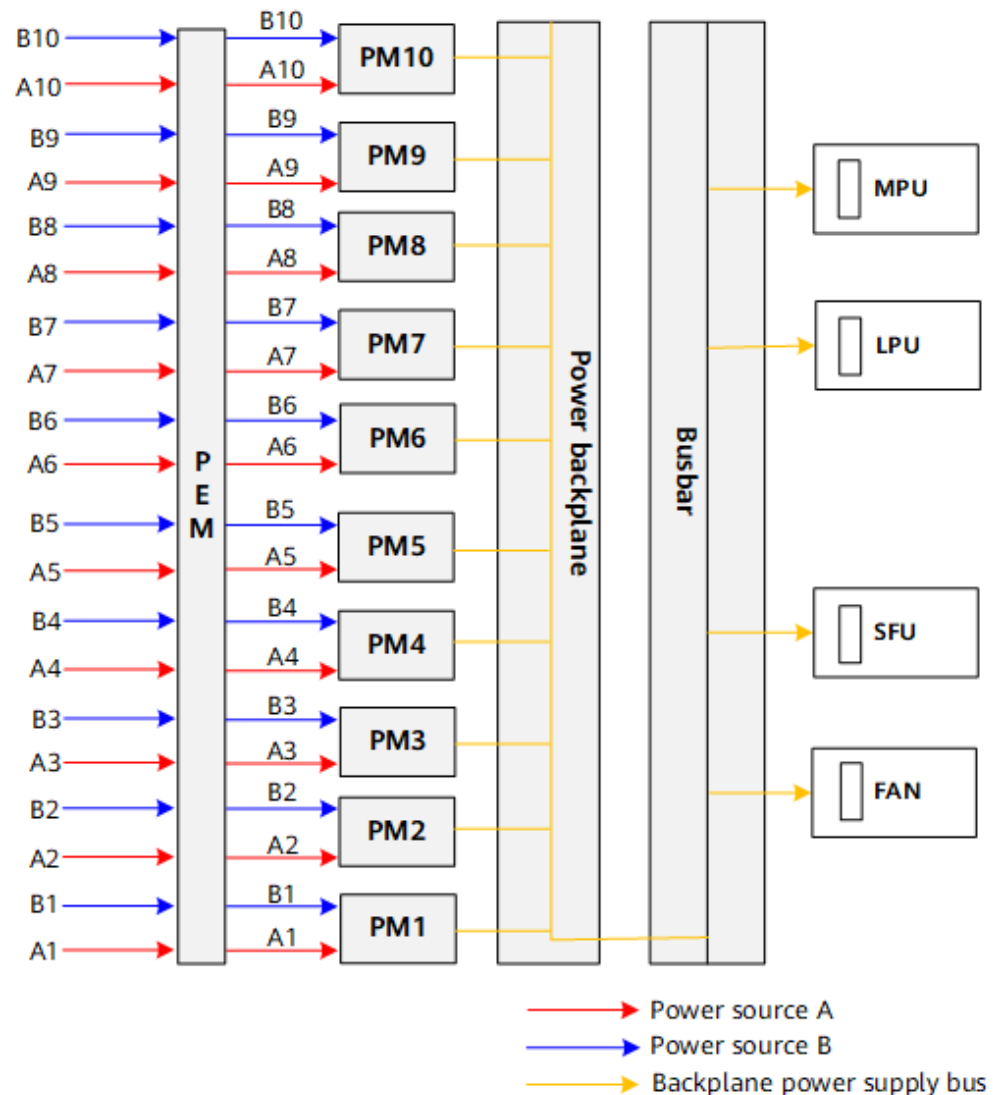
When no PM is installed in a PM slot, reserve the filler panel.

Power Supply Architecture

- DC power supply

The DC power supply system of the CloudEngine 16808 must be equipped with one DC PEM. The PEM has 20 inputs, corresponding to 20 power sockets. Power sources A and B each provide 10 outputs. Each DC PM has two inputs. The two inputs are from power sources A and B in the equipment room, respectively. The DC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other. [Figure 5-11](#) shows the DC power supply architecture of the CloudEngine 16808.

Figure 5-11 DC power supply architecture of the CloudEngine 16808



- AC & HVDC power supply

The AC & HVDC power supply system of the CloudEngine 16808 must be equipped with one AC & HVDC PEM. The PEM has 20 inputs, corresponding to 20 power sockets: 10 inputs of power source A and 10 inputs of power source B. Each AC & HVDC PM has two inputs. The two inputs are from power sources A and B in the equipment room, respectively. The AC & HVDC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other.

The AC & HVDC power supply system of the CloudEngine 16808 can be directly connected to a power supply or power distribution unit (PDU). Connecting the power supply system to a power supply is similar to the DC power supply mode, and is not described here. The following describes the power supply architecture in the scenario where the power supply system is connected to a PDU.

The CloudEngine 16800 series switches provide the following types of PDUs:

- AC PDU: Each AC PDU provides one input and six outputs.

- HVDC PDU: Each HVDC PDU provides two inputs and eight outputs. Each input corresponds to four outputs.

Power sources A and B can provide AC outputs, HVDC outputs, or both AC & HVDC outputs. [Figure 5-12](#), [Figure 5-13](#), and [Figure 5-14](#) show the power supply architectures of the three modes.

Figure 5-12 AC power supply architecture of the CloudEngine 16808

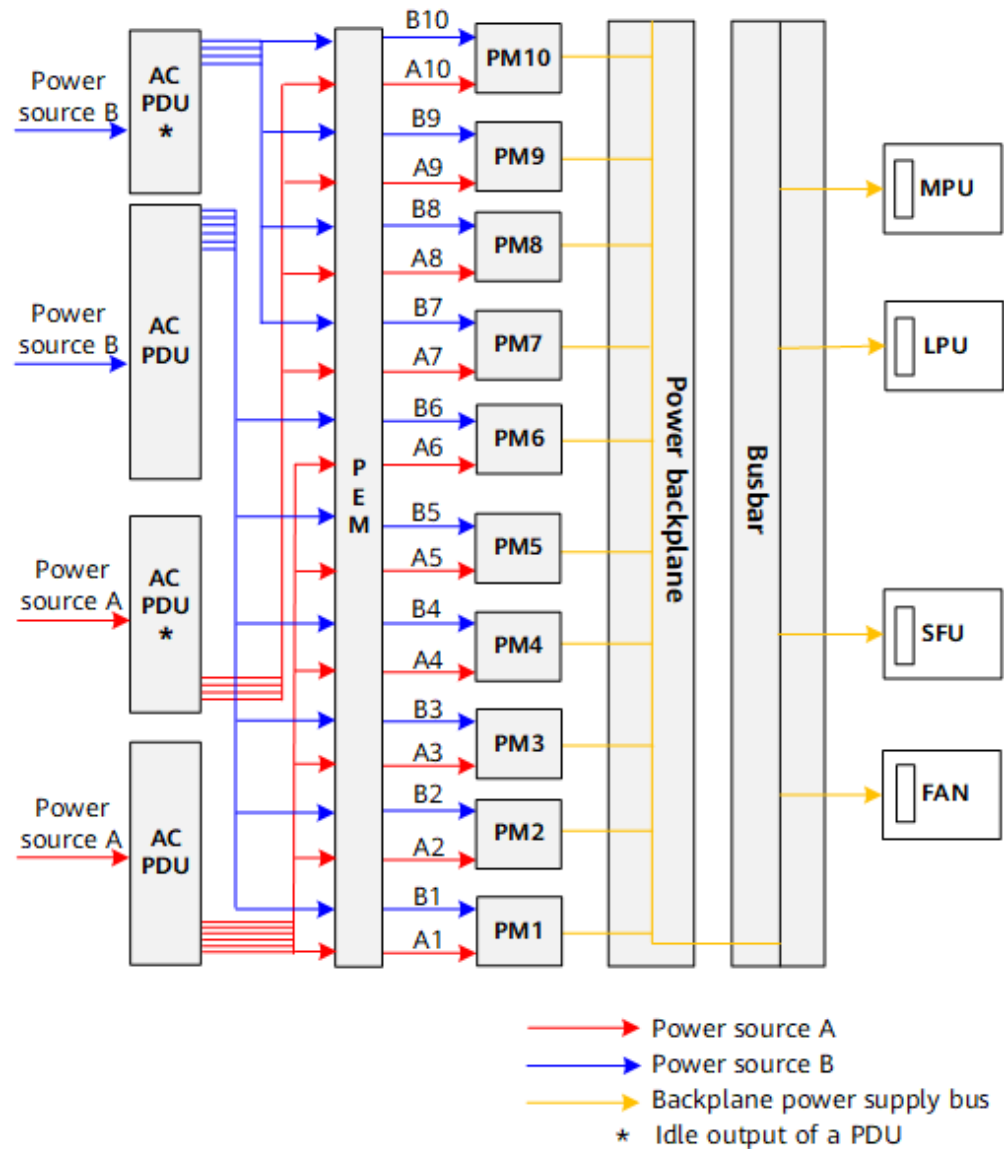


Figure 5-13 HVDC power supply architecture of the CloudEngine 16808

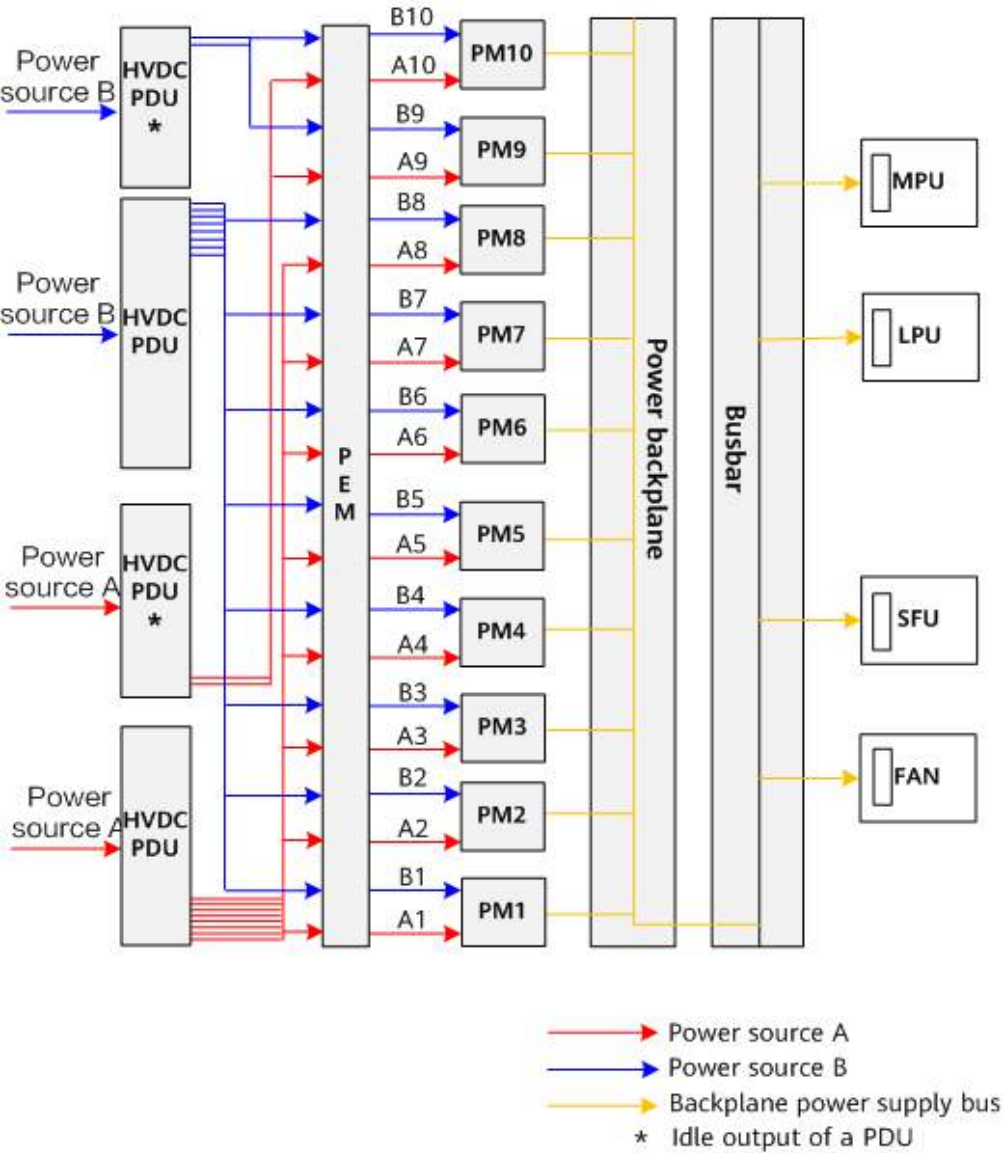
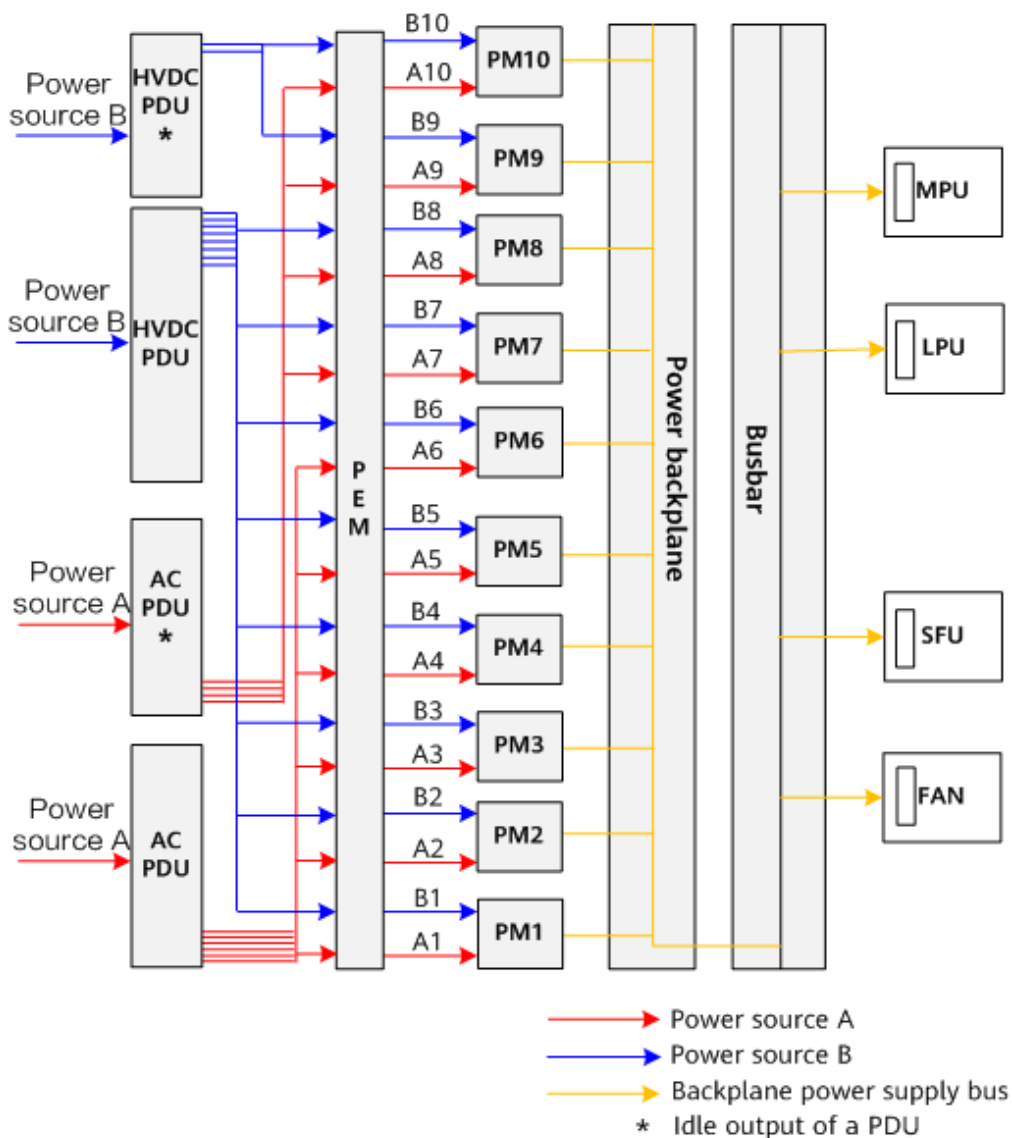


Figure 5-14 AC & HVDC hybrid power supply architecture of the CloudEngine 16808



5.5 CloudEngine 16808 AC&High-Voltage DC PEM

Overview

The CloudEngine 16808 AC&high-voltage DC PEM connects the CloudEngine 16808 to AC&high-voltage DC power sources. The PEM is pre-installed by default, and cannot be removed, installed, or replaced onsite.

CAUTION

The PEM is integrated into the chassis and does not need to be configured onsite. Do not remove or install the PEM onsite.

Appearance

Figure 5-15 Appearance of the CloudEngine 16808 AC&high-voltage DC PEM



Panel

Figure 5-16 Power sockets on the CloudEngine 16808 AC&high-voltage DC PEM

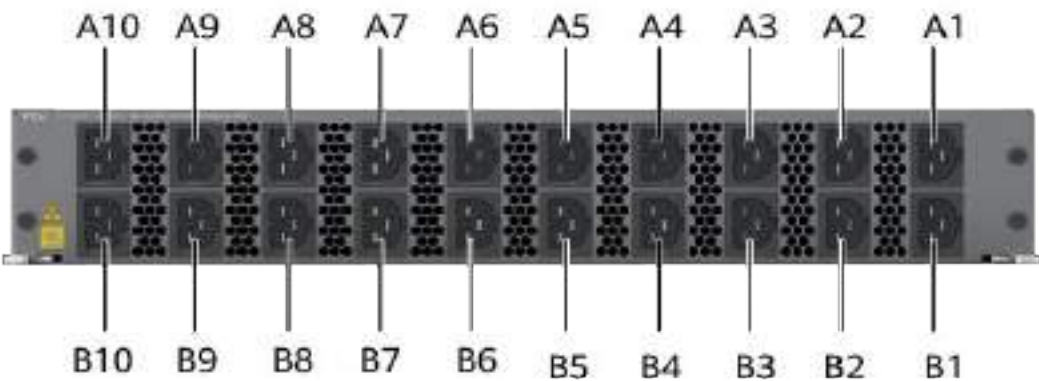


Table 5-13 describes the location mapping between CloudEngine 16808 PEM power sockets and power modules.

Table 5-13 Location mapping between PEM power sockets and power modules

PEM Power Socket	A1/B1	A2/B2	A3/B3	...	A9/B9	A10/B10
Power Module	1	2	3	...	9	10

Table 5-14 Power socket description

Name	Description	Power Cable Terminal
Power socket	200 V AC to 240 V AC/240 V DC/380 V DC power socket	Dedicated high-voltage DC terminal: Power Connector,3Pin,450V,16A, Straight Female,1.5mm^2 or 14AWG

Input Requirements

- In an AC power distribution scenario, the 220 V AC input is supported.
- 240 V/380 V high-voltage DC input is supported in high-voltage DC power distribution scenarios.
- Protection components must be provided for power distribution frames (PDFs). When the AC and high-voltage DC power is from an uninterruptible power supply (UPS), ensure that the rated current on each line is not less than 16 A. When the AC and high-voltage DC power is from the mains power grid, ensure that the rated current on each line is not less than 20 A.

NOTICE

Note the derating requirements of protection components when using protection components such as fuses and circuit breakers.

NOTE

The input circuit breakers of a power distribution frame (PDF) must match the output circuit breakers.

- If the front-end PDF does not provide enough power inputs, configure a power distribution box (PDB) that meets the following requirements:
 - When the AC power is from a UPS, the output current on each line of the AC PDB is not less than 16 A. When the AC power is from the mains power grid, the output current on each line of the AC PDB is not less than 20 A.
 - The PDB supports double-line backup.
 - The input current of the PDB is not less than the sum of current of its connected output lines.

5.6 CloudEngine 16808 DC PEM

Overview

The CloudEngine 16808 DC PEM connects the CloudEngine 16808 to DC power sources. The PEM is pre-installed by default, and cannot be removed, installed, or replaced onsite.

CAUTION

The PEM is integrated into the chassis and does not need to be installed, configured, or removed onsite.

Appearance

Figure 5-17 Appearance of the CloudEngine 16808 DC PEM



Panel

Figure 5-18 Terminal blocks on the CloudEngine 16808 DC PEM

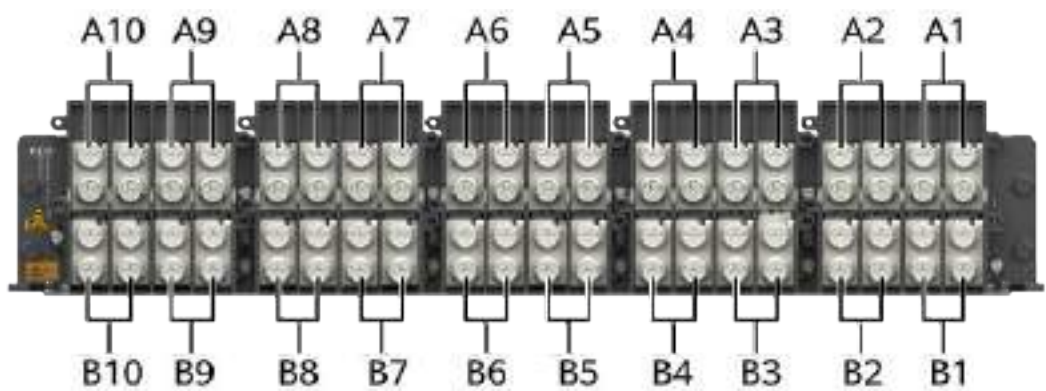


Table 5-15 describes the location mapping between CloudEngine 16808 PEM terminal blocks and PMs.

Table 5-15 Location mapping between PEM terminal blocks and PMs

PEM Terminal Block	A1/B1	A2/B2	A3/B3	...	A9/B9	A10/B10
PM	1	2	3	...	9	10

Table 5-16 Terminal block description

Name	Description	Power Cable Terminal
NEG (-)	-48 V/-60 V terminal block	- Naked Crimping Connector,JG2,25mm^2,M6,80A,Right angle_Galaxy - Naked Crimping Connector,JG2,35mm^2,M6,160A,Right angle_Galaxy NOTICE Do not bend terminals to 90 degrees onsite.
RTN (+)	BGND terminal block	

5.7 CloudEngine 16816 Power Supply System

Overview

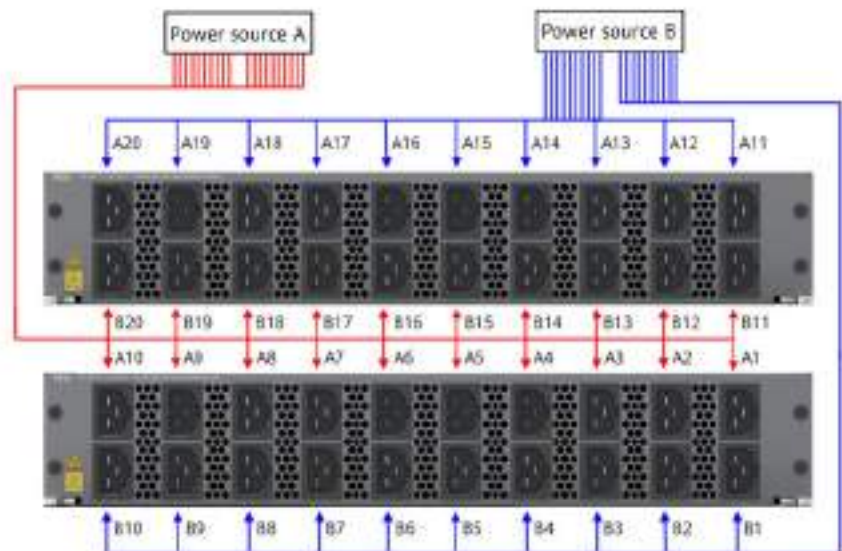
The CloudEngine 16816 supports DC, AC, and HVDC power supplies. The device is connected to an external power distribution system through power entry modules (PEMs) and supports power sources A and B. Each power module (PM) has two inputs. The PM converts the power supplied to the PEMs to power the entire device.

NOTE

- If there is only one power source in the equipment room, connect the power source to all A power sockets (A1 to A20) or all B power sockets (B1 to B20).
- If two power sources are available in the equipment room:
 - In AC or HVDC scenarios, power source A is used by default. If power source A fails, the system automatically switches to power source B. To prevent power supply fluctuation caused by dual power sources, the system does not immediately switch back to power source A after the fault is rectified, doing so only after a delay of 4.5 hours. It is advised to connect one power source to all A power sockets (A1 to A20) and the other power source to all B power sockets (B1 to B20).

If the two power sources need to work in load-balancing mode, connect them to power sockets as follows:

- Connect power source A to one half of A power sockets (A1 to A10) and one half of B power sockets (B11 to B20).
- Similarly, connect power source B to the other half of A power sockets (A11 to A20) and the other half of B power sockets (B1 to B10).



- In DC scenarios, you are advised to connect one power source to A power sockets (A1 to A20) and the other power source to B power sockets (B1 to B20). If the two power sources have different voltages, the power source with a higher voltage provides a higher current.
- If AC and HVDC inputs are available to both power sources A and B, the 220 V AC input is used by default. If the 220 V AC input is faulty, the system automatically switches to the HVDC input.
- Requirements for the power supply system of a customer's cabinet: The output power of two power sources must be greater than the maximum input power of the device.

Components

As shown in [Figure 5-19](#), the power supply system of the CloudEngine 16816 is located at the top of the chassis. It consists of PMs and power switches at the front of the chassis, and PEMs at the rear of the chassis. [Table 5-17](#) describes the functions of each module.

Figure 5-19 Key components in the power supply system of the CloudEngine 16816

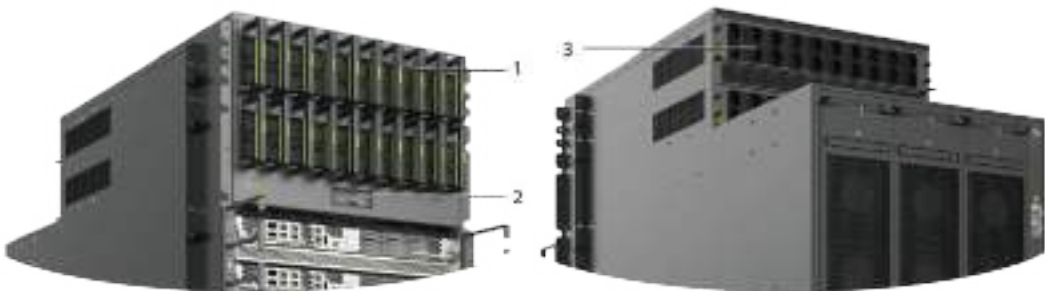


Table 5-17 Functions of key components in the power supply system of the CloudEngine 16816

No.	Component	Description
1	PM	Converts the power supplied to a PEM to power the entire device through the power backplane and busbar.
2	Power switch	Controls the power output of PMs. The two power switches back up each other. When one or both of them are in ON state, the power output of PMs is enabled. NOTICE To ensure reliability, both power switches must be turned on when the device is running.
3	PEM	Connects to an external power distribution system. PEMs can be divided into power supply areas A and B, corresponding to two power distribution systems with different sources, respectively. This design implements 1+1 backup of external power distribution systems. If a power supply system in the equipment room fails, the device is not powered off. This ensures the reliability of the entire power supply system.

PM and PEM Applications

The CloudEngine 16816 supports multiple power supply scenarios, which vary according to the installed PEMs and PMs. [Table 5-18](#) describes the power supply scenarios and PEM/PM applications.

Table 5-18 Power supply scenarios and PEM/PM applications on the CloudEngine 16816

Power Supply Scenario	PEM	PM
-48 V DC power supply	CloudEngine 16816 DC PEM	2200 W DC PM

Power Supply Scenario	PEM	PM
AC power supply	CloudEngine 16816 AC & HVDC PEM	3000 W dual-input AC & HVDC power module
240 V HVDC power supply	CloudEngine 16816 AC & HVDC PEM	3000 W dual-input AC & HVDC power module
380 V HVDC power supply	CloudEngine 16816 AC & HVDC PEM	3000 W dual-input AC & HVDC power module

PM and PEM Application Principles

PEMs are integrated into the chassis and do not need to be configured onsite. Do not remove or install PEMs onsite. PMs can be configured based on the overall power consumption of the device to flexibly supply power.

DC PMs and AC & HVDC PMs of the CloudEngine 16816 support N+1 backup. N indicates the number of PMs configured based on the actual power consumption of the chassis. The maximum power supply capability of the system is equal to the total maximum output power of N PMs. In N+1 backup mode, N PMs provide power for the chassis, and one PM works as a backup for power redundancy.

The following uses a 2200 W DC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to the number of required PMs), the value of $N \times 2200 \text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-19](#).

Table 5-19 DC power distribution examples of the CloudEngine 16816

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Redundancy
$P < 2200 \text{ W}$	1, 2	A1, B1, A2, B2	4	63A	1+1
$2200 \text{ W} \leq P < 4400 \text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$4400 \text{ W} \leq P < 6600 \text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
...	...	...	...		...
$39600 \text{ W} \leq P < 41800 \text{ W}$	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	A1, B1, A2, B2, A3, B3 ... A20, B20	40		19+1

The following uses a 3000 W AC & HVDC PM as an example. If PMs are configured according to the N+1 backup mechanism (that is, one PM is added to the number of required PMs), the value of $N \times 3000 \text{ W}$ must be greater than the power consumption of the entire system, as described in [Table 5-20](#).

Table 5-20 AC & HVDC power module application examples of the CloudEngine 16816

Overall Power Consumption	PM Connection	PEM Connection	Number of Power Inputs	Rated Current of Each Input (Recommended Value)	N+1 Redundancy
$P < 3000 \text{ W}$	1, 2	A1, B1, A2, B2	4	AC: 16 A/20 A HVDC: 16 A	1+1
$3000 \text{ W} \leq P < 6000 \text{ W}$	1, 2, 3	A1, B1, A2, B2, A3, B3	6		2+1
$6000 \text{ W} \leq P < 9000 \text{ W}$	1, 2, 3, 4	A1, B1, A2, B2, A3, B3, A4, B4	8		3+1
...	...	...	...		...
$54000 \text{ W} \leq P < 57000 \text{ W}$	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	A1, B1, A2, B2, A3, B3 ... A20, B20	40		19+1

NOTE

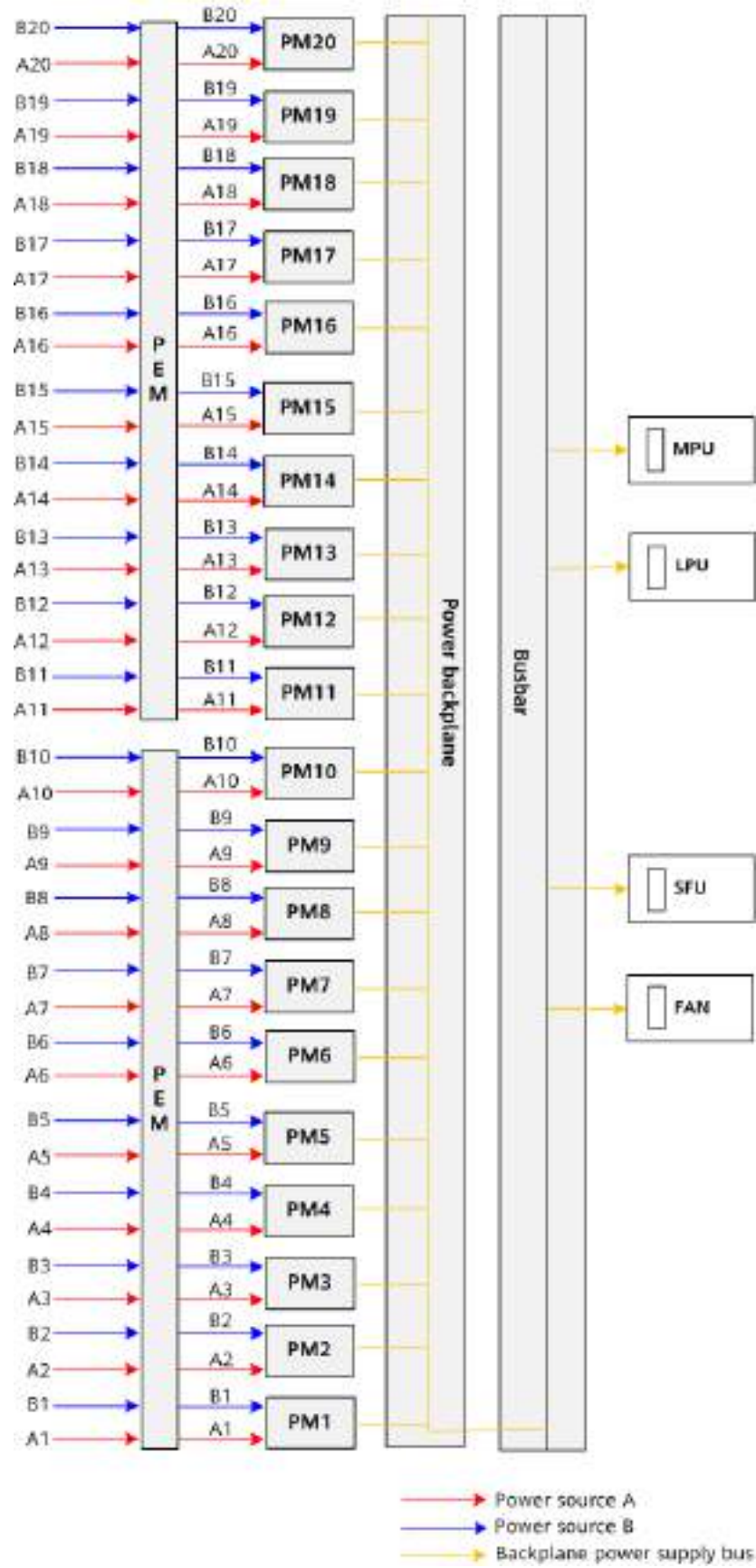
If PMs are not fully configured, connect power cables only to power sockets of PEMs with corresponding PMs installed, and retain filler panels in PM slots without PMs installed.

Power Supply Architecture

- DC power supply

The DC power supply system of the CloudEngine 16816 must be equipped with two DC PEMs. Each PEM has 20 inputs, corresponding to 20 power sockets. Power supplies A and B provide 10 inputs, respectively. Each DC PM has two inputs. The two inputs are from power sources A and B in the equipment room, respectively. The DC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other. [Figure 5-20](#) shows the DC power supply architecture of the CloudEngine 16816.

Figure 5-20 DC power supply architecture of the CloudEngine 16816



- AC & HVDC power supply

The AC & HVDC power supply system of the CloudEngine 16816 must be equipped with two AC & HVDC PEMs. Each PEM has 20 inputs, corresponding to 20 power sockets. Power sources A and B each provide 10 outputs. Each AC & HVDC PM has two inputs. The two inputs are from power sources A and B in the equipment room, respectively. The AC & HVDC PM converts the power and outputs it to the power backplane. In this manner, power sources A and B back up each other.

The AC & HVDC power supply system of the CloudEngine 16816 can be directly connected to a power supply or PDU. Connecting the power supply system to a power source is similar to the DC power supply mode, and is not described here. The following describes the power supply architecture in the scenario where the power supply system is connected to a PDU.

The CloudEngine 16800 series switches provide the following types of PDUs:

- AC PDU: Each AC PDU provides one input and six outputs.
- HVDC PDU: Each HVDC PDU provides two inputs and eight outputs. Each input corresponds to four outputs.

Power sources A and B can provide AC outputs, HVDC outputs, or both AC and HVDC outputs. [Figure 5-21](#), [Figure 5-22](#), and [Figure 5-23](#) show the power supply architectures of the three modes.

Figure 5-21 AC power supply architecture of the CloudEngine 16816

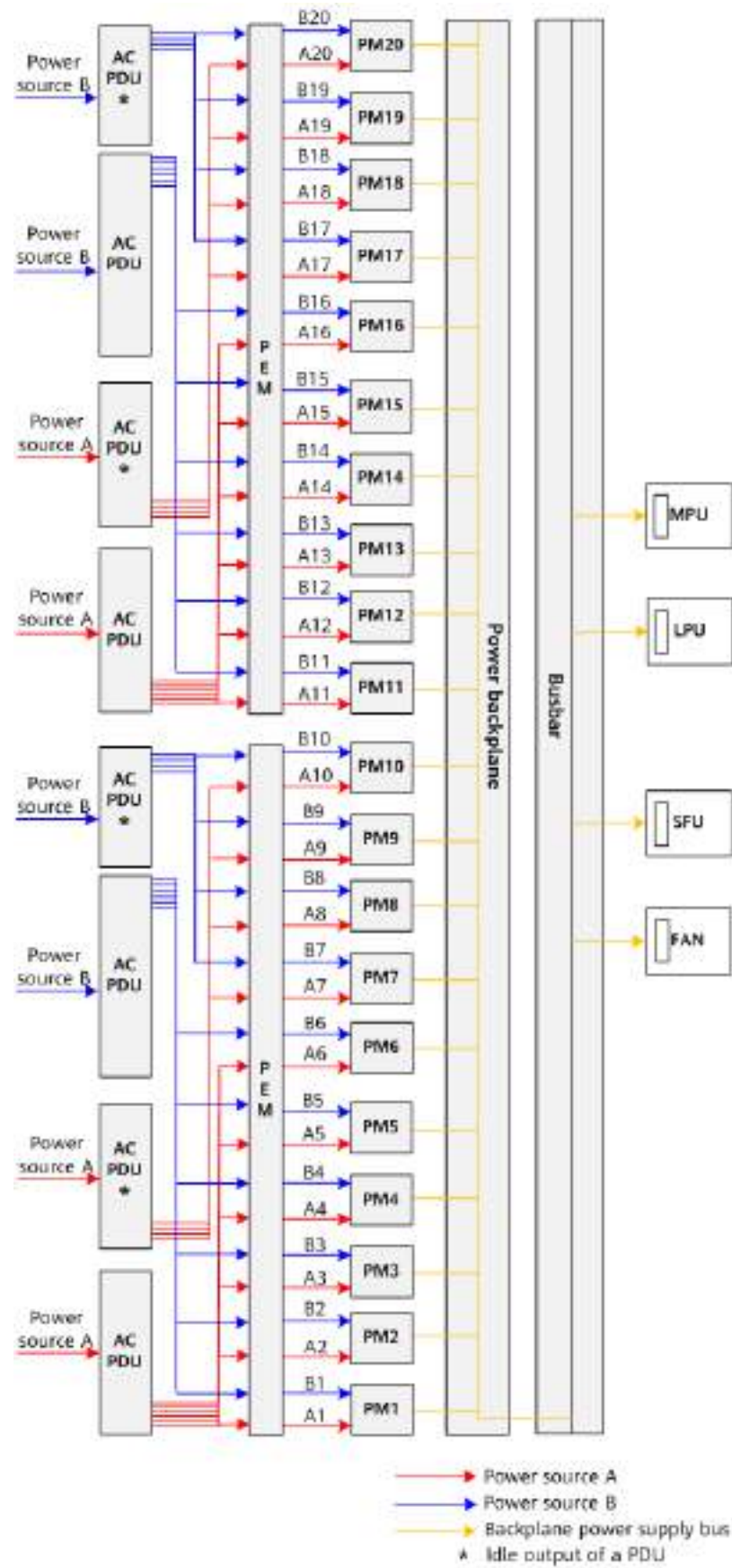


Figure 5-22 HVDC power supply architecture of the CloudEngine 16816

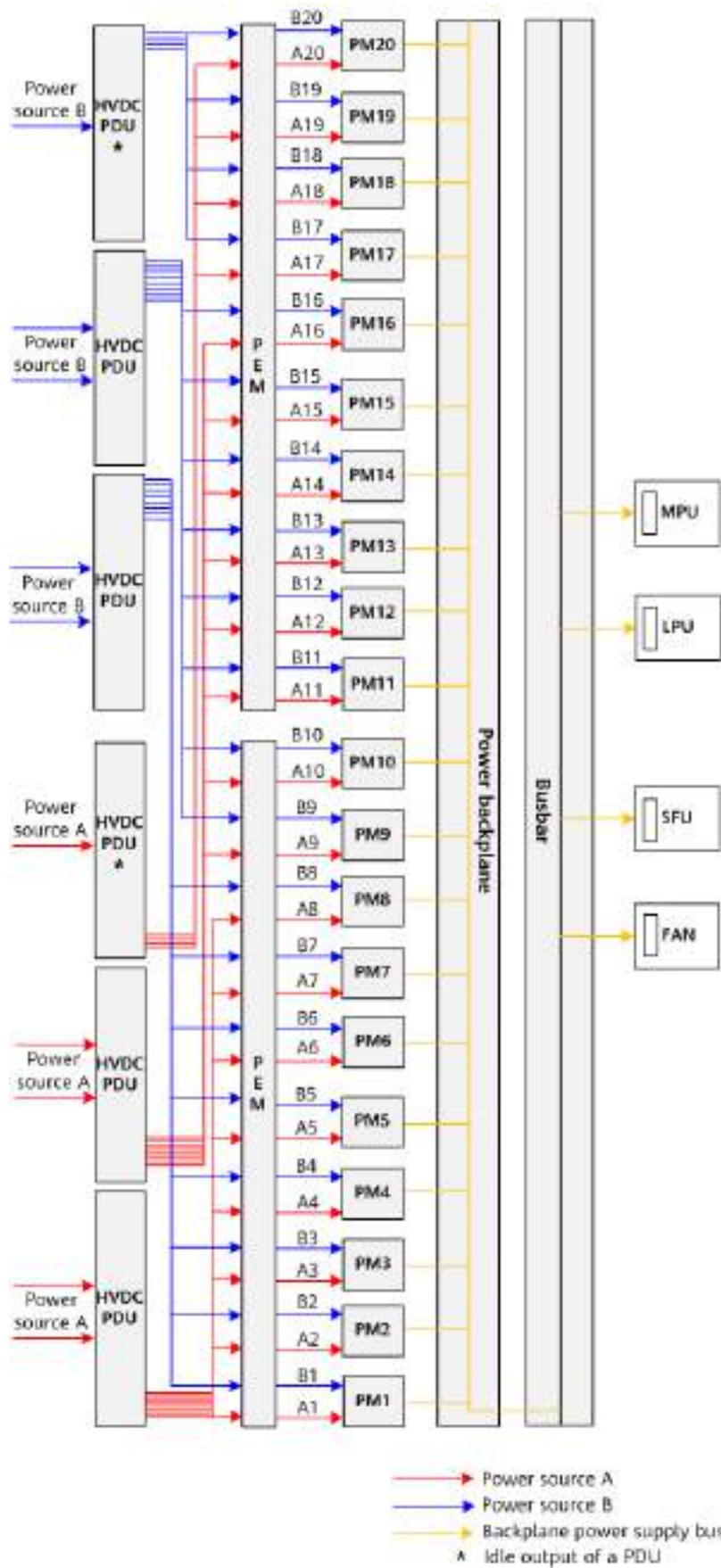
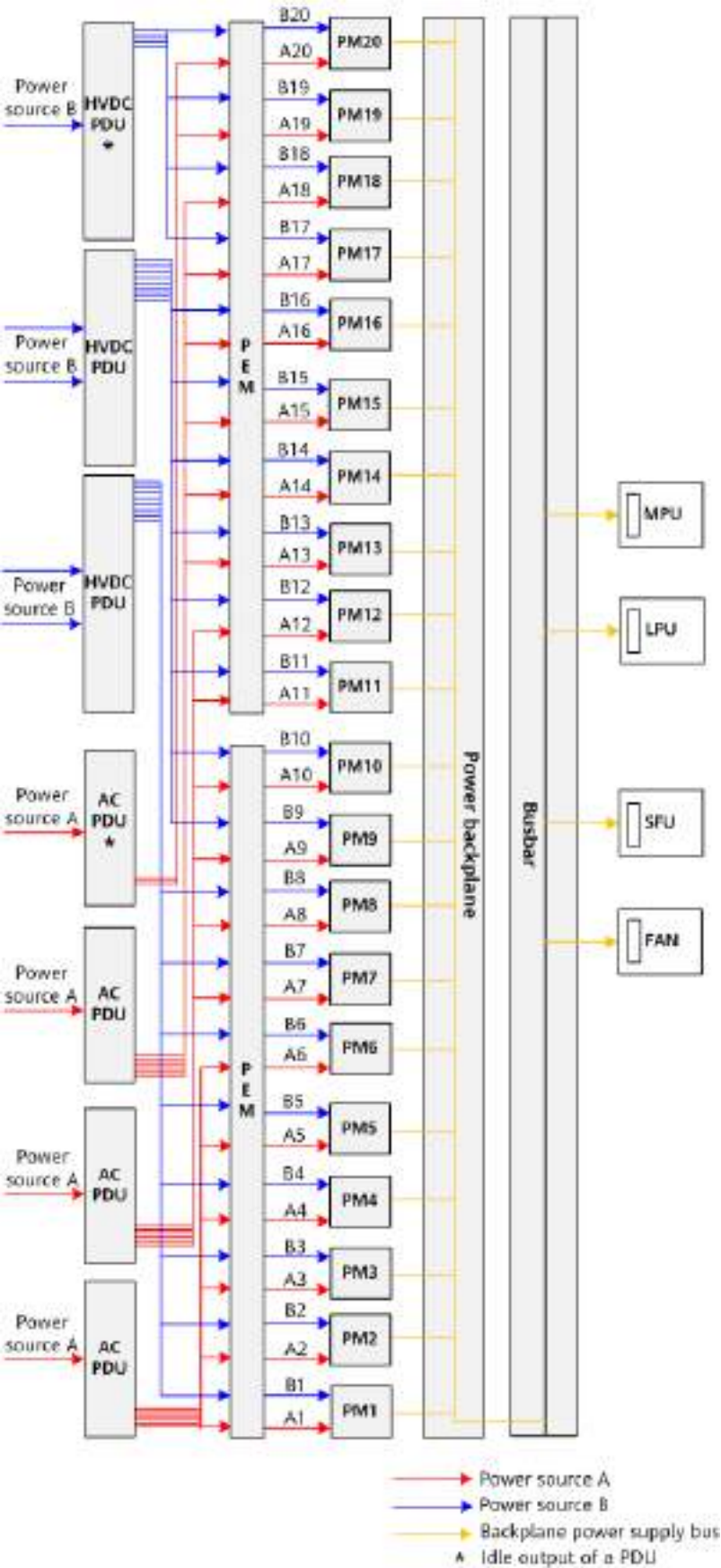


Figure 5-23 AC and HVDC hybrid power supply architecture of the CloudEngine 16816



5.8 CloudEngine 16816 AC&High-Voltage DC PEM

Overview

CloudEngine 16816 AC&high-voltage DC PEMs connect the CloudEngine 16816 to AC&high-voltage DC power sources. The PEMs are pre-installed by default, and cannot be removed, installed, or replaced onsite.

The CloudEngine 16816 and CloudEngine 16808 use the same AC&high-voltage DC PEM. The difference is that the CloudEngine 16816 needs to be configured with two AC&high-voltage DC PEMs.

Appearance

Figure 5-24 Appearance of CloudEngine 16816 AC&high-voltage DC PEMs



Panel

Figure 5-25 Power sockets on CloudEngine 16816 AC&high-voltage DC PEMs

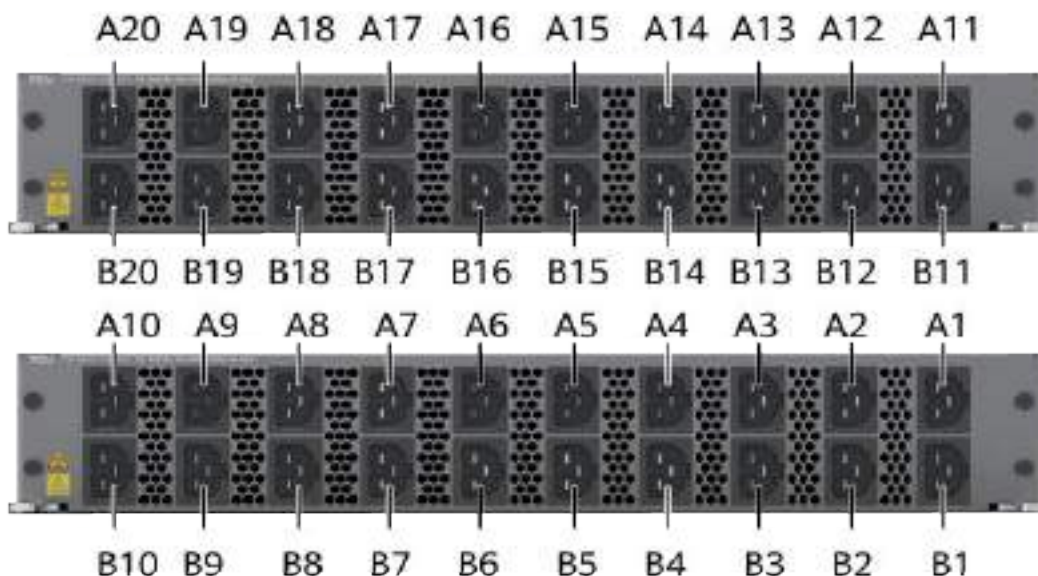


Table 5-21 describes the location mapping between CloudEngine 16816 PEM power sockets and power modules.

Table 5-21 Location mapping between PEM power sockets and power modules

PEM Power Socket	A1/B1	A2/B2	A3/B3	...	A19/B19	A20/B20
Power Module	1	2	3	...	19	20

Table 5-22 Power socket description

Name	Description	Power Cable Terminal
Power socket	200 V AC to 240 V AC/240 V DC/380 V DC power socket	Power Connector, 3Pin, 450V, 16A, Straight Female, 1.5mm ² or 14AWG

Input Requirements

- In an AC power distribution scenario, the 220 V AC input is supported.
- 240 V/380 V high-voltage DC input is supported in high-voltage DC power distribution scenarios.
- Protection components must be provided for power distribution frames (PDFs). When the AC and high-voltage DC power is from an uninterruptible power supply (UPS), ensure that the rated current on each line is not less than 16 A. When the AC and high-voltage DC power is from the mains power grid, ensure that the rated current on each line is not less than 20 A.

NOTICE

Note the derating requirements of protection components when using protection components such as fuses and circuit breakers.

NOTE

The input circuit breakers of a power distribution frame (PDF) must match the output circuit breakers.

- If the front-end PDF does not provide enough power inputs, configure a power distribution box (PDB) that meets the following requirements:
 - When the AC power is from a UPS, the output current on each line of the AC PDB is not less than 16 A. When the AC power is from the mains

power grid, the output current on each line of the AC PDB is not less than 20 A.

- The PDB supports double-line backup.
- The input current of the PDB is not less than the sum of current of its connected output lines.

5.9 CloudEngine 16816 DC PEM

Overview

CloudEngine 16816 DC PEMs connect the CloudEngine 16816 to DC power sources. The PEMs are pre-installed by default, and cannot be removed, installed, or replaced onsite.

The CloudEngine 16816 and CloudEngine 16808 use the same DC PEM. The difference is that the CloudEngine 16816 needs to be configured with two DC PEMs.

Appearance

Figure 5-26 Appearance of CloudEngine 16816 DC PEMs



Panel

Figure 5-27 Terminal blocks on CloudEngine 16816 DC PEMs

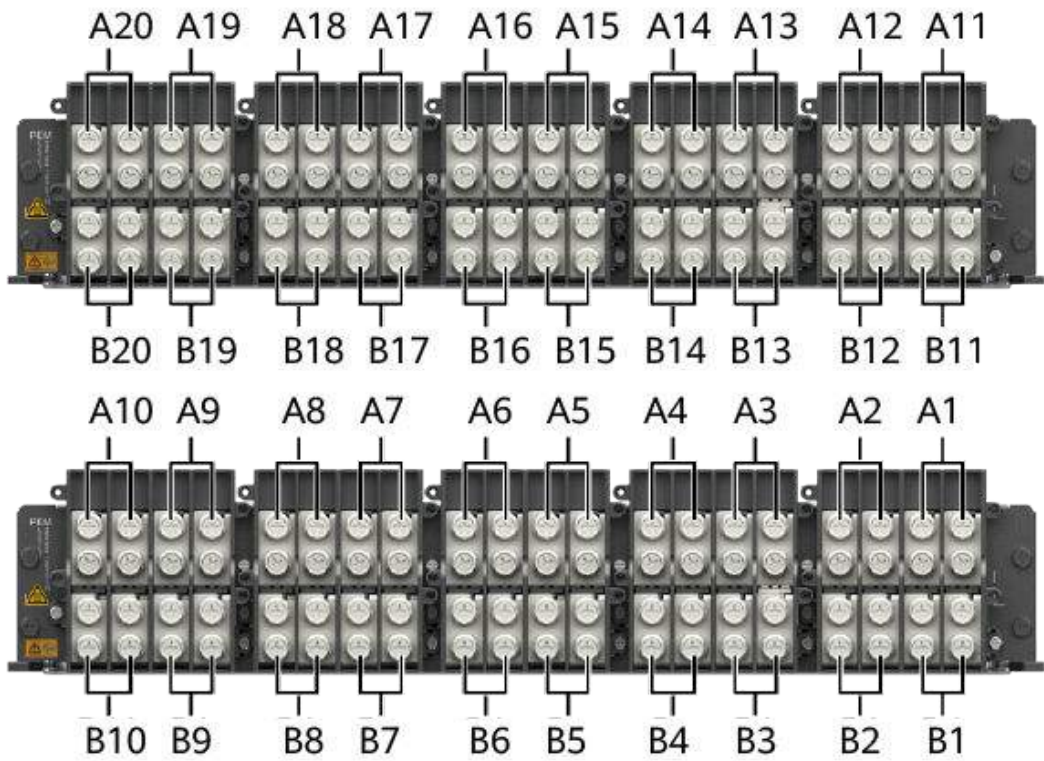


Table 5-23 describes the location mapping between CloudEngine 16816 PEM terminal blocks and PMs.

Table 5-23 Location mapping between PEM terminal blocks and PMs

PEM Terminal Block	A1/B1	A2/B2	A3/B3	...	A19/B19	A20/B20
PM	1	2	3	...	19	20

Table 5-24 Terminal block description

Name	Description	Power Cable Terminal
NEG (-)	-48 V/-60 V terminal block	- Naked Crimping Connector,JG2,25mm^2,M6,80A,Right angle_Galaxy - Naked Crimping Connector,JG2,35mm^2,M6,160A,Right angle_Galaxy NOTICE Do not bend terminals to 90 degrees onsite.
RTN (+)	BGND terminal block	

5.10 Power Switch

A switch controls the power output of power modules (PMs). The device has two switches that back up each other. When one or both switches are in the ON state, the power output of PMs is enabled.

Appearance

Figure 5-28 shows the appearance of the switches.

Figure 5-28 Switches of the power system



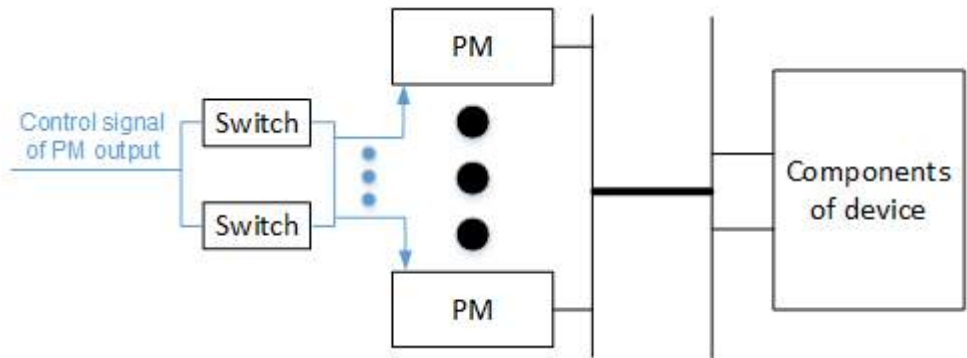
Panel

Table 5-25 Switch description

Name	Description
OFF/ON	Controls device power-on and power-off.

Functions

When one or both switches are turned on, the PMs output power, and the device is powered on. When both switches are turned off, the PMs do not output power, and the device is powered off.



NOTICE

Ensure that both switches are turned on when the device is running for reliability purposes.

5.11 PAH-3000WA (3000W Dual Inputs AC&HVDC Power Module)

Overview

Table 5-26 Basic information about the PAH-3000WA

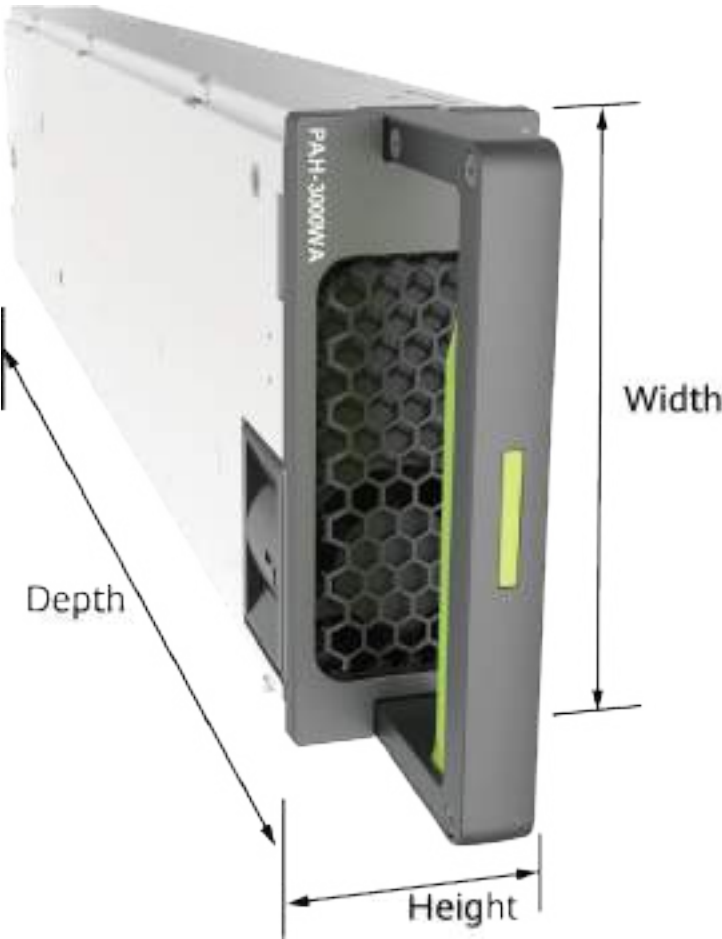
Item	Details
Description	3000W Dual Inputs AC&HVDC Power Module
Part Number	02312JEA
Model	PAH-3000WA

NOTE

PMs can be configured based on the overall power consumption of the device to flexibly supply power.

Appearance

Figure 5-29 Appearance of the PAH-3000WA



Version Mapping

Table 5-27 Mappings between PAH-3000WA and product models

Product	Product Model	First Supported Version	Limitations	IDs of Available Slots
CloudEngine 16804	CE16804-AH	V200R005C20	-	PM1-PM6
CloudEngine 16808	CE16808-AH	V200R005C20	-	PM1-PM10
CloudEngine 16816	CE16816-AH	V200R005C20	-	PM1-PM20

Panel

Figure 5-30 Panel of the PAH-3000WA



Table 5-28 Indicators on the PAH-3000WA

Silkscreen	Name	Color	Status	Description
Input	Power indicator	Green	Steady on	The input power is in the normal range.
		Green	Fast blinking (4 Hz)	The input voltage is out of the normal range.
		Green	Off	The power module receives no input power.
Output	Alarm indicator	Green	Steady on	The output power is in the normal range.
		Green	Off	The power module has no output power or the output power is abnormal.
Alarm	Fault indicator	Red	Off	The power module is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The power module has failed. Possible causes include internal short circuit, fan failure, output overvoltage, overtemperature-triggered shutdown, communication failure, and severely uneven current.

Functions and Features

Table 5-29 Functions and features of the PAH-3000WA

Functions and Features	Description
Input undervoltage protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from input undervoltage, the power module can automatically resume power supply. The Input indicator blinks green.
Input overvoltage protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from input overvoltage, the power module can automatically resume power supply. The Input indicator blinks green.
Input overcurrent protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from input overcurrent, the power module cannot automatically resume power supply. The Input indicator is off.

Functions and Features	Description
Output overvoltage protection	- If output overvoltage is caused by the power module itself, the power module is turned off and stops supplying power. The power module cannot automatically resume power supply. The Output indicator is off. - If output overvoltage is caused by increase of the input voltage received from the external power source, the power module is turned off and stops supplying power. When the system recovers from output overvoltage, the power module can automatically resume power supply. The Output indicator is steady green.
Output current limiting protection	In this protection state, the output current is limited to a certain value. When the system recovers from output overcurrent, the power module can automatically resume power supply. The Output indicator is steady green. When the output voltage is lower than 44 V DC, the Output indicator is off.
Output short-circuit protection	In this protection state, the power module supplies power intermittently, and the output current is limited within a range. When the system recovers from output short-circuit, the power module can automatically resume power supply. The Output indicator is off.
Overtemperature protection	- When the temperature of the power module ranges from 55°C to 75°C, the output current reduces linearly as the temperature increases. The Output indicator is steady green. - When the temperature exceeds 75°C, the power module stops supplying power. When the temperature falls below 75°C, the power module can automatically resume power supply. The Output indicator is off.

Functions and Features	Description
Heat dissipation mode	The power module has its own fans, which use a front-to-back airflow design.
Dual-input power supply switchover	In the case of dual-input power supply, if power supply A fails, power supply B is automatically used. To prevent fluctuation of the two power supplies, the system does not switch back to power supply A immediately after power supply A recovers. Instead, it switches back to power supply A 4.5 hours later.
Hot swapping	The power module is hot swappable.

 **CAUTION**

- If a power module stops supplying power in protection state and cannot automatically resume power supply after the triggering condition of the protection state is removed, remove the power module from the chassis and reinstall it at least 30s later. The power module then can work normally.
- When a power module enters overtemperature protection state, take measures to lower the temperature of the chassis. The power module can automatically resume power supply when the temperature falls within the normal range.

Technical Specifications

Table 5-30 Technical specifications of the PAH-3000WA

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	41 mm x 106.5 mm x 485 mm (1.61 in. x 4.19 in. x 19.09 in.)
Weight without packaging [kg (lb)]	2.8 kg (6.17 lb)
Number of inputs	2
Number of outputs	1
Rated input voltage [V]	200 V AC to 240 V AC (220 V power grid) 240 V DC/380 V DC
Input voltage range [V]	• AC: 176–290 V AC; 45–65 Hz • High-voltage DC: 188–288 V DC/260–400 V DC

Item	Specification
Rated input current [A]	14 A
Maximum input current [A]	16 A
Rated output voltage [V]	53.5 V
Rated output current [A]	56 A
Rated output power [W]	3000 W (Note: The AC power supply system supports 200 V AC to 240 V AC power and the input voltage range of 176 V to 290 V. If the input voltage is lower than or equal to 175 V AC, for example, the input voltage is 110 V AC, the rated output power reduces to 1500 W.)

5.12 PDC-2200WC (PM2200W DC Power Supply)

Overview

Table 5-31 Basic information about the PDC-2200WC

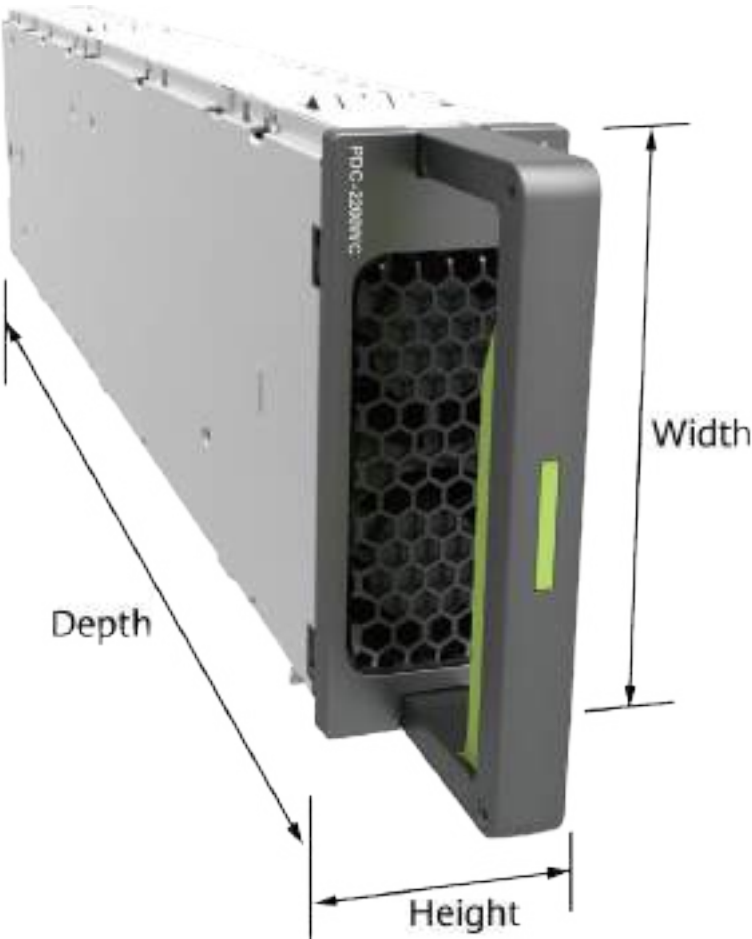
Item	Details
Description	PM2200W DC Power Supply
Part Number	02312GDC
Model	PDC-2200WC

 NOTE

PMs can be configured based on the overall power consumption of the device to flexibly supply power.

Appearance

Figure 5-31 Appearance of the PDC-2200WC



Version Mapping

Table 5-32 Mappings between PDC-2200WC and product models

Product	Product Model	First Supported Version	Limitations	IDs of Available Slots
CloudEngine 16804	CE16804-DC	V200R005C20	-	PM1-PM6
CloudEngine 16808	CE16808-DC	V200R005C20	-	PM1-PM10
CloudEngine 16816	CE16816-DC	V200R005C20	-	PM1-PM20

Panel

Figure 5-32 Panel of the PDC-2200WC



Table 5-33 Indicators on the PDC-2200WC

Silkscreen	Name	Color	Status	Description
INPUT	Power indicator	Green	Steady on	The input power is in the normal range.
		Green	Blinking	The input voltage is out of the normal range.
		Green	Off	The power module receives no input power.
OUTPUT	Alarm indicator	Green	Steady on	The output power is in the normal range.
		Green	Off	The power module has no output power or the output power is abnormal.
ALARM	Fault indicator	Red	Off	The power module is working properly.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The power module has failed. Possible causes include internal short circuit, fan failure, output overvoltage, overtemperature-triggered shutdown, communication failure, and severely uneven current.

Functions and Features

Table 5-34 Functions and features of the PDC-2200WC

Functions and Features	Description
Input undervoltage protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from input undervoltage, the power module can automatically resume power supply. The INPUT indicator blinks green.
Input overcurrent protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from input overcurrent, the power module cannot automatically resume power supply. The INPUT indicator is off.
Output overvoltage protection	In this protection state, the power module is turned off and stops supplying power. When the system recovers from output overvoltage, the power module cannot automatically resume power supply. The OUTPUT indicator is off.

Functions and Features	Description
Output current limiting protection	In this protection state, the output current is limited to a certain value. When the system recovers from output overcurrent, the power module can automatically resume power supply. The OUTPUT indicator is steady green.
Output short-circuit protection	In this protection state, the power module supplies power intermittently, and the output current is limited within a range. When the system recovers from output short-circuit, the power module can automatically resume power supply. The OUTPUT indicator is off.
Overtemperature protection	When the temperature exceeds 55°C (131°F), the power module stops supplying power. When the temperature falls below 55°C (131°F), the power module automatically resumes power supply. The OUTPUT indicator is off.
Heat dissipation mode	The power module has its own fans, which use a front-to-back airflow design.
Hot swapping	The power module is hot swappable.

 CAUTION

- If a power module stops supplying power in protection state and cannot automatically resume power supply after the triggering condition of the protection state is removed, remove the power module from the chassis and reinstall it at least 30s later. The power module then can work normally.
- When a power module enters overtemperature protection state, take measures to lower the temperature of the chassis. The power module can automatically resume power supply when the temperature falls within the normal range.

Technical Specifications

Table 5-35 Technical specifications of the PDC-2200WC

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	41 mm x 106.5 mm x 485 mm (1.61 in. x 4.19 in. x 19.09 in.)
Weight without packaging [kg (lb)]	2.4 kg (5.29 lb)
Number of inputs	2
Number of outputs	1
Rated input voltage [V]	-48 V to -60 V
Input voltage range [V]	-40 V to -72 V
Rated input current [A]	48 A
Maximum input current [A]	58.5 A
Rated output voltage [V]	53.5 V
Rated output current [A]	46 A
Rated output power [W]	2200 W

5.13 CR8MP2K3PHC0

Overview

The CR8MP2K3PHC0 is a 380 V three-phase AC power distribution unit (PDU).

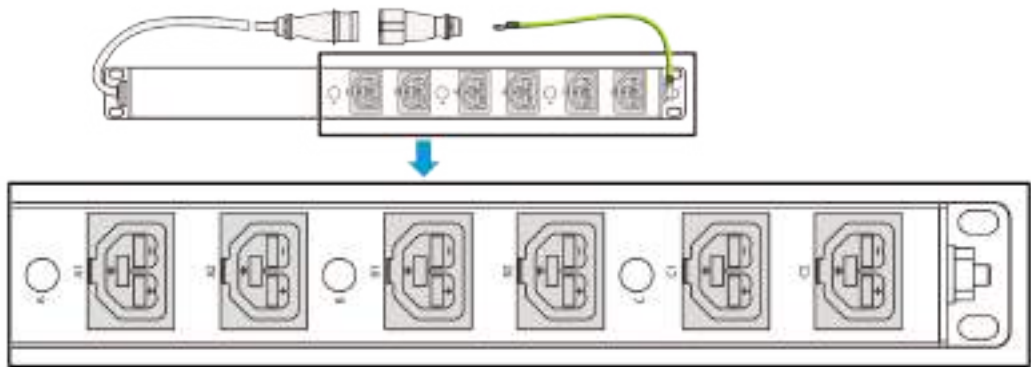
Table 5-36 Basic information about the CR8MP2K3PHC0

Item	Details
Description	Three-phase AC Power Distribution Unit
Part number	02122537
Model	CR8MP2K3PHC0
Remarks	-

Appearance

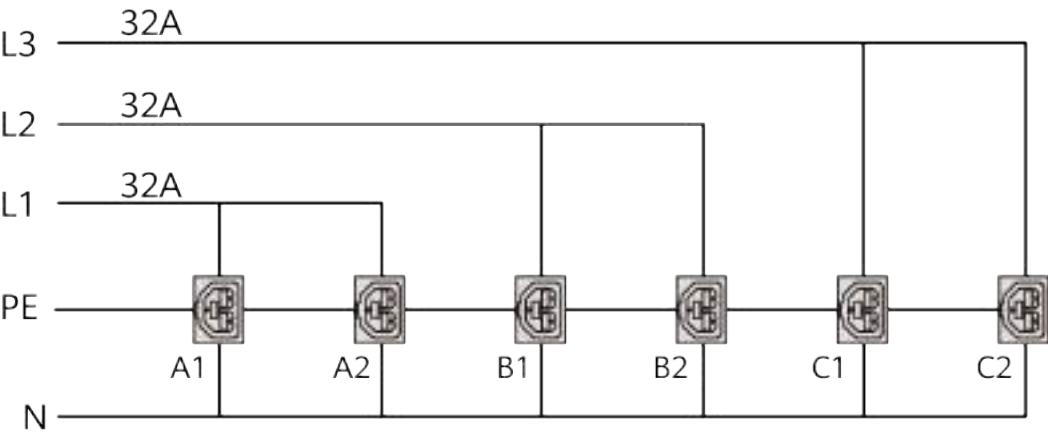
Figure 5-33 shows the appearance of the three-phase PDU.

Figure 5-33 Appearance of the three-phase PDU



Functions

A 380 V three-phase PDU supports three-phase 346 V to 415 V AC power input and provides three groups of power outputs A, B, and C. Each group controls two HVDC-3Z-03 sockets. When the groups are connected to a chassis, the total current of all sockets in each group cannot exceed 32 A.



Technical Specifications

Table 5-37 lists technical specifications of the three-phase PDU.

Table 5-37 Technical specifications of the three-phase PDU

Item	Details
Dimensions (H x W x D)	- With mounting brackets: 43.6 mm x 484 mm x 60 mm (1.72 in. x 19.06 in. x 2.36 in.) - Without mounting brackets: 43.6 mm x 442 mm x 60 mm (1.72 in. x 17.40 in. x 2.36 in.)
Hole distance for installing mounting brackets	465 mm

Item	Details
Installation mode	The 1U PDU can be installed horizontally or vertically.
Weight	≤ 5 kg
Rated input voltage range	346 V AC to 415 V AC, 50/60 Hz
Rated input current	Three-phase 32 A
Rated output voltage range	200 V AC to 240 V AC, 50/60 Hz
Rated output current	Max.6*16A
Input connector	Three-phase five-pin industrial plug (IEC 60309/250 V AC/32 A)
Power input cable of the PDU	Cross-sectional area: 6 mm ² x 5 cores; length: 3 m
Power input cable from the EoR to the PDU	Cross-sectional area: 6 mm ² x 5 cores. Each PDU is equipped with one cable and the cable length is determined based on the site survey result. Part number of the cable: 25030828 Part number of the terminal package: 02230XYB
Number of outputs and output mode	Six HVDC-3Z-03 sockets for power output

5.14 CR8MP2KHDCC0

Overview

The CR8MP2KHDCC0 is a 380 V high-voltage DC PDU and is used to supply power to a device through a 240 V high-voltage DC, 380 V high-voltage DC, or 220 V AC power distribution system.

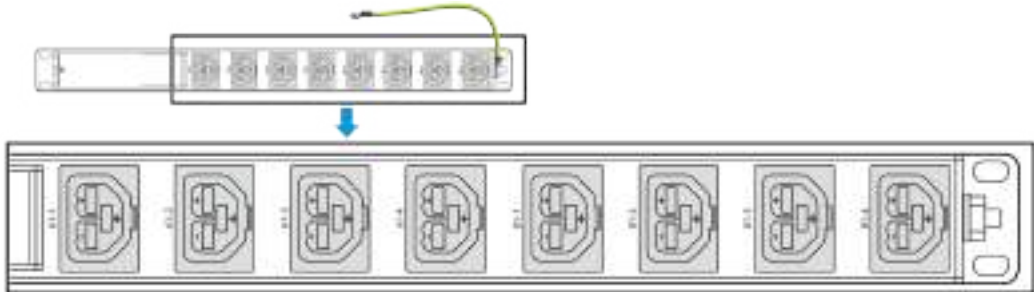
Table 5-38 Basic information about the CR8MP2KHDCC0

Item	Details
Description	380VDC HVDC and 220V AC Power Distribution Unit
Part number	02122538
Model	CR8MP2KHDCC0
Remarks	-

Appearance

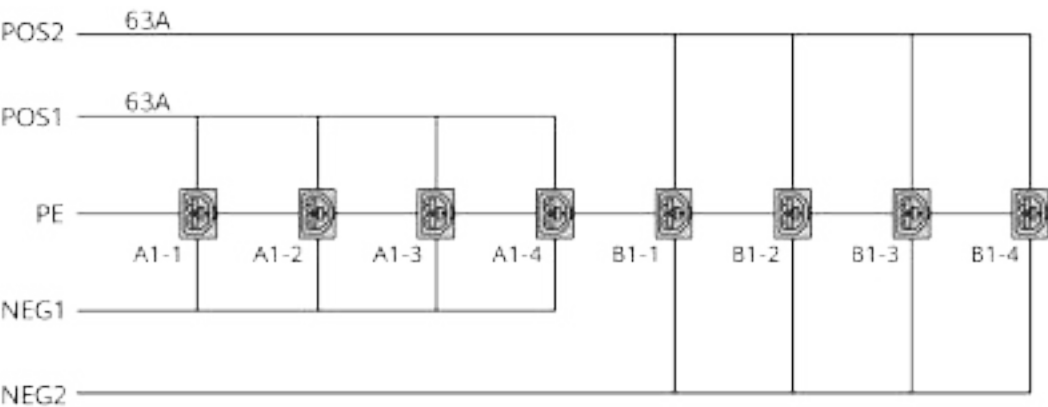
Figure 5-34 shows the appearance of the 380 V high-voltage DC PDU.

Figure 5-34 Appearance of the 380 V high-voltage DC PDU



Functions

A 380 V high-voltage DC PDU supports 240 V/380 V DC and 220V AC power input and provides two output groups A1 and B1. Each group controls four HVDC-3Z-03 sockets. When the groups are connected to a chassis, the total current of all sockets in each group cannot exceed 63 A.



Technical Specifications

Table 5-39 lists technical specifications of the 380 V high-voltage DC PDU.

Table 5-39 Technical specifications of the 380 V high-voltage DC PDU

Item	Details
Dimensions (H x W x D)	- With mounting brackets: 43.6 mm x 484 mm x 60 mm (1.72 in. x 19.06 in. x 2.36 in.) (excluding the raised terminal) - Without mounting brackets: 43.6 mm x 442 mm x 60 mm (1.72 in. x 17.40 in. x 2.36 in.) (excluding the raised terminal)

Item	Details
Hole distance for installing mounting brackets	456 mm
Installation mode	The 1U PDU can be installed horizontally or vertically.
Weight	≤ 5 kg
Rated input voltage	- HVDC: 240 V/380 V - AC: 220 V
Maximum input current	2 x 63 A
Rated output voltage	- HVDC: 240 V/380 V - AC: 220 V
Rated output current	Maximum: 8 x 16 A
Input connector	Customized, 63 A three-row terminal block
Power input cable from the EoR to the PDU	Cross-sectional area: 16 mm ² x 3 cores. Each PDU is equipped with two cables and the length of each cable is determined based on the site survey result. Part number of the cable: 25031887 Part number of the terminal package: 02233EBW
Number of outputs and output mode	Eight HVDC-3Z-03 sockets for power output

5.15 PDU2000-32-11/2-B1-3M

Overview

The PDU2000-32-11/2-B1-3M is a 220 V single-phase AC PDU.

Table 5-40 Basic information about the PDU2000-32-11/2-B1-3M

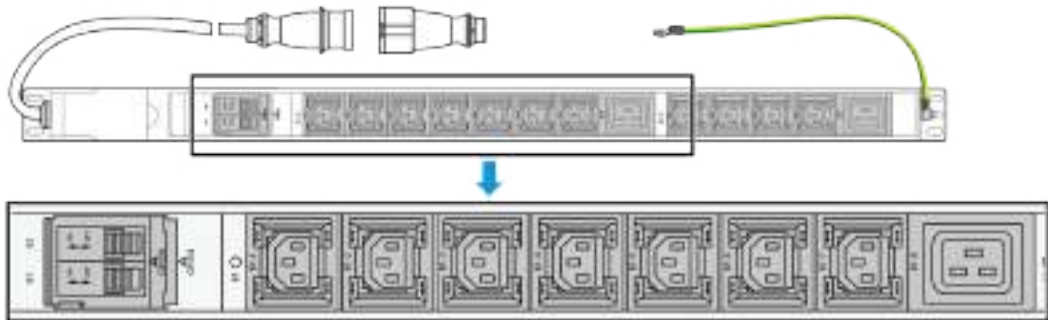
Item	Details
Description	PDU,null,PDU2000-32-1PH-11/2-B1,13,7.0kVA,3meters length of input cord
Part number	02121616
Model	PDU2000-32-11/2-B1-3M

Item	Details
Remarks	-

Appearance

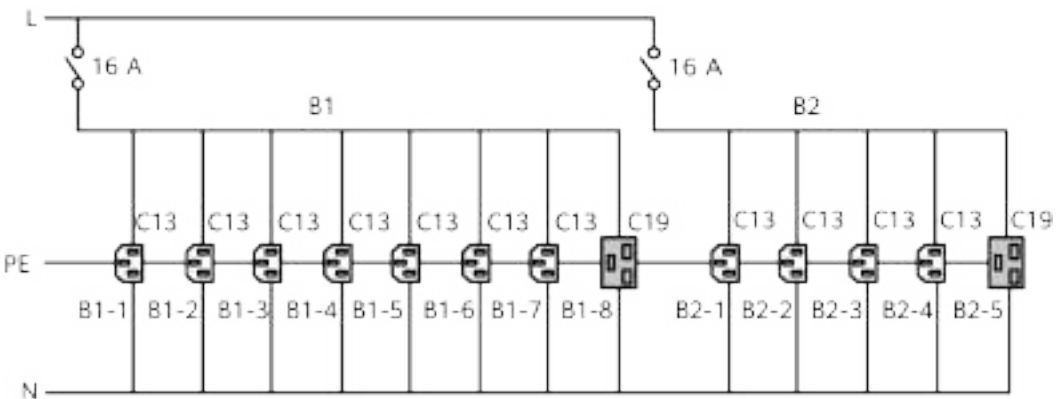
Figure 5-35 shows the appearance of the single-phase PDU.

Figure 5-35 Appearance of the single-phase PDU



Functions

A 220 V single-phase PDU supports single-phase 200 V to 240 V AC power input and provides two groups of power outputs B1 and B2. Group B1 controls seven C13 sockets and one C19 socket. Group B2 controls four C13 sockets and one C19 socket. Each group is controlled by a 16 A circuit breaker. When the groups are connected to a chassis, the total current of all sockets in each group cannot exceed 16 A.



Technical Specifications

Table 5-41 lists technical specifications of the single-phase PDU.

Table 5-41 Technical specifications of the single-phase PDU

Item	Details
Dimensions (H x W x D)	- With mounting brackets: 44.5 mm x 876 mm x 50 mm (1.75 in. x 34.49 in. x 1.97 in.) (excluding the circuit breaker) - Without mounting brackets: 44.5 mm x 818 mm x 50 mm (1.75 in. x 32.20 in. x 1.97 in.) (excluding the circuit breaker)
Hole distance for installing mounting brackets	852 mm
Installation mode	Vertical installation
Weight	≤ 5 kg
Rated input voltage range	220 V AC to 240 V AC, 50/60 Hz
Rated input current	Max. 32 A
Rated output voltage range	200 V AC to 240 V AC, 50/60 Hz
Rated output current	Max. 16 A x 2
Input connector	Three-pin industrial plug (IEC 60309/220 V AC/32 A)
Input power cable specifications	Diameter: 3 x 6 mm ² ; length: 3 m
Number of outputs and output mode	11 C13 outputs and two C19 outputs

5.16 PDU2000-32-3PH-1

Overview

The PDU2000-32-3PH-1 is a 380 V three-phase AC PDU.

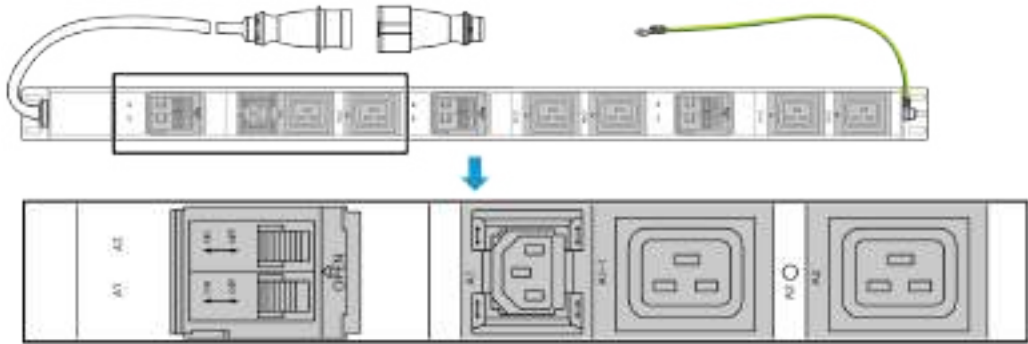
Table 5-42 Basic information about the PDU2000-32-3PH-1

Item	Details
Description	PDU,basic,PDU2000-32-3PH-1/6-B1,7,21kVA,3 meters length of input cord
Part number	02121900
Model	PDU2000-32-3PH-1
Remarks	-

Appearance

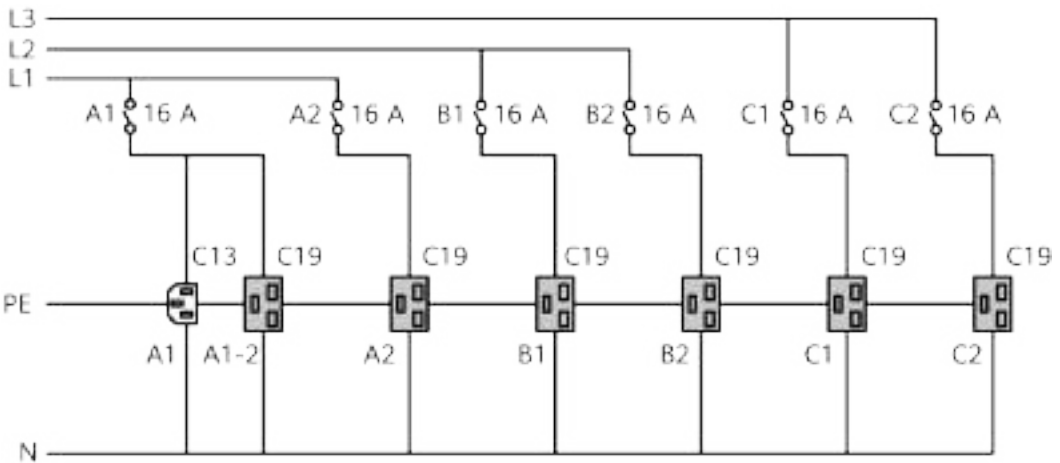
Figure 5-36 shows the appearance of the three-phase PDU.

Figure 5-36 Appearance of the three-phase PDU



Functions

A 380 V three-phase PDU supports three-phase (L1, L2, and L3) 380 V to 415 V AC input, and provides two groups of outputs on each phase line. L1 provides outputs A1 and A2; L2 provides outputs B1 and B2; L3 provides outputs C1 and C2. A1 controls one C13 socket and one C19 socket. B1, C1, A2, B2, and C2 each control one C19 socket. Each group is controlled by a 16 A circuit breaker. When the groups are connected to a chassis, the total current of all sockets in each group cannot exceed 16 A.



Technical Specifications

Table 5-43 lists technical specifications of the three-phase PDU.

Table 5-43 Technical specifications of the three-phase PDU

Item	Details
Dimensions (H x W x D)	- With mounting brackets: 44.5 mm x 876 mm x 50 mm (1.75 in. x 34.49 in. x 1.97 in.) (excluding the circuit breaker) - Without mounting brackets: 44.5 mm x 818 mm x 50 mm (1.75 in. x 32.20 in. x 1.97 in.) (excluding the circuit breaker)
Hole distance for installing mounting brackets	847 mm
Installation mode	Vertical installation
Weight	≤ 5 kg
Rated input voltage range	380 V AC to 415 V AC, 50/60Hz
Rated input current	Max. 32 A
Rated output voltage range	200 V AC to 240 V AC, 50/60Hz
Rated output current	Max. 32 A per phase
Input connector	Three-phase five-pin industrial plug (IEC 60309/250 V AC/32 A)
Input power cable specifications	Diameter: 5 x 6 mm ² ; length: 3 m
Number of outputs and output mode	One C13 output and six C19 outputs

5.17 PDU2000-32-HVDC-6/4-B1

Overview

The PDU2000-32-HVDC-6/4-B1 is a 240 V high-voltage DC PDU and is used to supply power to a device through a 240 V high-voltage DC power distribution system.

Table 5-44 Basic information about the PDU2000-32-HVDC-6/4-B1

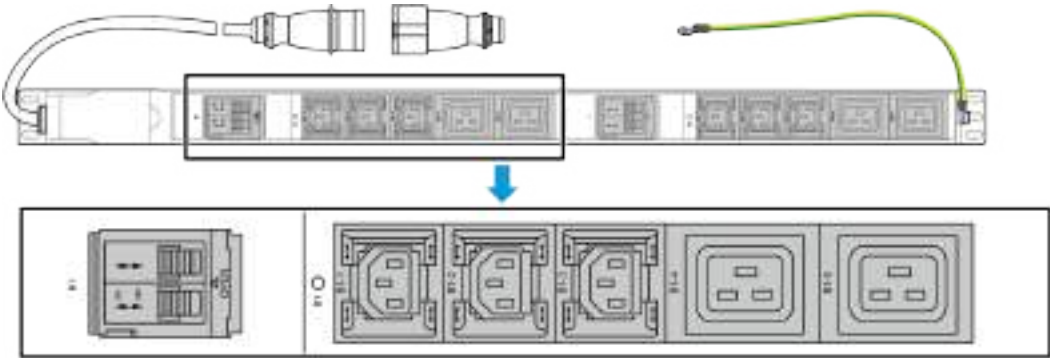
Item	Details
Description	Rack Power Distribution Unit,Basic Type,PDU2000-32-HVDC-6/4-B1,Output Outlet 6*C13+4*C19,7.6KVA
Part number	02121106
Model	PDU2000-32-HVDC-6/4-B1

Item	Details
Remarks	-

Appearance

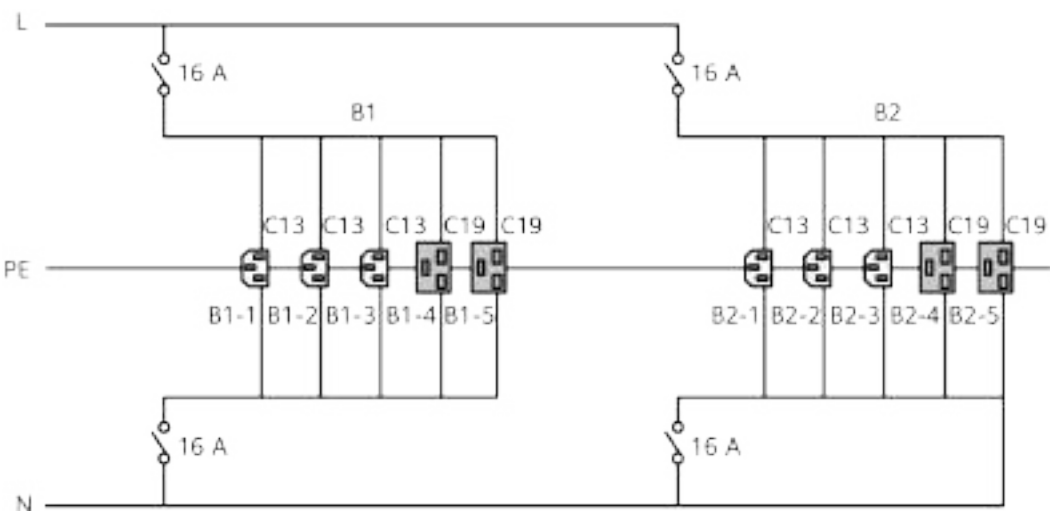
Figure 5-37 shows the appearance of the 240 V high-voltage DC PDU.

Figure 5-37 Appearance of the 240 V high-voltage DC PDU



Functions

A 240 V high-voltage DC PDU provides two output groups B1 and B2. Each group controls three C13 sockets and two C19 sockets. Each group is controlled by a 16 A circuit breaker on the L line and a 16 A circuit breaker on the N line. The circuit breakers for the same group are closed or open at the same time. When the groups are connected to a chassis, the total current of all sockets in each group cannot exceed 16 A.



Technical Specifications

Table 5-45 lists technical specifications of the 240 V high-voltage DC PDU.

Table 5-45 Technical specifications of the 240 V high-voltage DC PDU

Item	Details
Dimensions (H x W x D)	- With mounting brackets: 44.5 mm x 876 mm x 50 mm (1.75 in. x 34.49 in. x 1.97 in.) (excluding the circuit breaker) - Without mounting brackets: 44.5 mm x 818 mm x 50 mm (1.75 in. x 32.20 in. x 1.97 in.) (excluding the circuit breaker)
Hole distance for installing mounting brackets	852 mm
Installation mode	Vertical installation
Weight	≤ 5 kg
Rated input voltage	240 V DC
Maximum input current	32 A
Rated output voltage	240 V DC
Rated output current	Max. 16 A x 2
Input connector	Three-pin industrial plug (IEC 60309/240 V DC/32 A)
Input power cable specifications	Diameter: 3 x 6 mm ² ; length: 3 m
Number of outputs and output mode	Six C13 outputs and four C19 outputs

5.18 APD32-6-24

Overview

The APD32-6-24 is a 220 V single-phase AC power distribution box (PDB).

Table 5-46 Basic information about the APD32-6-24

Item	Details
Description	Power Distribution Box,FusionServer 9032,BC1E01PDU A,380V,3PH,32A,24 paths,Double 12 paths, (12*16A /12*16A),Compatible with 220V single phase, Black
Part number	02121723
Model	APD32-6-24

Item	Details
Remarks	-

Appearance

Figure 5-38 shows the appearance of the single-phase PDB.

Figure 5-38 Appearance of the single-phase PDB



Functions

A 220 V single-phase PDB supports single-phase 176 V to 276 V AC power input and provides two groups of power outputs A and B. Group A and B control twelve cord end terminal outputs respectively. When the groups are connected to a chassis, the total current of all cord end terminals in each group cannot exceed 16 A.



Technical Specifications

Table 5-47 lists technical specifications of the single-phase PDB.

Table 5-47 Technical specifications of the single-phase PDB

Item	Details
Dimensions (H x W x D)	• With mounting brackets: 128.2 mm x 482.6 mm x 665 mm (5.05 in. x 19 in. x 26.18 in.) (excluding the circuit breaker) • Without mounting brackets: 128.2 mm x 436 mm x 665 mm (5.05 in. x 17.17 in. x 26.18 in.) (excluding the circuit breaker)
Hole distance for installing mounting brackets	465.9 mm
Installation mode	The 3U PDB can be installed horizontally
Weight	6 kg
Rated input voltage range	176 V AC to 276 V AC, 47/63 Hz
Rated input current	Max. 32 A
Rated output voltage range	176 V AC to 276 V AC, 50/60 Hz
Rated output current	Max. 16 A x 12
Specifications of the input cables delivered with the PDB	Cross-sectional area: 1.5 mm ² x 3 cores; length: 2.5 m
Specifications of the input power cable from the EoS to the PDB	Cross-sectional area: 6 mm ² x 3 cores. Each PDU is equipped with six cables and the cables length is determined based on the site survey result. Part number of the cable: 25031888, 25032798 Part number of the terminal package: 02230BSC
Number of outputs and output mode	12 cord end terminal outputs

5.19 CR5MDPD12500

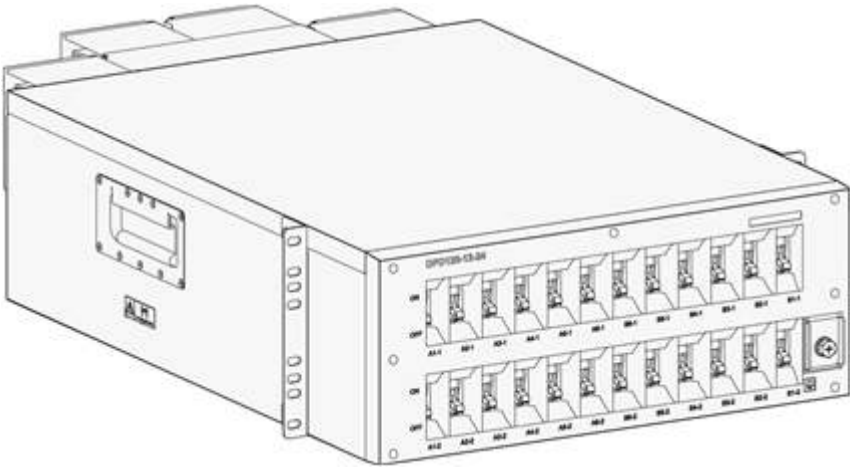
Overview

Table 5-48 Basic information about the CR5MDPD12500

Item	Details
Description	DPD125-12-24 DC Power Distribution Box
Part Number	02122145
Model	CR5MDPD12500

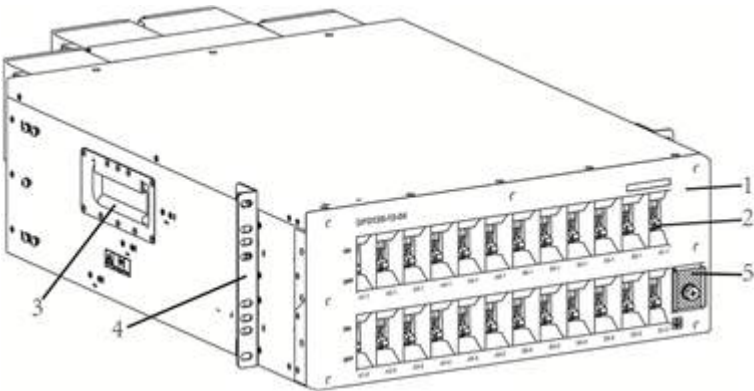
Appearance

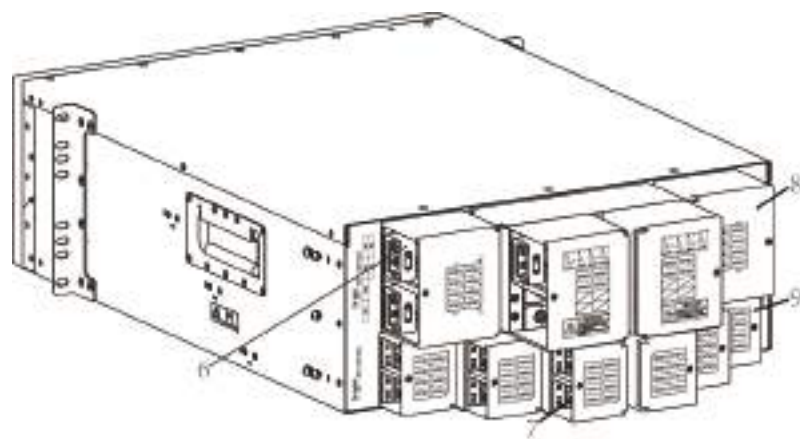
Figure 5-39 Appearance of the CR5MDPD12500



Components

Figure 5-40 Components of the CR5MDPD12500





NOTE

The fourth channel of the input terminal cannot be wiring.

Table 5-49 list the CR5MDPD12500 components and quantities.

Table 5-49 Components and Quantities

No.	Component	Quantity
1	Plastic panel	1
2	Circuit breaker	24
3	Handle	2
4	Mounting ear	2
5	Grounding terminal	1
6	Power input terminal	12
7	Power output terminal	24
8	Protective plate for power input terminals	4
9	Protective plate for power output terminals	6

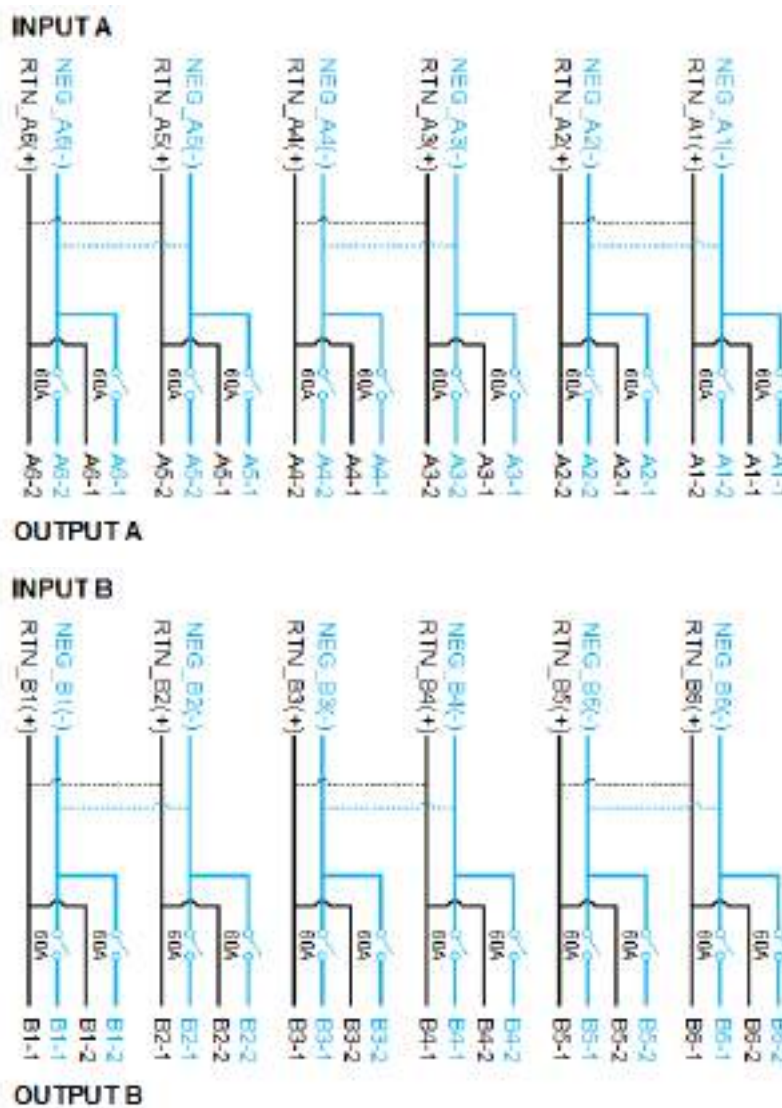
Functions and Features

The power distribution box can convert 6 or 12 DC input lines into 24 output lines. It can be used in equipment room that cannot provide enough power lines.

NOTE

- You can change the power system of the product to 6-channel input and 24-channel output by copper fitting NEG A1(-) and NEG A2(-), NEG A3(-) and NEG A4(-), NEG A5(-) and NEG A6(-), RTN A1(+) and RTN A2(+), RTN A3(+) and RTN A4(+), RTN A5(+) and RTN A6(+), NEG B1(-) and NEG B2(-), NEG B3(-) and NEG B4(-), NEG B5(-) and NEG B6(-), RTN B1(+) and RTN B1(+) and RTN B2(+), RTN B3(+) and RTN B4(+), and RTN B5(+) and RTN B6(+).
- The changed output power must be greater than or equal to the rated power of the device that is powered by the power distribution box.

Figure 5-41 Circuit principle



Technical Specifications

Table 5-50 Technical specifications of the CR5MDPD12500

Item	Specification
Dimensions without packaging (H x W x D) [mm(in.)]	175 mm x 442 mm x 740 mm (6.89 in. x 17.4 in. x 29.13 in.)
Installation mode	The 4U PDB can be installed horizontally.
Weight without packaging [kg(lb)]	30 kg (60.15 lb)
Rated input voltage [V]	-48 V/-60 V
Input voltage range [V]	-40V DC to -72V DC
Rated output voltage [V]	-48 V/-60 V
Maximum input current [A]	6 x 125 A+6 x 125 A (no copper fitting) 3 x 250 A+3 x 250 A (with copper fitting)
Maximum output current [A]	12 x 60 A + 12 x 60 A
Number of inputs	12 (no copper fitting) 6 (with copper fitting)
Number of outputs	24
Circuit breaker [A]	60 A
Input terminal	OT terminal
Output terminal	OT terminal

6 Heat Dissipation System

- [6.1 Overview of the Heat Dissipation System](#)
- [6.2 FAN-240SA-B \(FAN-240SA-B Fan Module\)](#)
- [6.3 FAN-480SA-B \(FAN-480SA-B Fan Module\)](#)
- [6.4 FAN-960SA-B \(FAN-960SA-B Fan Module\)](#)
- [6.5 CloudEngine 16804 Chassis Door](#)
- [6.6 CloudEngine 16808 Chassis Door](#)
- [6.7 CloudEngine 16816 Chassis Door](#)

6.1 Overview of the Heat Dissipation System

The heat dissipation system of CloudEngine 16800 series switches consists of fan modules and a chassis door.

- Fan modules installed at the rear of the device cool the MPUs, LPUs, and SFUs in the chassis through front-to-back airflow (from the perspective of the cabinet), ensuring a normal operating temperature range for the chassis.
Power modules have their own fans, which take the heat generated by power modules out of the chassis through front-to-back airflow. These fans ensure that the power modules work in a normal temperature range.
- The chassis door is optional and installed at the front of the chassis to shield electromagnetic noise. The door also can prevent dust from entering the chassis with airflows so that the chassis can work properly.

NOTICE

Cover vacant power module and card slots with filler panels to ensure efficient heat dissipation and EMC compliance.

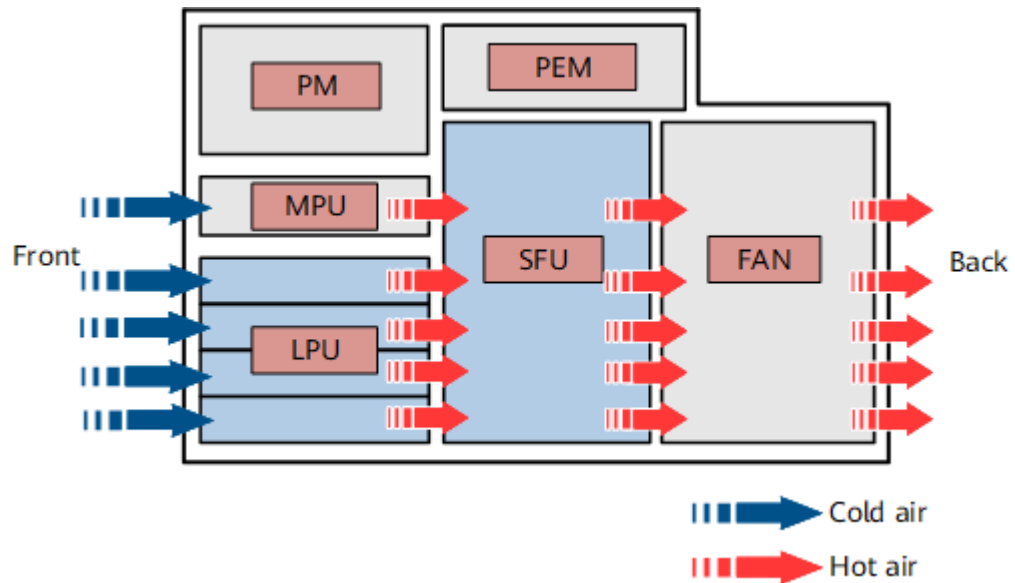
Heat Dissipation of Cards

The MPUs, LPUs, and SFUs use a front-to-back airflow design.

1. Front panels of the MPUs and LPUs have air holes. Cold air flows from these air holes into the MPUs and LPUs.
2. Cool air flows through the MPUs and LPUs to the SFUs.
3. Hot air flows through the SFUs and is exhausted from the chassis by fan modules.

Figure 6-1 shows the airflow for heat dissipation of the MPUs, LPUs, and SFUs.

Figure 6-1 Airflow for heat dissipation of the MPUs, LPUs, and SFUs (side view)



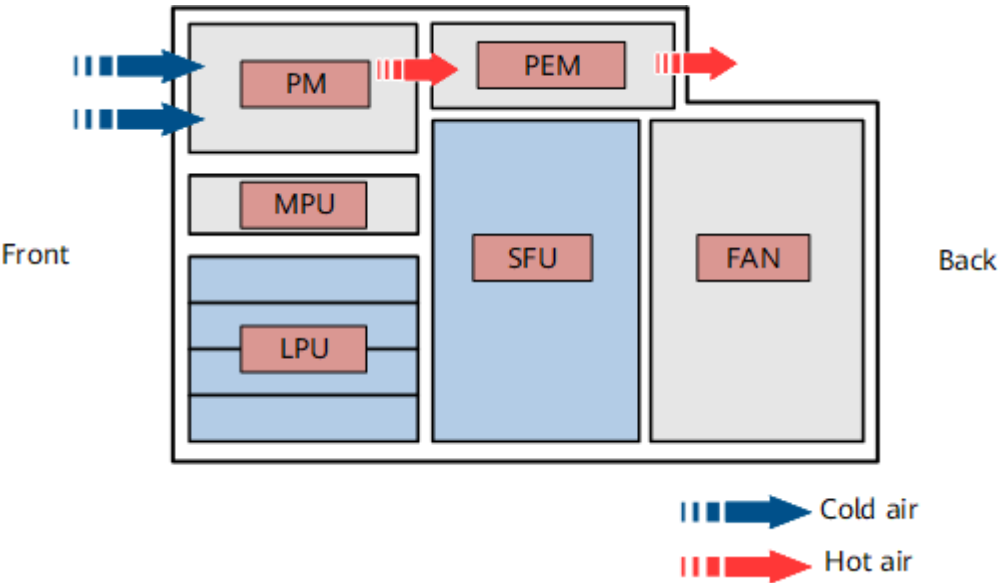
Heat Dissipation of Power Modules

The power modules use a front-to-back airflow design.

1. There are air holes on power module panels. Cold air flows into the power modules from these holes and is blown through the power module area by built-in fans of the power modules.
2. Cold air becomes hot air, passes through the power backplane, and goes out of the chassis from holes on the PEM.

Figure 6-2 shows the airflow for heat dissipation of power modules.

Figure 6-2 Airflow for heat dissipation of power modules (side view)



6.2 FAN-240SA-B (FAN-240SA-B Fan Module)

Overview

Table 6-1 Basic information about the FAN-240SA-B

Item	Details
Description	FAN-240SA-B Fan Module
Part Number	02122487
Model	FAN-240SA-B

Appearance

The FAN-240SA-B fan module consists of a fan tray, fans, and a Fan Monitoring Unit (FMU). [Figure 6-3](#) shows the appearance of the FAN-240SA-B fan module.

Figure 6-3 Appearance of the FAN-240SA-B



Version Mapping

Table 6-2 Mappings between FAN-240SA-B and product

Product	First Supported Version	Unsupported Model	Limitations	IDs of Available Slots
CloudEngine 16804	V200R005C20	-	-	FAN1–FAN3

Table 6-3 Mappings between FAN-240SA-B and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R021C00 SPC100	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10 SPC600	-
CloudEngine 16804	CE-SFU04F1-G	03059671	V200R019C10	-
CloudEngine 16804	CE-SFU04G-A	03050EGY	V200R021C00 SPC100	-
CloudEngine 16804	CE-SFU04G-A	03050EGY-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10 SPC600	-
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R019C10	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10 SPC600	-

Panel

Figure 6-4 Panel of the FAN-240SA-B



1. Captive screw	2. Status indicators	3. Handle	4. Air exhaust vent
------------------	----------------------	-----------	---------------------

Table 6-4 Indicators on the FAN-240SA-B

Silkscreen	Name	Color	Status	Description
-	Fan status indicator	Green	Fast blinking (4 Hz)	The fan module has not established communication with the MPU or communication loss occurs.
		Green	Slow blinking (0.5 Hz)	The fan module is working and communicating with the MPU properly.
		Red	Slow blinking (0.5 Hz)	An alarm is generated, and the fan module is faulty.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The fan module is abnormal. The possible cause is that the software of the FMU on the fan module is abnormal or the fan module fails.
SFU	SFU status indicator	Green	Steady on	The SFU is working properly.
		Red	Steady on	A fault that affects services has occurred. The fault cannot be rectified automatically and requires manual intervention.
		Yellow	Steady on	The fan module software is not loaded, or the communication fails, and the SFU indicator cannot be properly displayed.
		-	Steady off	The corresponding SFUs are not in position.

Functions and Features

Table 6-5 Functions and features of the FAN-240SA-B

Functions and Features	Description
Basic function	The fan module dissipates heat for the chassis so that the chassis can work properly and efficiently within the operating temperature range.
Noise reduction	When the fan module is powered on, it rotates at 48% of the full speed for at most 5 minutes. After the fan module communicates with the MPU normally, the MPU controls running of the fan module.
Automatic fan speed adjustment	After the fan module communicates with the MPU normally, the MPU controls the speed of the fan module according to temperature of cards in the chassis.
Alarm report	The fan module can report alarms on fan module failures and communication failures.
Electronic label	The MPU reads and loads the electronic label of the fan module through RS485.
Hot swapping	Supported

Technical Specifications

Table 6-6 Technical specifications of the FAN-240SA-B

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	361 mm x 129 mm x 143 mm (14.21 in. x 5.08 in. x 5.63 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Number of fans	2
Typical power consumption [W]	18 W
Maximum power consumption [W]	180 W
Maximum airflow [CFM]	580 CFM
Maximum wind pressure [Pa]	652.8 Pa

Item	Specification
Airflow direction	Air exhaust from the panel side of the fan module

6.3 FAN-480SA-B (FAN-480SA-B Fan Module)

Overview

Table 6-7 Basic information about the FAN-480SA-B

Item	Details
Description	FAN-480SA-B Fan Module
Part Number	02122486
Model	FAN-480SA-B

Appearance

The FAN-480SA-B fan module consists of a fan tray, fans, and a Fan Monitoring Unit (FMU). [Figure 6-5](#) shows the appearance of the FAN-480SA-B fan module.

Figure 6-5 Appearance of the FAN-480SA-B



Version Mapping

Table 6-8 Mappings between FAN-480SA-B and product

Product	First Supported Version	Unsupported Model	Limitations	IDs of Available Slots
CloudEngine 16808	V200R005C20	-	-	FAN1-FAN3

Table 6-9 Mappings between FAN-480SA-B and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-SFU08F-A	03050ECD	V200R021C00 SPC100	-
CloudEngine 16808	CE-SFU08F-A	03050ECD-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10 SPC600	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C00	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10 SPC600	-
CloudEngine 16808	CE-SFU08F1-G	03059672	V200R019C10	-
CloudEngine 16808	CE-SFU08G-A	03050EQK	V200R021C10 SPC600	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R005C20	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10 SPC600	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R019C10	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10 SPC600	-

Panel

Figure 6-6 Panel of the FAN-480SA-B



1. Captive screw	2. Status indicators	3. Handle	4. Air exhaust vent
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Table 6-10 Indicators on the FAN-480SA-B

Silkscreen	Name	Color	Status	Description
-	Fan status indicator	Green	Fast blinking (4 Hz)	The fan module has not established communication with the MPU or communication loss occurs.
		Green	Slow blinking (0.5 Hz)	The fan module is working and communicating with the MPU properly.
		Red	Slow blinking (0.5 Hz)	An alarm is generated, and the fan module is faulty.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The fan module is abnormal. The possible cause is that the software of the FMU on the fan module is abnormal or the fan module fails.
SFU	SFU status indicator	Green	Steady on	The SFU is working properly.
		Red	Steady on	A fault that affects services has occurred. The fault cannot be rectified automatically and requires manual intervention.
		Yellow	Steady on	The fan module software is not loaded, or the communication fails, and the SFU indicator cannot be properly displayed.
		-	Steady off	The corresponding SFUs are not in position.

Functions and Features

Table 6-11 Functions and features of the FAN-480SA-B

Functions and Features	Description
Basic function	The fan module dissipates heat for the chassis so that the chassis can work properly and efficiently within the operating temperature range.
Noise reduction	When the fan module is powered on, it rotates at 48% of the full speed for at most 5 minutes. After the fan module communicates with the MPU normally, the MPU controls running of the fan module.
Automatic fan speed adjustment	After the fan module communicates with the MPU normally, the MPU controls the speed of the fan module according to temperature of cards in the chassis.
Alarm report	The fan module can report alarms on fan module failures and communication failures.
Electronic label	The MPU reads and loads the electronic label of the fan module through RS485.
Hot swapping	Supported

Technical Specifications

Table 6-12 Technical specifications of the FAN-480SA-B

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	605 mm x 129 mm x 143 mm (23.82 in. x 5.08 in. x 5.63 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Number of fans	4
Typical power consumption [W]	36 W
Maximum power consumption [W]	360 W
Maximum airflow [CFM]	1160 CFM
Maximum wind pressure [Pa]	652.8 Pa

Item	Specification
Airflow direction	Air exhaust from the panel side of the fan module

6.4 FAN-960SA-B (FAN-960SA-B Fan Module)

Overview

Table 6-13 Basic information about the FAN-960SA-B

Item	Details
Description	FAN-960SA-B Fan Module
Part Number	02122582
Model	FAN-960SA-B

Appearance

The FAN-960SA-B fan module consists of a fan tray, fans, and a Fan Monitoring Unit (FMU). [Figure 6-7](#) shows the appearance of the FAN-960SA-B fan module.

Figure 6-7 Appearance of the FAN-960SA-B



Version Mapping

Table 6-14 Mappings between FAN-960SA-B and product

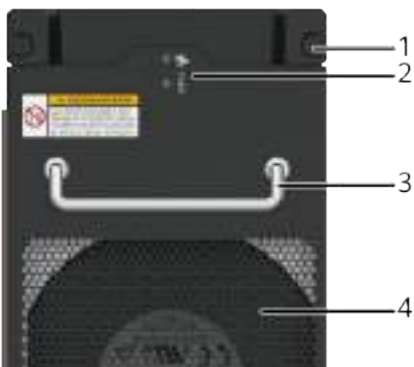
Product	First Supported Version	Unsupported Model	Limitations	IDs of Available Slots
CloudEngine 16816	V200R005C20	-	-	FAN1-FAN3

Table 6-15 Mappings between FAN-960SA-B and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-SFU16F-G	03058885	V200R005C20	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10 SPC600	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-
CloudEngine 16816	CE-SFU16G-A	03050EQJ	V200R021C10 SPC600	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R005C20	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10 SPC600	-
CloudEngine 16816	CE-SFU16G-GK	03059632	V200R019C10	-
CloudEngine 16816	CE-SFU16G-GK	03059632-002	V200R021C10 SPC600	-

Panel

Figure 6-8 Panel of the FAN-960SA-B



1. Captive screw	2. Status indicators	3. Handle	4. Air exhaust vent
------------------	----------------------	-----------	---------------------

Table 6-16 Indicators on the FAN-960SA-B

Silkscreen	Name	Color	Status	Description
-	Fan status indicator	Green	Fast blinking (4 Hz)	The fan module has not established communication with the MPU or communication loss occurs.
		Green	Slow blinking (0.5 Hz)	The fan module is working and communicating with the MPU properly.
		Red	Slow blinking (0.5 Hz)	An alarm is generated, and the fan module is faulty.
		Yellow	Steady on	The fan module is abnormal. The possible cause is that the software of the FMU on the fan module is abnormal or the fan module fails.
SFU	SFU status indicator	Green	Steady on	The SFU is working properly.
		Red	Steady on	A fault that affects services has occurred. The fault cannot be rectified automatically and requires manual intervention.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The fan module software is not loaded, or the communication fails, and the SFU indicator cannot be properly displayed.
		-	Steady off	The corresponding SFUs are not in position.

Functions and Features

Table 6-17 Functions and features of the FAN-960SA-B

Functions and Features	Description
Basic function	The fan module dissipates heat for the chassis so that the chassis can work properly and efficiently within the operating temperature range.
Noise reduction	When the fan module is powered on, it rotates at 48% of the full speed for at most 5 minutes. After the fan module communicates with the MPU normally, the MPU controls running of the fan module.
Automatic fan speed adjustment	After the fan module communicates with the MPU normally, the MPU controls the speed of the fan module according to temperature of cards in the chassis.
Alarm report	The fan module can report alarms on fan module failures and communication failures.
Electronic label	The MPU reads and loads the electronic label of the fan module through RS485.

Functions and Features	Description
Hot swapping	Supported

Technical Specifications

Table 6-18 Technical specifications of the FAN-960SA-B

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	1193 mm x 129 mm x 143 mm (46.97 in. x 5.08 in. x 5.63 in.)
Weight without packaging [kg (lb)]	10.4 kg (22.93 lb)
Number of fans	8
Typical power consumption [W]	72 W
Maximum power consumption [W]	720 W
Maximum airflow [CFM]	2320 CFM
Maximum wind pressure [Pa]	652.8 Pa
Airflow direction	Air exhaust from the panel side of the fan module

6.5 CloudEngine 16804 Chassis Door

Overview

The chassis door is optional and is installed at the front side of the chassis. It is used to shield electromagnetic noise and can protect the air intake vents of MPUs and LPUs from dust.

- Install a door on a chassis when the chassis is installed in a dusty environment or a cabinet without doors.
- A chassis door must be configured for a device that complies with FCC standards.

Table 6-19 Basic attributes of the CloudEngine 16804 chassis door

Item	Details
Description	Chassis door
Part number	02352TPK

Appearance

Figure 6-9 Appearance of the CloudEngine 16804 chassis door



Components

Figure 6-10 Structure of the CloudEngine 16804 chassis door

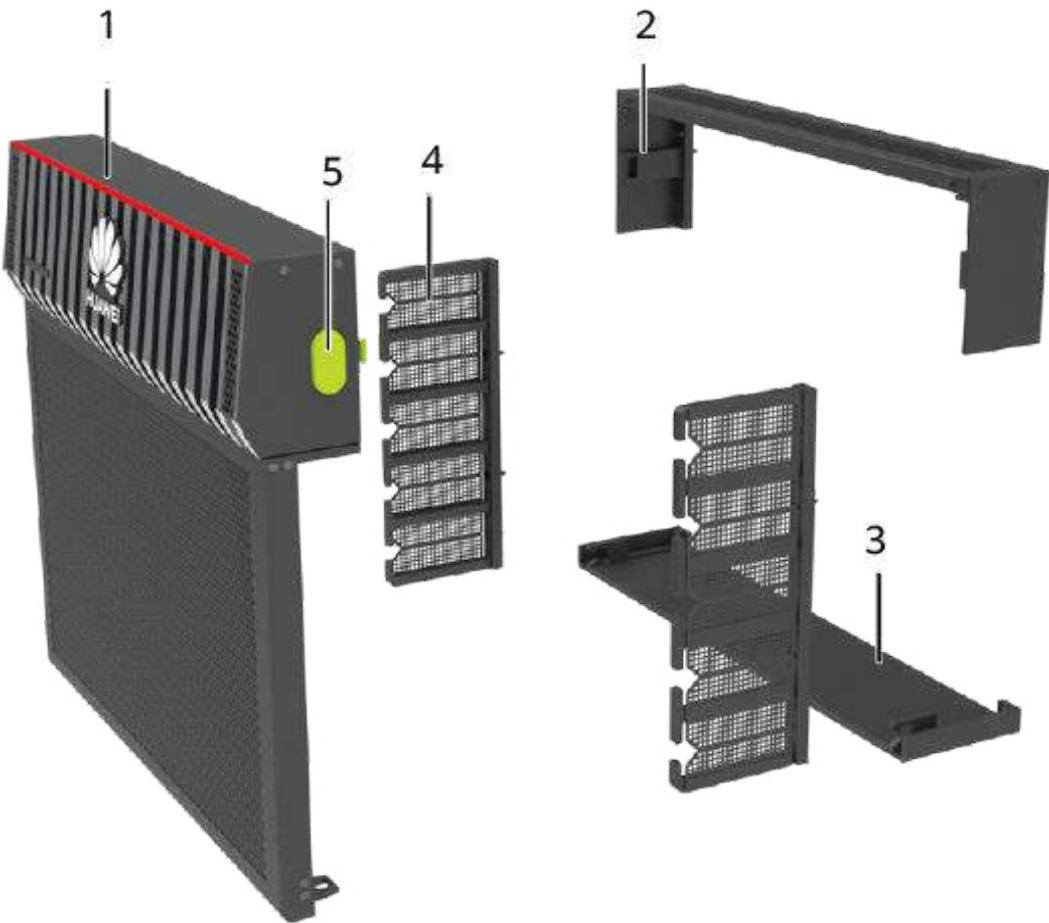


Table 6-20 Components

1. Chassis door body	2. Upper enclosure frame	3. Lower enclosure frame	4. Cable management frame (including the rubber part)	5. Release button
----------------------	--------------------------	--------------------------	---	-------------------

Technical Specifications

Table 6-21 Technical specifications of the CloudEngine 16804 chassis door

Item	Description
Dimensions (H x W x D)	437.5 mm x 482.6 mm x 135 mm (17.22 in. x 19.00 in. x 5.31 in.)
Weight	4.0 kg (8.82 lb)

6.6 CloudEngine 16808 Chassis Door

Overview

The chassis door is optional and is installed at the front side of the chassis. It is used to shield electromagnetic noise and can protect the air intake vents of MPUs and LPUs from dust.

- Install a door on a chassis when the chassis is installed in a dusty environment or a cabinet without doors.
- A chassis door must be configured for a device that complies with FCC standards.

Table 6-22 Basic attributes of the CloudEngine 16808 chassis door

Item	Details
Description	Chassis door
Part number	02352TPL

Appearance

Figure 6-11 Appearance of the CloudEngine 16808 chassis door



Components

Figure 6-12 Structure of the CloudEngine 16808 chassis door

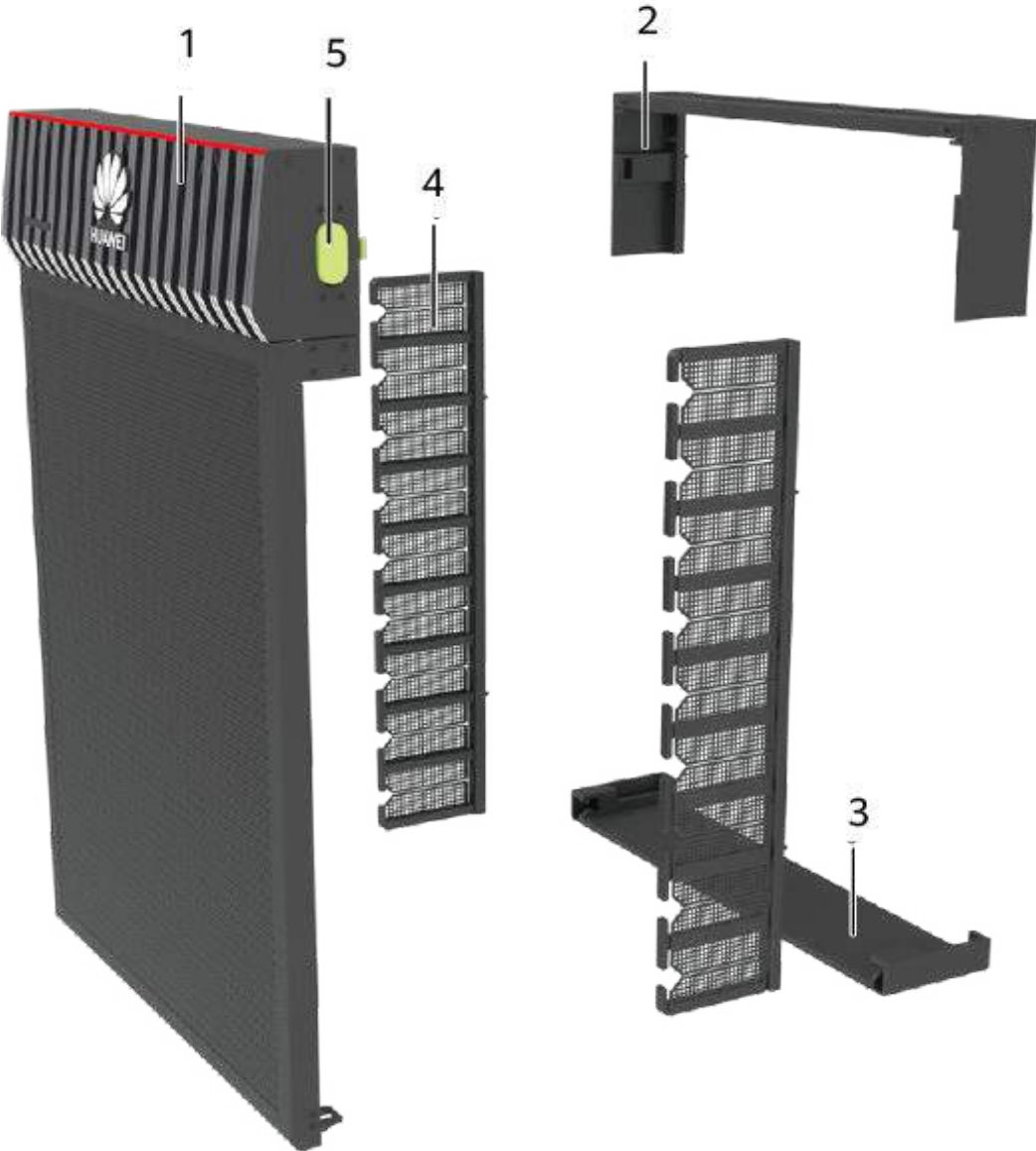


Table 6-23 Components

1. Chassis door body	2. Upper enclosure frame	3. Lower enclosure frame	4. Cable management frame (including the rubber part)	5. Release button
----------------------	--------------------------	--------------------------	---	-------------------

Technical Specifications

Table 6-24 Technical specifications of the CloudEngine 16808 chassis door

Item	Description
Dimensions (H x W x D)	703.6 mm x 482.6 mm x 135 mm (27.70 in. x 19.00 in. x 5.31 in.)
Weight	6.7 kg (14.77 lb)

6.7 CloudEngine 16816 Chassis Door

Overview

The chassis door is optional and is installed at the front side of the chassis. It is used to shield electromagnetic noise and can protect the air intake vents of MPUs and LPUs from dust.

- Install a door on a chassis when the chassis is installed in a dusty environment or a cabinet without doors.
- A chassis door must be configured for a device that complies with FCC standards.

Table 6-25 Basic attributes of the CloudEngine 16816 chassis door

Item	Description
Description	Chassis door
Part number	02352TPM

Appearance

Figure 6-13 Appearance of the CloudEngine 16816 chassis door



Components

Figure 6-14 Structure of the CloudEngine 16816 chassis door

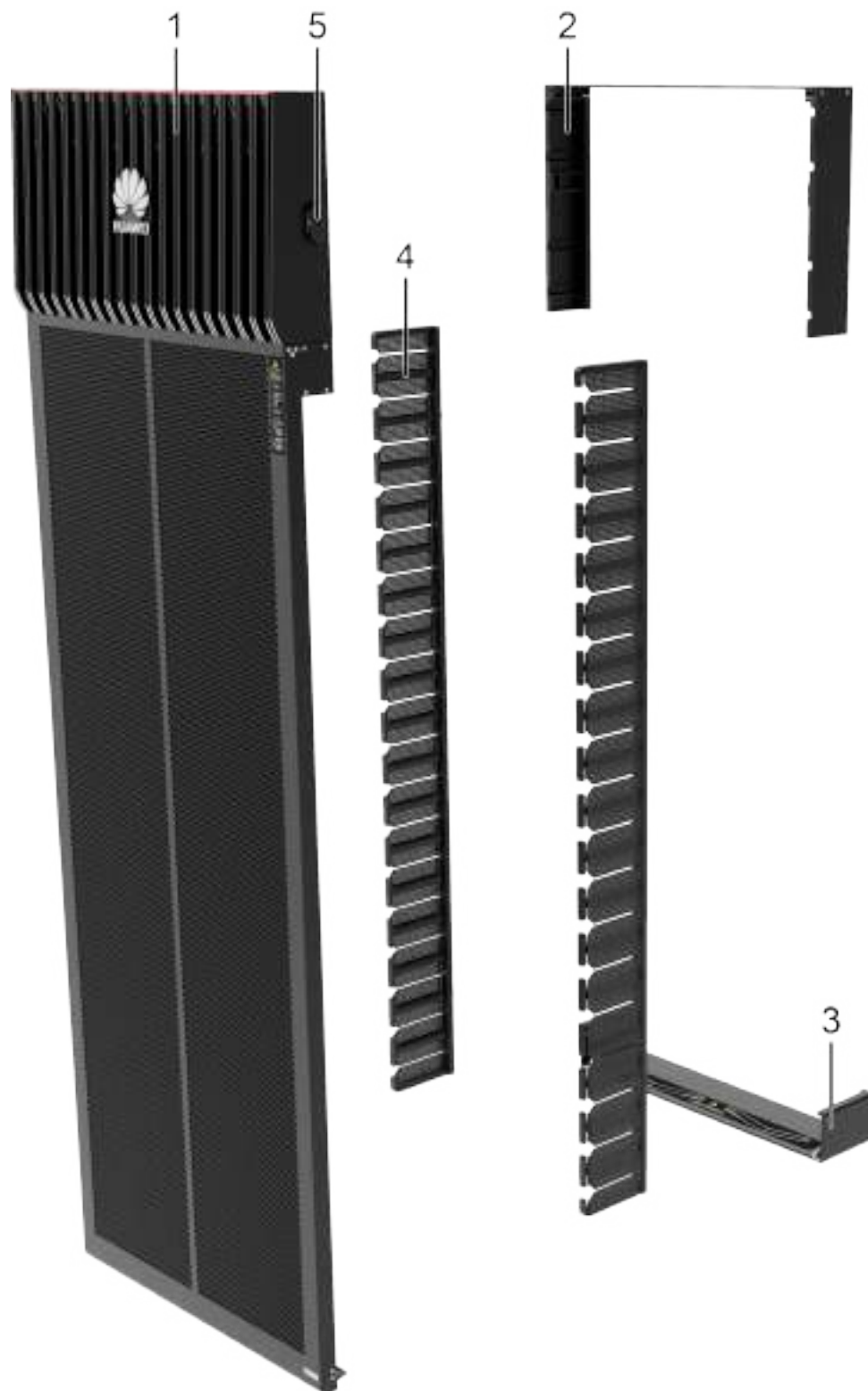


Table 6-26 Components

1. Chassis door body	2. Upper enclosure frame	3. Lower enclosure frame	4. Cable management frame (including the rubber part)	5. Release button
----------------------	--------------------------	--------------------------	---	-------------------

Technical Specifications

Table 6-27 Technical specifications of the CloudEngine 16816 chassis door

Item	Description
Dimensions (H x W x D)	1435.7 mm x 482.6 mm x 135 mm (56.52 in. x 19.00 in. x 5.31 in.)
Weight	8.0 kg (17.64 lb)

7 Cards

- [7.1 Card Introduction](#)
- [7.2 MPU](#)
- [7.3 SFU](#)
- [7.4 10GE Interface Card](#)
- [7.5 40GE Interface Card](#)
- [7.6 100GE Interface Card](#)

7.1 Card Introduction

7.1.1 Card Classification

NOTICE

- This section describes all the cards supported by the device. The available cards are specified in the product change notices (PCNs). For more details, contact the product manager at your local Huawei office.
- **The MPUs installed in the active and standby MPU slots must be of the same type. For details about the MPU types supported by this version, see "MPU".**
- **Some cards are attached with labels of different colors near the ejector levers to identify their types. LPUs or SFUs of different types cannot be used together. Otherwise, they will be damaged.**
- For detailed mapping between cards, see the "Version Mapping" tabs for the corresponding cards.

Depending on functions, the cards of CloudEngine 16800 series switches devices are classified into the following types: MPU, LPU, and SFU. [Table 7-1](#) describes these cards.

The LPUs are available in G, GK, and A series, as listed in [Table 7-2](#). The SFUs are available in G, GK, and A series, as listed in [Table 7-3](#).

Table 7-1 Cards supported by the CloudEngine 16800 series switches devices

Card Type	Full Name	Function
MPU	Main Processing Unit	Responsible for system control, management, and monitoring
SFU	Switch Fabric Unit	Responsible for line-rate data switching in the data plane
LPU	Line Processing Unit	Responsible for data packet processing and traffic management

Table 7-2 LPUs supported by the CloudEngine 16800 series switches

Series	Name
G series	CEL48XSFD-G, CEL48XSFD1-G, CEL24LQFD-G, CEL24LQFD1-G, CEL36LQFD-G, CEL36LQFD1-G, CEL18CQFD-G, CEL36CQFD-G
GK series	CEL48XSFD-GK, CEL24LQFD-GK, CEL36LQFD-GK, CEL18CQFD-GK, CEL36CQFD-GK
A series	CEL48XSFD-A, CEL36LQFD-A, CEL18CQFD-A, CEL36CQFD-A

Table 7-3 SFUs supported by the CloudEngine 16800 series switches

Series	Name
G series	CE-SFU04F-G, CE-SFU04F1-G, CE-SFU04G-G, CE-SFU08F-G, CE-SFU08F1-G, CE-SFU08G-G, CE-SFU16F-G, CE-SFU16G-G, CE-SFU16F-G1
GK series	CE-SFU04G-GK, CE-SFU08G-GK, CE-SFU08F-GK, CE-SFU16G-GK
A series	CE-SFU04F-A, CE-SFU04G-A, CE-SFU08F-A, CE-SFU08G-A, CE-SFU16G-A

7.1.2 Card Naming Conventions

MPU Naming Conventions

Figure 7-1 MPU naming conventions

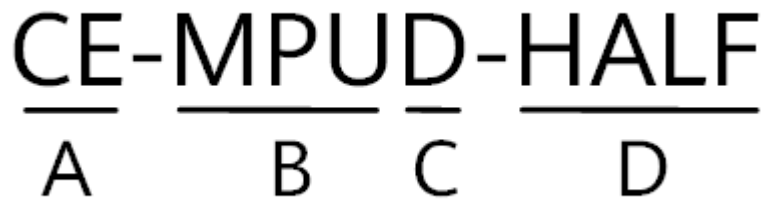


Table 7-4 MPU naming conventions

Identifier	Meaning	Description
A	Brand name	This field is fixed at CE, representing CloudEngine.
B	MPU	The card is an MPU.
C	Version	This field indicates the MPU version.
D	Extended field	• HALF/HALF2/HALF2E/HALF3E: half-width MPU • FULL/FULL2/FULL2E/FULL3E: full-width MPU • HK/H2EK: half-width MPU that has passed the certification of the Security and Reliability Technology Industry Alliance • FK/F2K: full-width MPU that has passed the certification of the Security and Reliability Technology Industry Alliance

SFU Naming Conventions

Figure 7-2 SFU naming conventions

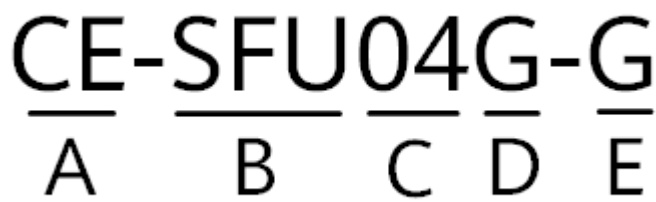


Table 7-5 SFU naming conventions

Identifier	Meaning	Description
A	Brand name	This field is fixed at CE, representing CloudEngine.
B	SFU	The card is an SFU.
C	Chassis model	The last two digits (number of slots) of the chassis name indicate the chassis model.
D	Version	This field indicates the SFU version.
E	Extended field	This field indicates SFUs with different types of chips, helping you quickly learn the mapping between LPUs and SFUs. K indicates an SFU that has passed the certification of the Security and Reliability Technology Industry Alliance.

LPU Naming Conventions

Figure 7-3 LPU naming conventions



Table 7-6 LPU naming conventions

Identifier	Meaning	Description
A	Brand name	This field is fixed at CE, representing CloudEngine.
B	LPU type	• L: LPU • F: FPIC
C	Number of ports	For an LPU with different types of ports, this field indicates the number of downlink ports. It has two digits. The first digit is 0 if the card has fewer than 10 ports.

Identifier	Meaning	Description
D	Port rate	• G: GE port • X: 10GE/25GE port • L: 40GE/50GE port • C: 100GE port • D: 400GE port
E	Port type	• T: BASE-T port • S: SFP/SFP+/SFP28 port • X: XFP port • Q: QSFP+/QSFP28/QSFP-DD port • F: CFP/CFP2/CFP4 port • C: CXP port
F	LPU specifications	• FD/FD1/HG: enhanced specifications • HGA: enhanced specifications with combo ports
G	Extended field	This field indicates the interface card of a specific chip type: • K: LPU that has passed the certification of the Security and Reliability Technology Industry Alliance. • SAN: LPU specifically used to connect to a storage device.

7.1.3 Card Structure and Dimensions

Card Structure

Figure 7-4 shows the structure of a card.

Figure 7-4 Structure of a card

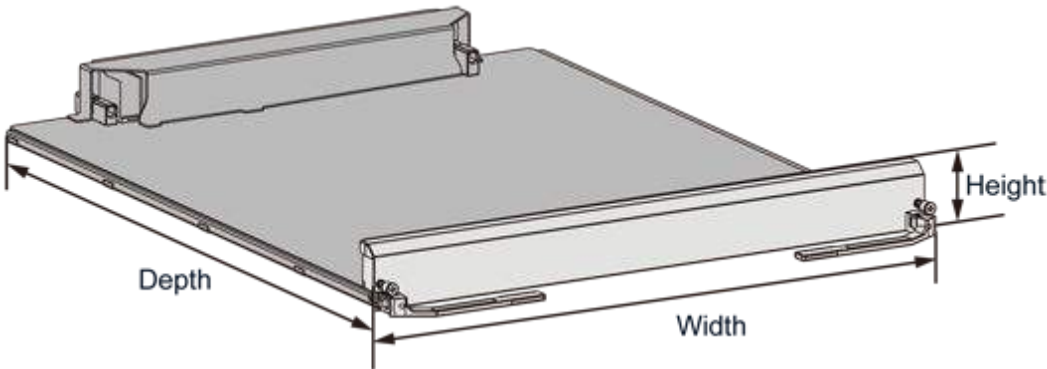


1. Card name label	2. Barcode label	3. Indicator
4. Captive screw NOTE This component is used to fix a card into the chassis.	5. Ejector lever NOTE This component allows you to insert and remove a card.	6. Front panel plate NOTE The front panel plate contains the card name label, indicators, and ports. It is also attached with additional labels like the barcode label and laser label.
7. Ports	8. Printed circuit board (PCB) NOTE - The PCB contains all the functional modules of the card and is the core component of the card. The PCB provides indicators, buttons, and ports on the front panel plate. - Different cards provide different indicators, buttons, and ports. For details, see the description of specific cards.	-

Card Dimensions

Figure 7-5 shows how the height, width, and depth of a card are measured.

Figure 7-5 Card dimensions



NOTE

The card dimensions are defined as follows:

- Height: refers to the height of the front panel.
- Width: refers to the longest distance between the tops of two ejector levers.
- Depth: refers to the distance between the front panel and the card connector.

7.1.4 Port Numbering of Cards

On the CloudEngine 16800 series switches, ports are numbered in the format of stack member ID/slot ID/subcard ID/port number.

- Stack member ID: A port number does not contain the stack member ID if the device is not in a stack.
- Slot ID: identifies in which slot an LPU is installed.

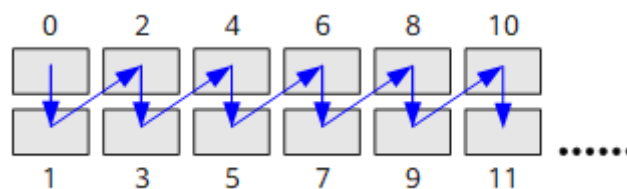
Depending on the switch model, the slot ID range is as follows:

- CloudEngine 16804: 1–4
- CloudEngine 16808: 1–8
- CloudEngine 16816: 1–16

- Subcard ID: Cards on the CloudEngine 16800 series switches do not support subcards, so the subcard ID is always 0.
- Port number: indicates the number of each port on a card.

Generally, there are two rows of ports on a card of the CloudEngine 16800 series switches. These ports are numbered from top to bottom and left to right starting from 0. For details about the port numbers, see the appearance of the corresponding card.

Figure 7-6 Port numbers (for reference only)



For example, if a CEL48XSFD-G card is installed in slot 3 of the CloudEngine 16800 series switch, the first 10GE port is numbered 10GE 3/0/0.

40GE, and 100GE ports can be split. Converted ports are numbered using the dimension numbering rule. For example, when a 40GE port numbered 40GE x/y/n is split into four 10GE ports, the converted 10GE ports are numbered 40GE x/y/n:1, 40GE x/y/n:2, 40GE x/y/n:3, and 40GE x/y/n:4.

7.1.5 Port Rate Description

The port rate is the nominal rate at which a port transmits data. To meet various transmission rate requirements, pluggable modules with different rates are provided, including 100GE, 40GE, 10GE, and GE optical modules, and GE copper modules. Generally, the rate supported by a port corresponds to the rate of the

pluggable module on the port. However, the rate may be affected by the following factors.

Port Rate Auto-Sensing

Typically, port rate auto-sensing allows a port to automatically work at the rate of the medium that is installed, and you do not need to run a command to change the port rate.

Port Auto-Negotiation

Unlike auto-sensing, auto-negotiation enables connected ports to exchange information. When two ports are connected through media that support rate auto-negotiation, they can negotiate some port attribute settings and automatically configure their transmission capabilities. The duplex mode, rate, and FEC of the two ports are negotiated. Ports at both ends of a physical link automatically select the same operating parameter settings by exchanging information. In this way, the ports can operate at their maximum supported transmission capability.

Port Split

Port split allows a high-bandwidth physical port on the switch to be split into multiple independent lower-bandwidth ports. With the port split function, ports on a card can connect to various types of ports on the remote device, allowing for flexible networking and lowering hardware costs. For example, the 40GE port on the CEL36LQFD-G can be split into four 10GE ports.

Port Rate Configuration

In addition to the preceding methods, you can also run the command to change the interface rate. For details, see Configuration Guide > Interface Management > Configuring the Interface Rate.

7.2 MPU

7.2.1 CE-MPUD-HALF (CE16800 Main Processing Unit D (half-width))

7.2.1.1 CE-MPUD-HALF (03058876: CE16800 Main Processing Unit D (half-width))

Overview

Table 7-7 Basic information about the CE-MPUD-HALF

Item	Details
Description	CE16800 Main Processing Unit D (half-width)
Part Number	03058876
Model	CE-MPUD-HALF
Silkscreen	CE-MPUD-HALF
Card Type	MPU

Appearance

Figure 7-7 Appearance of the CE-MPUD-HALF



Version Mapping

Table 7-8 Mappings between CE-MPUD-HALF and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	5-6	-
CloudEngine 16808	V200R005C20	-	9-10	-

Table 7-9 Mappings between CE-MPUD-HALF and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C00 SPC100	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R005C20	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R020C10	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R005C20	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R020C10	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R005C20	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-

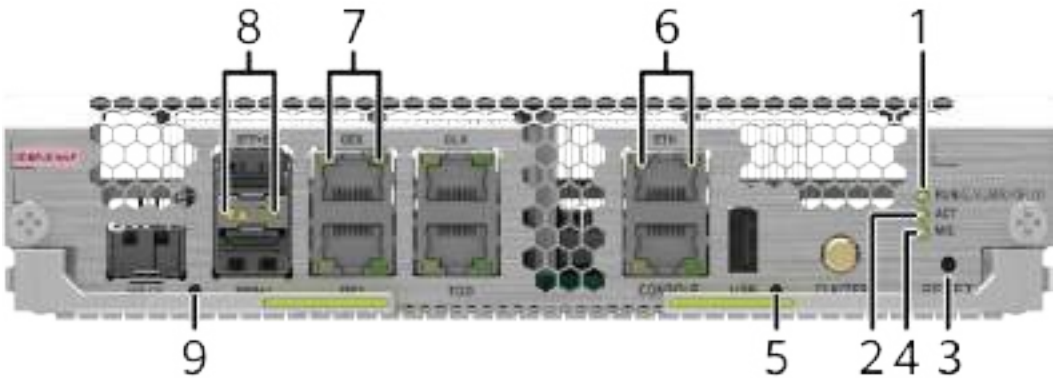
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C00 SPC100	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R005C20	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C00 SPC100	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R005C20	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R019C10	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C00 SPC100	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R005C20	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R020C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R005C20	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R020C10	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R005C20	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C00 SPC100	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R005C20	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C00 SPC100	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R005C20	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R019C10	-

Indicators and Buttons

Figure 7-8 Indicators and buttons on the CE-MPUD-HALF



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
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6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-
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 NOTE

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-10 Indicators on the CE-MPUD-HALF

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.

Silkscreen	Name	Color	Status	Description
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.

Silkscreen	Name	Color	Status	Description
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.

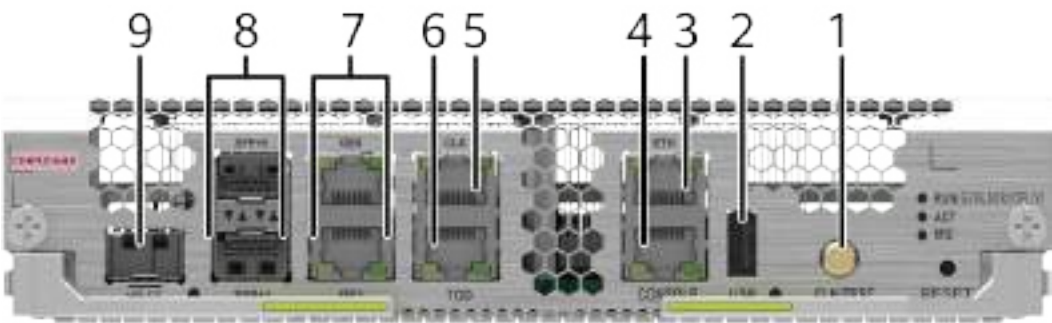
Silkscreen	Name	Color	Status	Description
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-11 Buttons on the CE-MPUD-HALF

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-9 Ports on the CE-MPUD-HALF



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

 NOTE

The device does not support the HP-GE port.

Table 7-12 Ports on the CE-MPUD-HALF

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP+1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default. NOTE This function is available only in V200 and is reserved in V300.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default. NOTE This function is available only in V200 and is reserved in V300.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-13 Functions and features of the CE-MPUD-HALF

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-14 Technical specifications of the CE-MPUD-HALF

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	3.1 kg (6.83 lb)
Weight with packaging [kg (lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.1.2 CE-MPUD-HALF (03058876-001: CE16800 Main Processing Unit D (half-width))

Overview

Table 7-15 Basic information about the CE-MPUD-HALF

Item	Details
Description	CE16800 Main Processing Unit D (half-width)
Part Number	03058876-001
Model	CE-MPUD-HALF
Silkscreen	CE-MPUD-HALF
Card Type	MPU

Appearance

Figure 7-10 Appearance of the CE-MPUD-HALF



Version Mapping

Table 7-16 Mappings between CE-MPUD-HALF and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	5-6	-
CloudEngine 16808	V200R021C10 SPC600	-	9-10	-

Table 7-17 Mappings between CE-MPUD-HALF and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 2	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 3	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-

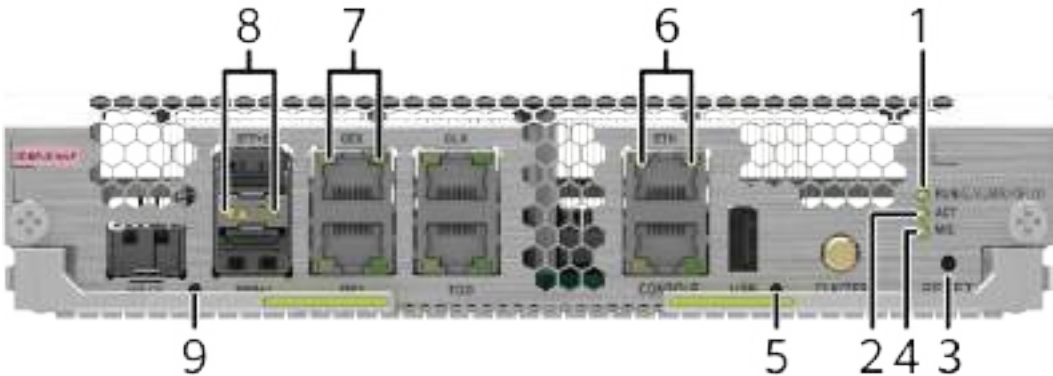
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-11 Indicators and buttons on the CE-MPUD-HALF



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-18 Indicators on the CE-MPUD-HALF

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.

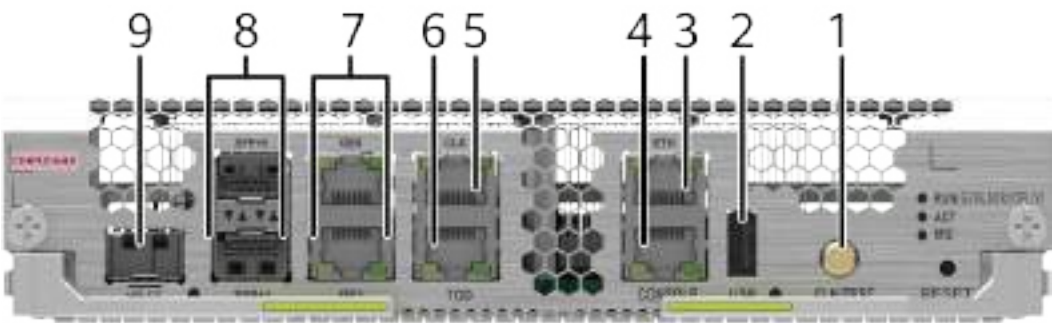
Silkscreen	Name	Color	Status	Description
		Yellow	Off	The port is not transmitting or receiving data.
-	Two single-color indicators for each SFP+ optical port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical module indicator	Green	Steady on	A link has been established on the port.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-19 Buttons on the CE-MPUD-HALF

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-12 Ports on the CE-MPUD-HALF



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

 NOTE

The device does not support the HP-GE port.

Table 7-20 Ports on the CE-MPUD-HALF

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB flash drive
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP+1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-21 Functions and features of the CE-MPUD-HALF

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-22 Technical specifications of the CE-MPUD-HALF

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	3.1 kg (6.83 lb)
Weight with packaging [kg (lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.2 CE-MPUD-FULL (CE16800 Main Processing Unit D (full-width))

7.2.2.1 CE-MPUD-FULL (03058877: CE16800 Main Processing Unit D (full-width))

Overview

Table 7-23 Basic information about the CE-MPUD-FULL

Item	Details
Description	CE16800 Main Processing Unit D (full-width)
Part Number	03058877
Model	CE-MPUD-FULL
Silkscreen	CE-MPUD-FULL
Card Type	MPU

Appearance

Figure 7-13 Appearance of the CE-MPUD-FULL



Version Mapping

Table 7-24 Mappings between CE-MPUD-FULL and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R005C20	-	17-18	-

Table 7-25 Mappings between CE-MPUD-FULL and Interface Card

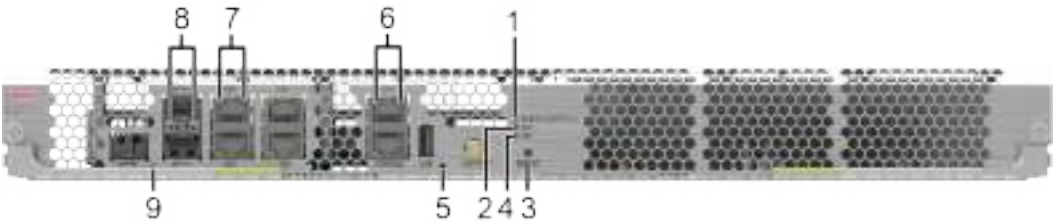
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R005C20	-
CloudEngine 16816	CEL18CQFD-G	03058870-00 1	V200R020C10	-
CloudEngine 16816	CEL18CQFD-G	03058870-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R005C20	-
CloudEngine 16816	CEL24LQFD-G	03058872-00 1	V200R020C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R005C20	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R005C20	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL48XSFD-G	03058873	V200R005C20	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R019C10	-

Indicators and Buttons

Figure 7-14 Indicators and buttons on the CE-MPUD-FULL



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-26 Indicators on the CE-MPUD-FULL

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

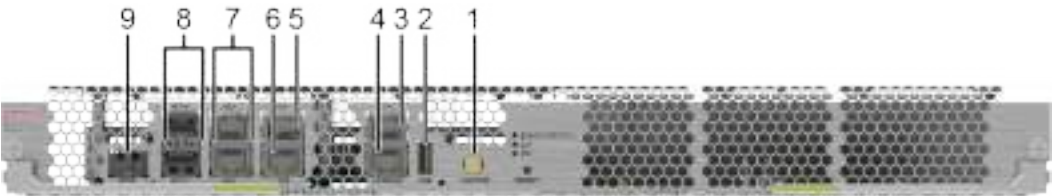
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-27 Buttons on the CE-MPUD-FULL

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-15 Ports on the CE-MPUD-FULL



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

 **NOTE**

The device does not support the HP-GE port.

Table 7-28 Ports on the CE-MPUD-FULL

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default. NOTE This function is available only in V200 and is reserved in V300.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default. NOTE This function is available only in V200 and is reserved in V300.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-29 Functions and features of the CE-MPUD-FULL

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-30 Technical specifications of the CE-MPUD-FULL

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)

Item	Specification
Weight without packaging [kg (lb)]	4.3 kg (9.48 lb)
Weight with packaging [kg (lb)]	7.3 kg (16.09 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.2.2 CE-MPUD-FULL (03058877-001: CE16800 Main Processing Unit D (full-width))

Overview

Table 7-31 Basic information about the CE-MPUD-FULL

Item	Details
Description	CE16800 Main Processing Unit D (full-width)
Part Number	03058877-001
Model	CE-MPUD-FULL
Silkscreen	CE-MPUD-FULL
Card Type	MPU

Appearance

Figure 7-16 Appearance of the CE-MPUD-FULL



Version Mapping

Table 7-32 Mappings between CE-MPUD-FULL and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	17-18	-

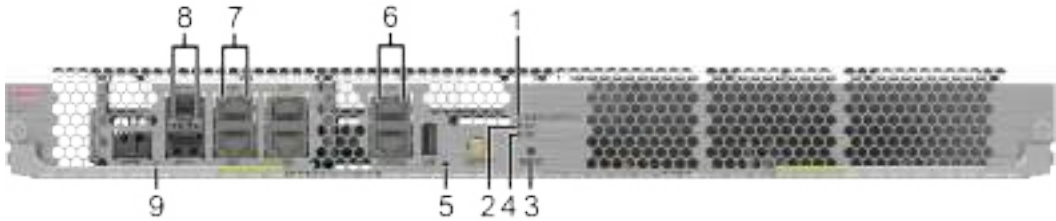
Table 7-33 Mappings between CE-MPUD-FULL and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-17 Indicators and buttons on the CE-MPUD-FULL



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

 **NOTE**

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-34 Indicators on the CE-MPUD-FULL

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

Silkscreen	Name	Color	Status	Description
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	Two single-color indicators for each SFP+ optical port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.

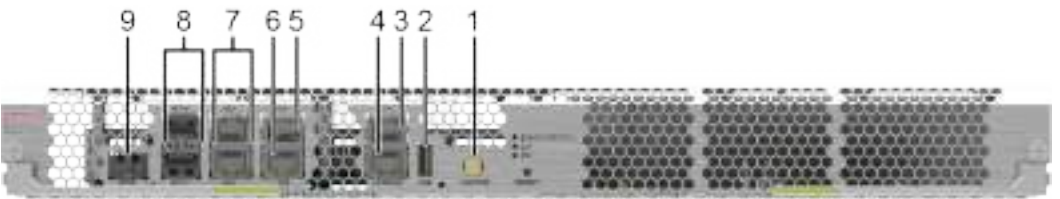
Silkscreen	Name	Color	Status	Description
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical module indicator	Green	Steady on	A link has been established on the port.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-35 Buttons on the CE-MPUD-FULL

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-18 Ports on the CE-MPUD-FULL



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

 **NOTE**

The device does not support the HP-GE port.

Table 7-36 Ports on the CE-MPUD-FULL

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB flash drive

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP+1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-37 Functions and features of the CE-MPUD-FULL

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-38 Technical specifications of the CE-MPUD-FULL

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)

Item	Specification
Weight without packaging [kg (lb)]	4.3 kg (9.48 lb)
Weight with packaging [kg (lb)]	7.3 kg (16.09 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.3 CE-MPUD-HK (CE16800 Main Processing Unit D-HK (half-width))

7.2.3.1 CE-MPUD-HK (03059590: CE16800 Main Processing Unit D-HK (half-width))

Overview

Table 7-39 Basic information about the CE-MPUD-HK

Item	Details
Description	CE16800 Main Processing Unit D-HK (half-width)
Part Number	03059590
Model	CE-MPUD-HK
Silkscreen	CE-MPUD-HK
Card Type	MPU

Appearance

Figure 7-19 Appearance of the CE-MPUD-HK



Version Mapping

Table 7-40 Mappings between CE-MPUD-HK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	5-6	-
CloudEngine 16808	V200R019C10	-	9-10	-

Table 7-41 Mappings between CE-MPUD-HK and Interface Card

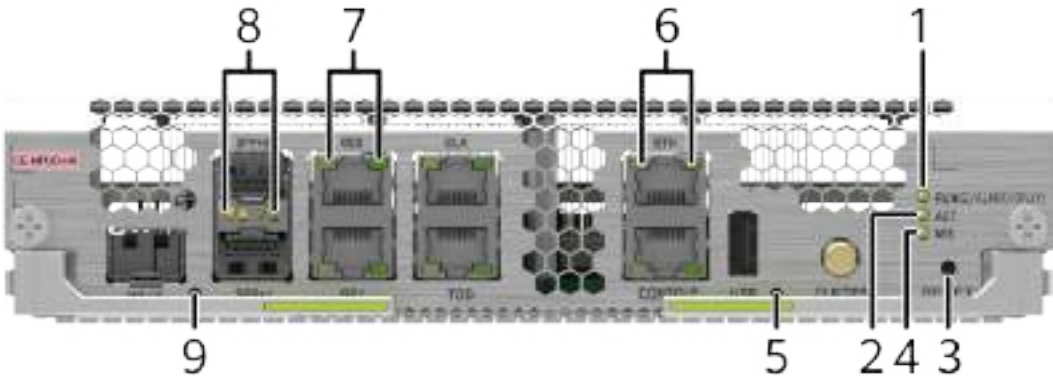
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-GK	03059591	V200R019C10	-
CloudEngine 16804	CEL18CQFD-GK	03059591-001	V200R020C10	-
CloudEngine 16804	CEL18CQFD-GK	03059591-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-GK	03059593	V200R019C10	-
CloudEngine 16804	CEL24LQFD-GK	03059593-001	V200R020C10	-
CloudEngine 16804	CEL24LQFD-GK	03059593-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36CQFD-GK	03059589	V200R019C10	-
CloudEngine 16804	CEL36CQFD-GK	03059589-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-GK	03059592	V200R019C10	-
CloudEngine 16804	CEL36LQFD-GK	03059592-001	V200R020C10	-
CloudEngine 16804	CEL36LQFD-GK	03059592-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-GK	03059594	V200R019C10	-
CloudEngine 16804	CEL48XSFD-GK	03059594-001	V200R020C10	-
CloudEngine 16804	CEL48XSFD-GK	03059594-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-GK	03059591	V200R019C10	-
CloudEngine 16808	CEL18CQFD-GK	03059591-001	V200R020C10	-
CloudEngine 16808	CEL18CQFD-GK	03059591-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-GK	03059593	V200R019C10	-
CloudEngine 16808	CEL24LQFD-GK	03059593-001	V200R020C10	-
CloudEngine 16808	CEL24LQFD-GK	03059593-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-GK	03059589	V200R019C10	-
CloudEngine 16808	CEL36CQFD-GK	03059589-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-GK	03059592	V200R019C10	-
CloudEngine 16808	CEL36LQFD-GK	03059592-001	V200R020C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-GK	03059592-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-GK	03059594	V200R019C10	-
CloudEngine 16808	CEL48XSFD-GK	03059594-001	V200R020C10	-
CloudEngine 16808	CEL48XSFD-GK	03059594-002	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-20 Indicators and buttons on the CE-MPUD-HK



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

NOTE

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-42 Indicators on the CE-MPUD-HK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

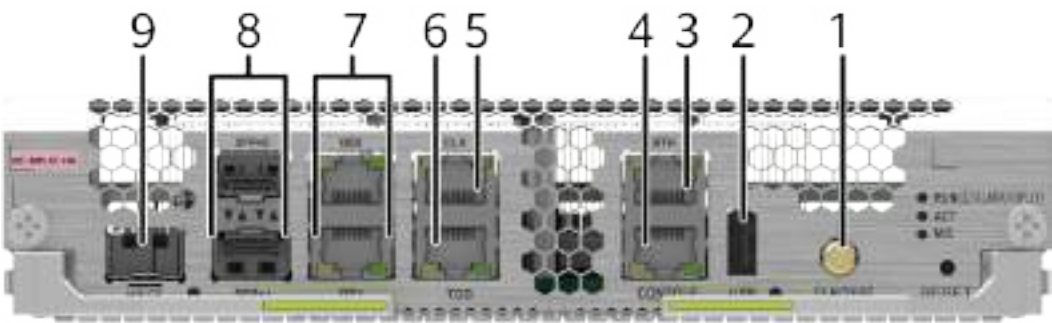
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-43 Buttons on the CE-MPUD-HK

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-21 Ports on the CE-MPUD-HK



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

 NOTE

The device does not support the HP-GE port.

Table 7-44 Ports on the CE-MPUD-HK

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP+1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-45 Functions and features of the CE-MPUD-HK

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-46 Technical specifications of the CE-MPUD-HK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	3.1 kg (6.83 lb)
Weight with packaging [kg (lb)]	5.1 kg (11.24 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.3.2 CE-MPUD-HK (03059590-001: CE16800 Main Processing Unit D-HK (half-width))

Overview

Table 7-47 Basic information about the CE-MPUD-HK

Item	Details
Description	CE16800 Main Processing Unit D-HK (half-width)
Part Number	03059590-001
Model	CE-MPUD-HK
Silkscreen	CE-MPUD-HK
Card Type	MPU

Appearance

Figure 7-22 Appearance of the CE-MPUD-HK



Version Mapping

Table 7-48 Mappings between CE-MPUD-HK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R022C10	-	5-6	-
CloudEngine 16808	V200R022C10	-	9-10	-

Table 7-49 Mappings between CE-MPUD-HK and Interface Card

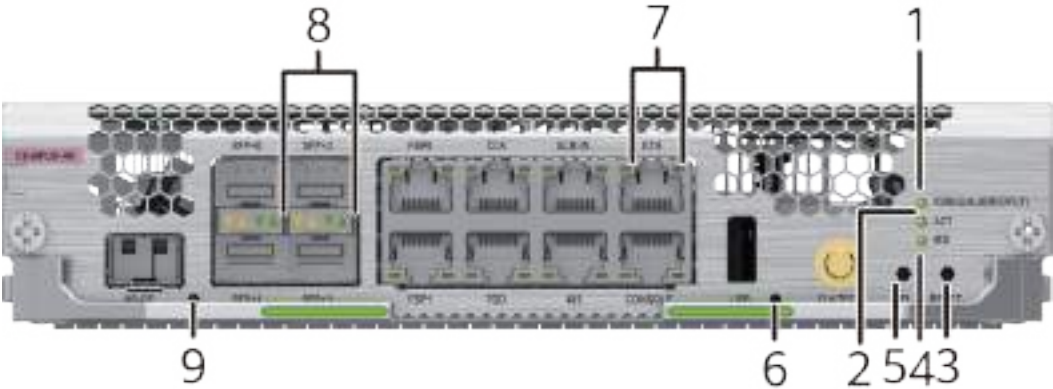
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-GK	03059591	V200R022C10	-
CloudEngine 16804	CEL18CQFD-GK	03059591-001	V200R022C10	-
CloudEngine 16804	CEL18CQFD-GK	03059591-002	V200R022C10	-
CloudEngine 16804	CEL24LQFD-GK	03059593	V200R022C10	-
CloudEngine 16804	CEL24LQFD-GK	03059593-001	V200R022C10	-
CloudEngine 16804	CEL24LQFD-GK	03059593-002	V200R022C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36CQFD-GK	03059589	V200R022C10	-
CloudEngine 16804	CEL36CQFD-GK	03059589-001	V200R022C10	-
CloudEngine 16804	CEL36LQFD-GK	03059592	V200R022C10	-
CloudEngine 16804	CEL36LQFD-GK	03059592-001	V200R022C10	-
CloudEngine 16804	CEL36LQFD-GK	03059592-002	V200R022C10	-
CloudEngine 16804	CEL48XSFD-GK	03059594	V200R022C10	-
CloudEngine 16804	CEL48XSFD-GK	03059594-001	V200R022C10	-
CloudEngine 16804	CEL48XSFD-GK	03059594-002	V200R022C10	-
CloudEngine 16808	CEL18CQFD-GK	03059591	V200R022C10	-
CloudEngine 16808	CEL18CQFD-GK	03059591-001	V200R022C10	-
CloudEngine 16808	CEL18CQFD-GK	03059591-002	V200R022C10	-
CloudEngine 16808	CEL24LQFD-GK	03059593	V200R022C10	-
CloudEngine 16808	CEL24LQFD-GK	03059593-001	V200R022C10	-
CloudEngine 16808	CEL24LQFD-GK	03059593-002	V200R022C10	-
CloudEngine 16808	CEL36CQFD-GK	03059589	V200R022C10	-
CloudEngine 16808	CEL36CQFD-GK	03059589-001	V200R022C10	-
CloudEngine 16808	CEL36LQFD-GK	03059592	V200R022C10	-
CloudEngine 16808	CEL36LQFD-GK	03059592-001	V200R022C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-GK	03059592-002	V200R022C10	-
CloudEngine 16808	CEL48XSFD-GK	03059594	V200R022C10	-
CloudEngine 16808	CEL48XSFD-GK	03059594-001	V200R022C10	-
CloudEngine 16808	CEL48XSFD-GK	03059594-002	V200R022C10	-

Indicators and Buttons

Figure 7-23 Indicators and buttons on the CE-MPUD-HK



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-50 Indicators on the CE-MPUD-HK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

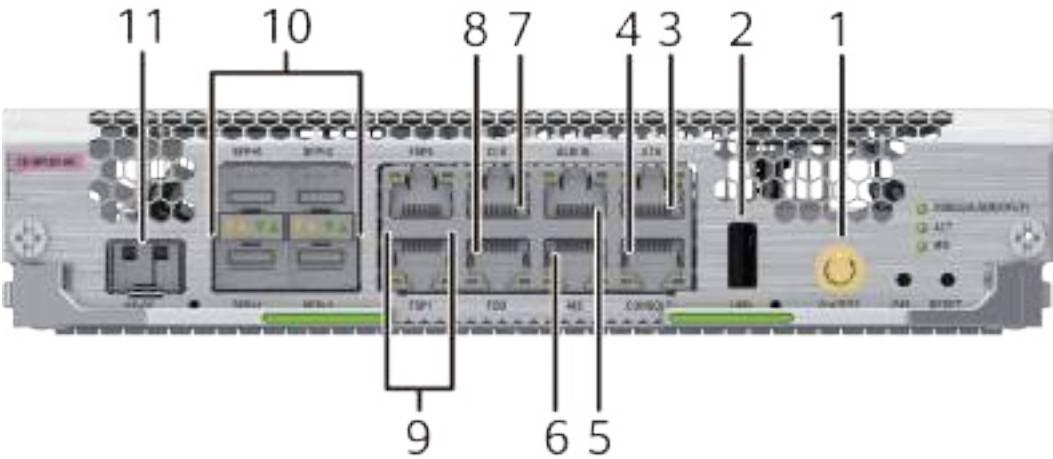
Table 7-51 Buttons on the CE-MPUD-HK

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Silkscreen	Name	Description
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-24 Ports on the CE-MPUD-HK



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

NOTE

The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.

Table 7-52 Ports on the CE-MPUD-HK

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-53 Functions and features of the CE-MPUD-HK

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-54 Technical specifications of the CE-MPUD-HK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	4.2 kg (9.26 lb)
Weight with packaging [kg (lb)]	6.2 kg (13.67 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.4 CE-MPUD-FK (CE16800 Main Processing Unit D-FK (full-width))

Overview

Table 7-55 Basic information about the CE-MPUD-FK

Item	Details
Description	CE16800 Main Processing Unit D-FK (full-width)
Part Number	03059636
Model	CE-MPUD-FK
Silkscreen	CE-MPUD-FK
Card Type	MPU

Appearance

Figure 7-25 Appearance of the CE-MPUD-FK



Version Mapping

Table 7-56 Mappings between CE-MPUD-FK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R019C10	-	17-18	-

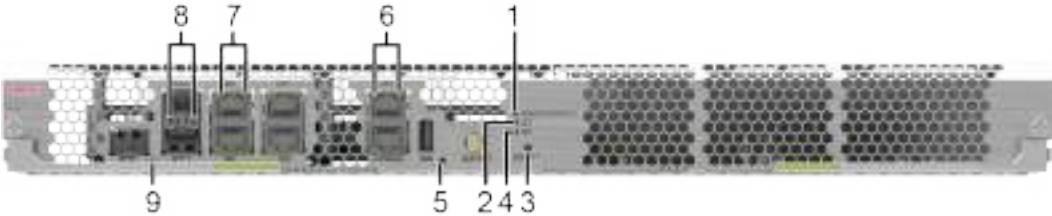
Table 7-57 Mappings between CE-MPUD-FK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-GK	03059591	V200R019C10	-
CloudEngine 16816	CEL18CQFD-GK	03059591-001	V200R020C10	-
CloudEngine 16816	CEL18CQFD-GK	03059591-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-GK	03059593	V200R019C10	-
CloudEngine 16816	CEL24LQFD-GK	03059593-001	V200R020C10	-
CloudEngine 16816	CEL24LQFD-GK	03059593-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-GK	03059589	V200R019C10	-
CloudEngine 16816	CEL36CQFD-GK	03059589-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36LQFD-GK	03059592	V200R019C10	-
CloudEngine 16816	CEL36LQFD-GK	03059592-001	V200R020C10	-
CloudEngine 16816	CEL36LQFD-GK	03059592-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-GK	03059594	V200R019C10	-
CloudEngine 16816	CEL48XSFD-GK	03059594-001	V200R020C10	-
CloudEngine 16816	CEL48XSFD-GK	03059594-002	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-26 Indicators and buttons on the CE-MPUD-FK



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. USB-based deployment indicator
6. ETH port indicator	7. GE electrical port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

NOTE

The indicators on CONSOLE, CLK, and TOD ports are not in use.

Table 7-58 Indicators on the CE-MPUD-FK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	GE electrical port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

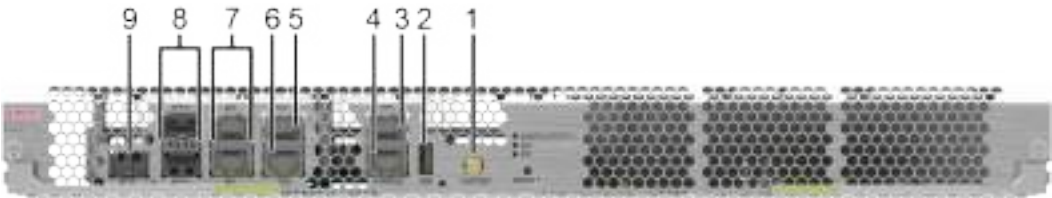
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-59 Buttons on the CE-MPUD-FK

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Ports

Figure 7-27 Ports on the CE-MPUD-FK



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. External synchronization port CLK (CLK)
6. External synchronization port TOD (TOD)	7. GE electrical port (GE0 and GE1)	8. SFP+ optical port (SFP+0 and SFP+1)	9. Clock SFP port (HP-GE)	-

NOTE

The device does not support the HP-GE port.

Table 7-60 Ports on the CE-MPUD-FK

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable

Port	Connector Type	Description	Available Components
GE electrical port	RJ45	GE0 and SFP+0, and GE1 and SFP+1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Cable

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+/SFP	GE0 and SFP+0, and GE1 and SFP +1 form a SIP port respectively. The two SIP ports are used to establish stack management links. Each SIP port is a combo port that consists of a GE electrical port and an SFP+ optical port. If the GE electrical port and the SFP+ optical port of a combo port do not establish links at the same time, the one that establishes a link first is used. If the two ports establish links at the same time, the optical port is used by default.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-61 Functions and features of the CE-MPUD-FK

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses flash to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-62 Technical specifications of the CE-MPUD-FK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)

Item	Specification
Weight without packaging [kg (lb)]	4.3 kg (9.48 lb)
Weight with packaging [kg (lb)]	7.3 kg (16.09 lb)
Typical power consumption [W]	58 W
Typical heat dissipation [BTU/hour]	198 BTU/hour
Static power consumption [W]	55 W
Static heat dissipation [BTU/hour]	188 BTU/hour
Maximum power consumption [W]	82 W
Maximum heat dissipation [BTU/hour]	280 BTU/hour
CPU	16-core, 1.85 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	4 GB

7.2.5 CE-MPUD-HALF2 (CE16800 Main Processing Unit D-2 (half-width))

7.2.5.1 CE-MPUD-HALF2 (03050EQL: CE16800 Main Processing Unit D-2 (half-width))

Overview

Table 7-63 Basic information about the CE-MPUD-HALF2

Item	Details
Description	CE16800 Main Processing Unit D-2 (half-width)
Part Number	03050EQL
Model	CE-MPUD-HALF2
Silkscreen	CE-MPUD-HALF2
Card Type	MPU

Appearance

Figure 7-28 Appearance of the CE-MPUD-HALF2



Version Mapping

Table 7-64 Mappings between CE-MPUD-HALF2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C01	-	5-6	-
CloudEngine 16808	V200R021C01	-	9-10	-

Table 7-65 Mappings between CE-MPUD-HALF2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C01	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R021C01	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R021C01	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL24LQFD-G	03058872	V200R021C01	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R021C01	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R021C01	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R021C01	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C01	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R021C01	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R021C01	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R021C01	-

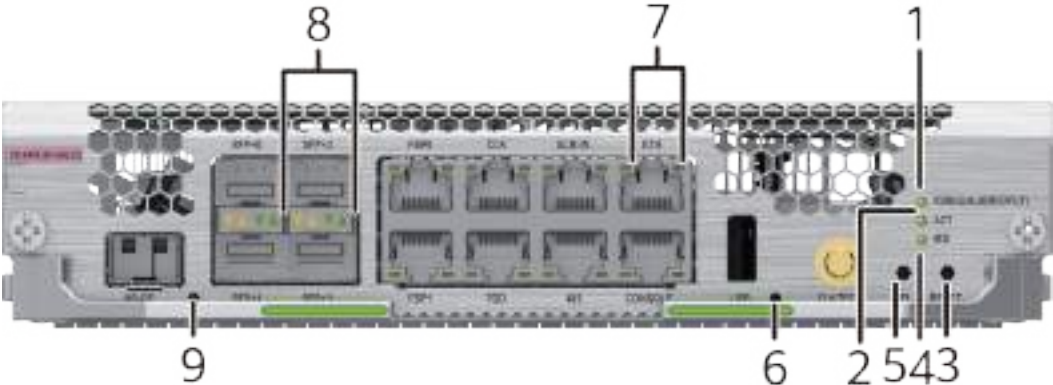
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C01	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R021C01	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R021C01	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R021C01	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C01	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R021C01	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R021C01	-
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R021C01	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R021C01	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R021C01	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C01	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R021C01	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R021C01	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R021C01	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C01	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R021C01	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R021C01	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R021C01	-

Indicators and Buttons

Figure 7-29 Indicators and buttons on the CE-MPUD-HALF2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the indicator on USB port is not in use.

Table 7-66 Indicators on the CE-MPUD-HALF2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

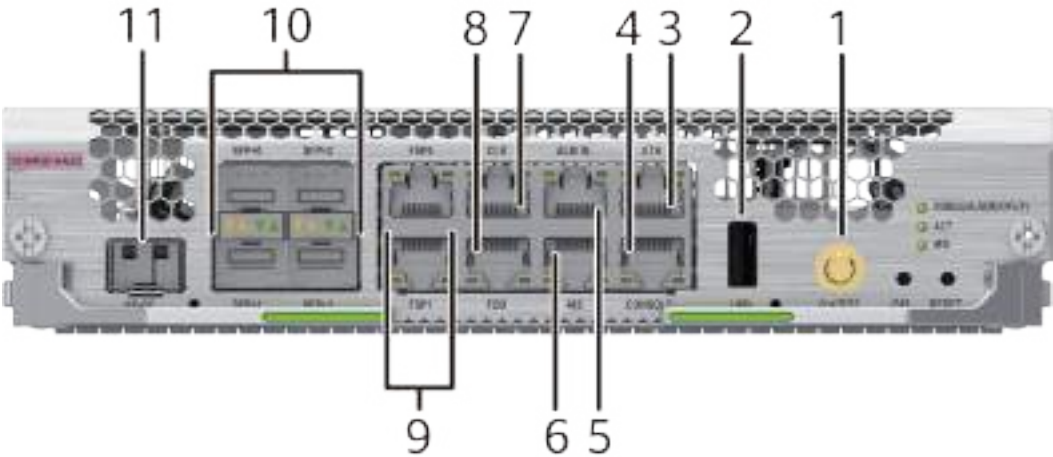
Table 7-67 Buttons on the CE-MPUD-HALF2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.

Silkscreen	Name	Description
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-30 Ports on the CE-MPUD-HALF2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the device does not support the USB port.

Table 7-68 Ports on the CE-MPUD-HALF2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-69 Functions and features of the CE-MPUD-HALF2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-70 Technical specifications of the CE-MPUD-HALF2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	4.2 kg (9.26 lb)
Weight with packaging [kg (lb)]	6.2 kg (13.67 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.5.2 CE-MPUD-HALF2 (03050EQL-001: CE16800 Main Processing Unit D-2 (half-width))

Overview

Table 7-71 Basic information about the CE-MPUD-HALF2

Item	Details
Description	CE16800 Main Processing Unit D-2 (half-width)
Part Number	03050EQL-001
Model	CE-MPUD-HALF2
Silkscreen	CE-MPUD-HALF2
Card Type	MPU

Appearance

Figure 7-31 Appearance of the CE-MPUD-HALF2



Version Mapping

Table 7-72 Mappings between CE-MPUD-HALF2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	5-6	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	9-10	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-73 Mappings between CE-MPUD-HALF2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-

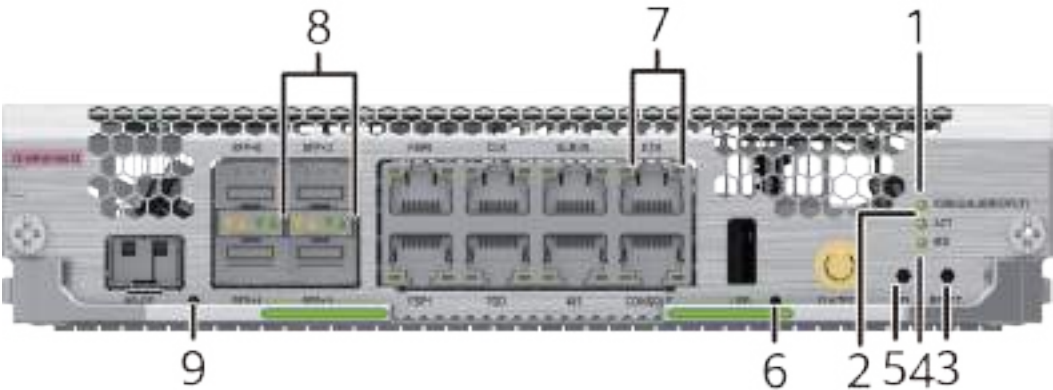
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-32 Indicators and buttons on the CE-MPUD-HALF2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
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6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-
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 NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R022C00 and V200R022C10, the indicator on USB port is not in use.

Table 7-74 Indicators on the CE-MPUD-HALF2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.

Silkscreen	Name	Color	Status	Description
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.

Silkscreen	Name	Color	Status	Description
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

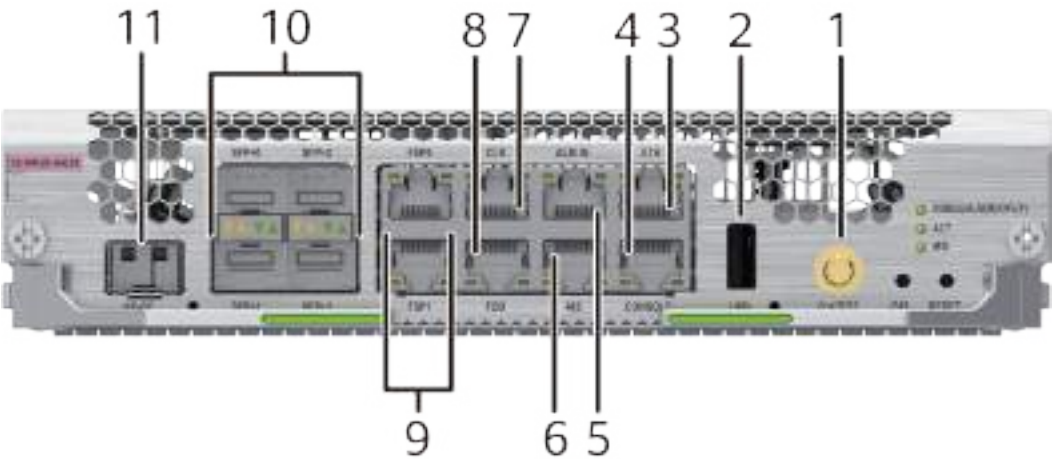
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-75 Buttons on the CE-MPUD-HALF2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-33 Ports on the CE-MPUD-HALF2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R022C00 and V200R022C10, the device does not support the USB port.

Table 7-76 Ports on the CE-MPUD-HALF2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-77 Functions and features of the CE-MPUD-HALF2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-78 Technical specifications of the CE-MPUD-HALF2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)
Weight without packaging [kg (lb)]	4.2 kg (9.26 lb)
Weight with packaging [kg (lb)]	6.2 kg (13.67 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.5.3 CE-MPUD-HALF2 (03050EQL-002: CE16800 Main Processing Unit D-2 (half-width))

Overview

Table 7-79 Basic information about the CE-MPUD-HALF2

Item	Details
Description	CE16800 Main Processing Unit D-2 (half-width)
Part Number	03050EQL-002
Model	CE-MPUD-HALF2
Silkscreen	CE-MPUD-HALF2
Card Type	MPU

Appearance

Figure 7-34 Appearance of the CE-MPUD-HALF2



Version Mapping

Table 7-80 Mappings between CE-MPUD-HALF2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R022C00	-	5-6	-
CloudEngine 16808	V200R022C00	-	9-10	-

Table 7-81 Mappings between CE-MPUD-HALF2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R022C00	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R022C00	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R022C00	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R022C00	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R022C00	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R022C00	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R022C00	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R022C00	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R022C00	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R022C00	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R022C00	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R022C00	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R022C00	-

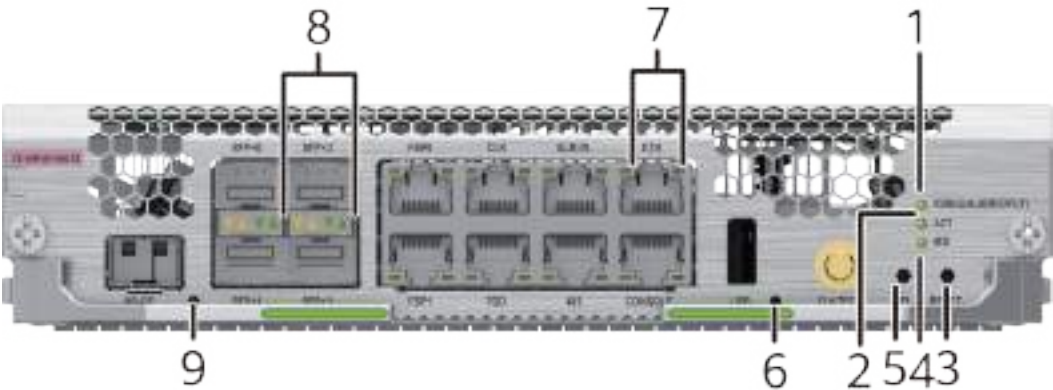
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R022C00	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R022C00	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R022C00	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R022C00	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R022C00	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R022C00	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R022C00	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R022C00	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R022C00	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R022C00	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R022C00	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R022C00	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R022C00	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R022C00	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R022C00	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R022C00	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R022C00	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R022C00	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R022C00	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R022C00	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R022C00	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R022C00	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R022C00	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R022C00	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R022C00	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R022C00	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R022C00	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R022C00	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R022C00	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R022C00	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R022C00	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R022C00	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R022C00	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R022C00	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R022C00	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R022C00	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R022C00	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R022C00	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R022C00	-

Indicators and Buttons

Figure 7-35 Indicators and buttons on the CE-MPUD-HALF2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
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6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-
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 NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R022C00 and V200R022C10, the indicator on USB port is not in use.

Table 7-82 Indicators on the CE-MPUD-HALF2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.

Silkscreen	Name	Color	Status	Description
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.

Silkscreen	Name	Color	Status	Description
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

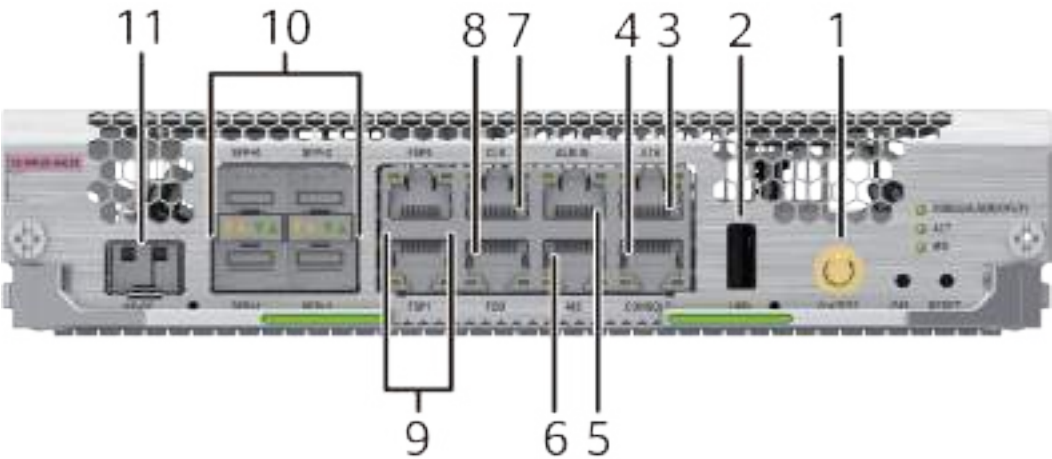
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-83 Buttons on the CE-MPUD-HALF2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-36 Ports on the CE-MPUD-HALF2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R022C00 and V200R022C10, the device does not support the USB port.

Table 7-84 Ports on the CE-MPUD-HALF2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-85 Functions and features of the CE-MPUD-HALF2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-86 Technical specifications of the CE-MPUD-HALF2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)
Weight without packaging [kg (lb)]	4.2 kg (9.26 lb)
Weight with packaging [kg (lb)]	6.2 kg (13.67 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.6 CE-MPUD-FULL2 (CE16800 Main Processing Unit D-2 (full-width))

7.2.6.1 CE-MPUD-FULL2 (03050EQP: CE16800 Main Processing Unit D-2 (full-width))

Overview

Table 7-87 Basic information about the CE-MPUD-FULL2

Item	Details
Description	CE16800 Main Processing Unit D-2 (full-width)
Part Number	03050EQP
Model	CE-MPUD-FULL2
Silkscreen	CE-MPUD-FULL2
Card Type	MPU

Appearance

Figure 7-37 Appearance of the CE-MPUD-FULL2



Version Mapping

Table 7-88 Mappings between CE-MPUD-FULL2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C01	-	17-18	-

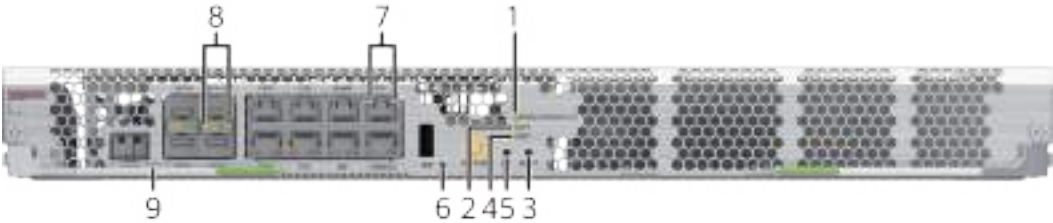
Table 7-89 Mappings between CE-MPUD-FULL2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C01	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C01	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R021C01	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R021C01	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R021C01	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R021C01	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R021C01	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R021C01	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R021C01	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R021C01	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R021C01	-
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R021C01	-

Indicators and Buttons

Figure 7-38 Indicators and buttons on the CE-MPUD-FULL2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

 NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the indicator on USB port is not in use.

Table 7-90 Indicators on the CE-MPUD-FULL2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

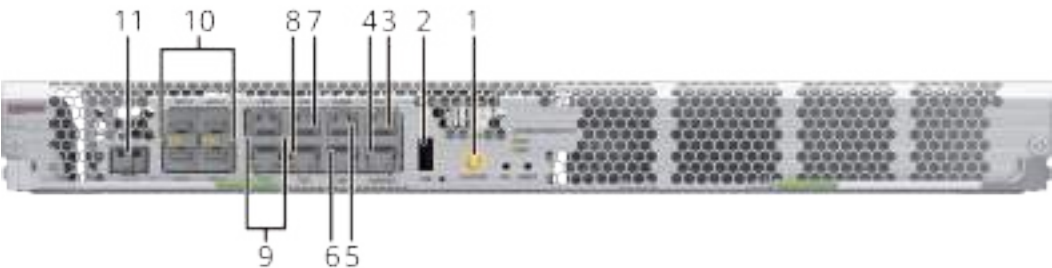
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-91 Buttons on the CE-MPUD-FULL2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-39 Ports on the CE-MPUD-FULL2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the device does not support the USB port.

Table 7-92 Ports on the CE-MPUD-FULL2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-93 Functions and features of the CE-MPUD-FULL2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-94 Technical specifications of the CE-MPUD-FULL2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Weight with packaging [kg (lb)]	8.5 kg (18.74 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.6.2 CE-MPUD-FULL2 (03050EQP-001: CE16800 Main Processing Unit D-2 (full-width))

Overview

Table 7-95 Basic information about the CE-MPUD-FULL2

Item	Details
Description	CE16800 Main Processing Unit D-2 (full-width)
Part Number	03050EQP-001
Model	CE-MPUD-FULL2
Silkscreen	CE-MPUD-FULL2
Card Type	MPU

Appearance

Figure 7-40 Appearance of the CE-MPUD-FULL2



Version Mapping

Table 7-96 Mappings between CE-MPUD-FULL2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	17-18	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

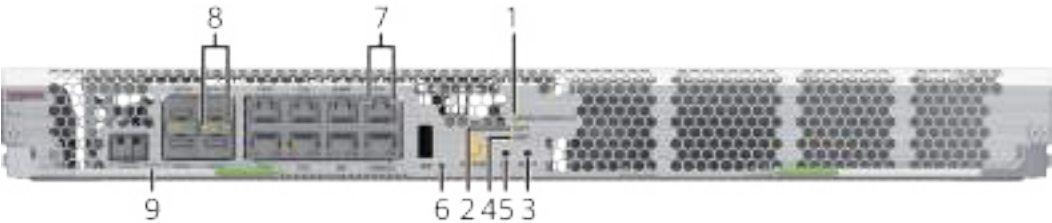
Table 7-97 Mappings between CE-MPUD-FULL2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-41 Indicators and buttons on the CE-MPUD-FULL2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R022C00 and V200R022C10, the indicator on USB port is not in use.

Table 7-98 Indicators on the CE-MPUD-FULL2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

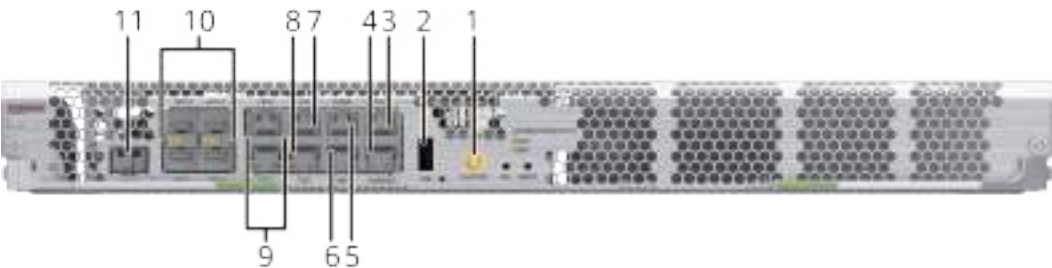
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-99 Buttons on the CE-MPUD-FULL2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-42 Ports on the CE-MPUD-FULL2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R022C00 and V200R022C10, the device does not support the USB port.

Table 7-100 Ports on the CE-MPUD-FULL2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-101 Functions and features of the CE-MPUD-FULL2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-102 Technical specifications of the CE-MPUD-FULL2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Weight with packaging [kg (lb)]	8.5 kg (18.74 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.6.3 CE-MPUD-FULL2 (03050EQP-002: CE16800 Main Processing Unit D-2 (full-width))

Overview

Table 7-103 Basic information about the CE-MPUD-FULL2

Item	Details
Description	CE16800 Main Processing Unit D-2 (full-width)
Part Number	03050EQP-002
Model	CE-MPUD-FULL2
Silkscreen	CE-MPUD-FULL2
Card Type	MPU

Appearance

Figure 7-43 Appearance of the CE-MPUD-FULL2



Version Mapping

Table 7-104 Mappings between CE-MPUD-FULL2 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R022C00	-	17-18	-

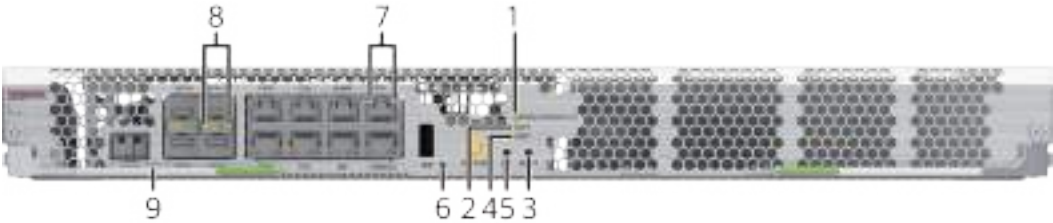
Table 7-105 Mappings between CE-MPUD-FULL2 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R022C00	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R022C00	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R022C00	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R022C00	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R022C00	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R022C00	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R022C00	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R022C00	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R022C00	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R022C00	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R022C00	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R022C00	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R022C00	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R022C00	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R022C00	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R022C00	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R022C00	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R022C00	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R022C00	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R022C00	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-001	V200R022C00	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R022C00	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R022C00	-
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R022C00	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R022C00	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R022C00	-

Indicators and Buttons

Figure 7-44 Indicators and buttons on the CE-MPUD-FULL2



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

 NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R022C00 and V200R022C10, the indicator on USB port is not in use.

Table 7-106 Indicators on the CE-MPUD-FULL2

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

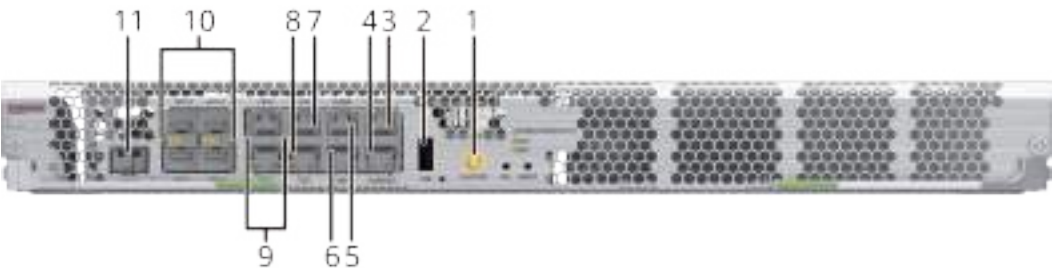
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-107 Buttons on the CE-MPUD-FULL2

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-45 Ports on the CE-MPUD-FULL2



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R022C00 and V200R022C10, the device does not support the USB port.

Table 7-108 Ports on the CE-MPUD-FULL2

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-109 Functions and features of the CE-MPUD-FULL2

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-110 Technical specifications of the CE-MPUD-FULL2

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Weight with packaging [kg (lb)]	8.5 kg (18.74 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.7 CE-MPUD-F2K (CE16800 Main Processing Unit D-2K (full-width))

7.2.7.1 CE-MPUD-F2K (03050EQN: CE16800 Main Processing Unit D-2K (full-width))

Overview

Table 7-111 Basic information about the CE-MPUD-F2K

Item	Details
Description	CE16800 Main Processing Unit D-2K (full-width)
Part Number	03050EQN
Model	CE-MPUD-F2K
Silkscreen	CE-MPUD-F2K
Card Type	MPU

Appearance

Figure 7-46 Appearance of the CE-MPUD-F2K



Version Mapping

Table 7-112 Mappings between CE-MPUD-F2K and product

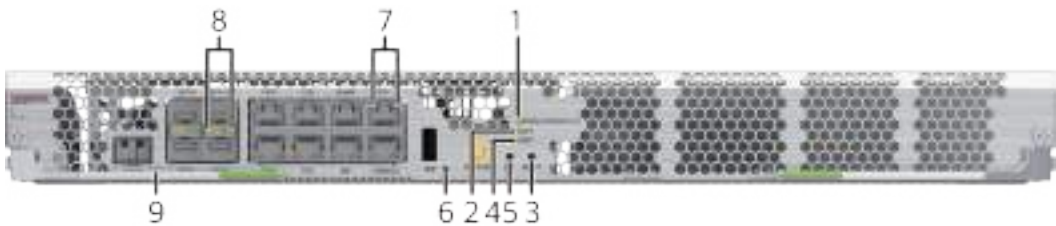
Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C01	-	17-18	-

Table 7-113 Mappings between CE-MPUD-F2K and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-GK	03059591	V200R021C01	-
CloudEngine 16816	CEL18CQFD-GK	03059591-001	V200R021C01	-
CloudEngine 16816	CEL18CQFD-GK	03059591-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-GK	03059593	V200R021C01	-
CloudEngine 16816	CEL24LQFD-GK	03059593-001	V200R021C01	-
CloudEngine 16816	CEL24LQFD-GK	03059593-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-GK	03059589	V200R021C01	-
CloudEngine 16816	CEL36CQFD-GK	03059589-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-GK	03059592	V200R021C01	-
CloudEngine 16816	CEL36LQFD-GK	03059592-001	V200R021C01	-
CloudEngine 16816	CEL36LQFD-GK	03059592-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-GK	03059594	V200R021C01	-
CloudEngine 16816	CEL48XSFD-GK	03059594-001	V200R021C01	-
CloudEngine 16816	CEL48XSFD-GK	03059594-002	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-47 Indicators and buttons on the CE-MPUD-F2K



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the indicator on USB port is not in use.

Table 7-114 Indicators on the CE-MPUD-F2K

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

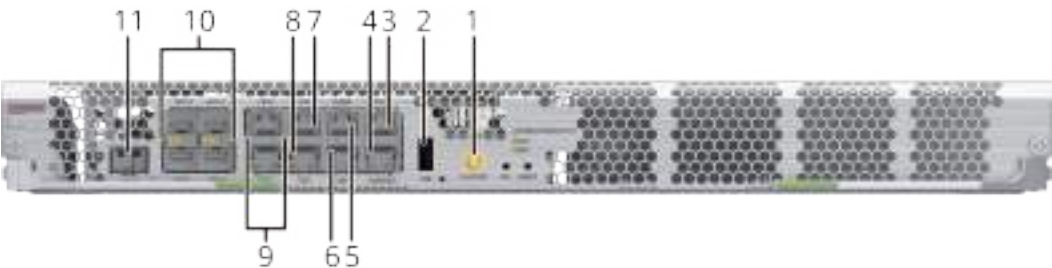
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-115 Buttons on the CE-MPUD-F2K

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-48 Ports on the CE-MPUD-F2K



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R021C10SPC500, V200R022C00, and V200R022C10, the device does not support the USB port.

Table 7-116 Ports on the CE-MPUD-F2K

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-117 Functions and features of the CE-MPUD-F2K

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-118 Technical specifications of the CE-MPUD-F2K

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Weight with packaging [kg (lb)]	8.5 kg (18.74 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.7.2 CE-MPUD-F2K (03050EQN-001: CE16800 Main Processing Unit D-2K (full-width))

Overview

Table 7-119 Basic information about the CE-MPUD-F2K

Item	Details
Description	CE16800 Main Processing Unit D-2K (full-width)
Part Number	03050EQN-001
Model	CE-MPUD-F2K
Silkscreen	CE-MPUD-F2K
Card Type	MPU

Appearance

Figure 7-49 Appearance of the CE-MPUD-F2K



Version Mapping

Table 7-120 Mappings between CE-MPUD-F2K and product

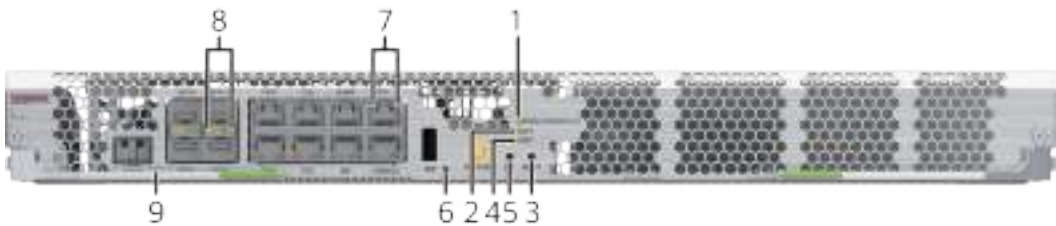
Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	17-18	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-121 Mappings between CE-MPUD-F2K and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-GK	03059591	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-GK	03059591-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-GK	03059591-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-GK	03059593	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-GK	03059593-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-GK	03059593-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-GK	03059589	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-GK	03059589-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-GK	03059592	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-GK	03059592-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-GK	03059592-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-GK	03059594	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-GK	03059594-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-GK	03059594-00 2	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-50 Indicators and buttons on the CE-MPUD-F2K



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

- The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.
- In V200R022C00 and V200R022C10, the indicator on USB port is not in use.

Table 7-122 Indicators on the CE-MPUD-F2K

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

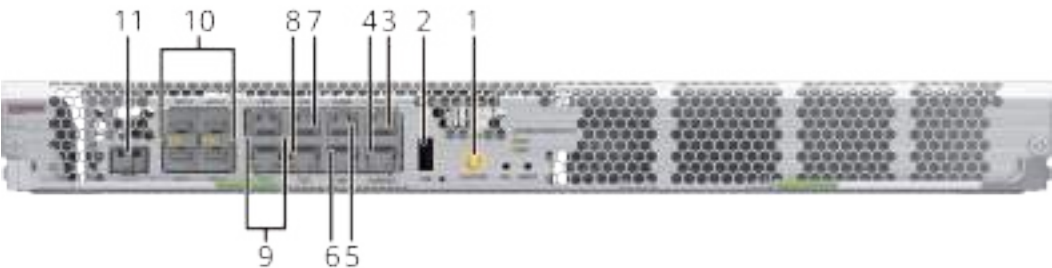
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-123 Buttons on the CE-MPUD-F2K

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-51 Ports on the CE-MPUD-F2K



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

- The device does not support the ALMIN, 485, FSP0, FSP1 and HP-GE ports.
- In V200R022C00 and V200R022C10, the device does not support the USB port.

Table 7-124 Ports on the CE-MPUD-F2K

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Connects to an external clock.	50-ohm clock cable
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Inputs or outputs 2-Mbit/s clock signals, 2-MHz clock signals, or 1 pps time signals.	120-ohm clock cable
External synchronization port TOD	RJ45	Inputs or outputs 1 pps and ASCII time signals or DCLS time signals.	120-ohm clock cable
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE SFP+2 and SFP+3 optical ports do not support the stacking function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-125 Functions and features of the CE-MPUD-F2K

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.

Functions and Features	Description
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-126 Technical specifications of the CE-MPUD-F2K

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.5 kg (12.13 lb)
Weight with packaging [kg (lb)]	8.5 kg (18.74 lb)
Typical power consumption [W]	88 W
Typical heat dissipation [BTU/hour]	300 BTU/hour
Static power consumption [W]	67 W
Static heat dissipation [BTU/hour]	229 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	8-core, 1.5 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.8 CE-MPUE-HALF (CE16800 Main Processing Unit E (half-width))

Overview

Table 7-127 Basic information about the CE-MPUE-HALF

Item	Details
Description	CE16800 Main Processing Unit E (half-width)
Part Number	03059364
Model	CE-MPUE-HALF
Silkscreen	CE-MPUE-HALF
Card Type	MPU

Appearance

Figure 7-52 Appearance of the CE-MPUE-HALF



Version Mapping

Table 7-128 Mappings between CE-MPUE-HALF and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	5-6	-
CloudEngine 16808	V200R019C10	-	9-10	-

Table 7-129 Mappings between CE-MPUE-HALF and Interface Card

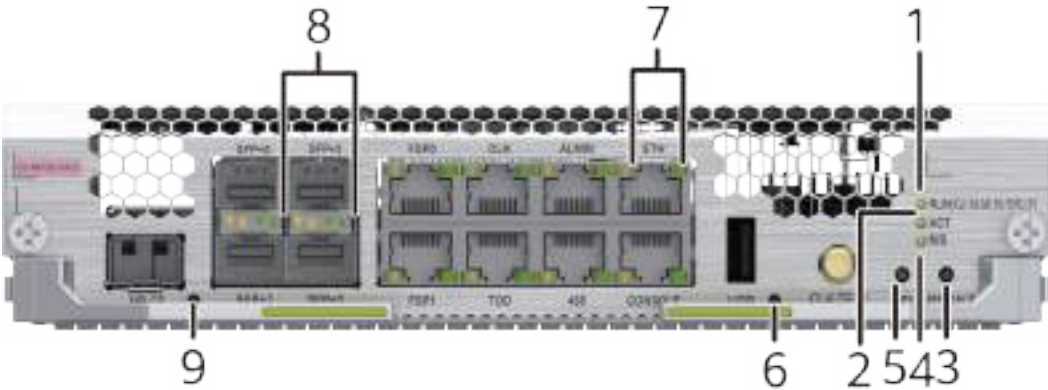
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-G	03058870	V200R019C10	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R020C10	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R019C10	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R020C10	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R019C10	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R019C10	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R019C10	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R019C10	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R019C10	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R020C10	-
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R019C10	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R020C10	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R019C10	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R019C10	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R019C10	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R019C10	-

Indicators and Buttons

Figure 7-53 Indicators and buttons on the CE-MPUE-HALF



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

 NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-130 Indicators on the CE-MPUE-HALF

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

Silkscreen	Name	Color	Status	Description
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.

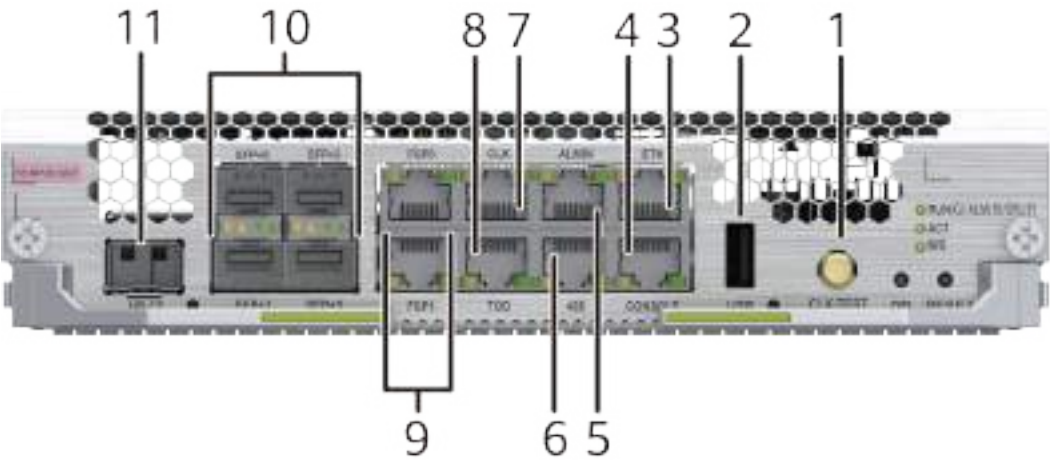
Silkscreen	Name	Color	Status	Description
		Green	Off	The link on the port is disconnected.

Table 7-131 Buttons on the CE-MPUE-HALF

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-54 Ports on the CE-MPUE-HALF



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1, SFP+0, SFP+1, SFP+2, SFP+3 and HP-GE ports.

Table 7-132 Ports on the CE-MPUE-HALF

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+/SFP	Reserved function. This function is not enabled.	-
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-133 Functions and features of the CE-MPUE-HALF

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.

Functions and Features	Description
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.
AI feature	The main control board has a built-in AI module to implement the AI function.

Technical Specifications

Table 7-134 Technical specifications of the CE-MPUE-HALF

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)
Weight without packaging [kg (lb)]	4.0 kg (8.82 lb)
Weight with packaging [kg (lb)]	6 kg (13.23 lb)
Typical power consumption [W]	131 W
Typical heat dissipation [BTU/hour]	447 BTU/hour
Static power consumption [W]	114 W
Static heat dissipation [BTU/hour]	389 BTU/hour
Maximum power consumption [W]	181 W
Maximum heat dissipation [BTU/hour]	618 BTU/hour
CPU	24-core, 2.5 GHz clock speed
Memory	32 GB (standard configuration)
Storage	32 GB SSD

7.2.9 CE-MPUE-FULL (CE16800 Main Processing Unit E (full-width))

Overview

Table 7-135 Basic information about the CE-MPUE-FULL

Item	Details
Description	CE16800 Main Processing Unit E (full-width)
Part Number	03059365
Model	CE-MPUE-FULL
Silkscreen	CE-MPUE-FULL
Card Type	MPU

Appearance

Figure 7-55 Appearance of the CE-MPUE-FULL



Version Mapping

Table 7-136 Mappings between CE-MPUE-FULL and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R019C10	-	17-18	-

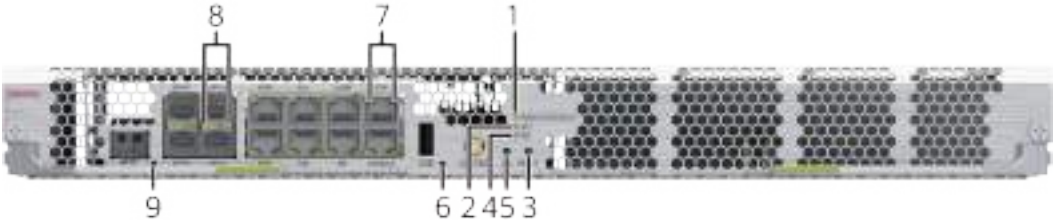
Table 7-137 Mappings between CE-MPUE-FULL and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-G	03058870	V200R019C10	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R020C10	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R019C10	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R020C10	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R019C10	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R019C10	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R019C10	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R020C10	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R019C10	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R019C10	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R020C10	-
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R019C10	-

Indicators and Buttons

Figure 7-56 Indicators and buttons on the CE-MPUE-FULL



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-138 Indicators on the CE-MPUE-FULL

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.

Silkscreen	Name	Color	Status	Description
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

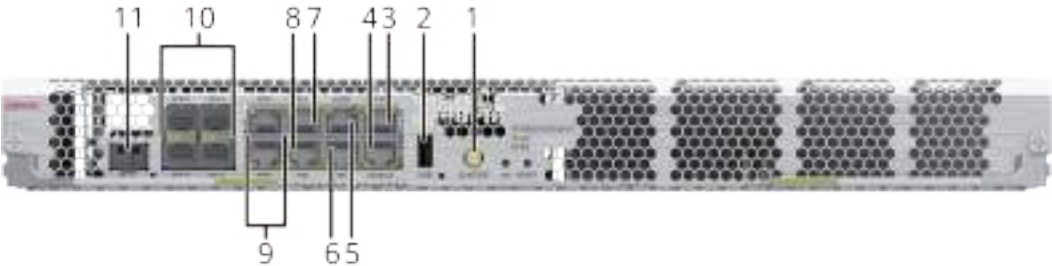
Silkscreen	Name	Color	Status	Description
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-139 Buttons on the CE-MPUE-FULL

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-57 Ports on the CE-MPUE-FULL



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1, SFP+0, SFP+1, SFP+2, SFP+3 and HP-GE ports.

Table 7-140 Ports on the CE-MPUE-FULL

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+/SFP	Reserved function. This function is not enabled.	-
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-141 Functions and features of the CE-MPUE-FULL

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.

Functions and Features	Description
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.
AI feature	The main control board has a built-in AI module to implement the AI function.

Technical Specifications

Table 7-142 Technical specifications of the CE-MPUE-FULL

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.1 kg (11.24 lb)
Weight with packaging [kg (lb)]	8.1 kg (17.86 lb)
Typical power consumption [W]	131 W
Typical heat dissipation [BTU/hour]	447 BTU/hour
Static power consumption [W]	114 W
Static heat dissipation [BTU/hour]	389 BTU/hour
Maximum power consumption [W]	181 W
Maximum heat dissipation [BTU/hour]	618 BTU/hour
CPU	24-core, 2.5 GHz clock speed
Memory	32 GB (standard configuration)
Storage	32 GB SSD

7.2.10 CE-MPUF-HALF (CE16800 Main Processing Unit F (half-width))

7.2.10.1 CE-MPUF-HALF (03050AVA: CE16800 Main Processing Unit F (half-width))

Overview

Table 7-143 Basic information about the CE-MPUF-HALF

Item	Details
Description	CE16800 Main Processing Unit F (half-width)
Part Number	03050AVA
Model	CE-MPUF-HALF
Silkscreen	CE-MPUF-HALF
Card Type	MPU

Appearance

Figure 7-58 Appearance of the CE-MPUF-HALF



Version Mapping

Table 7-144 Mappings between CE-MPUF-HALF and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	5-6	-
CloudEngine 16808	V200R021C00 SPC100	-	9-10	-

Table 7-145 Mappings between CE-MPUF-HALF and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C00 SPC100	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R021C00 SPC100	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 1	V200R021C00 SPC100	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 2	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-00 3	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R021C00 SPC100	-
CloudEngine 16804	CEL24LQFD-G	03058872-00 1	V200R021C00 SPC100	-
CloudEngine 16804	CEL24LQFD-G	03058872-00 2	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-00 3	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R021C00 SPC100	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-

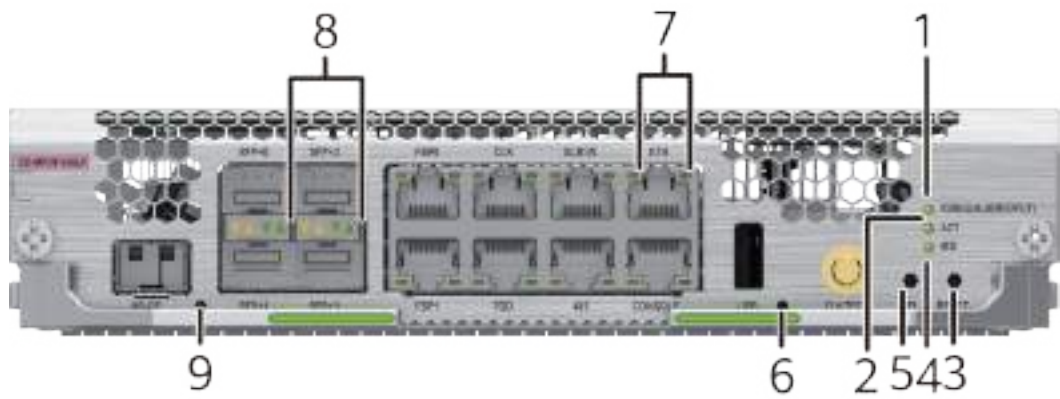
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R021C00 SPC100	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C00 SPC100	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R021C00 SPC100	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R021C00 SPC100	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R021C00 SPC100	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C00 SPC100	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R021C00 SPC100	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R021C00 SPC100	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R021C00 SPC100	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C00 SPC100	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R021C00 SPC100	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R021C00 SPC100	-
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R021C00 SPC100	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R021C00 SPC100	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R021C00 SPC100	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R021C00 SPC100	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C00 SPC100	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R021C00 SPC100	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R021C00 SPC100	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R021C00 SPC100	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C00 SPC100	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R021C00 SPC100	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R021C00 SPC100	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R021C00 SPC100	-

Indicators and Buttons

Figure 7-59 Indicators and buttons on the CE-MPUF-HALF



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-

NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-146 Indicators on the CE-MPUF-HALF

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

Silkscreen	Name	Color	Status	Description
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.

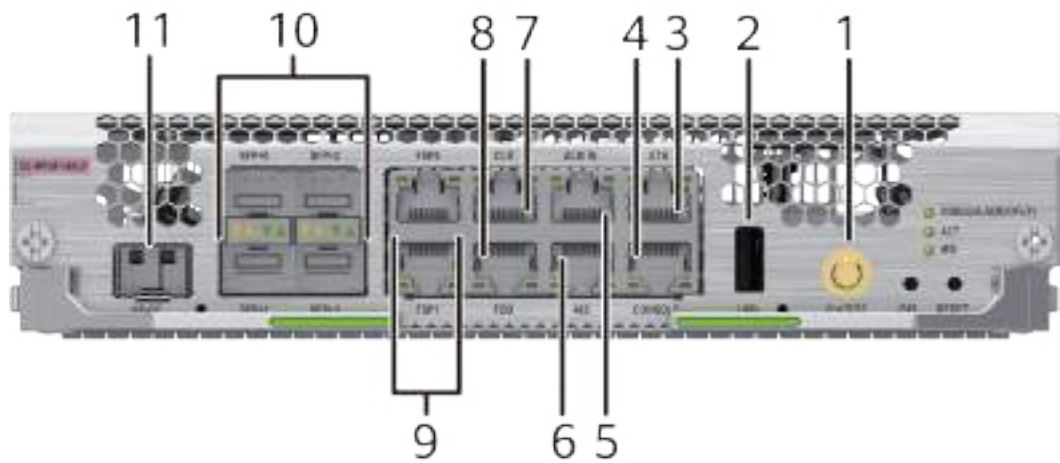
Silkscreen	Name	Color	Status	Description
		Green	Off	The link on the port is disconnected.

Table 7-147 Buttons on the CE-MPUF-HALF

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-60 Ports on the CE-MPUF-HALF



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1 and HP-GE ports.

Table 7-148 Ports on the CE-MPUF-HALF

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE - The SFP+2 and SFP+3 optical ports are reserved for stacking, and are not in use. - In V200R021C10 and later versions, SFP+0 and SFP+1 optical ports support the stack function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-149 Functions and features of the CE-MPUF-HALF

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.

Functions and Features	Description
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-150 Technical specifications of the CE-MPUF-HALF

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)

Item	Specification
Weight without packaging [kg (lb)]	3.8 kg (8.38 lb)
Weight with packaging [kg (lb)]	5.8 kg (12.79 lb)
Typical power consumption [W]	117 W
Typical heat dissipation [BTU/hour]	400 BTU/hour
Static power consumption [W]	100 W
Static heat dissipation [BTU/hour]	342 BTU/hour
Maximum power consumption [W]	186 W
Maximum heat dissipation [BTU/hour]	635 BTU/hour
CPU	8-core, 2.3 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.10.2 CE-MPUF-HALF (03050AVA-001: CE16800 Main Processing Unit F (half-width))

Overview

Table 7-151 Basic information about the CE-MPUF-HALF

Item	Details
Description	CE16800 Main Processing Unit F (half-width)
Part Number	03050AVA-001
Model	CE-MPUF-HALF
Silkscreen	CE-MPUF-HALF
Card Type	MPU

Appearance

Figure 7-61 Appearance of the CE-MPUF-HALF



Version Mapping

Table 7-152 Mappings between CE-MPUF-HALF and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	5-6	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	9-10	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.

Table 7-153 Mappings between CE-MPUF-HALF and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16804	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-

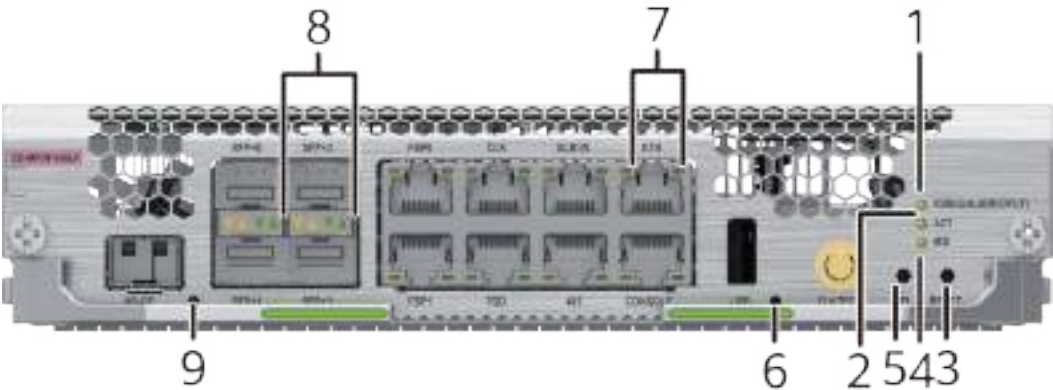
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16804	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16808	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16808	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-62 Indicators and buttons on the CE-MPUF-HALF



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
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6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical module indicator	-
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 NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-154 Indicators on the CE-MPUF-HALF

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.

Silkscreen	Name	Color	Status	Description
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.

Silkscreen	Name	Color	Status	Description
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).
-	Two single-color indicators for each ETH port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

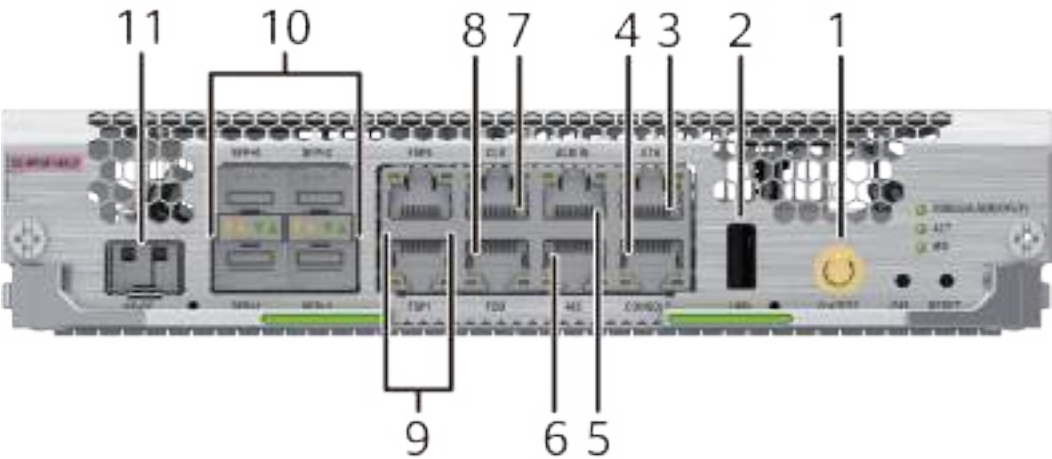
Silkscreen	Name	Color	Status	Description
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-155 Buttons on the CE-MPUF-HALF

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-63 Ports on the CE-MPUF-HALF



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1 and HP-GE ports.

Table 7-156 Ports on the CE-MPUF-HALF

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE The SFP+2 and SFP+3 optical ports are reserved for stacking, and are not in use.	Optical fiber

Port	Connector Type	Description	Available Components
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-157 Functions and features of the CE-MPUF-HALF

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported

Functions and Features	Description
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-158 Technical specifications of the CE-MPUF-HALF

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 215 mm x 522.5 mm (1.80 in. x 8.46 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 385 mm x 780 mm (7.28 in. x 15.16 in. x 30.71 in.)
Weight without packaging [kg (lb)]	3.8 kg (8.38 lb)
Weight with packaging [kg (lb)]	5.8 kg (12.79 lb)
Typical power consumption [W]	117 W
Typical heat dissipation [BTU/hour]	400 BTU/hour
Static power consumption [W]	100 W
Static heat dissipation [BTU/hour]	342 BTU/hour
Maximum power consumption [W]	186 W
Maximum heat dissipation [BTU/hour]	635 BTU/hour
CPU	8-core, 2.3 GHz clock speed
Memory	16 GB (standard configuration)
Storage	32 GB SSD

7.2.11 CE-MPUF-FULL (CE16800 Main Processing Unit F (full-width))

7.2.11.1 CE-MPUF-FULL (03050AUY: CE16800 Main Processing Unit F (full-width))

Overview

Table 7-159 Basic information about the CE-MPUF-FULL

Item	Details
Description	CE16800 Main Processing Unit F (full-width)
Part Number	03050AUY
Model	CE-MPUF-FULL
Silkscreen	CE-MPUF-FULL
Card Type	MPU

Appearance

Figure 7-64 Appearance of the CE-MPUF-FULL



Version Mapping

Table 7-160 Mappings between CE-MPUF-FULL and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C00 SPC100	-	17-18	-

Table 7-161 Mappings between CE-MPUF-FULL and Interface Card

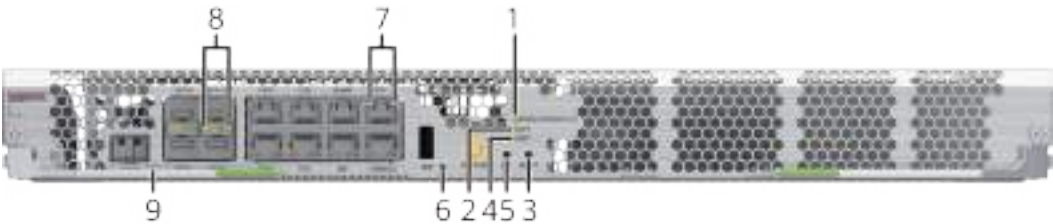
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C00 SPC100	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C00 SPC100	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R021C00 SPC100	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R021C00 SPC100	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R021C00 SPC100	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC500	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R021C00 SPC100	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL36LQFD-G	03058871	V200R021C00 SPC100	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 1	V200R021C00 SPC100	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R021C00 SPC100	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-00 1	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R021C00 SPC100	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 1	V200R021C00 SPC100	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 2	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-00 3	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R021C00 SPC100	-

Indicators and Buttons

Figure 7-65 Indicators and buttons on the CE-MPUF-FULL



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

 NOTE

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-162 Indicators on the CE-MPUF-FULL

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.
		Green	Off	The card is the standby MPU.

Silkscreen	Name	Color	Status	Description
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

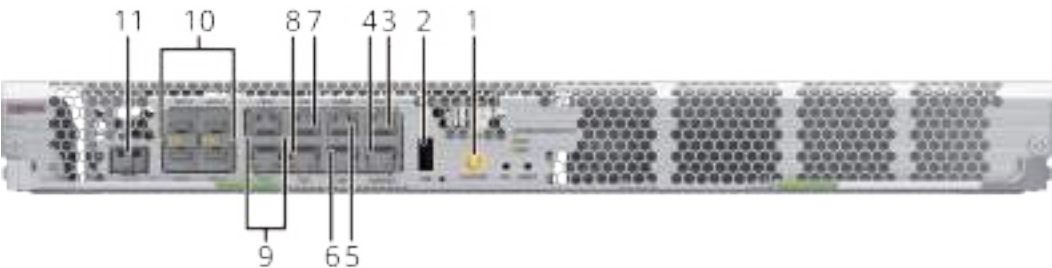
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-163 Buttons on the CE-MPUF-FULL

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-66 Ports on the CE-MPUF-FULL



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1 and HP-GE ports.

Table 7-164 Ports on the CE-MPUF-FULL

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-
FSP serial port	RJ45	Reserved function. This function is not enabled.	-

Port	Connector Type	Description	Available Components
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE • The SFP+2 and SFP+3 optical ports are reserved for stacking, and are not in use. • In V200R021C10 and later versions, SFP+0 and SFP+1 optical ports support the stack function.	Optical fiber
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-165 Functions and features of the CE-MPUF-FULL

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.

Functions and Features	Description
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-166 Technical specifications of the CE-MPUF-FULL

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.7 kg (12.57 lb)
Weight with packaging [kg (lb)]	8.7 kg (19.18 lb)
Typical power consumption [W]	117 W
Typical heat dissipation [BTU/hour]	400 BTU/hour
Static power consumption [W]	100 W
Static heat dissipation [BTU/hour]	342 BTU/hour

Item	Specification
Maximum power consumption [W]	186 W
Maximum heat dissipation [BTU/hour]	635 BTU/hour
CPU	8-core, 2.3 GHz clock speed
Memory	Standard configuration: 16 GB
Storage	32 GB SSD

7.2.11.2 CE-MPUF-FULL (03050AUY-001: CE16800 Main Processing Unit F (full-width))

Overview

Table 7-167 Basic information about the CE-MPUF-FULL

Item	Details
Description	CE16800 Main Processing Unit F (full-width)
Part Number	03050AUY-001
Model	CE-MPUF-FULL
Silkscreen	CE-MPUF-FULL
Card Type	MPU

Appearance

Figure 7-67 Appearance of the CE-MPUF-FULL



Version Mapping

Table 7-168 Mappings between CE-MPUF-FULL and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	17-18	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.

Table 7-169 Mappings between CE-MPUF-FULL and Interface Card

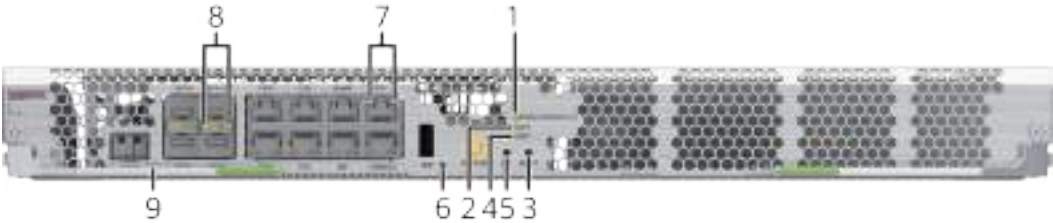
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL18CQFD-A	03050ECC	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-A	03050ECC-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL18CQFD-G	03058870-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL24LQFD-G	03058872-003	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL24LQFD1-G	03059669	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-A	03050EPY-001	V200R022C00	-
CloudEngine 16816	CEL36CQFD-G	03050EPU	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36CQFD-G	03058869-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-A	03050ECB-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD-G	03058871-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL36LQFD1-G	03059668	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-A	03050ECA-001	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CEL48XSFD-G	03058873-002	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD-G	03058873-003	V200R021C10 SPC600	-
CloudEngine 16816	CEL48XSFD1-G	03059670	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-68 Indicators and buttons on the CE-MPUF-FULL



1. Running status indicator	2. Active/ Standby status indicator	3. Reset button	4. Stack status indicator	5. OFL button
6. USB-based deployment indicator	7. ETH port indicator	8. SFP+ optical port indicator	9. High-precision clock optical transceiver indicator	-

 **NOTE**

The indicators on CONSOLE, ALMIN, 485, CLK, TOD, and FSP ports are not in use.

Table 7-170 Indicators on the CE-MPUF-FULL

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	A fault that affects services has occurred and it cannot be rectified automatically (critical alarm about hardware), or the card has generated an alarm because the memory size is not equal to the standard specification.
		Red	Fast blinking (4 Hz)	The system power is insufficient.
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
ACT	Active/ Standby status indicator	Green	Steady on	The card is the active MPU.

Silkscreen	Name	Color	Status	Description
		Green	Off	The card is the standby MPU.
M/S	Stack status indicator NOTE You can run the dfs-master led enable command to enable the stack status indicator to display the DFS group master and backup status. After this function is enabled, the stack status indicator on the DFS master device is steady on and that on the DFS backup device is off.	Green	Steady on	The stacking function is enabled, and the card is the active MPU of the stack.
		Green	Slow blinking (0.5 Hz)	The stacking function is enabled, and the card is not the active MPU of the stack.
		Green	Off	The stacking function is not enabled.
-	USB-based deployment indicator	Green	Steady on	USB-based deployment has been completed.
		Green	Blinking (4 Hz)	The system is reading data from a USB flash drive.
		Red	Steady on	USB-based deployment fails.
		-	Off	USB-based deployment is disabled (default state).

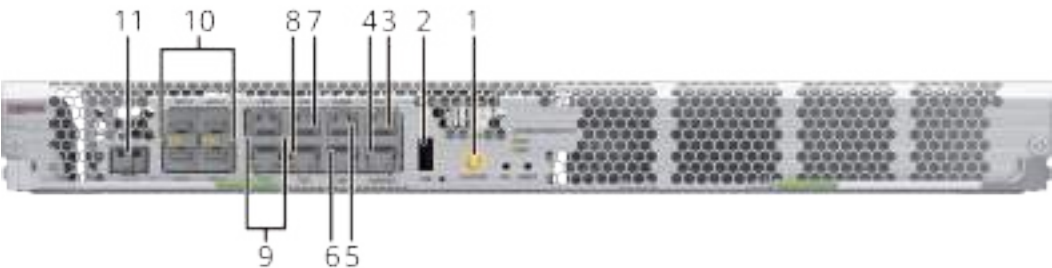
Silkscreen	Name	Color	Status	Description
-	ETH port indicator, Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	SFP+ optical port indicator Two single-color indicators for each port: Green: LINK indicator Yellow: ACT indicator	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.
-	High-precision clock optical transceiver indicator	Green	Steady on	The link on the port is connected.
		Green	Blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.

Table 7-171 Buttons on the CE-MPUF-FULL

Silkscreen	Name	Description
RESET	Reset button	The button is used to manually reset an MPU. If the device has only one MPU, pressing this button will cause the device to restart. If the device has double MPUs, the following situations may occur: Pressing the reset button on the active MPU will trigger an active/standby switchover. Pressing the reset button on the standby MPU will reset the standby MPU, which does not affect running of the system.
OFL	OFL button	The button is used to power off the MPU and collect logs. To implement this, you need to press and hold down the OFL button for about 6 seconds. NOTE The OFL button is valid only for the standby MPU.

Ports

Figure 7-69 Ports on the CE-MPUF-FULL



1. Clock SMA port (CLK/TEST)	2. USB port (USB)	3. ETH management port (ETH)	4. Console port (CONSOLE)	5. Boolean alarm input port (ALMIN)
6. RS485 port (485)	7. External synchronization port CLK (CLK)	8. External synchronization port TOD (TOD)	9. FSP serial port (FSP0 and FSP1)	10. SFP+ optical port (SFP+0, SFP+1, SFP+2, and SFP+3)
11. Clock SFP port (HP-GE)	-	-	-	-

 NOTE

The device does not support the CLK/TEST, ALMIN, 485, CLK, TOD, FSP0, FSP1 and HP-GE ports.

Table 7-172 Ports on the CE-MPUF-FULL

Port	Connector Type	Description	Available Components
Clock SMA port	SMA	Reserved function. This function is not enabled.	-
USB port	USB 2.0 Type A	Is used for USB-based deployment. It is recommended that the dimensions of the USB flash drive be less than 10 mm x 40 mm x 20 mm (H x W x D, 0.39 in. x 1.57 in. x 0.79 in.). If the dimensions of a USB flash drive are larger than this specification, an extension cable is required.	USB

Port	Connector Type	Description	Available Components
ETH management port	RJ45	Connects to the NMS workstation. This port can work at the rates of 10 Mbit/s, 100 Mbit/s, and 1000 Mbit/s.	Super category 5 shielded twisted pair
Console port	RJ45	Connected to the console for onsite system configuration. The default baud rate is 9600 baud, and the baud rate is configurable.	8-core shielded cable
Boolean input port	RJ45	Reserved function. This function is not enabled.	-
RS485 port	RJ45	Reserved function. This function is not enabled.	-
External synchronization port CLK	RJ45	Reserved function. This function is not enabled.	-
External synchronization port TOD	RJ45	Reserved function. This function is not enabled.	-
FSP serial port	RJ45	Reserved function. This function is not enabled.	-
SFP+ optical port	SFP+	SFP+0, SFP+1, SFP+2, and SFP+3 are four mutually independent 10GE optical ports that are dedicated to establishing stack management links. NOTE The SFP+2 and SFP+3 optical ports are reserved for stacking, and are not in use.	Optical fiber

Port	Connector Type	Description	Available Components
Clock SFP port	SFP	Reserved function. This function is not enabled.	-

Functions and Features

Table 7-173 Functions and features of the CE-MPUF-FULL

Functions and Features	Description
Device management and maintenance	The MPU provides management ports for managing and maintaining the system.
Device monitoring	The MPU integrates a monitoring module. The monitoring module provides the monitoring plane, which allows administrators to remotely power on, power off, and reset the card, upgrade firmware, monitor card temperature, voltage, and power, manage asset information, and diagnose system faults.
Out-of-band communication between cards	The MPU integrates a LAN switch module that provides out-of-band communication between cards. The LAN switch module completes control, monitoring, maintenance, and message exchange between SFUs and LPUs.
Route calculation	The MPU processes all routing protocol packets, which are sent from the forwarding engine. The MPU broadcasts and filters routing protocol packets, and downloads routing policies from the policy server.
Data configuration	The MPU stores system configuration data, startup files, upgrade software, and system logs.
Data saving	The MPU uses SATA to save data files.
Hot swapping	Supported

Functions and Features	Description
Restrictions and remarks	In the same chassis, the MPU cannot be installed together with an MPU of a different model.

Technical Specifications

Table 7-174 Technical specifications of the CE-MPUF-FULL

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	45.7 mm x 433 mm x 522.5 mm (1.80 in. x 17.05 in. x 20.57 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 550 mm x 720 mm (7.28 in. x 21.65 in. x 28.35 in.)
Weight without packaging [kg (lb)]	5.7 kg (12.57 lb)
Weight with packaging [kg (lb)]	8.7 kg (19.18 lb)
Typical power consumption [W]	117 W
Typical heat dissipation [BTU/hour]	400 BTU/hour
Static power consumption [W]	100 W
Static heat dissipation [BTU/hour]	342 BTU/hour
Maximum power consumption [W]	186 W
Maximum heat dissipation [BTU/hour]	635 BTU/hour
CPU	8-core, 2.3 GHz clock speed
Memory	16 GB (standard configuration)
Storage	32 GB SSD

7.3 SFU

7.3.1 CE-SFU04G-G (CE16804 Switch Fabric Unit G)

7.3.1.1 CE-SFU04G-G (03058880: CE16804 Switch Fabric Unit G)

Overview

Table 7-175 Basic information about the CE-SFU04G-G

Item	Details
Description	CE16804 Switch Fabric Unit G
Part Number	03058880
Model	CE-SFU04G-G
Silkscreen	CE-SFU04G-G
Card Type	SFU

Appearance

Figure 7-70 Appearance of the CE-SFU04G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-176 Mappings between CE-SFU04G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	SFU3-SFU8	-

Table 7-177 Mappings between CE-SFU04G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQF D-G	03058870	V200R005C20	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872	V200R005C20	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R005C20	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R005C20	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R005C20	-	-

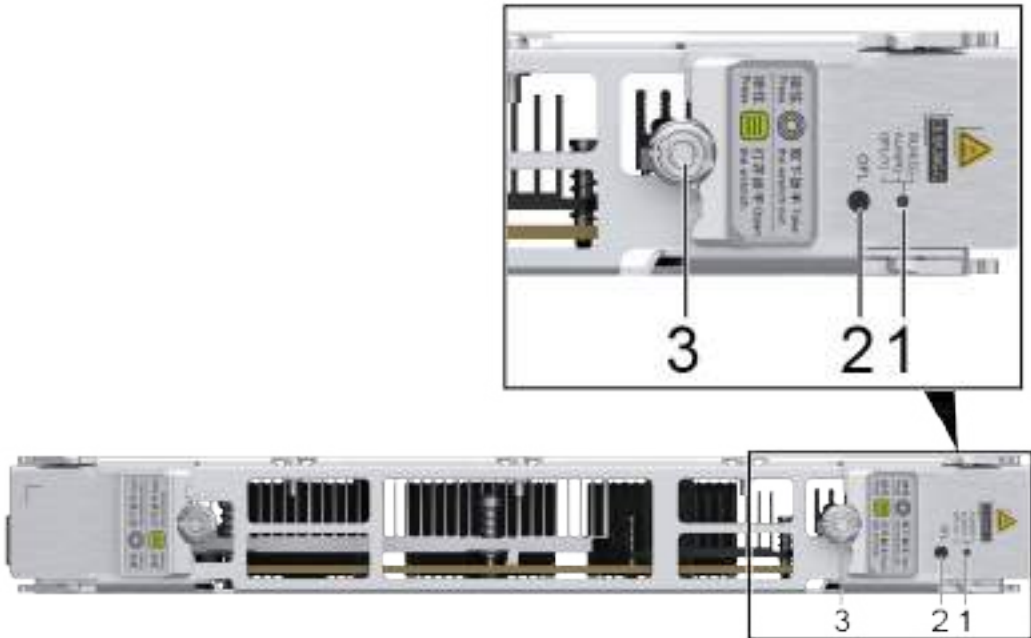
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-178 Mappings between CE-SFU04G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R005C20	-

Indicators and Buttons

Figure 7-71 Indicators and buttons on the CE-SFU04G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-179 Indicators on the CE-SFU04G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-180 Buttons on the CE-SFU04G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-181 Functions and features of the CE-SFU04G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-182 Technical specifications of the CE-SFU04G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.1.2 CE-SFU04G-G (03058880-001: CE16804 Switch Fabric Unit G)

Overview

Table 7-183 Basic information about the CE-SFU04G-G

Item	Details
Description	CE16804 Switch Fabric Unit G
Part Number	03058880-001
Model	CE-SFU04G-G
Silkscreen	CE-SFU04G-G
Card Type	SFU

Appearance

Figure 7-72 Appearance of the CE-SFU04G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-184 Mappings between CE-SFU04G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH007 or a later version.

Table 7-185 Mappings between CE-SFU04G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-G	03058870	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R020C10	-	-

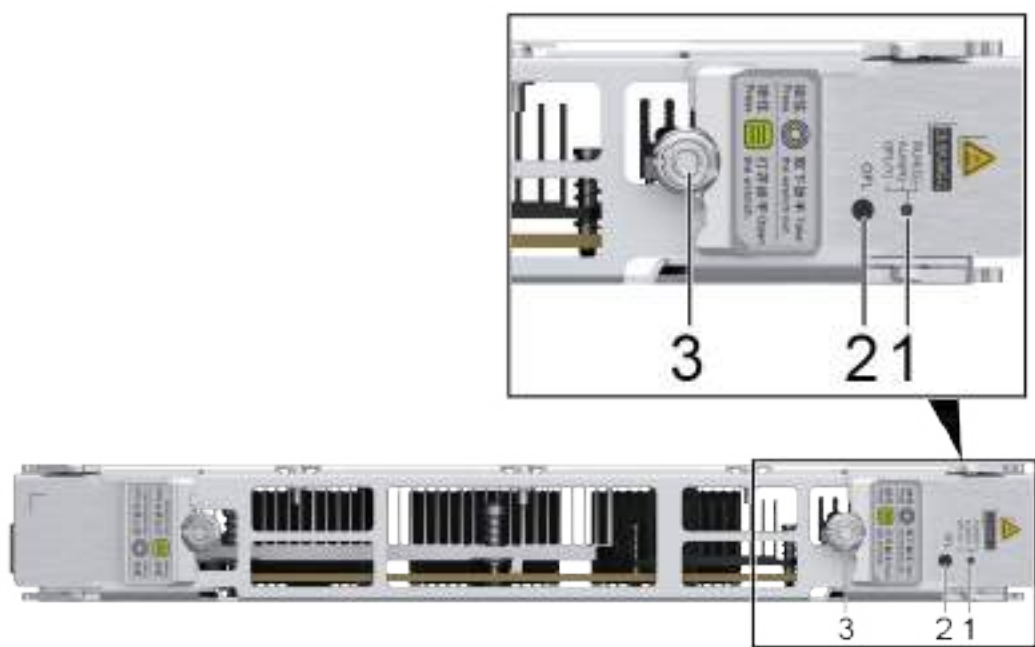
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872	V200R020C10	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R020C10	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-186 Mappings between CE-SFU04G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R020C10	-

Indicators and Buttons

Figure 7-73 Indicators and buttons on the CE-SFU04G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-187 Indicators on the CE-SFU04G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-188 Buttons on the CE-SFU04G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.

Silkscreen	Name	Description
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-189 Functions and features of the CE-SFU04G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-190 Technical specifications of the CE-SFU04G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)

Item	Specification
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.1.3 CE-SFU04G-G (03058880-002: CE16804 Switch Fabric Unit G)

Overview

Table 7-191 Basic information about the CE-SFU04G-G

Item	Details
Description	CE16804 Switch Fabric Unit G
Part Number	03058880-002
Model	CE-SFU04G-G
Silkscreen	CE-SFU04G-G
Card Type	SFU

Appearance

Figure 7-74 Appearance of the CE-SFU04G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-192 Mappings between CE-SFU04G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-193 Mappings between CE-SFU04G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQF D-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-

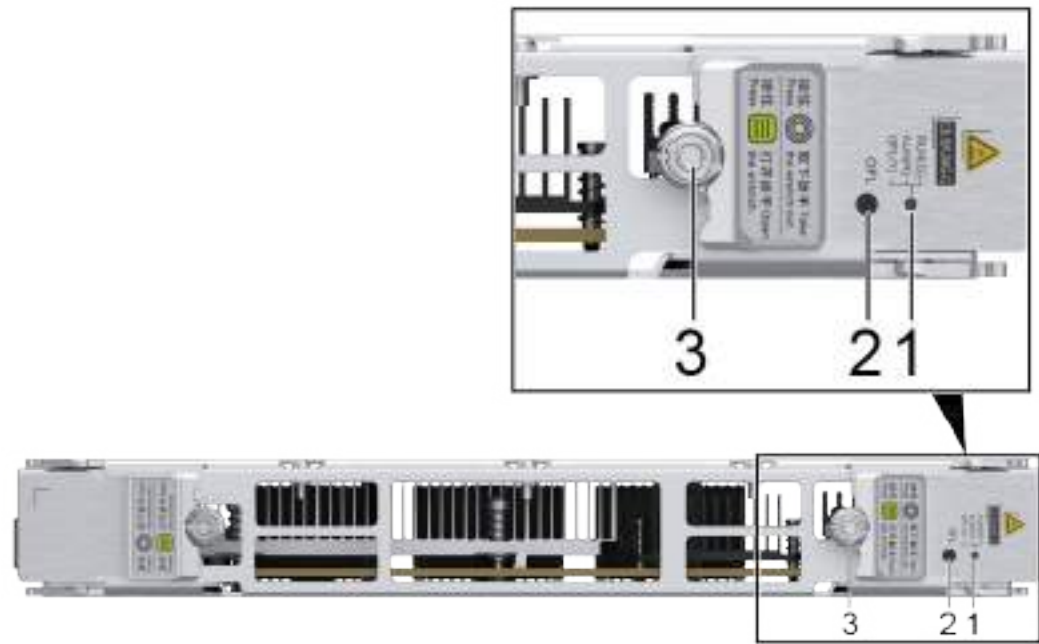
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-194 Mappings between CE-SFU04G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-75 Indicators and buttons on the CE-SFU04G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-195 Indicators on the CE-SFU04G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-196 Buttons on the CE-SFU04G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.

Silkscreen	Name	Description
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-197 Functions and features of the CE-SFU04G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-198 Technical specifications of the CE-SFU04G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)

Item	Specification
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.2 CE-SFU04G-GK (CE16804 Switch Fabric Unit GK)

7.3.2.1 CE-SFU04G-GK (03059596: CE16804 Switch Fabric Unit GK)

Overview

Table 7-199 Basic information about the CE-SFU04G-GK

Item	Details
Description	CE16804 Switch Fabric Unit GK
Part Number	03059596
Model	CE-SFU04G-GK
Silkscreen	CE-SFU04G-GK
Card Type	SFU

Appearance

Figure 7-76 Appearance of the CE-SFU04G-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-200 Mappings between CE-SFU04G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	SFU3-SFU8	-

Table 7-201 Mappings between CE-SFU04G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQF D-GK	03059591	V200R019C10	-	-
CloudEngine 16804	CEL18CQF D-GK	03059591-001	V200R020C10	-	-
CloudEngine 16804	CEL18CQF D-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-GK	03059593	V200R019C10	-	-
CloudEngine 16804	CEL24LQF D-GK	03059593-001	V200R020C10	-	-
CloudEngine 16804	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-GK	03059589	V200R019C10	-	-
CloudEngine 16804	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-GK	03059592	V200R019C10	-	-
CloudEngine 16804	CEL36LQF D-GK	03059592-001	V200R020C10	-	-

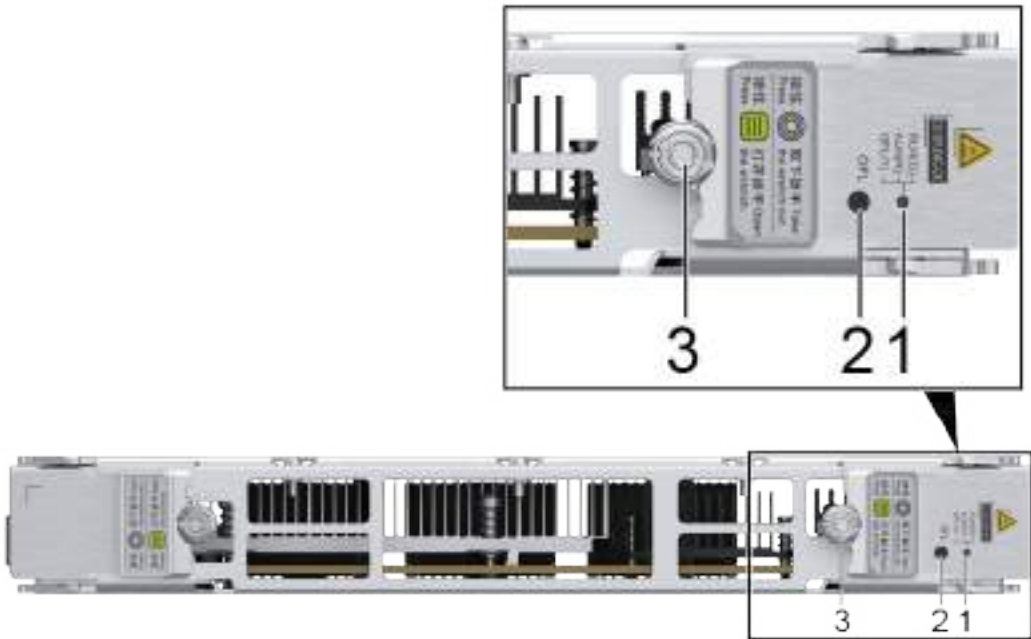
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL36LQFD-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594	V200R019C10	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-202 Mappings between CE-SFU04G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R019C10	-

Indicators and Buttons

Figure 7-77 Indicators and buttons on the CE-SFU04G-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-203 Indicators on the CE-SFU04G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-204 Buttons on the CE-SFU04G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-205 Functions and features of the CE-SFU04G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-206 Technical specifications of the CE-SFU04G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.2.2 CE-SFU04G-GK (03059596-001: CE16804 Switch Fabric Unit GK)

Overview

Table 7-207 Basic information about the CE-SFU04G-GK

Item	Details
Description	CE16804 Switch Fabric Unit GK
Part Number	03059596-001
Model	CE-SFU04G-GK
Silkscreen	CE-SFU04G-GK
Card Type	SFU

Appearance

Figure 7-78 Appearance of the CE-SFU04G-GK



Version Mapping

Table 7-208 Mappings between CE-SFU04G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH007 or a later version.

Table 7-209 Mappings between CE-SFU04G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-GK	03059591	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-GK	03059591-001	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-GK	03059593	V200R020C10	-	-
CloudEngine 16804	CEL24LQFD-GK	03059593-001	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-GK	03059589	V200R020C10	-	-
CloudEngine 16804	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-GK	03059592	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-GK	03059592-001	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-210 Mappings between CE-SFU04G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R020C10	-

Indicators and Buttons

Figure 7-79 Indicators and buttons on the CE-SFU04G-GK

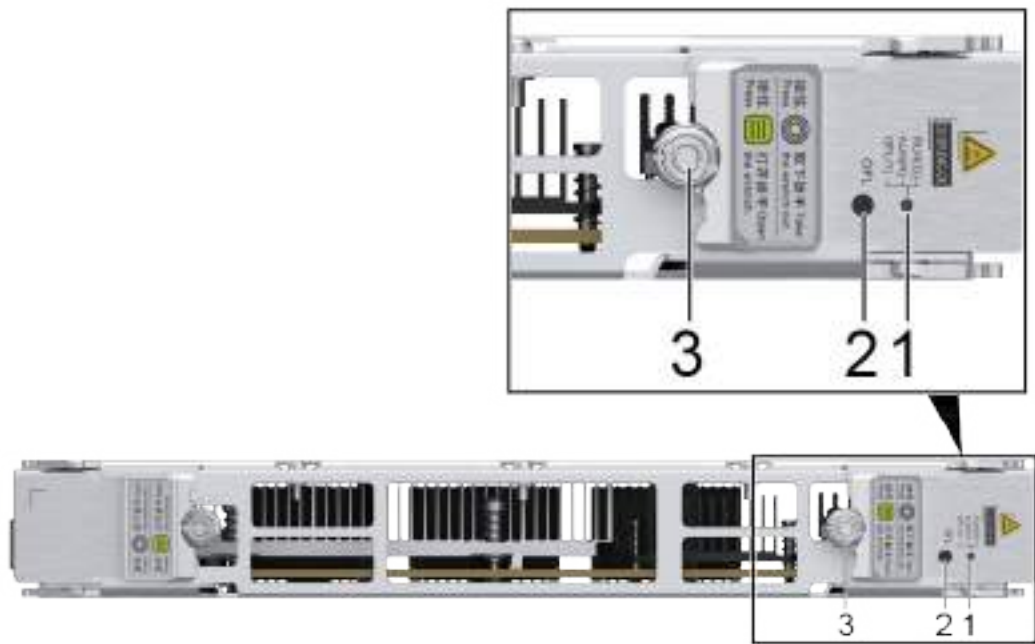


Table 7-211 Indicators on the CE-SFU04G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-212 Buttons on the CE-SFU04G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-213 Functions and features of the CE-SFU04G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-214 Technical specifications of the CE-SFU04G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.2.3 CE-SFU04G-GK (03059596-002: CE16804 Switch Fabric Unit GK)

Overview

Table 7-215 Basic information about the CE-SFU04G-GK

Item	Details
Description	CE16804 Switch Fabric Unit GK
Part Number	03059596-002
Model	CE-SFU04G-GK
Silkscreen	CE-SFU04G-GK
Card Type	SFU

Appearance

Figure 7-80 Appearance of the CE-SFU04G-GK



Version Mapping

Table 7-216 Mappings between CE-SFU04G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH020 or a later version.

Table 7-217 Mappings between CE-SFU04G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-GK	03059591	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-GK	03059591-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-GK	03059593	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-GK	03059593-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-GK	03059589	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQFD-GK	03059592	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQFD-GK	03059592-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL36LQFD-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-218 Mappings between CE-SFU04G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-81 Indicators and buttons on the CE-SFU04G-GK

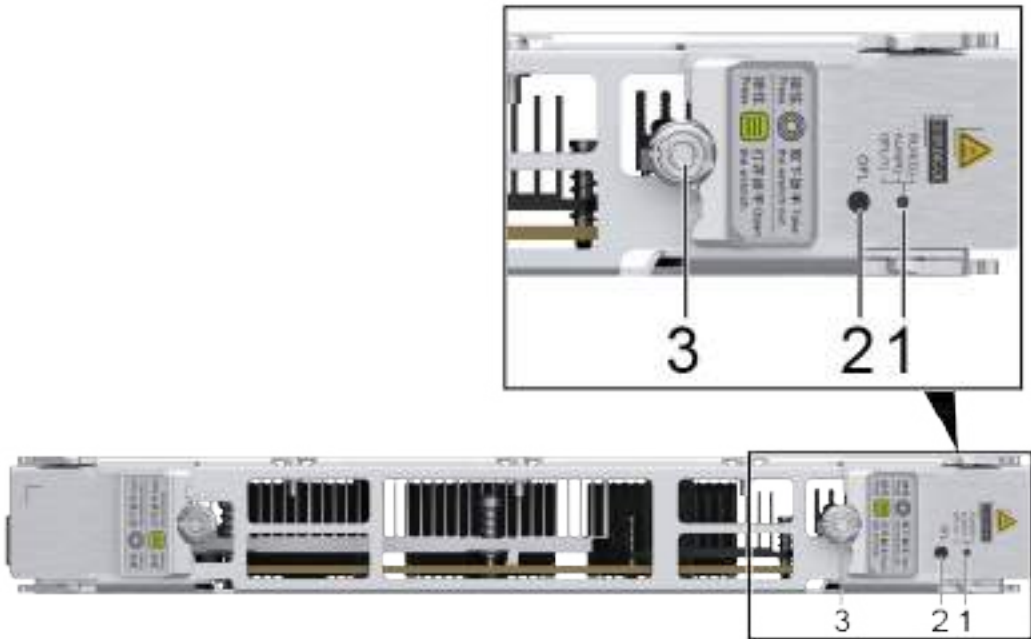


Table 7-219 Indicators on the CE-SFU04G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-220 Buttons on the CE-SFU04G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-221 Functions and features of the CE-SFU04G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-222 Technical specifications of the CE-SFU04G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	87 W
Typical heat dissipation [BTU/hour]	297 BTU/hour
Static power consumption [W]	74 W
Static heat dissipation [BTU/hour]	253 BTU/hour
Maximum power consumption [W]	128 W
Maximum heat dissipation [BTU/hour]	437 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.3 CE-SFU04F-G (CE16804 Switch Fabric Unit F)

7.3.3.1 CE-SFU04F-G (03058881: CE16804 Switch Fabric Unit F)

Overview

Table 7-223 Basic information about the CE-SFU04F-G

Item	Details
Description	CE16804 Switch Fabric Unit F
Part Number	03058881
Model	CE-SFU04F-G
Silkscreen	CE-SFU04F-G
Card Type	SFU

Appearance

Figure 7-82 Appearance of the CE-SFU04F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-224 Mappings between CE-SFU04F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	SFU3-SFU8	-

Table 7-225 Mappings between CE-SFU04F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-G	03058870	V200R005C20	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R005C20	-	-

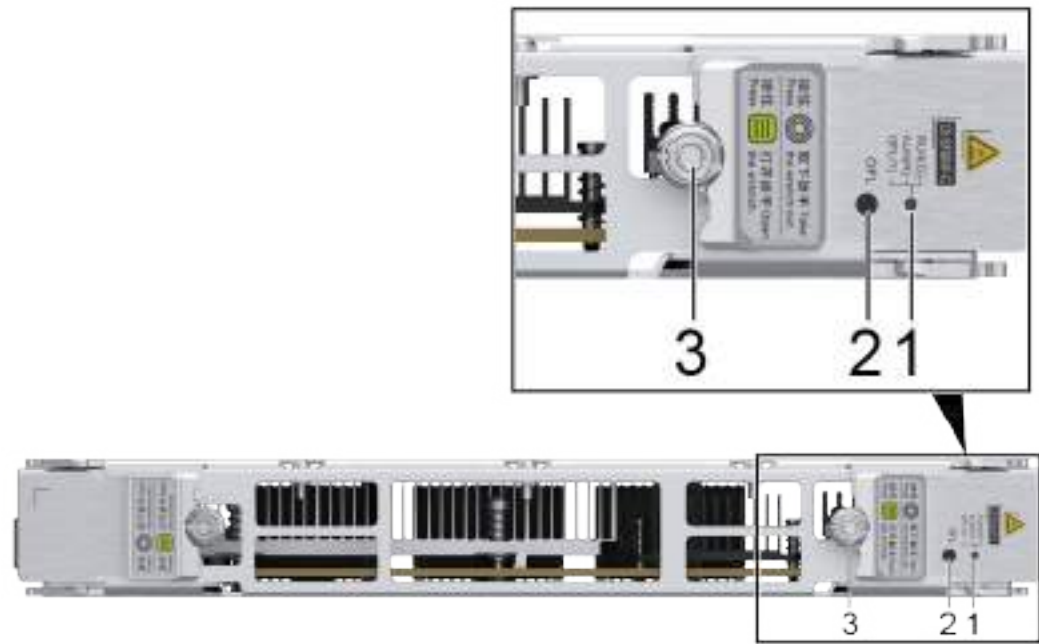
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R005C20	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R005C20	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R005C20	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-226 Mappings between CE-SFU04F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R005C20	-

Indicators and Buttons

Figure 7-83 Indicators and buttons on the CE-SFU04F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-227 Indicators on the CE-SFU04F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-228 Buttons on the CE-SFU04F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.

Silkscreen	Name	Description
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-229 Functions and features of the CE-SFU04F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-230 Technical specifications of the CE-SFU04F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)

Item	Specification
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	72 W
Typical heat dissipation [BTU/hour]	246 BTU/hour
Static power consumption [W]	62 W
Static heat dissipation [BTU/hour]	212 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.3.2 CE-SFU04F-G (03058881-001: CE16804 Switch Fabric Unit F)

Overview

Table 7-231 Basic information about the CE-SFU04F-G

Item	Details
Description	CE16804 Switch Fabric Unit F
Part Number	03058881-001
Model	CE-SFU04F-G
Silkscreen	CE-SFU04F-G
Card Type	SFU

Appearance

Figure 7-84 Appearance of the CE-SFU04F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-232 Mappings between CE-SFU04F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH007 or a later version.

Table 7-233 Mappings between CE-SFU04F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-G	03058870	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-001	V200R020C10	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-G	03058872	V200R020C10	-	-
CloudEngine 16804	CEL24LQFD-G	03058872-001	V200R020C10	-	-
CloudEngine 16804	CEL24LQFD-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQFD-G	03058872-003	V200R021C10SPC600	-	-

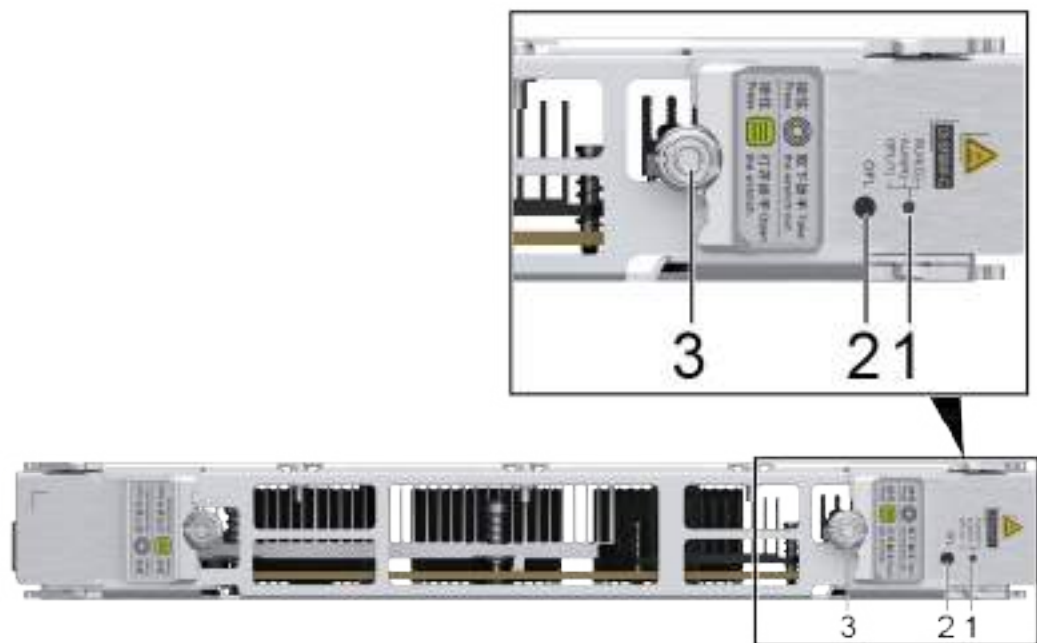
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R020C10	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-234 Mappings between CE-SFU04F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R020C10	-

Indicators and Buttons

Figure 7-85 Indicators and buttons on the CE-SFU04F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-235 Indicators on the CE-SFU04F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-236 Buttons on the CE-SFU04F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-237 Functions and features of the CE-SFU04F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-238 Technical specifications of the CE-SFU04F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	72 W
Typical heat dissipation [BTU/hour]	246 BTU/hour
Static power consumption [W]	62 W
Static heat dissipation [BTU/hour]	212 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.3.3 CE-SFU04F-G (03058881-002: CE16804 Switch Fabric Unit F)

Overview

Table 7-239 Basic information about the CE-SFU04F-G

Item	Details
Description	CE16804 Switch Fabric Unit F
Part Number	03058881-002
Model	CE-SFU04F-G
Silkscreen	CE-SFU04F-G
Card Type	SFU

Appearance

Figure 7-86 Appearance of the CE-SFU04F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-240 Mappings between CE-SFU04F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-241 Mappings between CE-SFU04F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQF D-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-

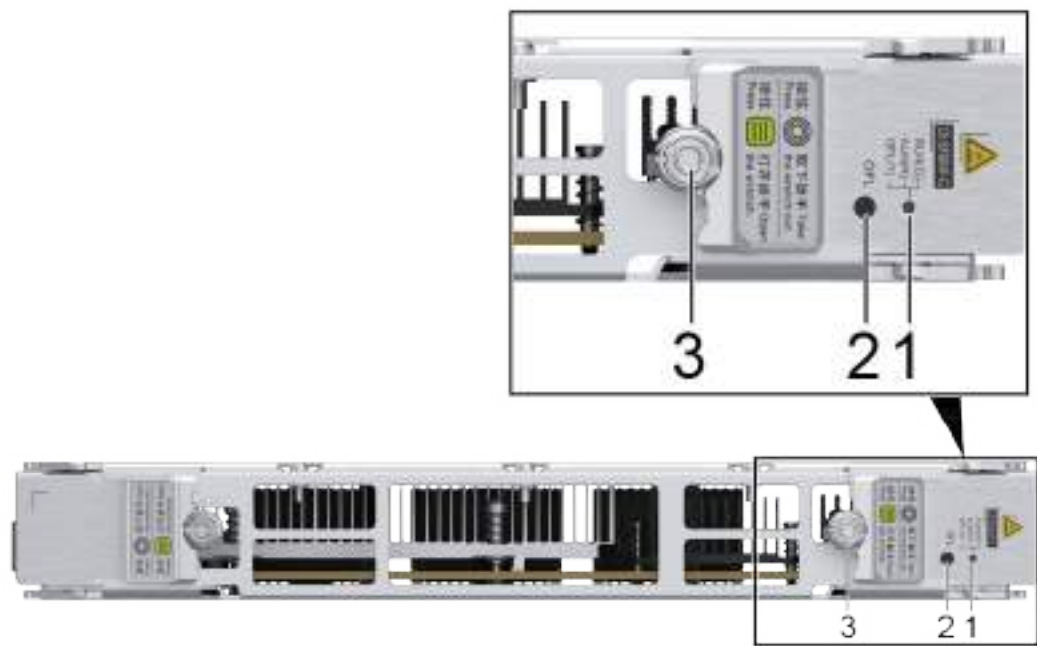
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-242 Mappings between CE-SFU04F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-87 Indicators and buttons on the CE-SFU04F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-243 Indicators on the CE-SFU04F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-244 Buttons on the CE-SFU04F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-245 Functions and features of the CE-SFU04F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-246 Technical specifications of the CE-SFU04F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	72 W
Typical heat dissipation [BTU/hour]	246 BTU/hour
Static power consumption [W]	62 W
Static heat dissipation [BTU/hour]	212 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.4 CE-SFU04F1-G (CE16804 Switch Fabric Unit F1)

Overview

Table 7-247 Basic information about the CE-SFU04F1-G

Item	Details
Description	CE16804 Switch Fabric Unit F1
Part Number	03059671
Model	CE-SFU04F1-G
Silkscreen	CE-SFU04F1-G
Card Type	SFU

Appearance

Figure 7-88 Appearance of the CE-SFU04F1-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-248 Mappings between CE-SFU04F1-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	SFU3-SFU8	-

Table 7-249 Mappings between CE-SFU04F1-G and Interface Card

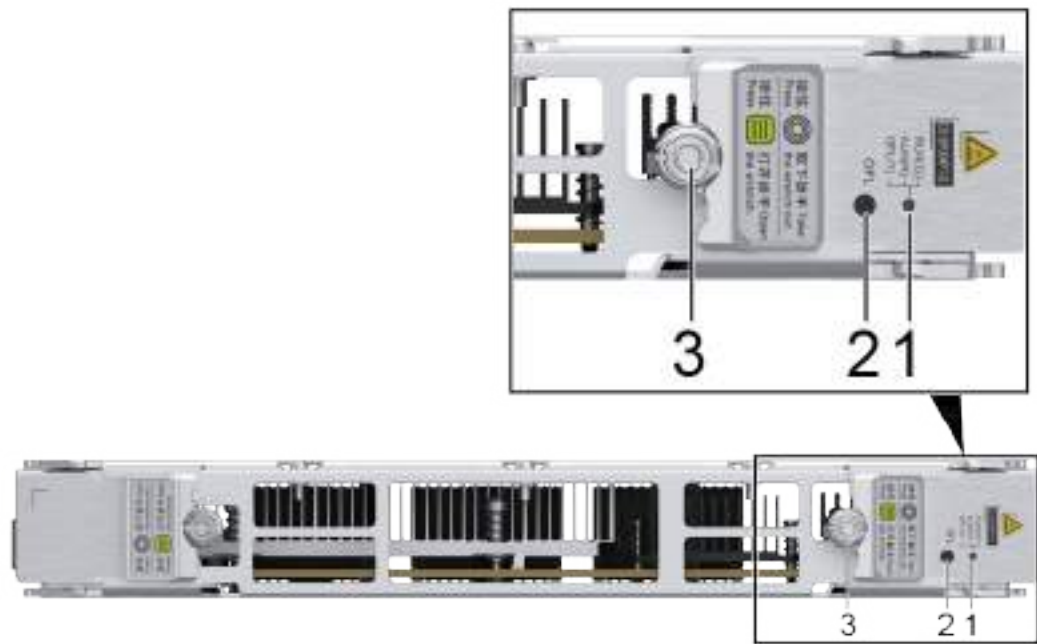
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL24LQF D1-G	03059669	V200R019C10	-	-
CloudEngine 16804	CEL36LQF D1-G	03059668	V200R019C10	-	-
CloudEngine 16804	CEL48XSF D1-G	03059670	V200R019C10	-	-

Table 7-250 Mappings between CE-SFU04F1-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R019C10	-

Indicators and Buttons

Figure 7-89 Indicators and buttons on the CE-SFU04F1-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-251 Indicators on the CE-SFU04F1-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-252 Buttons on the CE-SFU04F1-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-253 Functions and features of the CE-SFU04F1-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-254 Technical specifications of the CE-SFU04F1-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	72 W
Typical heat dissipation [BTU/hour]	246 BTU/hour
Static power consumption [W]	62 W
Static heat dissipation [BTU/hour]	212 BTU/hour
Maximum power consumption [W]	99 W
Maximum heat dissipation [BTU/hour]	338 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.5 CE-SFU04G-A (CE16804 Switch Fabric Unit G-A)

7.3.5.1 CE-SFU04G-A (03050EGY: CE16804 Switch Fabric Unit G-A)

Overview

Table 7-255 Basic information about the CE-SFU04G-A

Item	Details
Description	CE16804 Switch Fabric Unit G-A
Part Number	03050EGY
Model	CE-SFU04G-A
Silkscreen	CE-SFU04G-A
Card Type	SFU

Appearance

Figure 7-90 Appearance of the CE-SFU04G-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-256 Mappings between CE-SFU04G-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	SFU3-SFU8	-

Table 7-257 Mappings between CE-SFU04G-A and Interface Card

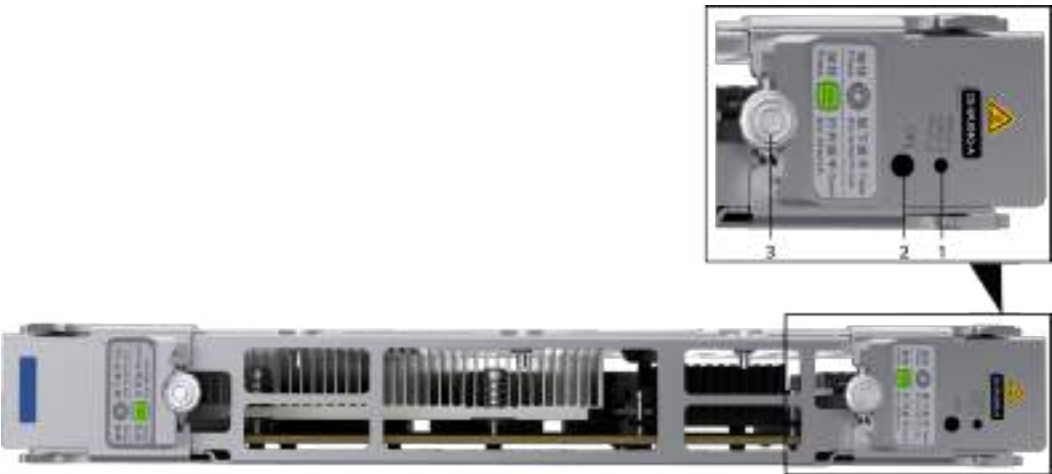
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C00SPC100	-	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C00SPC100	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C00SPC100	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-258 Mappings between CE-SFU04G-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C00 SPC100	-

Indicators and Buttons

Figure 7-91 Indicators and buttons on the CE-SFU04G-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-259 Indicators on the CE-SFU04G-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-260 Buttons on the CE-SFU04G-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-261 Functions and features of the CE-SFU04G-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-262 Technical specifications of the CE-SFU04G-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	351 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	144 W
Maximum heat dissipation [BTU/hour]	492 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed

Item	Specification
Memory	Standard 2 GB

7.3.5.2 CE-SFU04G-A (03050EGY-001: CE16804 Switch Fabric Unit G-A)

Overview

Table 7-263 Basic information about the CE-SFU04G-A

Item	Details
Description	CE16804 Switch Fabric Unit G-A
Part Number	03050EGY-001
Model	CE-SFU04G-A
Silkscreen	CE-SFU04G-A
Card Type	SFU

Appearance

Figure 7-92 Appearance of the CE-SFU04G-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-264 Mappings between CE-SFU04G-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH002 or a later version.

Table 7-265 Mappings between CE-SFU04G-A and Interface Card

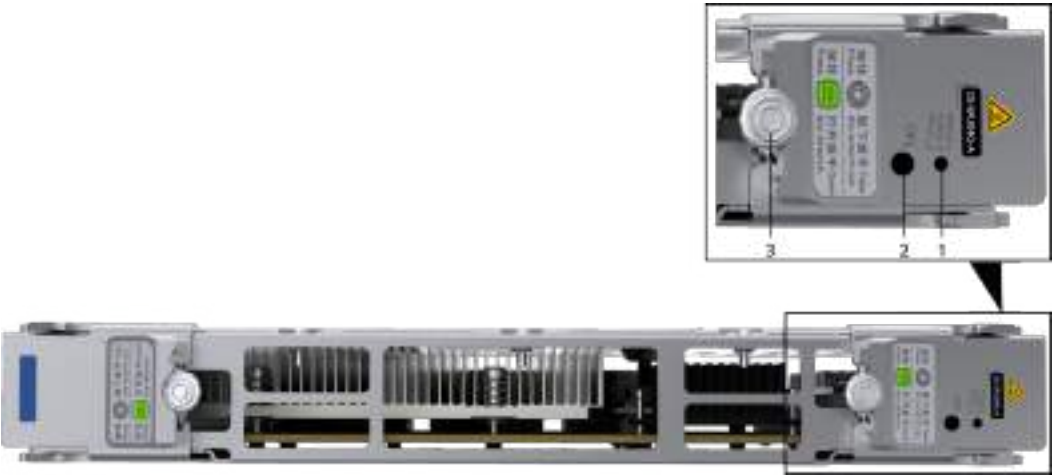
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-266 Mappings between CE-SFU04G-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-93 Indicators and buttons on the CE-SFU04G-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-267 Indicators on the CE-SFU04G-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-268 Buttons on the CE-SFU04G-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-269 Functions and features of the CE-SFU04G-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-270 Technical specifications of the CE-SFU04G-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	351 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	144 W
Maximum heat dissipation [BTU/hour]	492 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.6 CE-SFU04F-A (CE16804 Switch Fabric Unit F-A)

7.3.6.1 CE-SFU04F-A (03050ECE: CE16804 Switch Fabric Unit F-A)

Overview

Table 7-271 Basic information about the CE-SFU04F-A

Item	Details
Description	CE16804 Switch Fabric Unit F-A
Part Number	03050ECE
Model	CE-SFU04F-A
Silkscreen	CE-SFU04F-A
Card Type	SFU

Appearance

Figure 7-94 Appearance of the CE-SFU04F-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-272 Mappings between CE-SFU04F-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	SFU3-SFU8	-

Table 7-273 Mappings between CE-SFU04F-A and Interface Card

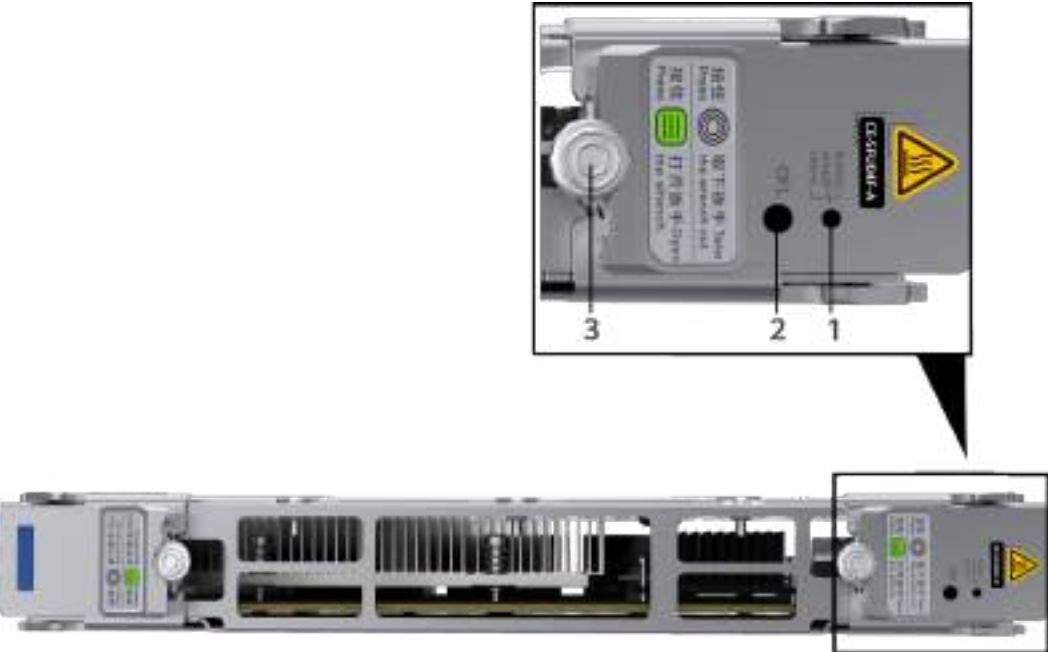
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C00SPC100	-	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C00SPC100	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA	V200R021C00SPC100	-	-
CloudEngine 16804	CEL48XSFD-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-274 Mappings between CE-SFU04F-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C00 SPC100	-

Indicators and Buttons

Figure 7-95 Indicators and buttons on the CE-SFU04F-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-275 Indicators on the CE-SFU04F-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-276 Buttons on the CE-SFU04F-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-277 Functions and features of the CE-SFU04F-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-278 Technical specifications of the CE-SFU04F-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	98 W
Typical heat dissipation [BTU/hour]	335 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	119 W
Maximum heat dissipation [BTU/hour]	406 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed

Item	Specification
Memory	Standard 2 GB

7.3.6.2 CE-SFU04F-A (03050ECE-001: CE16804 Switch Fabric Unit F-A)

Overview

Table 7-279 Basic information about the CE-SFU04F-A

Item	Details
Description	CE16804 Switch Fabric Unit F-A
Part Number	03050ECE-001
Model	CE-SFU04F-A
Silkscreen	CE-SFU04F-A
Card Type	SFU

Appearance

Figure 7-96 Appearance of the CE-SFU04F-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-280 Mappings between CE-SFU04F-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH0022 or a later version.

Table 7-281 Mappings between CE-SFU04F-A and Interface Card

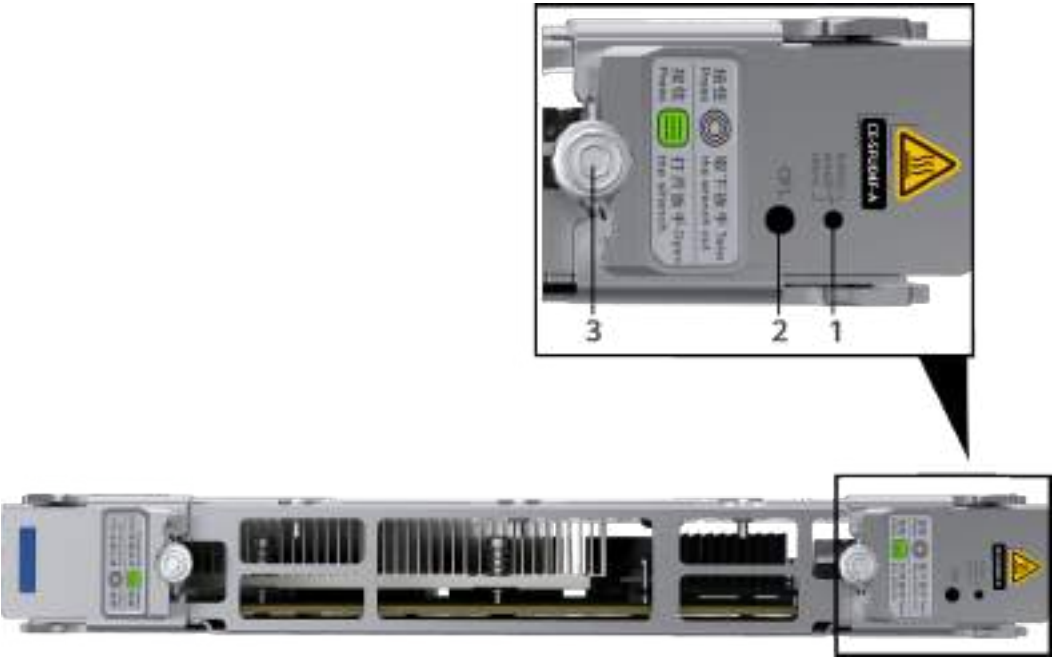
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CEL18CQFD-A	03050ECC	V200R021C10SPC600	-	-
CloudEngine 16804	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB	V200R021C10SPC600	-	-
CloudEngine 16804	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-A	03050ECA	V200R021C10SPC600	-	-
CloudEngine 16804	CEL48XSF D-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-282 Mappings between CE-SFU04F-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	FAN-240SA-B	02122487	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-97 Indicators and buttons on the CE-SFU04F-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-283 Indicators on the CE-SFU04F-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-284 Buttons on the CE-SFU04F-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.

Silkscreen	Name	Description
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-285 Functions and features of the CE-SFU04F-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-286 Technical specifications of the CE-SFU04F-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 316.5 mm x 233.6 mm (1.67 in. x 12.46 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	185 mm x 485 mm x 390 mm (7.28 in. x 19.09 in. x 15.35 in.)
Weight without packaging [kg (lb)]	3.3 kg (7.28 lb)

Item	Specification
Weight with packaging [kg (lb)]	4.7 kg (10.36 lb)
Typical power consumption [W]	98 W
Typical heat dissipation [BTU/hour]	335 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	119 W
Maximum heat dissipation [BTU/hour]	406 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.7 CE-SFU08G-G (CE16808 Switch Fabric Unit G)

7.3.7.1 CE-SFU08G-G (03058882: CE16808 Switch Fabric Unit G)

Overview

Table 7-287 Basic information about the CE-SFU08G-G

Item	Details
Description	CE16808 Switch Fabric Unit G
Part Number	03058882
Model	CE-SFU08G-G
Silkscreen	CE-SFU08G-G
Card Type	SFU

Appearance

Figure 7-98 Appearance of the CE-SFU08G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-288 Mappings between CE-SFU08G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R005C20	-	SFU3-SFU8	-

Table 7-289 Mappings between CE-SFU08G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-G	03058870	V200R005C20	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872	V200R005C20	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16808	CEL36CQF D-G	03058869	V200R005C20	-	-

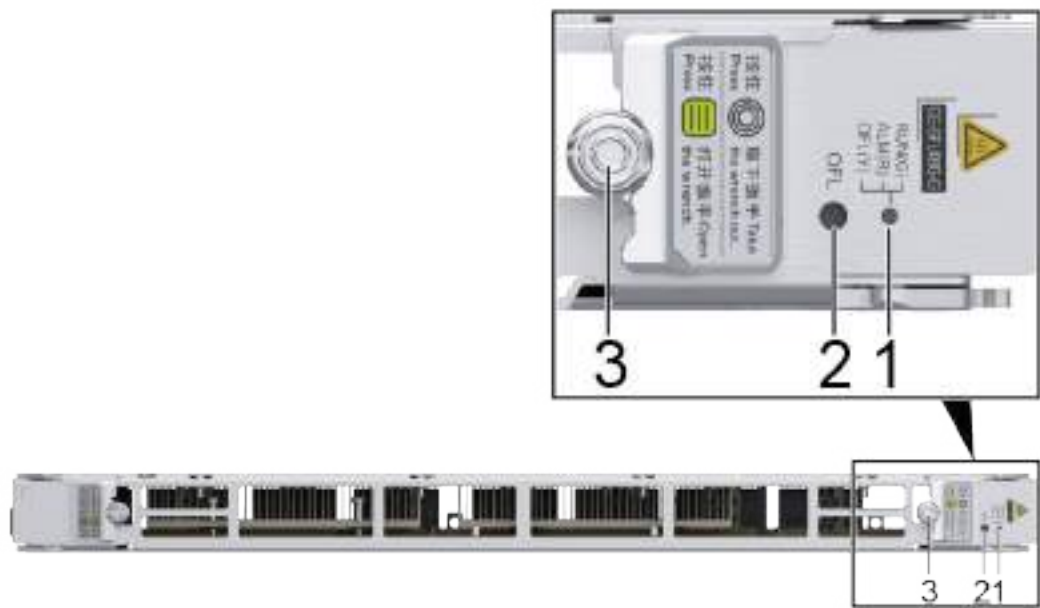
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871	V200R005C20	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873	V200R005C20	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-290 Mappings between CE-SFU08G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R005C20	-

Indicators and Buttons

Figure 7-99 Indicators and buttons on the CE-SFU08G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-291 Indicators on the CE-SFU08G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-292 Buttons on the CE-SFU08G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-293 Functions and features of the CE-SFU08G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-294 Technical specifications of the CE-SFU08G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.7.2 CE-SFU08G-G (03058882-001: CE16808 Switch Fabric Unit G)

Overview

Table 7-295 Basic information about the CE-SFU08G-G

Item	Details
Description	CE16808 Switch Fabric Unit G
Part Number	03058882-001
Model	CE-SFU08G-G
Silkscreen	CE-SFU08G-G
Card Type	SFU

Appearance

Figure 7-100 Appearance of the CE-SFU08G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-296 Mappings between CE-SFU08G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH007 or a later version.

Table 7-297 Mappings between CE-SFU08G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-G	03058870	V200R020C10	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16808	CEL36CQF D-G	03058869	V200R020C10	-	-

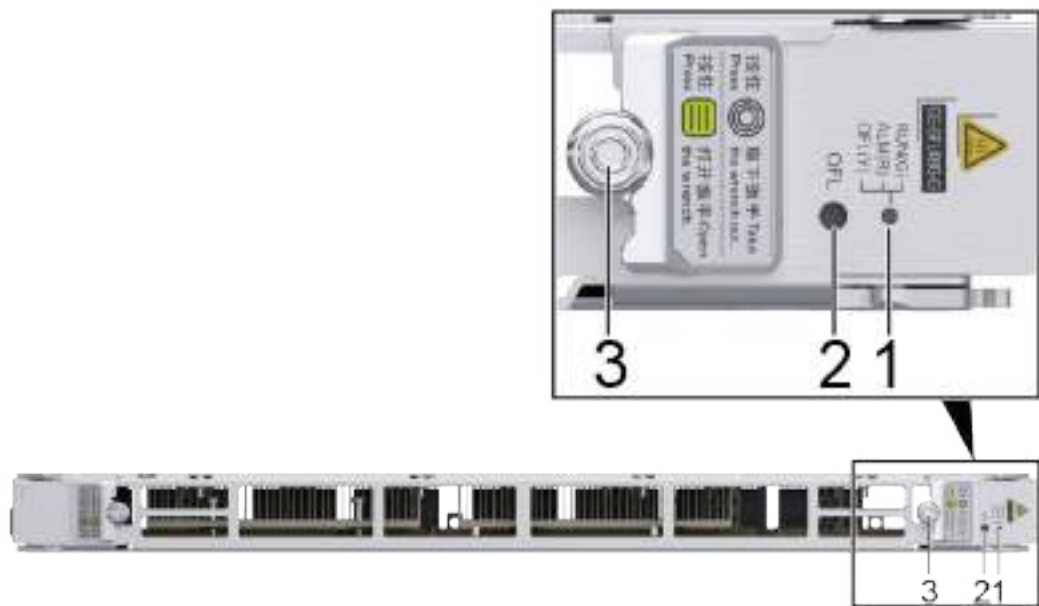
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-298 Mappings between CE-SFU08G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R020C10	-

Indicators and Buttons

Figure 7-101 Indicators and buttons on the CE-SFU08G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-299 Indicators on the CE-SFU08G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-300 Buttons on the CE-SFU08G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-301 Functions and features of the CE-SFU08G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-302 Technical specifications of the CE-SFU08G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.7.3 CE-SFU08G-G (03058882-002: CE16808 Switch Fabric Unit G)

Overview

Table 7-303 Basic information about the CE-SFU08G-G

Item	Details
Description	CE16808 Switch Fabric Unit G
Part Number	03058882-002
Model	CE-SFU08G-G
Silkscreen	CE-SFU08G-G
Card Type	SFU

Appearance

Figure 7-102 Appearance of the CE-SFU08G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-304 Mappings between CE-SFU08G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-305 Mappings between CE-SFU08G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-

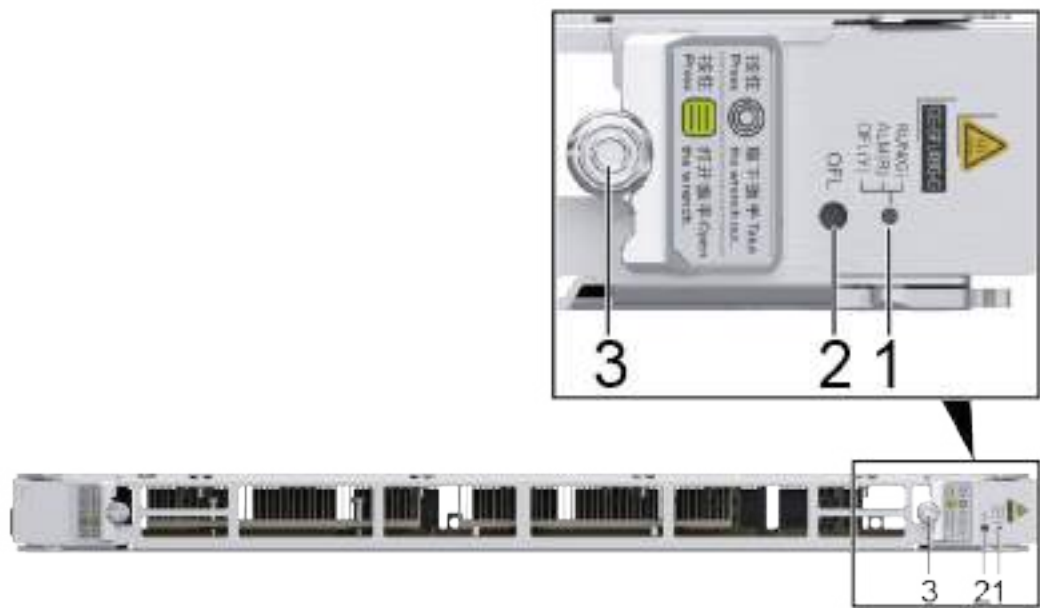
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-306 Mappings between CE-SFU08G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-103 Indicators and buttons on the CE-SFU08G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-307 Indicators on the CE-SFU08G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-308 Buttons on the CE-SFU08G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-309 Functions and features of the CE-SFU08G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-310 Technical specifications of the CE-SFU08G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.8 CE-SFU08G-GK (CE16808 Switch Fabric Unit GK)

7.3.8.1 CE-SFU08G-GK (03059595: CE16808 Switch Fabric Unit GK)

Overview

Table 7-311 Basic information about the CE-SFU08G-GK

Item	Details
Description	CE16808 Switch Fabric Unit GK
Part Number	03059595
Model	CE-SFU08G-GK
Silkscreen	CE-SFU08G-GK
Card Type	SFU

Appearance

Figure 7-104 Appearance of the CE-SFU08G-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-312 Mappings between CE-SFU08G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R019C10	-	SFU3-SFU8	-

Table 7-313 Mappings between CE-SFU08G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-GK	03059591	V200R019C10	-	-
CloudEngine 16808	CEL18CQF D-GK	03059591-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQF D-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593	V200R019C10	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-001	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589	V200R019C10	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592	V200R019C10	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594	V200R019C10	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-001	V200R020C10	-	-

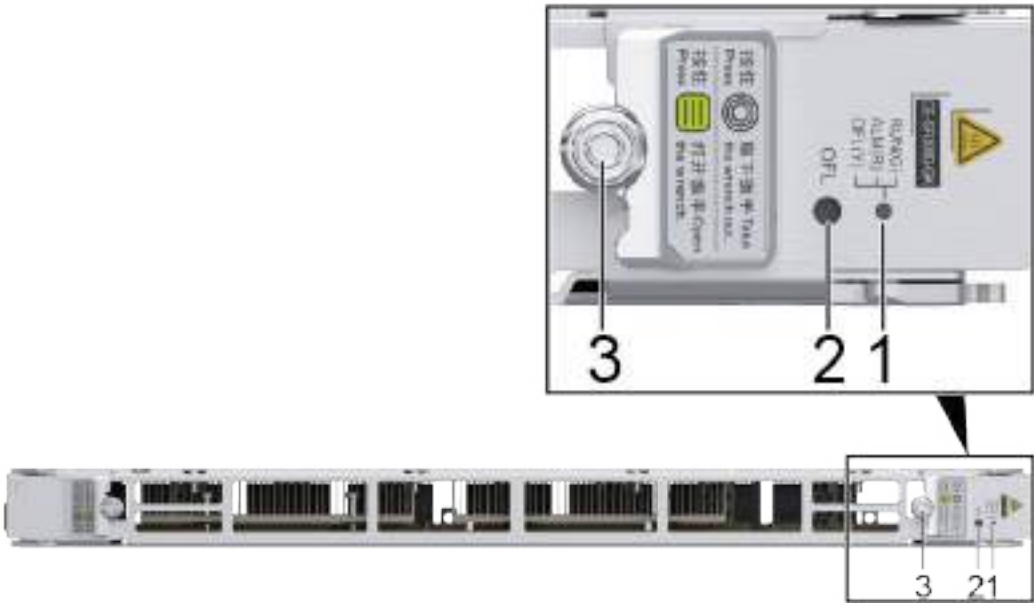
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-314 Mappings between CE-SFU08G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R019C10	-

Indicators and Buttons

Figure 7-105 Indicators and buttons on the CE-SFU08G-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-315 Indicators on the CE-SFU08G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-316 Buttons on the CE-SFU08G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-317 Functions and features of the CE-SFU08G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-318 Technical specifications of the CE-SFU08G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.8.2 CE-SFU08G-GK (03059595-001: CE16808 Switch Fabric Unit GK)

Overview

Table 7-319 Basic information about the CE-SFU08G-GK

Item	Details
Description	CE16808 Switch Fabric Unit GK
Part Number	03059595-001
Model	CE-SFU08G-GK
Silkscreen	CE-SFU08G-GK
Card Type	SFU

Appearance

Figure 7-106 Appearance of the CE-SFU08G-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-320 Mappings between CE-SFU08G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH007 or a later version.

Table 7-321 Mappings between CE-SFU08G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQFD-GK	03059591	V200R020C10	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-

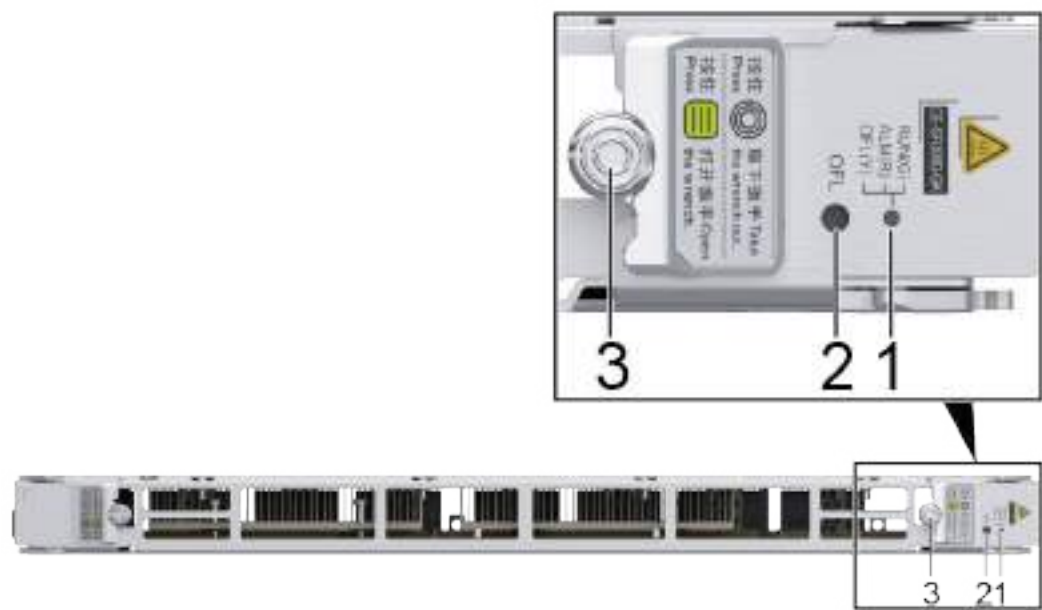
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL24LQF D-GK	03059593	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-001	V200R020C10	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589	V200R020C10	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-001	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-322 Mappings between CE-SFU08G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R020C10	-

Indicators and Buttons

Figure 7-107 Indicators and buttons on the CE-SFU08G-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-323 Indicators on the CE-SFU08G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-324 Buttons on the CE-SFU08G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-325 Functions and features of the CE-SFU08G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-326 Technical specifications of the CE-SFU08G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.8.3 CE-SFU08G-GK (03059595-002: CE16808 Switch Fabric Unit GK)

Overview

Table 7-327 Basic information about the CE-SFU08G-GK

Item	Details
Description	CE16808 Switch Fabric Unit GK
Part Number	03059595-002
Model	CE-SFU08G-GK
Silkscreen	CE-SFU08G-GK
Card Type	SFU

Appearance

Figure 7-108 Appearance of the CE-SFU08G-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-328 Mappings between CE-SFU08G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH020 or a later version.

Table 7-329 Mappings between CE-SFU08G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQFD-GK	03059591	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-GK	03059589	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQFD-GK	03059592	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQFD-GK	03059592-001	V200R021C10SPC600	-	-

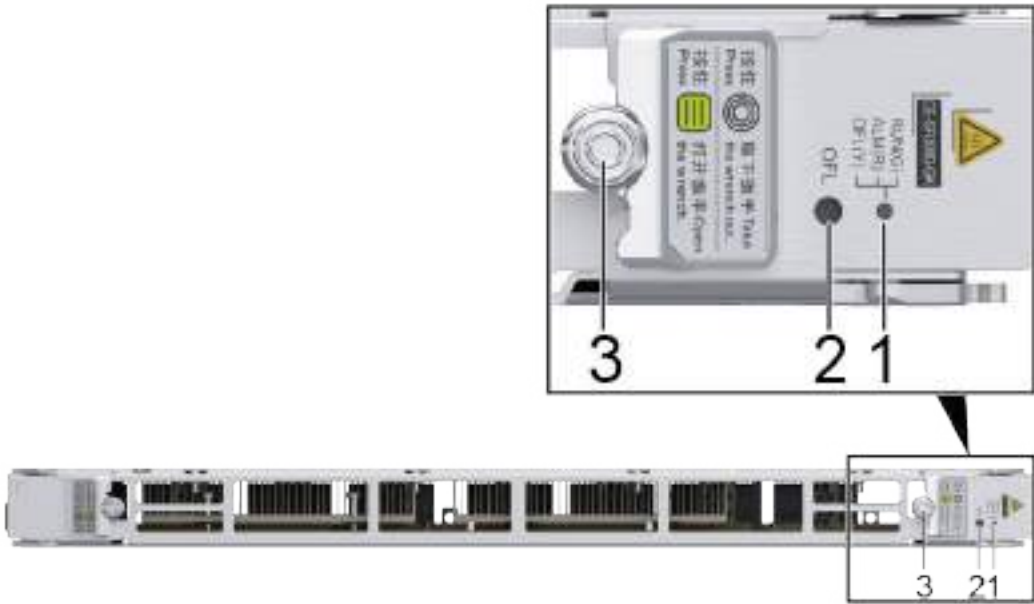
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL36LQFD-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-330 Mappings between CE-SFU08G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-109 Indicators and buttons on the CE-SFU08G-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-331 Indicators on the CE-SFU08G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-332 Buttons on the CE-SFU08G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-333 Functions and features of the CE-SFU08G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-334 Technical specifications of the CE-SFU08G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.8 kg (12.79 lb)
Weight with packaging [kg (lb)]	9.7 kg (21.38 lb)
Typical power consumption [W]	181 W
Typical heat dissipation [BTU/hour]	618 BTU/hour
Static power consumption [W]	148 W
Static heat dissipation [BTU/hour]	505 BTU/hour
Maximum power consumption [W]	261 W
Maximum heat dissipation [BTU/hour]	891 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.9 CE-SFU08F-G (CE16808 Switch Fabric Unit F)

7.3.9.1 CE-SFU08F-G (03058883: CE16808 Switch Fabric Unit F)

Overview

Table 7-335 Basic information about the CE-SFU08F-G

Item	Details
Description	CE16808 Switch Fabric Unit F
Part Number	03058883
Model	CE-SFU08F-G
Silkscreen	CE-SFU08F-G
Card Type	SFU

Appearance

Figure 7-110 Appearance of the CE-SFU08F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-336 Mappings between CE-SFU08F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R005C20	-	SFU3-SFU8	-

Table 7-337 Mappings between CE-SFU08F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-G	03058870	V200R005C20	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872	V200R005C20	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-001	V200R020C10	-	-

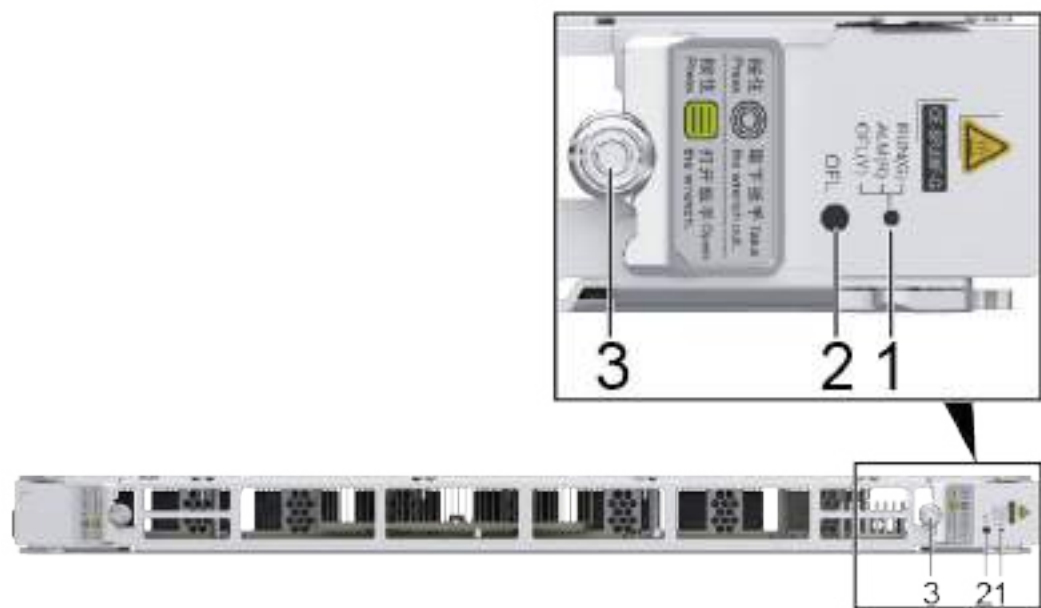
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16808	CEL36CQF D-G	03058869	V200R005C20	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871	V200R005C20	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873	V200R005C20	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-338 Mappings between CE-SFU08F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R005C20	-

Indicators and Buttons

Figure 7-111 Indicators and buttons on the CE-SFU08F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-339 Indicators on the CE-SFU08F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-340 Buttons on the CE-SFU08F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-341 Functions and features of the CE-SFU08F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-342 Technical specifications of the CE-SFU08F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.4 kg (11.9 lb)
Weight with packaging [kg (lb)]	9.3 kg (20.5 lb)
Typical power consumption [W]	105 W
Typical heat dissipation [BTU/hour]	359 BTU/hour
Static power consumption [W]	86 W
Static heat dissipation [BTU/hour]	294 BTU/hour
Maximum power consumption [W]	142 W
Maximum heat dissipation [BTU/hour]	485 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.9.2 CE-SFU08F-G (03058883-002: CE16808 Switch Fabric Unit F)

Overview

Table 7-343 Basic information about the CE-SFU08F-G

Item	Details
Description	CE16808 Switch Fabric Unit F
Part Number	03058883-002
Model	CE-SFU08F-G
Silkscreen	CE-SFU08F-G
Card Type	SFU

Appearance

Figure 7-112 Appearance of the CE-SFU08F-G



Version Mapping

Table 7-344 Mappings between CE-SFU08F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-345 Mappings between CE-SFU08F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-346 Mappings between CE-SFU08F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-113 Indicators and buttons on the CE-SFU08F-G

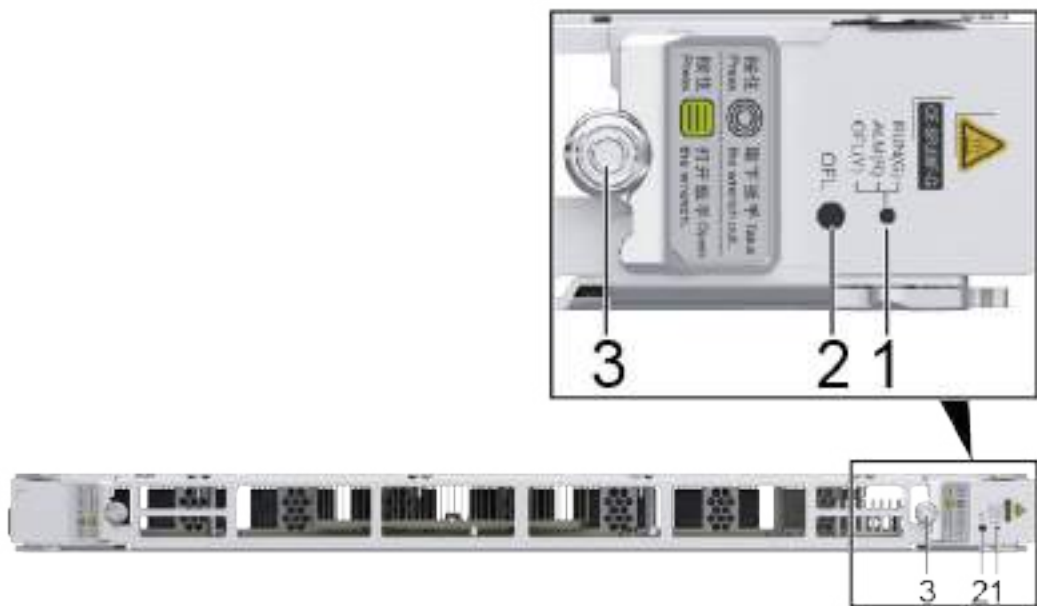


Table 7-347 Indicators on the CE-SFU08F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-348 Buttons on the CE-SFU08F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-349 Functions and features of the CE-SFU08F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-350 Technical specifications of the CE-SFU08F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.4 kg (11.9 lb)
Weight with packaging [kg (lb)]	9.3 kg (20.5 lb)
Typical power consumption [W]	105 W
Typical heat dissipation [BTU/hour]	359 BTU/hour
Static power consumption [W]	86 W
Static heat dissipation [BTU/hour]	294 BTU/hour
Maximum power consumption [W]	142 W
Maximum heat dissipation [BTU/hour]	485 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.10 CE-SFU08F1-G (CE16808 Switch Fabric Unit F1)

Overview

Table 7-351 Basic information about the CE-SFU08F1-G

Item	Details
Description	CE16808 Switch Fabric Unit F1
Part Number	03059672
Model	CE-SFU08F1-G
Silkscreen	CE-SFU08F1-G
Card Type	SFU

Appearance

Figure 7-114 Appearance of the CE-SFU08F1-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-352 Mappings between CE-SFU08F1-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R019C10	-	SFU3-SFU8	-

Table 7-353 Mappings between CE-SFU08F1-G and Interface Card

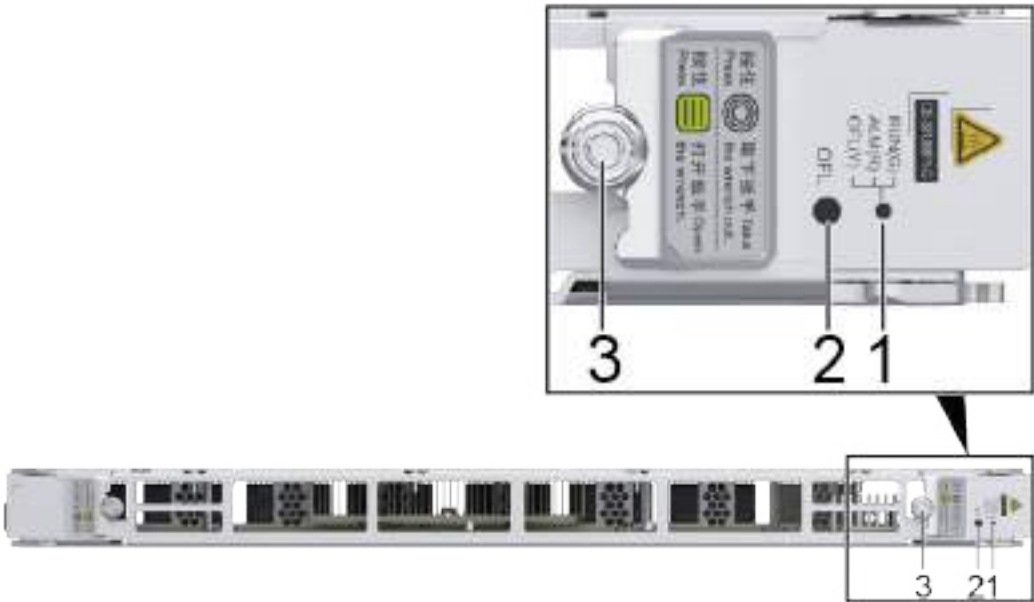
Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CEL24LQF D1-G	03059669	V200R019C 10	-	-
CloudEngin e 16808	CEL36LQF D1-G	03059668	V200R019C 10	-	-
CloudEngin e 16808	CEL48XSF D1-G	03059670	V200R019C 10	-	-

Table 7-354 Mappings between CE-SFU08F1-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R019C10	-

Indicators and Buttons

Figure 7-115 Indicators and buttons on the CE-SFU08F1-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-355 Indicators on the CE-SFU08F1-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-356 Buttons on the CE-SFU08F1-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-357 Functions and features of the CE-SFU08F1-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-358 Technical specifications of the CE-SFU08F1-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.4 kg (11.9 lb)
Weight with packaging [kg (lb)]	9.3 kg (20.5 lb)
Typical power consumption [W]	105 W
Typical heat dissipation [BTU/hour]	359 BTU/hour
Static power consumption [W]	86 W
Static heat dissipation [BTU/hour]	294 BTU/hour
Maximum power consumption [W]	142 W
Maximum heat dissipation [BTU/hour]	485 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.11 CE-SFU08F-GK (CE16808 Switch Fabric Unit F-GK)

7.3.11.1 CE-SFU08F-GK (03050BLU: CE16808 Switch Fabric Unit F-GK)

Overview

Table 7-359 Basic information about the CE-SFU08F-GK

Item	Details
Description	CE16808 Switch Fabric Unit F-GK
Part Number	03050BLU
Model	CE-SFU08F-GK
Silkscreen	CE-SFU08F-GK
Card Type	SFU

Appearance

Figure 7-116 Appearance of the CE-SFU08F-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-360 Mappings between CE-SFU08F-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R020C00	-	SFU3-SFU8	-

Table 7-361 Mappings between CE-SFU08F-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQFD-GK	03059591	V200R020C00	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-001	V200R020C10	-	-
CloudEngine 16808	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593	V200R020C00	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593-001	V200R020C10	-	-
CloudEngine 16808	CEL24LQFD-GK	03059593-002	V200R021C10SPC600	-	-

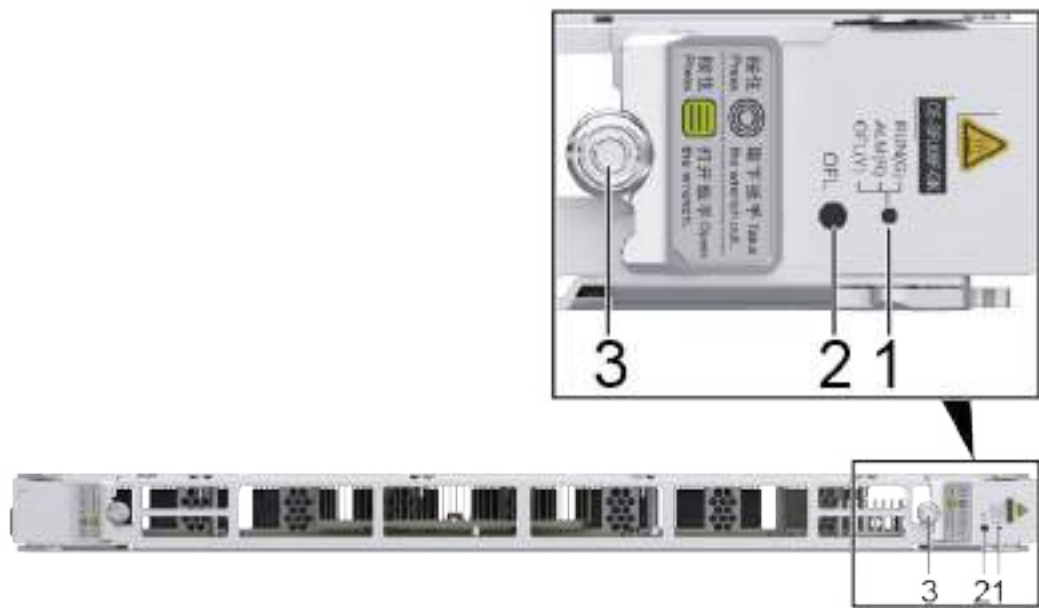
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL36CQF D-GK	03059589	V200R020C00	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592	V200R020C00	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-001	V200R020C10	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594	V200R020C00	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-001	V200R020C10	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-362 Mappings between CE-SFU08F-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R020C00	-

Indicators and Buttons

Figure 7-117 Indicators and buttons on the CE-SFU08F-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-363 Indicators on the CE-SFU08F-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-364 Buttons on the CE-SFU08F-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-365 Functions and features of the CE-SFU08F-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-366 Technical specifications of the CE-SFU08F-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.4 kg (11.9 lb)
Weight with packaging [kg (lb)]	9.3 kg (20.5 lb)
Typical power consumption [W]	105 W
Typical heat dissipation [BTU/hour]	359 BTU/hour
Static power consumption [W]	86 W
Static heat dissipation [BTU/hour]	294 BTU/hour
Maximum power consumption [W]	142 W
Maximum heat dissipation [BTU/hour]	485 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.11.2 CE-SFU08F-GK (03050BLU-002: CE16808 Switch Fabric Unit F-GK)

Overview

Table 7-367 Basic information about the CE-SFU08F-GK

Item	Details
Description	CE16808 Switch Fabric Unit F-GK
Part Number	03050BLU-002
Model	CE-SFU08F-GK
Silkscreen	CE-SFU08F-GK
Card Type	SFU

Appearance

Figure 7-118 Appearance of the CE-SFU08F-GK



Version Mapping

Table 7-368 Mappings between CE-SFU08F-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-369 Mappings between CE-SFU08F-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-GK	03059591	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-GK	03059591-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-370 Mappings between CE-SFU08F-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-119 Indicators and buttons on the CE-SFU08F-GK

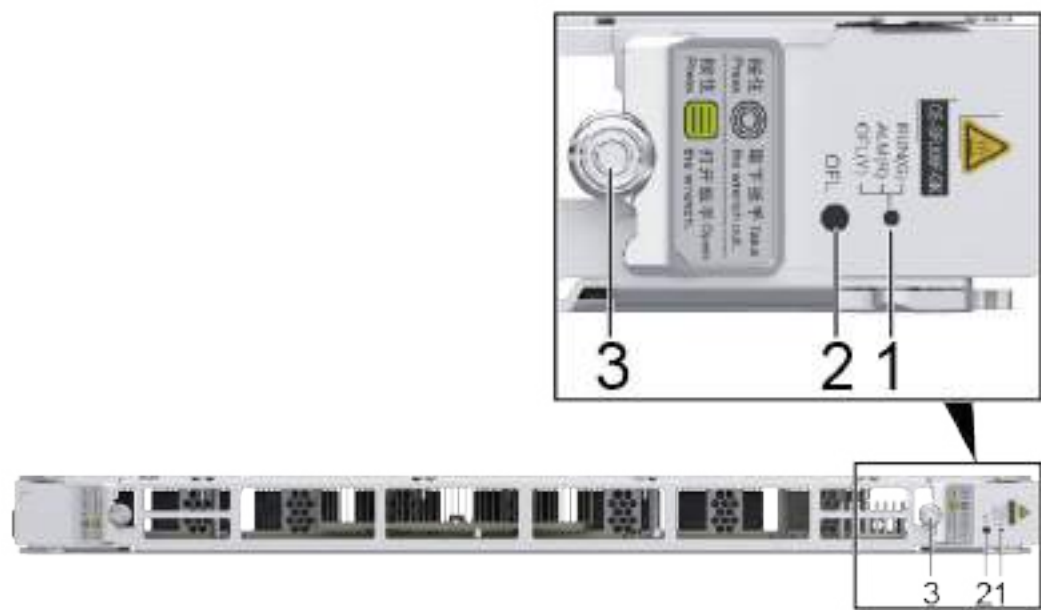


Table 7-371 Indicators on the CE-SFU08F-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-372 Buttons on the CE-SFU08F-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-373 Functions and features of the CE-SFU08F-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-374 Technical specifications of the CE-SFU08F-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.4 kg (11.9 lb)
Weight with packaging [kg (lb)]	9.3 kg (20.5 lb)
Typical power consumption [W]	105 W
Typical heat dissipation [BTU/hour]	359 BTU/hour
Static power consumption [W]	86 W
Static heat dissipation [BTU/hour]	294 BTU/hour
Maximum power consumption [W]	142 W
Maximum heat dissipation [BTU/hour]	485 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.12 CE-SFU08G-A (CE16808 Switch Fabric Unit G-A)

Overview

Table 7-375 Basic information about the CE-SFU08G-A

Item	Details
Description	CE16808 Switch Fabric Unit G-A
Part Number	03050EQK
Model	CE-SFU08G-A
Silkscreen	CE-SFU08G-A
Card Type	SFU

Appearance

Figure 7-120 Appearance of the CE-SFU08G-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-376 Mappings between CE-SFU08G-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	-

Table 7-377 Mappings between CE-SFU08G-A and Interface Card

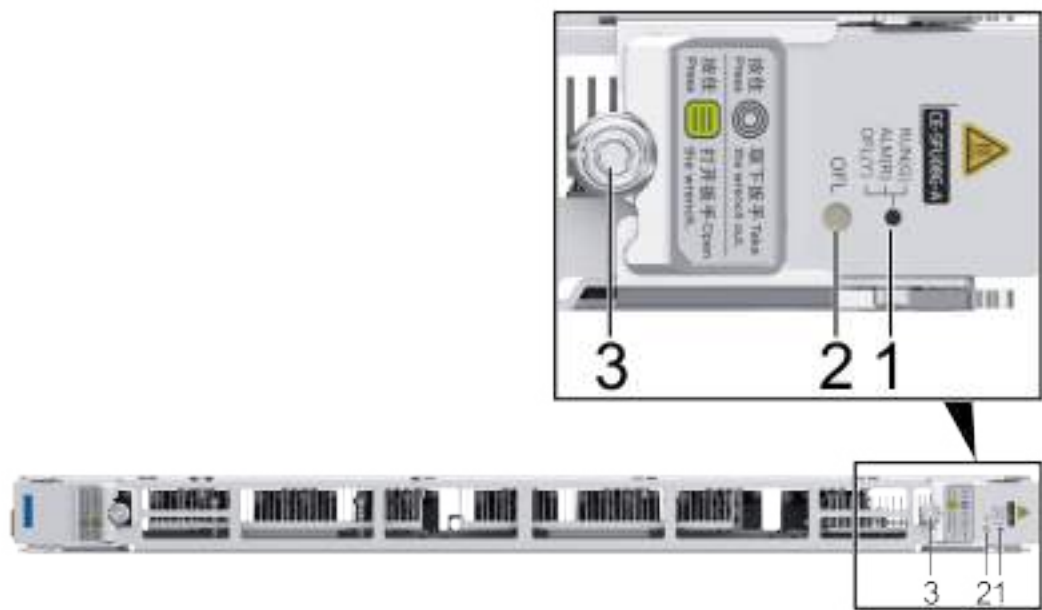
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQF D-A	03050ECC	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQF D-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQF D-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16808	CEL36LQF D-A	03050ECB	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQF D-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-A	03050ECA	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-378 Mappings between CE-SFU08G-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-121 Indicators and buttons on the CE-SFU08G-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-379 Indicators on the CE-SFU08G-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical hardware alarm).
		Yellow	Steady on	The card is not powered on. For example, the card has been forcibly powered off using the power off command or is in the pre-startup state.

Table 7-380 Buttons on the CE-SFU08G-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, press and hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	Supports removable wrenches. To remove the wrenches from an SFU, press and hold down the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-381 Functions and features of the CE-SFU08G-A

Functions and Features	Description
Line-rate data switching	Six SFUs form the switching core of the data plane and provide high-speed Serdes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-382 Technical specifications of the CE-SFU08G-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.6 kg (12.35 lb)
Weight with packaging [kg (lb)]	8.0 kg (17.64 lb)
Typical power consumption [W]	193 W
Typical heat dissipation [BTU/hour]	659 BTU/hour
Static power consumption [W]	166 W
Static heat dissipation [BTU/hour]	566 BTU/hour
Maximum power consumption [W]	252 W
Maximum heat dissipation [BTU/hour]	860 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.13 CE-SFU08F-A (CE16808 Switch Fabric Unit F-A)

7.3.13.1 CE-SFU08F-A (03050ECD: CE16808 Switch Fabric Unit F-A)

Overview

Table 7-383 Basic information about the CE-SFU08F-A

Item	Details
Description	CE16808 Switch Fabric Unit F-A
Part Number	03050ECD
Model	CE-SFU08F-A
Silkscreen	CE-SFU08F-A
Card Type	SFU

Appearance

Figure 7-122 Appearance of the CE-SFU08F-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-384 Mappings between CE-SFU08F-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C00 SPC100	-	SFU3-SFU8	-

Table 7-385 Mappings between CE-SFU08F-A and Interface Card

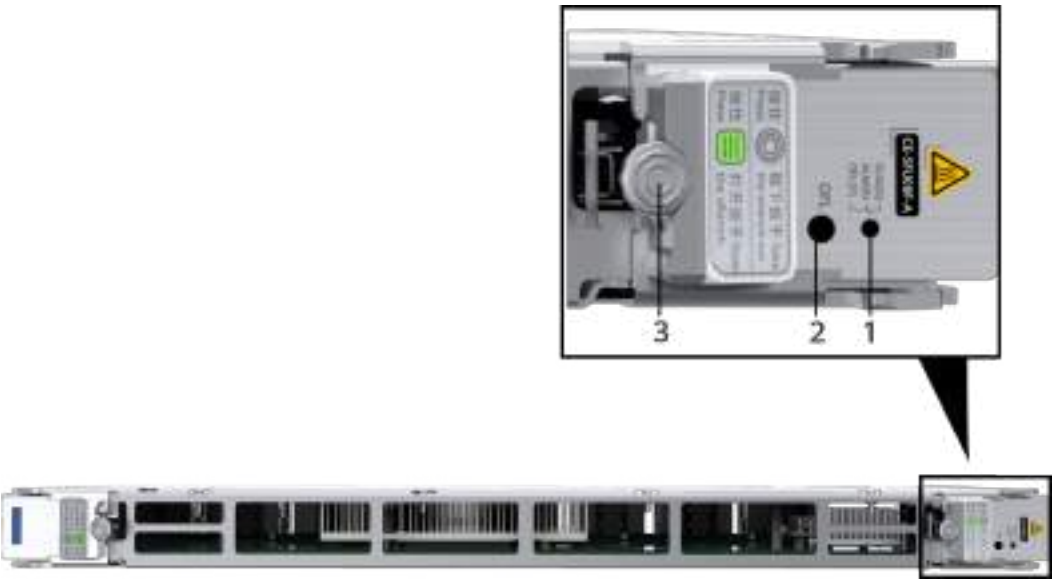
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C00SPC100	-	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C00SPC100	-	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSFD-A	03050ECA	V200R021C00SPC100	-	-
CloudEngine 16808	CEL48XSFD-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-386 Mappings between CE-SFU08F-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C00 SPC100	-

Indicators and Buttons

Figure 7-123 Indicators and buttons on the CE-SFU08F-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-387 Indicators on the CE-SFU08F-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-388 Buttons on the CE-SFU08F-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-389 Functions and features of the CE-SFU08F-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-390 Technical specifications of the CE-SFU08F-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.2 kg (11.46 lb)
Weight with packaging [kg (lb)]	7.6 kg (16.75 lb)
Typical power consumption [W]	99 W
Typical heat dissipation [BTU/hour]	338 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	144 W
Maximum heat dissipation [BTU/hour]	492 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed

Item	Specification
Memory	Standard 2 GB

7.3.13.2 CE-SFU08F-A (03050ECD-001: CE16808 Switch Fabric Unit F-A)

Overview

Table 7-391 Basic information about the CE-SFU08F-A

Item	Details
Description	CE16808 Switch Fabric Unit F-A
Part Number	03050ECD-001
Model	CE-SFU08F-A
Silkscreen	CE-SFU08F-A
Card Type	SFU

Appearance

Figure 7-124 Appearance of the CE-SFU08F-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-392 Mappings between CE-SFU08F-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH0022 or a later version.

Table 7-393 Mappings between CE-SFU08F-A and Interface Card

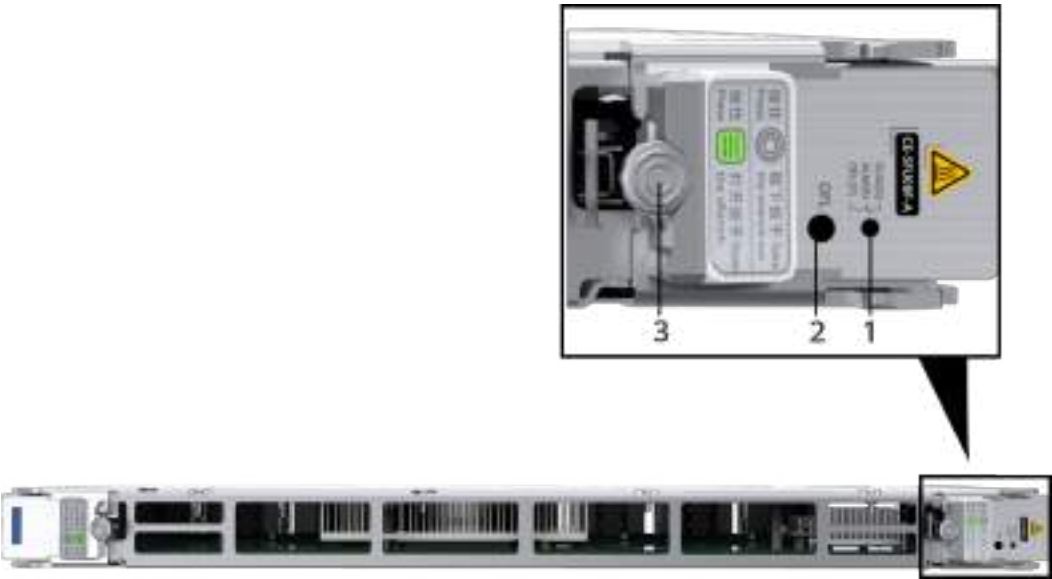
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CEL18CQFD-A	03050ECC	V200R021C10SPC600	-	-
CloudEngine 16808	CEL18CQFD-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36CQFD-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16808	CEL36LQFD-A	03050ECB	V200R021C10SPC600	-	-
CloudEngine 16808	CEL36LQFD-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-A	03050ECA	V200R021C10SPC600	-	-
CloudEngine 16808	CEL48XSF D-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-394 Mappings between CE-SFU08F-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	FAN-480SA-B	02122486	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-125 Indicators and buttons on the CE-SFU08F-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-395 Indicators on the CE-SFU08F-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-396 Buttons on the CE-SFU08F-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-397 Functions and features of the CE-SFU08F-A

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The system has N +1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-398 Technical specifications of the CE-SFU08F-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 559.9 mm x 233.6 mm (1.67 in. x 22.04 in. x 9.20 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	220 mm x 780 mm x 385 mm (8.66 in. x 30.71 in. x 15.16 in.)
Weight without packaging [kg (lb)]	5.2 kg (11.46 lb)
Weight with packaging [kg (lb)]	7.6 kg (16.75 lb)
Typical power consumption [W]	99 W
Typical heat dissipation [BTU/hour]	338 BTU/hour
Static power consumption [W]	88 W
Static heat dissipation [BTU/hour]	300 BTU/hour
Maximum power consumption [W]	144 W
Maximum heat dissipation [BTU/hour]	492 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.14 CE-SFU16F-G (CE16816 Switch Fabric Unit F)

7.3.14.1 CE-SFU16F-G (03058885: CE16816 Switch Fabric Unit F)

Overview

Table 7-399 Basic information about the CE-SFU16F-G

Item	Details
Description	CE16816 Switch Fabric Unit F
Part Number	03058885
Model	CE-SFU16F-G
Silkscreen	CE-SFU16F-G
Card Type	SFU

Appearance

Figure 7-126 Appearance of the CE-SFU16F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-400 Mappings between CE-SFU16F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R005C20	-	SFU3-SFU8	-

Table 7-401 Mappings between CE-SFU16F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870	V200R005C20	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R005C20	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R005C20	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R005C20	-	-

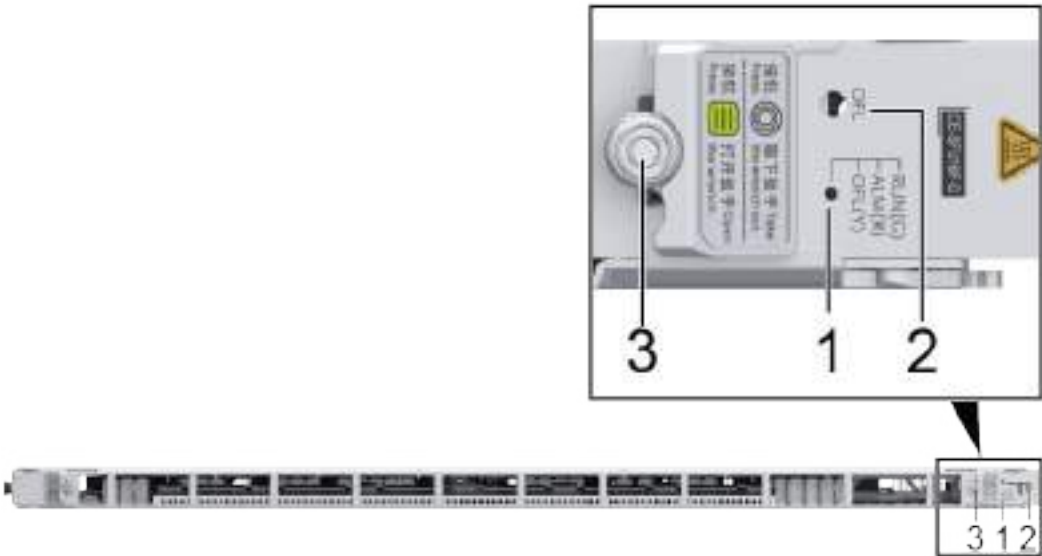
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R005C20	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-402 Mappings between CE-SFU16F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R005C20	-

Indicators and Buttons

Figure 7-127 Indicators and buttons on the CE-SFU16F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-403 Indicators on the CE-SFU16F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-404 Buttons on the CE-SFU16F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-405 Functions and features of the CE-SFU16F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU can be installed together with CE-SFU16F-G1 SFUs in the same chassis.

Technical Specifications

Table 7-406 Technical specifications of the CE-SFU16F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	19.3 kg (42.55 lb)
Weight with packaging [kg (lb)]	27.5 kg (60.63 lb)
Typical power consumption [W]	312 W
Typical heat dissipation [BTU/hour]	1065 BTU/hour
Static power consumption [W]	292 W
Static heat dissipation [BTU/hour]	997 BTU/hour
Maximum power consumption [W]	447 W
Maximum heat dissipation [BTU/hour]	1527 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.14.2 CE-SFU16F-G (03058885-001: CE16816 Switch Fabric Unit F)

Overview

Table 7-407 Basic information about the CE-SFU16F-G

Item	Details
Description	CE16816 Switch Fabric Unit F
Part Number	03058885-001
Model	CE-SFU16F-G
Silkscreen	CE-SFU16F-G
Card Type	SFU

Appearance

Figure 7-128 Appearance of the CE-SFU16F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-408 Mappings between CE-SFU16F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R020C10	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-409 Mappings between CE-SFU16F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQFD-G	03058870	V200R020C10	-	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R020C10	-	-
CloudEngine 16816	CEL18CQFD-G	03058870-002	V200R021C10SPC600	-	-

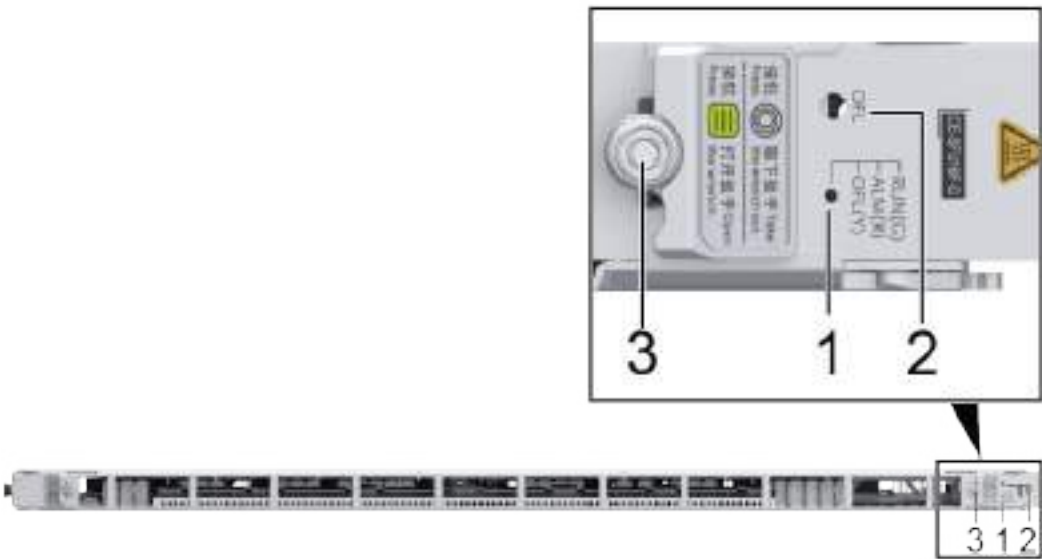
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R020C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R020C10	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R020C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R020C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-410 Mappings between CE-SFU16F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R020C10	-

Indicators and Buttons

Figure 7-129 Indicators and buttons on the CE-SFU16F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-411 Indicators on the CE-SFU16F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-412 Buttons on the CE-SFU16F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-413 Functions and features of the CE-SFU16F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU can be installed together with CE-SFU16F-G1 SFUs in the same chassis.

Technical Specifications

Table 7-414 Technical specifications of the CE-SFU16F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	19.3 kg (42.55 lb)
Weight with packaging [kg (lb)]	27.5 kg (60.63 lb)
Typical power consumption [W]	312 W
Typical heat dissipation [BTU/hour]	1065 BTU/hour
Static power consumption [W]	292 W
Static heat dissipation [BTU/hour]	997 BTU/hour
Maximum power consumption [W]	447 W
Maximum heat dissipation [BTU/hour]	1527 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.14.3 CE-SFU16F-G (03058885-002: CE16816 Switch Fabric Unit F)

Overview

Table 7-415 Basic information about the CE-SFU16F-G

Item	Details
Description	CE16816 Switch Fabric Unit F
Part Number	03058885-002
Model	CE-SFU16F-G
Silkscreen	CE-SFU16F-G
Card Type	SFU

Appearance

Figure 7-130 Appearance of the CE-SFU16F-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-416 Mappings between CE-SFU16F-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-417 Mappings between CE-SFU16F-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-

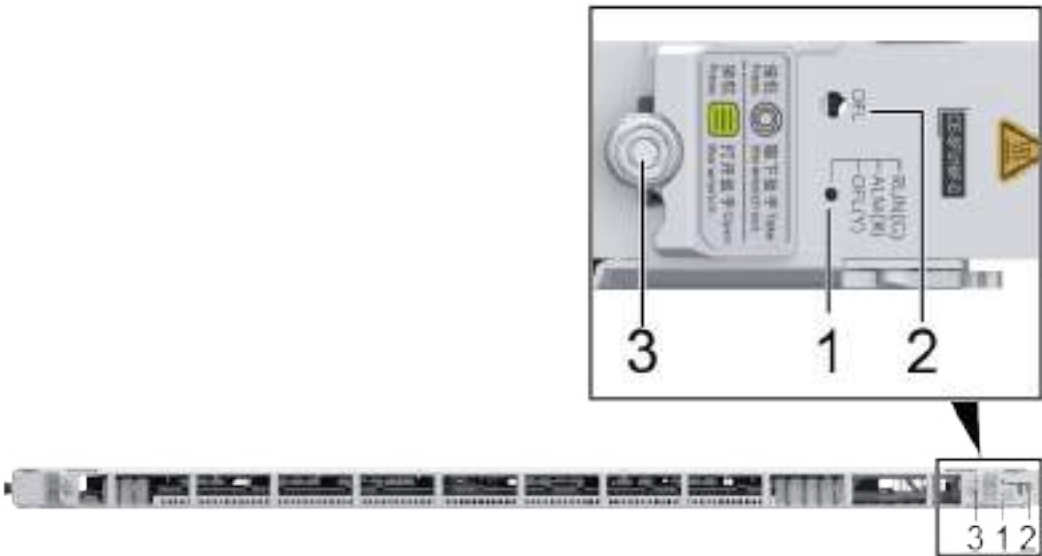
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-418 Mappings between CE-SFU16F-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-131 Indicators and buttons on the CE-SFU16F-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-419 Indicators on the CE-SFU16F-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-420 Buttons on the CE-SFU16F-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-421 Functions and features of the CE-SFU16F-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU can be installed together with CE-SFU16F-G1 SFUs in the same chassis.

Technical Specifications

Table 7-422 Technical specifications of the CE-SFU16F-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	19.3 kg (42.55 lb)
Weight with packaging [kg (lb)]	27.5 kg (60.63 lb)
Typical power consumption [W]	312 W
Typical heat dissipation [BTU/hour]	1065 BTU/hour
Static power consumption [W]	292 W
Static heat dissipation [BTU/hour]	997 BTU/hour
Maximum power consumption [W]	447 W
Maximum heat dissipation [BTU/hour]	1527 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.15 CE-SFU16G-G (CE16816 Switch Fabric Unit G)

7.3.15.1 CE-SFU16G-G (03058884: CE16816 Switch Fabric Unit G)

Overview

Table 7-423 Basic information about the CE-SFU16G-G

Item	Details
Description	CE16816 Switch Fabric Unit G
Part Number	03058884
Model	CE-SFU16G-G
Silkscreen	CE-SFU16G-G
Card Type	SFU

Appearance

Figure 7-132 Appearance of the CE-SFU16G-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-424 Mappings between CE-SFU16G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R005C20	-	SFU3-SFU8	-

Table 7-425 Mappings between CE-SFU16G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870	V200R005C20	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-001	V200R020C10	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R005C20	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R020C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R021C10SPC500	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R005C20	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R005C20	-	-

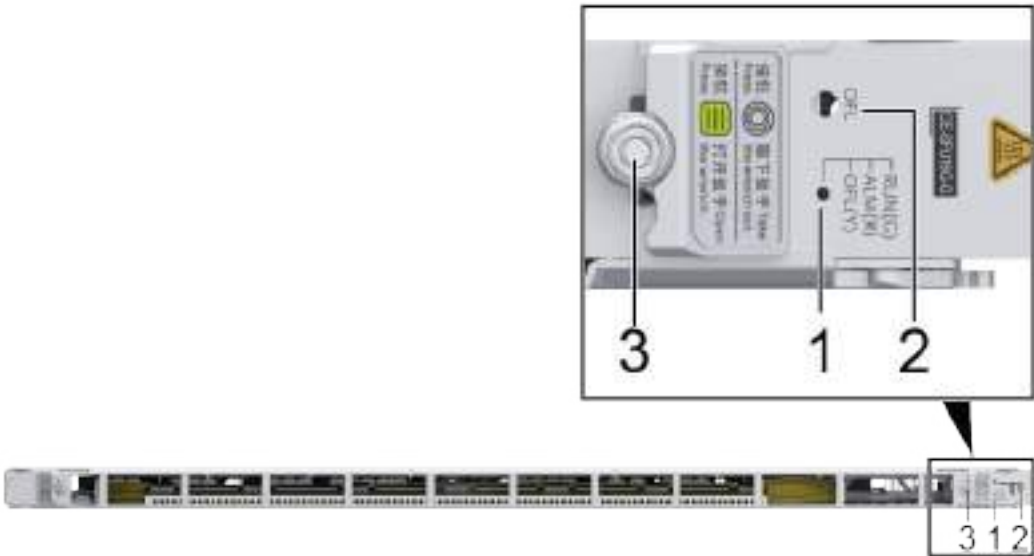
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R020C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R005C20	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R020C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-426 Mappings between CE-SFU16G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R005C20	-

Indicators and Buttons

Figure 7-133 Indicators and buttons on the CE-SFU16G-G



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-427 Indicators on the CE-SFU16G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-428 Buttons on the CE-SFU16G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-429 Functions and features of the CE-SFU16G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-430 Technical specifications of the CE-SFU16G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	22.4 kg (49.38 lb)
Weight with packaging [kg (lb)]	30.6 kg (67.46 lb)
Typical power consumption [W]	524 W
Typical heat dissipation [BTU/hour]	1789 BTU/hour
Static power consumption [W]	450 W
Static heat dissipation [BTU/hour]	1537 BTU/hour
Maximum power consumption [W]	661 W
Maximum heat dissipation [BTU/hour]	2257 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.15.2 CE-SFU16G-G (03058884-002: CE16816 Switch Fabric Unit G)

Overview

Table 7-431 Basic information about the CE-SFU16G-G

Item	Details
Description	CE16816 Switch Fabric Unit G
Part Number	03058884-002
Model	CE-SFU16G-G
Silkscreen	CE-SFU16G-G
Card Type	SFU

Appearance

Figure 7-134 Appearance of the CE-SFU16G-G



Version Mapping

Table 7-432 Mappings between CE-SFU16G-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-433 Mappings between CE-SFU16G-G and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQFD-G	03058870	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQFD-G	03058870-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R021C10SPC600	-	-

Table 7-434 Mappings between CE-SFU16G-G and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R021C10 SPC600	-

Indicators and Buttons

Figure 7-135 Indicators and buttons on the CE-SFU16G-G

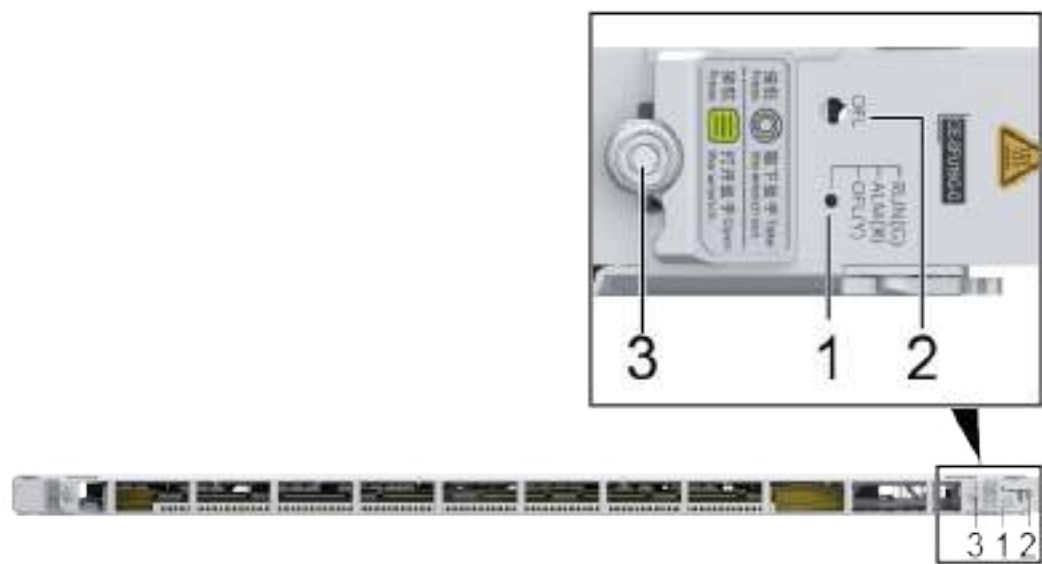


Table 7-435 Indicators on the CE-SFU16G-G

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-436 Buttons on the CE-SFU16G-G

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-437 Functions and features of the CE-SFU16G-G

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-438 Technical specifications of the CE-SFU16G-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	22.4 kg (49.38 lb)
Weight with packaging [kg (lb)]	30.6 kg (67.46 lb)
Typical power consumption [W]	524 W
Typical heat dissipation [BTU/hour]	1789 BTU/hour
Static power consumption [W]	450 W
Static heat dissipation [BTU/hour]	1537 BTU/hour
Maximum power consumption [W]	661 W
Maximum heat dissipation [BTU/hour]	2257 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.16 CE-SFU16G-GK (CE16816 Switch Fabric Unit GK)

7.3.16.1 CE-SFU16G-GK (03059632: CE16816 Switch Fabric Unit GK)

Overview

Table 7-439 Basic information about the CE-SFU16G-GK

Item	Details
Description	CE16816 Switch Fabric Unit GK
Part Number	03059632
Model	CE-SFU16G-GK
Silkscreen	CE-SFU16G-GK
Card Type	SFU

Appearance

Figure 7-136 Appearance of the CE-SFU16G-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-440 Mappings between CE-SFU16G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R019C10	-	SFU3-SFU8	-

Table 7-441 Mappings between CE-SFU16G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-GK	03059591	V200R019C10	-	-
CloudEngine 16816	CEL18CQF D-GK	03059591-001	V200R020C10	-	-
CloudEngine 16816	CEL18CQF D-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQF D-GK	03059593	V200R019C10	-	-
CloudEngine 16816	CEL24LQF D-GK	03059593-001	V200R020C10	-	-
CloudEngine 16816	CEL24LQF D-GK	03059593-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-GK	03059589	V200R019C10	-	-
CloudEngine 16816	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592	V200R019C10	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592-001	V200R020C10	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-GK	03059594	V200R019C10	-	-
CloudEngine 16816	CEL48XSF D-GK	03059594-001	V200R020C10	-	-

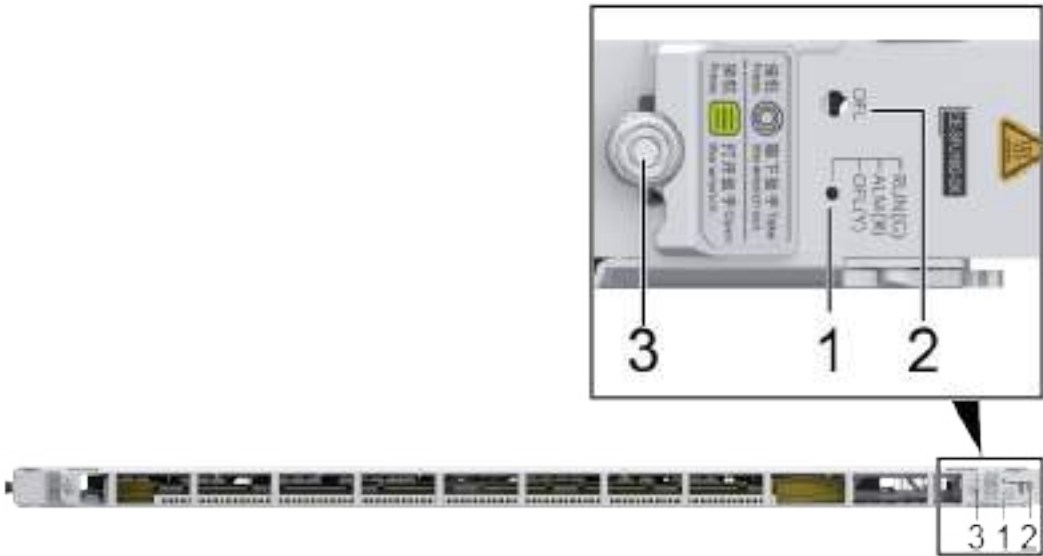
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-442 Mappings between CE-SFU16G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R019C10	-

Indicators and Buttons

Figure 7-137 Indicators and buttons on the CE-SFU16G-GK



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-443 Indicators on the CE-SFU16G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-444 Buttons on the CE-SFU16G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-445 Functions and features of the CE-SFU16G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-446 Technical specifications of the CE-SFU16G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	22.4 kg (49.38 lb)
Weight with packaging [kg (lb)]	30.6 kg (67.46 lb)
Typical power consumption [W]	524 W
Typical heat dissipation [BTU/hour]	1789 BTU/hour
Static power consumption [W]	450 W
Static heat dissipation [BTU/hour]	1537 BTU/hour
Maximum power consumption [W]	661 W
Maximum heat dissipation [BTU/hour]	2257 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.16.2 CE-SFU16G-GK (03059632-002: CE16816 Switch Fabric Unit GK)

Overview

Table 7-447 Basic information about the CE-SFU16G-GK

Item	Details
Description	CE16816 Switch Fabric Unit GK
Part Number	03059632-002
Model	CE-SFU16G-GK
Silkscreen	CE-SFU16G-GK
Card Type	SFU

Appearance

Figure 7-138 Appearance of the CE-SFU16G-GK



Version Mapping

Table 7-448 Mappings between CE-SFU16G-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	SFU3-SFU8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-449 Mappings between CE-SFU16G-GK and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQFD-GK	03059591	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQFD-GK	03059591-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQFD-GK	03059591-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQFD-GK	03059593	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQFD-GK	03059593-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL24LQFD-GK	03059593-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL36CQF D-GK	03059589	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-GK	03059589-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-GK	03059592-002	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-GK	03059594	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-GK	03059594-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-GK	03059594-002	V200R021C10SPC600	-	-

Table 7-450 Mappings between CE-SFU16G-GK and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-139 Indicators and buttons on the CE-SFU16G-GK

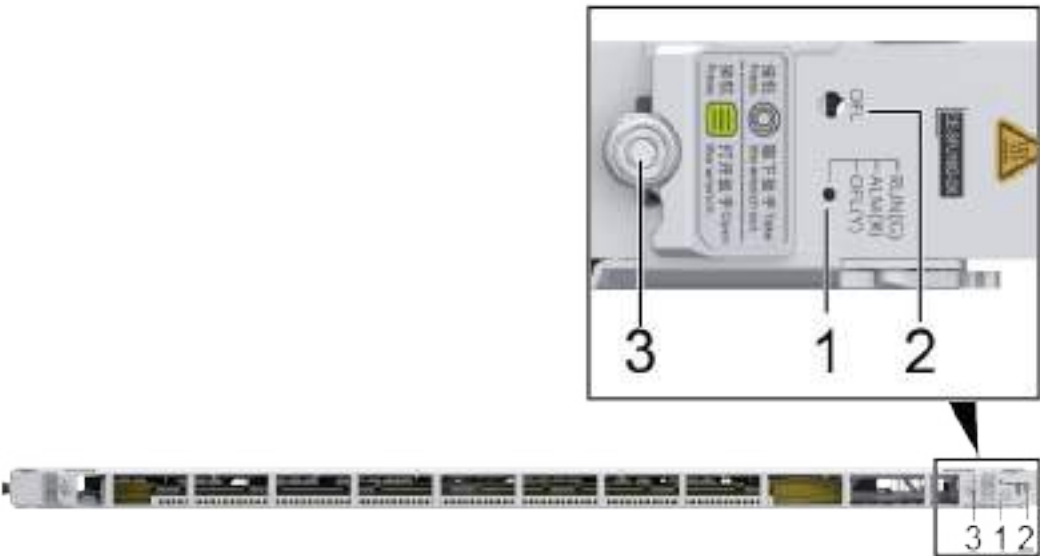


Table 7-451 Indicators on the CE-SFU16G-GK

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-452 Buttons on the CE-SFU16G-GK

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-453 Functions and features of the CE-SFU16G-GK

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.

Functions and Features	Description
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-454 Technical specifications of the CE-SFU16G-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	22.4 kg (49.38 lb)
Weight with packaging [kg (lb)]	30.6 kg (67.46 lb)
Typical power consumption [W]	524 W
Typical heat dissipation [BTU/hour]	1789 BTU/hour
Static power consumption [W]	450 W
Static heat dissipation [BTU/hour]	1537 BTU/hour
Maximum power consumption [W]	661 W
Maximum heat dissipation [BTU/hour]	2257 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.17 CE-SFU16G-A (CE16816 Switch Fabric Unit G-A)

Overview

Table 7-455 Basic information about the CE-SFU16G-A

Item	Details
Description	CE16816 Switch Fabric Unit G-A
Part Number	03050EQJ
Model	CE-SFU16G-A
Silkscreen	CE-SFU16G-A
Card Type	SFU

Appearance

Figure 7-140 Appearance of the CE-SFU16G-A



NOTICE

The blue label on the left side of this card is a mixed-card prevention label. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-456 Mappings between CE-SFU16G-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	SFU3-SFU8	-

Table 7-457 Mappings between CE-SFU16G-A and Interface Card

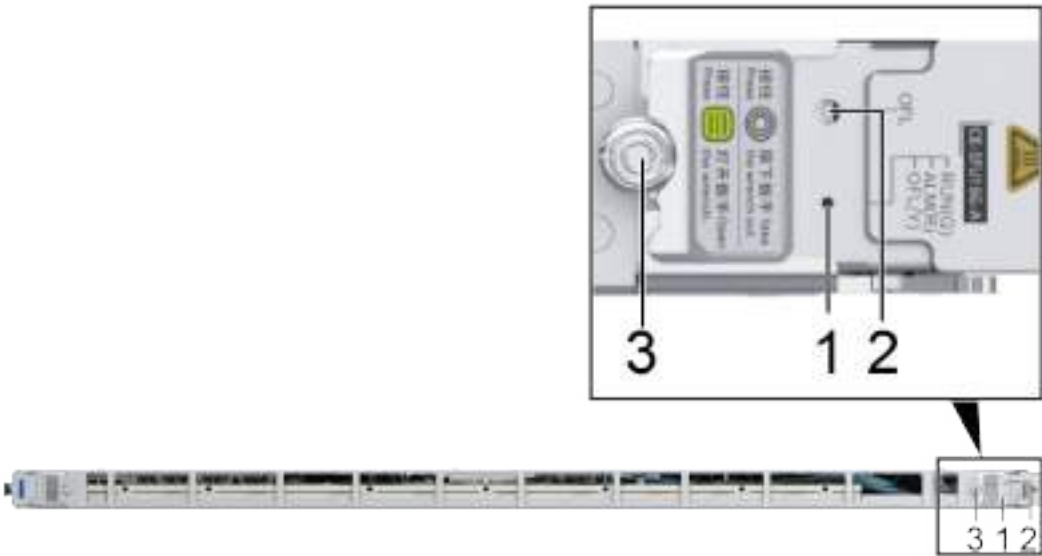
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-A	03050ECC	V200R021C10SPC600	-	-
CloudEngine 16816	CEL18CQF D-A	03050ECC-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-A	03050EPY	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36CQF D-A	03050EPY-001	V200R022C00	-	-
CloudEngine 16816	CEL36LQF D-A	03050ECB	V200R021C10SPC600	-	-
CloudEngine 16816	CEL36LQF D-A	03050ECB-001	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-A	03050ECA	V200R021C10SPC600	-	-
CloudEngine 16816	CEL48XSF D-A	03050ECA-001	V200R021C10SPC600	-	-

Table 7-458 Mappings between CE-SFU16G-A and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R021C10SPC600	-

Indicators and Buttons

Figure 7-141 Indicators and buttons on the CE-SFU16G-A



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-459 Indicators on the CE-SFU16G-A

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical hardware alarm).
		Yellow	Steady on	The card is not powered on. For example, the card has been forcibly powered off using the power off command or is in the pre-startup state.

Table 7-460 Buttons on the CE-SFU16G-A

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, press and hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	Supports removable wrenches. To remove the wrenches from an SFU, press and hold down the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-461 Functions and features of the CE-SFU16G-A

Functions and Features	Description
Line-rate data switching	Six SFUs form the switching core of the data plane and provide high-speed Serdes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU cannot be installed with the SFUs of other models in the same chassis.

Technical Specifications

Table 7-462 Technical specifications of the CE-SFU16G-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	16.6 kg (36.6 lb)
Weight with packaging [kg (lb)]	27.2 kg (59.97 lb)
Typical power consumption [W]	541 W
Typical heat dissipation [BTU/hour]	1846 BTU/hour
Static power consumption [W]	397 W
Static heat dissipation [BTU/hour]	1354 BTU/hour
Maximum power consumption [W]	719 W
Maximum heat dissipation [BTU/hour]	2453 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.3.18 CE-SFU16F-G1 (CE16816 Switch Fabric Unit F-G1)

Overview

Table 7-463 Basic information about the CE-SFU16F-G1

Item	Details
Description	CE16816 Switch Fabric Unit F-G1
Part Number	03050KHK
Model	CE-SFU16F-G1
Silkscreen	CE-SFU16F-G1
Card Type	SFU

Appearance

Figure 7-142 Appearance of the CE-SFU16F-G1



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-464 Mappings between CE-SFU16F-G1 and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R022C10	-	SFU3-SFU8	-

Table 7-465 Mappings between CE-SFU16F-G1 and Interface Card

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL18CQF D-G	03058870	V200R022C10	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-001	V200R022C10	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-002	V200R022C10	-	-
CloudEngine 16816	CEL18CQF D-G	03058870-003	V200R022C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872	V200R022C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-001	V200R022C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-002	V200R022C10	-	-
CloudEngine 16816	CEL24LQF D-G	03058872-003	V200R022C10	-	-
CloudEngine 16816	CEL36CQF D-G	03050EPU	V200R022C10	-	-
CloudEngine 16816	CEL36CQF D-G	03058869	V200R022C10	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-002	V200R022C10	-	-
CloudEngine 16816	CEL36CQF D-G	03058869-003	V200R022C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871	V200R022C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-001	V200R022C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-002	V200R022C10	-	-
CloudEngine 16816	CEL36LQF D-G	03058871-003	V200R022C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873	V200R022C10	-	-

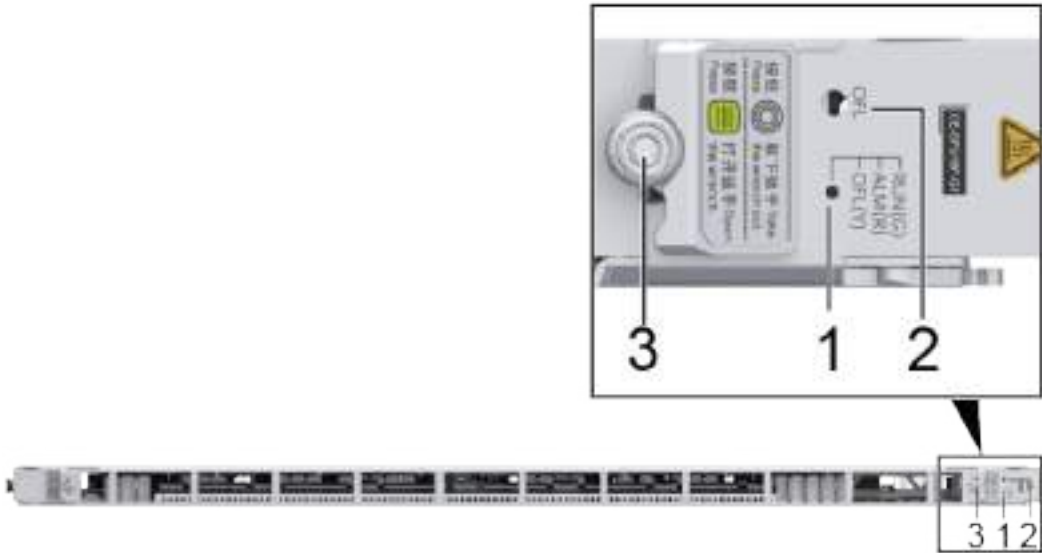
Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CEL48XSF D-G	03058873-001	V200R022C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-002	V200R022C10	-	-
CloudEngine 16816	CEL48XSF D-G	03058873-003	V200R022C10	-	-

Table 7-466 Mappings between CE-SFU16F-G1 and Fan Module

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	FAN-960SA-B	02122582	V200R022C10	-

Indicators and Buttons

Figure 7-143 Indicators and buttons on the CE-SFU16F-G1



1. Running status indicator	2. OFL button	3. Wrench removal button
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Table 7-467 Indicators on the CE-SFU16F-G1

Silkscreen	Name	Color	Status	Description
RUN(G)/ ALM(R)/ OFL(Y)	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Table 7-468 Buttons on the CE-SFU16F-G1

Silkscreen	Name	Description
OFL	OFL button	To remove an SFU, hold down the OFL button for 6s. You can remove the SFU until the running status indicator is steady yellow.
-	Wrench removal button	The SFU is installed and removed using removable wrenches. To remove the wrenches from an SFU, press and hold the round button (marked by number 3 in the preceding figure) on the SFU.

Functions and Features

Table 7-469 Functions and features of the CE-SFU16F-G1

Functions and Features	Description
Line-rate data switching	Six SFUs in a chassis constitute the switching core of the data plane and provide high-speed SerDes channels for LPUs.
Reliability	The device uses a single-stage multi-plane switch fabric to expand the switching capacity. The device has N+1 switching planes, which work in load balancing and redundancy mode to ensure nonstop service data switching.
Hot swapping	Supported
Restrictions and remarks	This SFU can be installed together with CE-SFU16F-G SFUs in the same chassis.

Technical Specifications

Table 7-470 Technical specifications of the CE-SFU16F-G1

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	42.4 mm x 1116.2 mm x 389.2 mm (1.67 in. x 43.94 in. x 15.32 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	285 mm x 1290 mm x 740 mm (11.22 in. x 50.79 in. x 29.13 in.)
Weight without packaging [kg (lb)]	19.3 kg (42.55 lb)
Weight with packaging [kg (lb)]	27.5 kg (60.63 lb)
Typical power consumption [W]	312 W
Typical heat dissipation [BTU/hour]	1065 BTU/hour
Static power consumption [W]	292 W
Static heat dissipation [BTU/hour]	997 BTU/hour
Maximum power consumption [W]	447 W
Maximum heat dissipation [BTU/hour]	1527 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 2 GB

7.4 10GE Interface Card

7.4.1 CEL48XSFD-G (48-Port 10GE Interface Card (FD-G, SFP+))

7.4.1.1 CEL48XSFD-G (03058873: 48-Port 10GE Interface Card (FD-G, SFP+))

Overview

Table 7-471 Basic information about the CEL48XSFD-G

Item	Details
Description	48-Port 10GE Interface Card (FD-G, SFP+)
Part Number	03058873

Item	Details
Model	CEL48XSFD-G
Silkscreen	CEL48XSFD-G
Card Type	Interface Card

Appearance

Figure 7-144 Appearance of the CEL48XSFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-472 Mappings between CEL48XSFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	1-4	-
CloudEngine 16808	V200R005C20	-	1-8	-
CloudEngine 16816	V200R005C20	-	1-16	-

Table 7-473 Mappings between CEL48XSFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R005C20	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-474 Mappings between CEL48XSFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16808	CE-SFU08G-G	03058882	V200R005C20	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R005C20	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R005C20	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators

Figure 7-145 Indicators on the CEL48XSFD-G

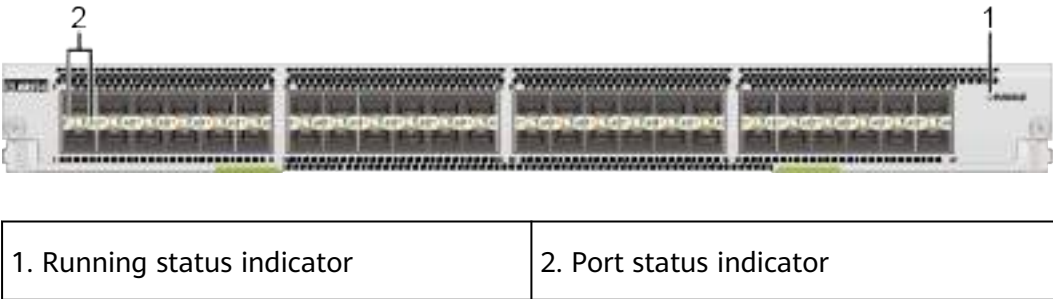


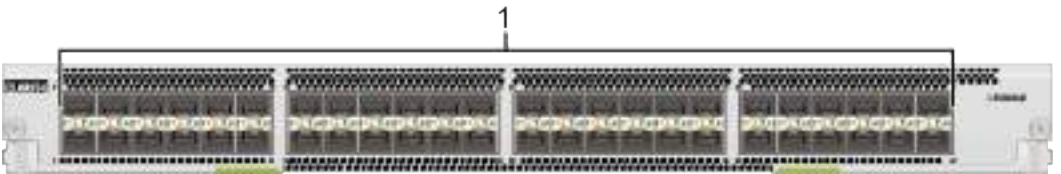
Table 7-475 Indicators on the CEL48XSFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-146 Ports on the CEL48XSFD-G



1. Forty-eight 10GE optical ports

Table 7-476 Ports on the CEL48XSFD-G

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-477 Functions and features of the CEL48XSFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported
Restrictions and remarks	In V200R021C00 and earlier versions, this LPU is used with the CE-SFU16F-G. When the LPU is installed in slot 14, SFUs cannot work in 4+2 backup mode and can only work in 5+1 backup mode when the device works in six-SFU mode; SFUs cannot work in 3+1 backup mode or any other backup mode when the device works in four-SFU mode.

Technical Specifications

Table 7-478 Technical specifications of the CEL48XSFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.1.2 CEL48XSFD-G (03058873-001: 48-Port 10GE Interface Card (FD-G, SFP+))

Overview

Table 7-479 Basic information about the CEL48XSFD-G

Item	Details
Description	48-Port 10GE Interface Card (FD-G, SFP+)
Part Number	03058873-001
Model	CEL48XSFD-G
Silkscreen	CEL48XSFD-G
Card Type	Interface Card

Appearance

Figure 7-147 Appearance of the CEL48XSFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-480 Mappings between CEL48XSFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-481 Mappings between CEL48XSFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R020C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R020C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

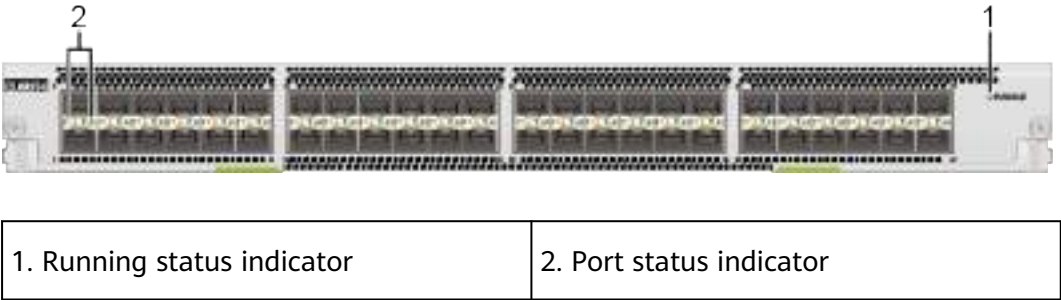
Table 7-482 Mappings between CEL48XSFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10 SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10 SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R020C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators

Figure 7-148 Indicators on the CEL48XSFD-G



1. Running status indicator	2. Port status indicator
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Table 7-483 Indicators on the CEL48XSFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-149 Ports on the CEL48XSFD-G

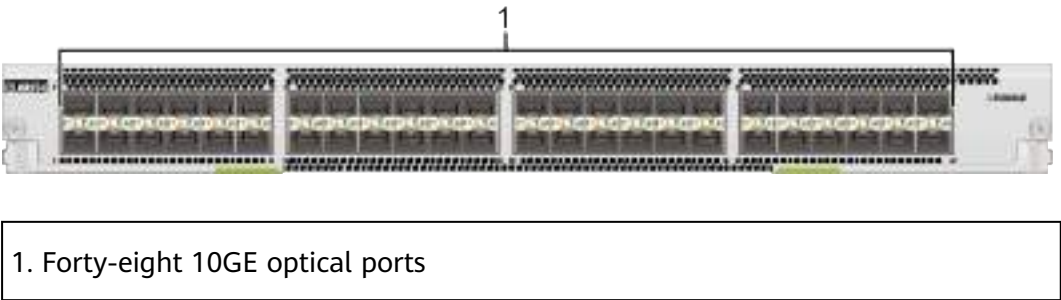


Table 7-484 Ports on the CEL48XSFD-G

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-485 Functions and features of the CEL48XSFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported
Restrictions and remarks	In V200R021C00 and earlier versions, this LPU is used with the CE-SFU16F-G. When the LPU is installed in slot 14, SFUs cannot work in 4+2 backup mode and can only work in 5+1 backup mode when the device works in six-SFU mode; SFUs cannot work in 3+1 backup mode or any other backup mode when the device works in four-SFU mode.

Technical Specifications

Table 7-486 Technical specifications of the CEL48XSFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.1.3 CEL48XSFD-G (03058873-002: 48-Port 10GE Interface Card (FD-G, SFP+))

Overview

Table 7-487 Basic information about the CEL48XSFD-G

Item	Details
Description	48-Port 10GE Interface Card (FD-G, SFP+)
Part Number	03058873-002
Model	CEL48XSFD-G
Silkscreen	CEL48XSFD-G
Card Type	Interface Card

Appearance

Figure 7-150 Appearance of the CEL48XSFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-488 Mappings between CEL48XSFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Table 7-489 Mappings between CEL48XSFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-490 Mappings between CEL48XSFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators

Figure 7-151 Indicators on the CEL48XSFD-G



1. Running status indicator	2. Port status indicator
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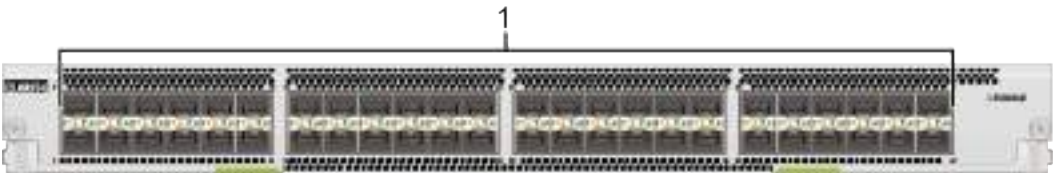
Table 7-491 Indicators on the CEL48XSFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-152 Ports on the CEL48XSFD-G



1. Forty-eight 10GE optical ports

Table 7-492 Ports on the CEL48XSFD-G

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-493 Functions and features of the CEL48XSFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported

Technical Specifications

Table 7-494 Technical specifications of the CEL48XSFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.1.4 CEL48XSFD-G (03058873-003: 48-Port 10GE Interface Card (FD-G, SFP+))

Overview

Table 7-495 Basic information about the CEL48XSFD-G

Item	Details
Description	48-Port 10GE Interface Card (FD-G, SFP+)
Part Number	03058873-003
Model	CEL48XSFD-G
Silkscreen	CEL48XSFD-G
Card Type	Interface Card

Appearance

Figure 7-153 Appearance of the CEL48XSFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-496 Mappings between CEL48XSFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-497 Mappings between CEL48XSFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-498 Mappings between CEL48XSFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators

Figure 7-154 Indicators on the CEL48XSFD-G



1. Running status indicator	2. Port status indicator
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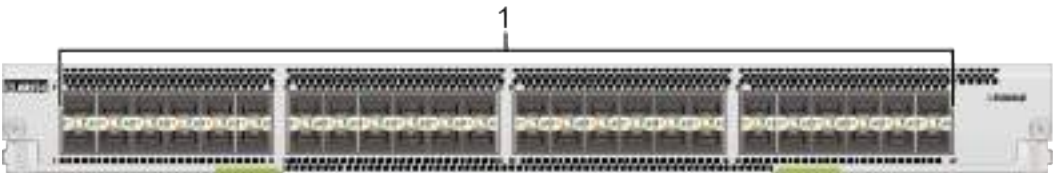
Table 7-499 Indicators on the CEL48XSFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of a top-row port, and the indicator with a down arrowhead shows the status of a bottom-row port.	Green	Steady on	A link has been established on the port.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-155 Ports on the CEL48XSFD-G



1. Forty-eight 10GE optical ports

Table 7-500 Ports on the CEL48XSFD-G

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-501 Functions and features of the CEL48XSFD-G

Functions and Features	Description
Basic function	Provides data packet processing and traffic management on 48 x 10GE optical ports.
Maximum port density	If all LPUs are such cards: • The CloudEngine 16804 chassis provides up to 192 x 10GE ports. • The CloudEngine 16808 chassis provides up to 384 x 10GE ports. • The CloudEngine 16816 chassis provides up to 768 x 10GE ports.

Functions and Features	Description
10GE port	When optical ports on the card work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When optical ports on the card work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. Such a port with a 1000BASE-T copper module installed can work at the rate of 100 Mbit/s or 1000 Mbit/s through auto-sensing.
Hot swapping	Supported

Technical Specifications

Table 7-502 Technical specifications of the CEL48XSFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour
CPU	4-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.2 CEL48XSFD-GK (48-port 10GE interface card (FD-GK, SFP+))

7.4.2.1 CEL48XSFD-GK (03059594: 48-port 10GE interface card (FD-GK, SFP+))

Overview

Table 7-503 Basic information about the CEL48XSFD-GK

Item	Details
Description	48-port 10GE interface card (FD-GK, SFP+)
Part Number	03059594
Model	CEL48XSFD-GK
Silkscreen	CEL48XSFD-GK
Card Type	Interface Card

Appearance

Figure 7-156 Appearance of the CEL48XSFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-504 Mappings between CEL48XSFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-505 Mappings between CEL48XSFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R019C10	-

Table 7-506 Mappings between CEL48XSFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R019C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C00	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R019C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632	V200R019C10	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632-002	V200R021C10SPC600	-	-

Indicators

Figure 7-157 Indicators on the CEL48XSFD-GK

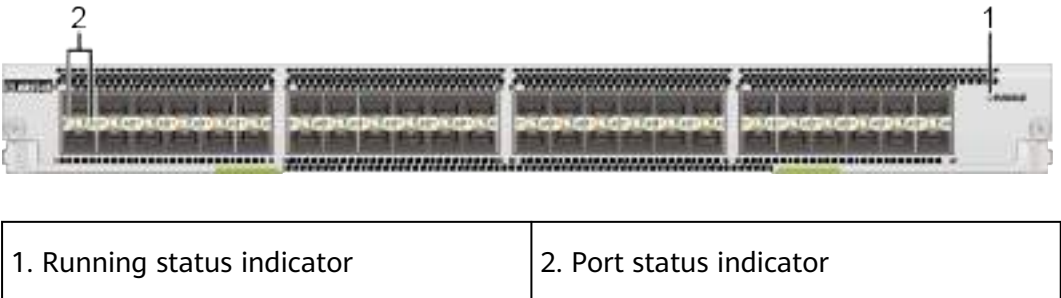


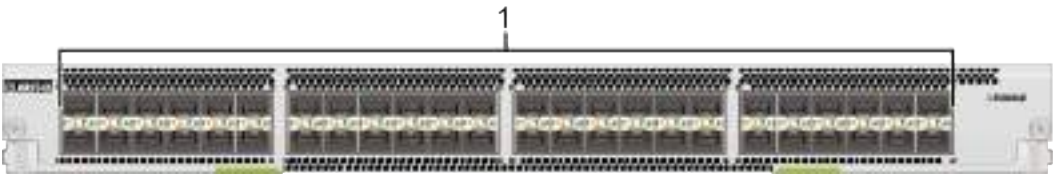
Table 7-507 Indicators on the CEL48XSFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-158 Ports on the CEL48XSFD-GK



1. Forty-eight 10GE optical ports

Table 7-508 Ports on the CEL48XSFD-GK

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-509 Functions and features of the CEL48XSFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported
Restrictions and remarks	In V200R021C00 and earlier versions, this LPU is used with the CE-SFU16F-G. When the LPU is installed in slot 14, SFUs cannot work in 4+2 backup mode and can only work in 5+1 backup mode when the device works in six-SFU mode; SFUs cannot work in 3+1 backup mode or any other backup mode when the device works in four-SFU mode.

Technical Specifications

Table 7-510 Technical specifications of the CEL48XSFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.2.2 CEL48XSFD-GK (03059594-001: 48-port 10GE interface card (FD-GK, SFP+))

Overview

Table 7-511 Basic information about the CEL48XSFD-GK

Item	Details
Description	48-port 10GE interface card (FD-GK, SFP+)
Part Number	03059594-001
Model	CEL48XSFD-GK
Silkscreen	CEL48XSFD-GK
Card Type	Interface Card

Appearance

Figure 7-159 Appearance of the CEL48XSFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-512 Mappings between CEL48XSFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-513 Mappings between CEL48XSFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R020C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R020C10	-

Table 7-514 Mappings between CEL48XSFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R020C 10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators

Figure 7-160 Indicators on the CEL48XSFD-GK



1. Running status indicator	2. Port status indicator
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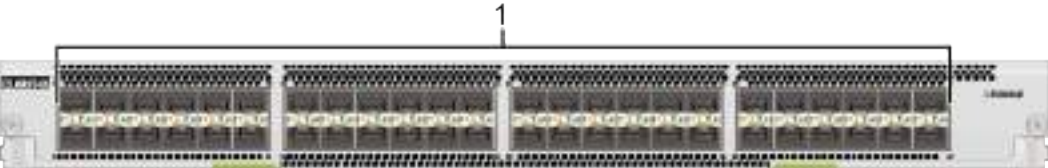
Table 7-515 Indicators on the CEL48XSFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-161 Ports on the CEL48XSFD-GK



1. Forty-eight 10GE optical ports

Table 7-516 Ports on the CEL48XSFD-GK

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-517 Functions and features of the CEL48XSFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. - The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. - The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported
Restrictions and remarks	In V200R021C00 and earlier versions, this LPU is used with the CE-SFU16F-G. When the LPU is installed in slot 14, SFUs cannot work in 4+2 backup mode and can only work in 5+1 backup mode when the device works in six-SFU mode; SFUs cannot work in 3+1 backup mode or any other backup mode when the device works in four-SFU mode.

Technical Specifications

Table 7-518 Technical specifications of the CEL48XSFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)

Item	Specification
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.2.3 CEL48XSFD-GK (03059594-002: 48-port 10GE interface card (FD-GK, SFP+))

Overview

Table 7-519 Basic information about the CEL48XSFD-GK

Item	Details
Description	48-port 10GE interface card (FD-GK, SFP+)
Part Number	03059594-002
Model	CEL48XSFD-GK
Silkscreen	CEL48XSFD-GK
Card Type	Interface Card

Appearance

Figure 7-162 Appearance of the CEL48XSFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-520 Mappings between CEL48XSFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1-8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-521 Mappings between CEL48XSFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R021C10 SPC600	-

Table 7-522 Mappings between CEL48XSFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators

Figure 7-163 Indicators on the CEL48XSFD-GK



1. Running status indicator	2. Port status indicator
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Table 7-523 Indicators on the CEL48XSFD-GK

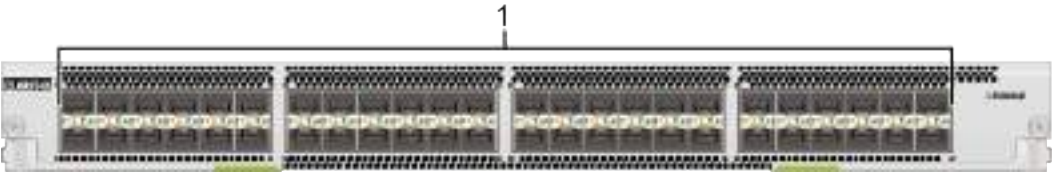
Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-164 Ports on the CEL48XSFD-GK



1. Forty-eight 10GE optical ports

Table 7-524 Ports on the CEL48XSFD-GK

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-525 Functions and features of the CEL48XSFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported

Technical Specifications

Table 7-526 Technical specifications of the CEL48XSFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.3 CEL48XSFD1-G (48-port 10GE interface card (FD1-G, SFP+))

Overview

Table 7-527 Basic information about the CEL48XSFD1-G

Item	Details
Description	48-port 10GE interface card (FD1-G, SFP+)
Part Number	03059670
Model	CEL48XSFD1-G
Silkscreen	CEL48XSFD1-G
Card Type	Interface Card

Appearance

Figure 7-165 Appearance of the CEL48XSFD1-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-528 Mappings between CEL48XSFD1-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-529 Mappings between CEL48XSFD1-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R019C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-530 Mappings between CEL48XSFD1-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F1-G	03059671	V200R019C10	-	-
CloudEngine 16808	CE-SFU08F1-G	03059672	V200R019C10	-	-

Indicators

Figure 7-166 Indicators on the CEL48XSFD1-G



1. Running status indicator	2. Port status indicator
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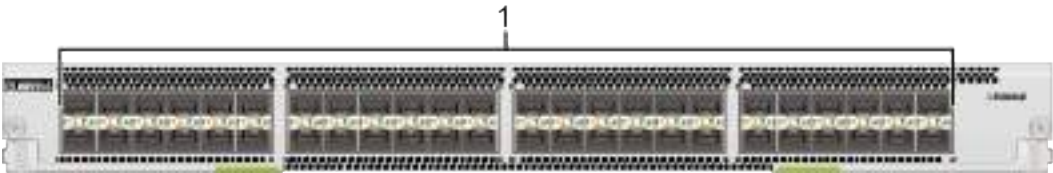
Table 7-531 Indicators on the CEL48XSFD1-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Fast blinking (4 Hz)	The card has been powered on but the system software is not running.
		Green	Steady on	The card is working properly.
		Green	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The link on the port is disconnected.
		Green	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Green	Off	The link on the port is disconnected.
RUN/ALM	Running status indicator	Yellow	Steady on	The port is transmitting or receiving data.
		Yellow	Slow blinking (0.5 Hz)	The port is not transmitting or receiving data.

Ports

Figure 7-167 Ports on the CEL48XSFD1-G



1. Forty-eight 10GE optical ports

Table 7-532 Ports on the CEL48XSFD1-G

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-533 Functions and features of the CEL48XSFD1-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 192 10GE ports. • The CloudEngine 16808 chassis provides a maximum of 384 10GE ports. • The CloudEngine 16816 chassis provides a maximum of 768 10GE ports.

Functions and Features	Description
10GE port	When the 10GE ports work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. When the 10GE ports work in GE mode, they support SFP GE optical modules and 1000BASE-T copper modules. When using a 1000BASE-T copper module, a port supports 100M/1000M auto-sensing.
Hot swapping	Supported
Restrictions and remarks	In V200R021C00 and earlier versions, this LPU is used with the CE-SFU16F-G. When the LPU is installed in slot 14, SFUs cannot work in 4+2 backup mode and can only work in 5+1 backup mode when the device works in six-SFU mode; SFUs cannot work in 3+1 backup mode or any other backup mode when the device works in four-SFU mode.

Technical Specifications

Table 7-534 Technical specifications of the CEL48XSFD1-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.7 kg (14.77 lb)
Weight with packaging [kg (lb)]	11.6 kg (25.57 lb)
Typical power consumption [W]	103 W
Typical heat dissipation [BTU/hour]	352 BTU/hour
Static power consumption [W]	85 W
Static heat dissipation [BTU/hour]	290 BTU/hour
Maximum power consumption [W]	193 W
Maximum heat dissipation [BTU/hour]	659 BTU/hour

Item	Specification
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.4 CEL48XSFD-A (48-port 10GE interface card (FD-A, SFP+))

7.4.4.1 CEL48XSFD-A (03050ECA: 48-port 10GE interface card (FD-A, SFP+))

Overview

Table 7-535 Basic information about the CEL48XSFD-A

Item	Details
Description	48-port 10GE interface card (FD-A, SFP +)
Part Number	03050ECA
Model	CEL48XSFD-A
Silkscreen	CEL48XSFD-A
Card Type	Interface Card

Appearance

Figure 7-168 Appearance of the CEL48XSFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-536 Mappings between CEL48XSFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	1-4	-
CloudEngine 16808	V200R021C00 SPC100	-	1-8	-
CloudEngine 16816	V200R021C10 SPC600	-	1-16	-

Table 7-537 Mappings between CEL48XSFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUD-HALF	03058876-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-00 2	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-00 1	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUD-HALF	03058876-00 1	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-00 1	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-538 Mappings between CEL48XSFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R021C00SPC100	-	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-A	03050EGY	V200R021C00SPC100	-	-
CloudEngine 16804	CE-SFU04G-A	03050EGY-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-A	03050ECD	V200R021C00SPC100	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators

Figure 7-169 Indicators on the CEL48XSFD-A



1. Running status indicator	2. Port status indicator
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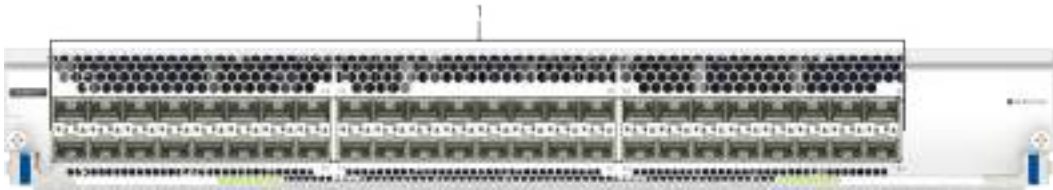
Table 7-539 Indicators on the CEL48XSFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-170 Ports on the CEL48XSFD-A



1. Forty-eight 10GE optical ports

Table 7-540 Ports on the CEL48XSFD-A

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-541 Functions and features of the CEL48XSFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are such cards: • The CloudEngine 16804 chassis provides up to 192 x 10GE ports. • The CloudEngine 16808 chassis provides up to 384 x 10GE ports. • The CloudEngine 16816 chassis provides up to 768 x 10GE ports.
10GE port	When optical ports on the card work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. NOTE • 10GE optical ports 20 to 31 do not support auto-negotiation when they have GE optical modules installed. To such a port to a port on another device, disable auto-negotiation on the peer port. Otherwise, one port may go Up and the other may go Down. Other ports support 1000BASE-X auto-negotiation. • 10GE optical ports 20 to 31 can only work at the rate of 1000 Mbit/s when they have copper modules installed.
Hot swapping	Supported

Technical Specifications

Table 7-542 Technical specifications of the CEL48XSFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.9 kg (15.21 lb)
Weight with packaging [kg (lb)]	10.2 kg (22.49 lb)
Typical power consumption [W]	180 W
Typical heat dissipation [BTU/hour]	615 BTU/hour
Static power consumption [W]	112 W
Static heat dissipation [BTU/hour]	382 BTU/hour

Item	Specification
Maximum power consumption [W]	242 W
Maximum heat dissipation [BTU/hour]	826 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.4.4.2 CEL48XSFD-A (03050ECA-001: 48-port 10GE interface card (FD-A, SFP +))

Overview

Table 7-543 Basic information about the CEL48XSFD-A

Item	Details
Description	48-port 10GE interface card (FD-A, SFP +)
Part Number	03050ECA-001
Model	CEL48XSFD-A
Silkscreen	CEL48XSFD-A
Card Type	Interface Card

Appearance

Figure 7-171 Appearance of the CEL48XSFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-544 Mappings between CEL48XSFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1-8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.

Table 7-545 Mappings between CEL48XSFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-546 Mappings between CEL48XSFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04F-A	03050ECE	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04F-A	03050ECE-001	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators

Figure 7-172 Indicators on the CEL48XSFD-A

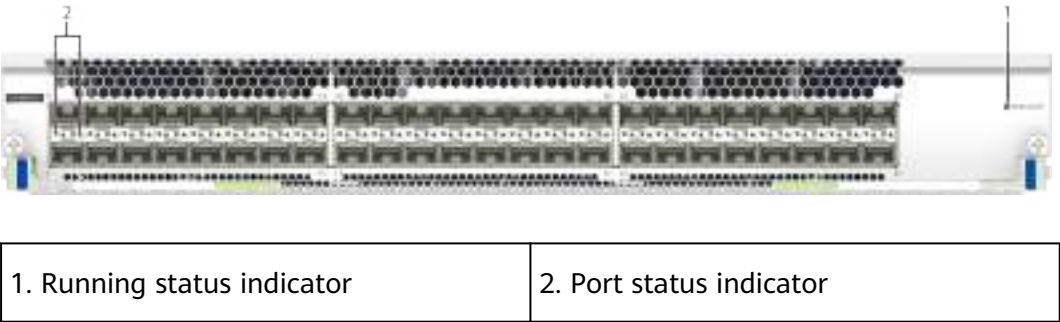


Table 7-547 Indicators on the CEL48XSFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator Two single-color indicators for each port Steady green: LINK indicator Blinking yellow: ACT indicator The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Off	The link on the port is disconnected.
		Yellow	Fast blinking (15 Hz)	The port is transmitting or receiving data.
		Yellow	Off	The port is not transmitting or receiving data.

Ports

Figure 7-173 Ports on the CEL48XSFD-A

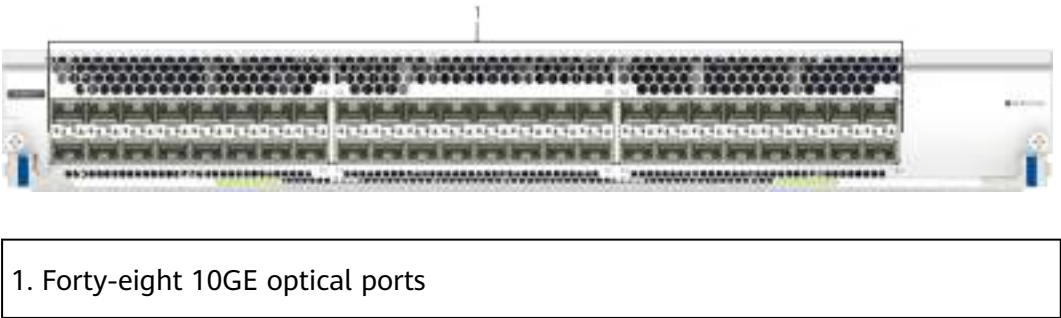


Table 7-548 Ports on the CEL48XSFD-A

Port	Connector Type	Description	Available Components
48 x 10GE optical ports	SFP+/SFP	Port for inputting and outputting 10GE/GE optical signals	• GE eSFP optical modules • GE SFP copper modules • 10GE SFP+ optical modules • SFP+ to SFP+ AOC cable • SFP+ to SFP+ high-speed cable

Functions and Features

Table 7-549 Functions and features of the CEL48XSFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 48 10GE optical ports.
Maximum port density	If all LPUs are such cards: • The CloudEngine 16804 chassis provides up to 192 x 10GE ports. • The CloudEngine 16808 chassis provides up to 384 x 10GE ports. • The CloudEngine 16816 chassis provides up to 768 x 10GE ports.

Functions and Features	Description
10GE port	When optical ports on the card work in 10GE mode, they support SFP+ 10GE optical modules, 10GE high-speed cables, and AOC cables. NOTE 10GE optical ports 20 to 31 do not support auto-negotiation when they have GE optical modules installed. To such a port to a port on another device, disable auto-negotiation on the peer port. Otherwise, one port may go Up and the other may go Down. Other ports support 1000BASE-X auto-negotiation. 10GE optical ports 20 to 31 can only work at the rate of 1000 Mbit/s when they have copper modules installed.
Hot swapping	Supported

Technical Specifications

Table 7-550 Technical specifications of the CEL48XSFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	6.9 kg (15.21 lb)
Weight with packaging [kg (lb)]	10.2 kg (22.49 lb)
Typical power consumption [W]	180 W
Typical heat dissipation [BTU/hour]	615 BTU/hour
Static power consumption [W]	112 W
Static heat dissipation [BTU/hour]	382 BTU/hour
Maximum power consumption [W]	242 W
Maximum heat dissipation [BTU/hour]	826 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5 40GE Interface Card

7.5.1 CEL24LQFD-G (24-Port 40GE Interface Card (FD-G, QSFP+))

7.5.1.1 CEL24LQFD-G (03058872: 24-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-551 Basic information about the CEL24LQFD-G

Item	Details
Description	24-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058872
Model	CEL24LQFD-G
Silkscreen	CEL24LQFD-G
Card Type	Interface Card

Appearance

Figure 7-174 Appearance of the CEL24LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-552 Mappings between CEL24LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	1-4	-
CloudEngine 16808	V200R005C20	-	1-8	-
CloudEngine 16816	V200R005C20	-	1-16	-

Table 7-553 Mappings between CEL24LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R005C20	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-554 Mappings between CEL24LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R005C20	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R005C20	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R005C20	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-175 Indicators and buttons on the CEL24LQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-555 Indicators on the CEL24LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-556 Buttons on the CEL24LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-176 Ports on the CEL24LQFD-G

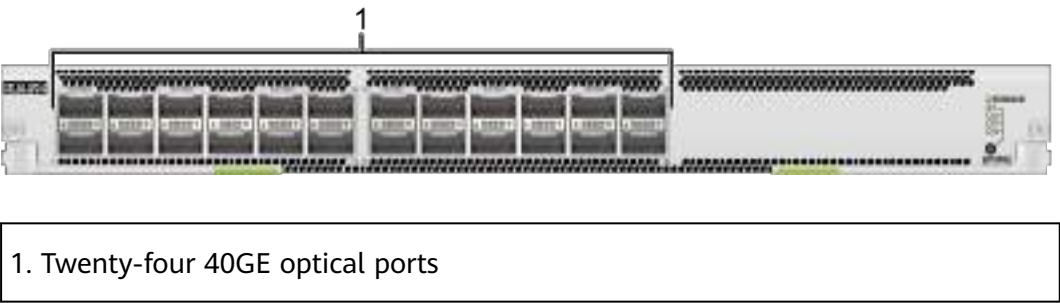


Table 7-557 Ports on the CEL24LQFD-G

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-558 Functions and features of the CEL24LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-559 Technical specifications of the CEL24LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.1.2 CEL24LQFD-G (03058872-001: 24-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-560 Basic information about the CEL24LQFD-G

Item	Details
Description	24-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058872-001
Model	CEL24LQFD-G
Silkscreen	CEL24LQFD-G
Card Type	Interface Card

Appearance

Figure 7-177 Appearance of the CEL24LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-561 Mappings between CEL24LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-562 Mappings between CEL24LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R020C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R020C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-563 Mappings between CEL24LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R020C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-178 Indicators and buttons on the CEL24LQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-564 Indicators on the CEL24LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-565 Buttons on the CEL24LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-179 Ports on the CEL24LQFD-G

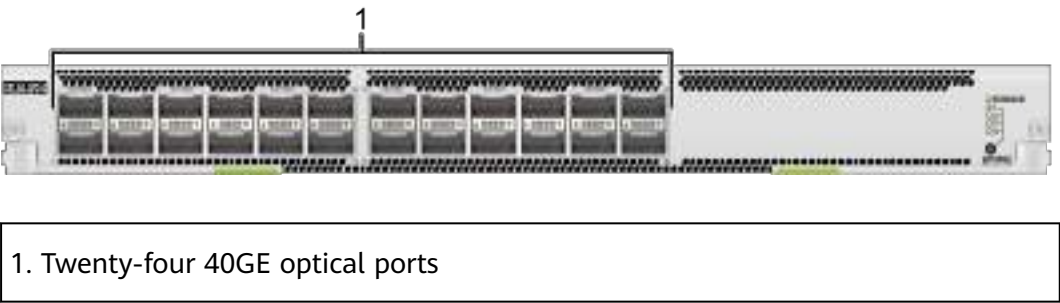


Table 7-566 Ports on the CEL24LQFD-G

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-567 Functions and features of the CEL24LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-568 Technical specifications of the CEL24LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.1.3 CEL24LQFD-G (03058872-002: 24-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-569 Basic information about the CEL24LQFD-G

Item	Details
Description	24-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058872-002
Model	CEL24LQFD-G
Silkscreen	CEL24LQFD-G
Card Type	Interface Card

Appearance

Figure 7-180 Appearance of the CEL24LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-570 Mappings between CEL24LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Table 7-571 Mappings between CEL24LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-572 Mappings between CEL24LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-181 Indicators and buttons on the CEL24LQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-573 Indicators on the CEL24LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-574 Buttons on the CEL24LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-182 Ports on the CEL24LQFD-G

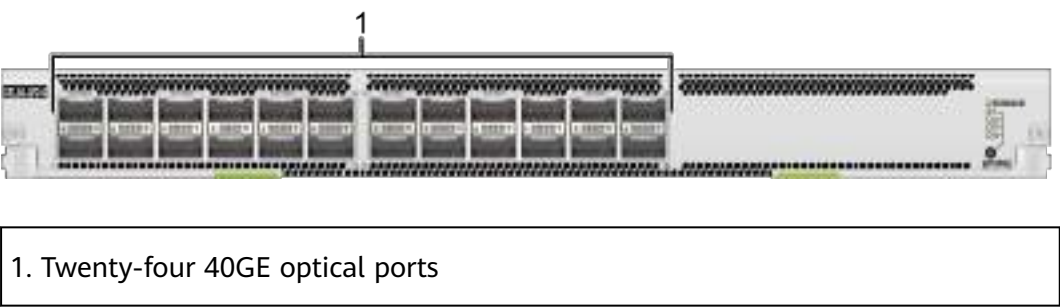


Table 7-575 Ports on the CEL24LQFD-G

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-576 Functions and features of the CEL24LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-577 Technical specifications of the CEL24LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.1.4 CEL24LQFD-G (03058872-003: 24-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-578 Basic information about the CEL24LQFD-G

Item	Details
Description	24-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058872-003
Model	CEL24LQFD-G
Silkscreen	CEL24LQFD-G
Card Type	Interface Card

Appearance

Figure 7-183 Appearance of the CEL24LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-579 Mappings between CEL24LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-580 Mappings between CEL24LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-581 Mappings between CEL24LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-184 Indicators and buttons on the CEL24LQFD-G

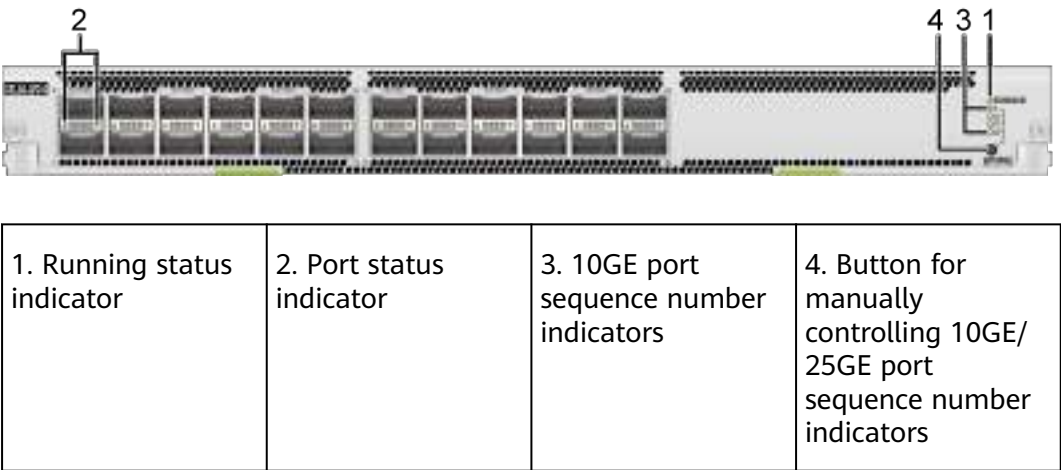


Table 7-582 Indicators on the CEL24LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-583 Buttons on the CEL24LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-185 Ports on the CEL24LQFD-G

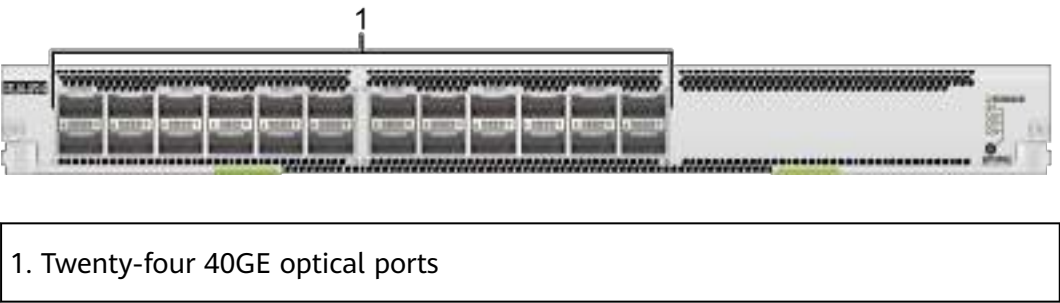


Table 7-584 Ports on the CEL24LQFD-G

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-585 Functions and features of the CEL24LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-586 Technical specifications of the CEL24LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.2 CEL24LQFD-GK (24-port 40GE interface card (FD-GK, QSFP+))

7.5.2.1 CEL24LQFD-GK (03059593: 24-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-587 Basic information about the CEL24LQFD-GK

Item	Details
Description	24-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059593
Model	CEL24LQFD-GK

Item	Details
Silkscreen	CEL24LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-186 Appearance of the CEL24LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-588 Mappings between CEL24LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-589 Mappings between CEL24LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R019C10	-

Table 7-590 Mappings between CEL24LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R019C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C00	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R019C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R020C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R019C 10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-187 Indicators and buttons on the CEL24LQFD-GK

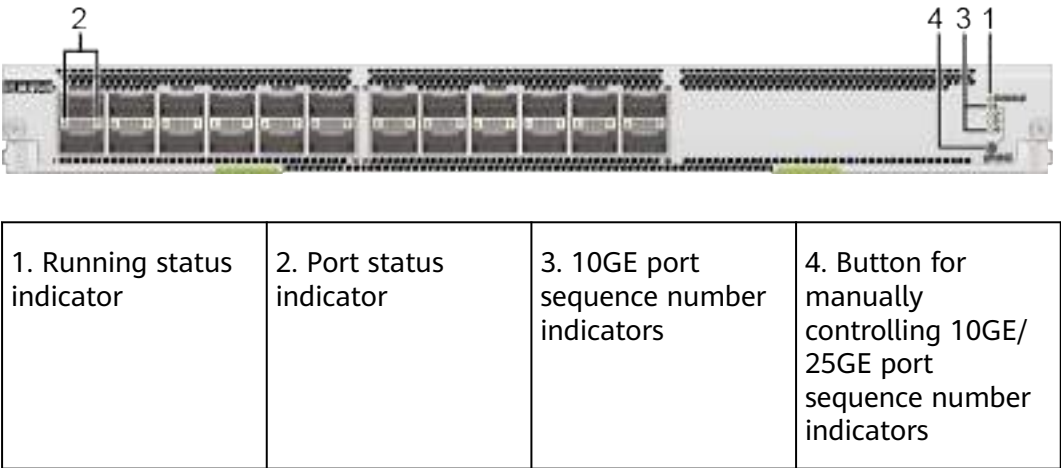


Table 7-591 Indicators on the CEL24LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-592 Buttons on the CEL24LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-188 Ports on the CEL24LQFD-GK

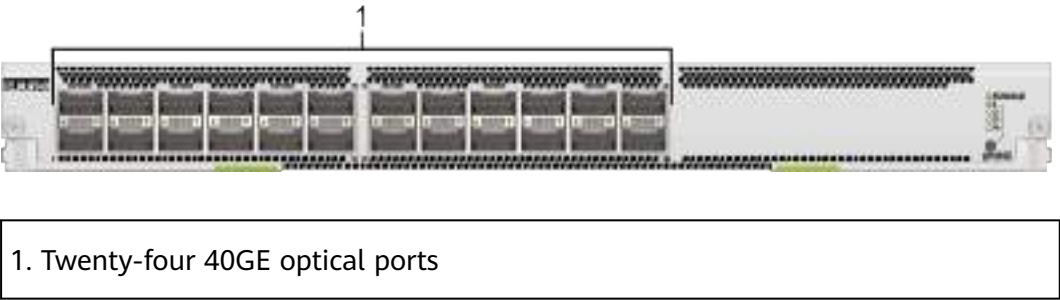


Table 7-593 Ports on the CEL24LQFD-GK

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-594 Functions and features of the CEL24LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-595 Technical specifications of the CEL24LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.2.2 CEL24LQFD-GK (03059593-001: 24-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-596 Basic information about the CEL24LQFD-GK

Item	Details
Description	24-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059593-001
Model	CEL24LQFD-GK
Silkscreen	CEL24LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-189 Appearance of the CEL24LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-597 Mappings between CEL24LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-598 Mappings between CEL24LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R020C10	-

Table 7-599 Mappings between CEL24LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632	V200R020C10	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-190 Indicators and buttons on the CEL24LQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-600 Indicators on the CEL24LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-601 Buttons on the CEL24LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-191 Ports on the CEL24LQFD-GK

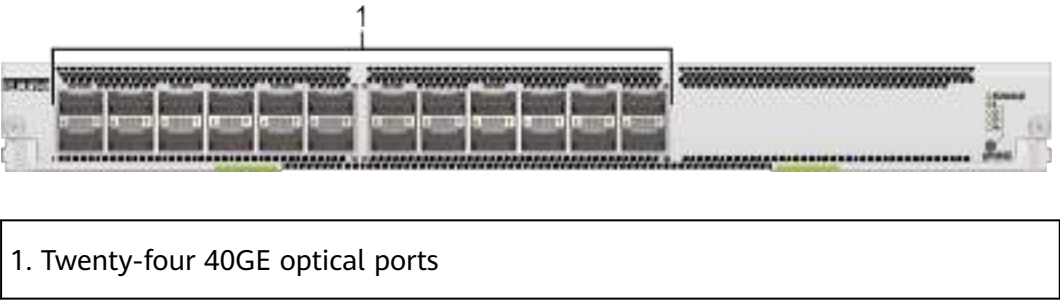


Table 7-602 Ports on the CEL24LQFD-GK

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-603 Functions and features of the CEL24LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-604 Technical specifications of the CEL24LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.2.3 CEL24LQFD-GK (03059593-002: 24-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-605 Basic information about the CEL24LQFD-GK

Item	Details
Description	24-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059593-002
Model	CEL24LQFD-GK
Silkscreen	CEL24LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-192 Appearance of the CEL24LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-606 Mappings between CEL24LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-607 Mappings between CEL24LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R021C10 SPC600	-

Table 7-608 Mappings between CEL24LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-193 Indicators and buttons on the CEL24LQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-609 Indicators on the CEL24LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-610 Buttons on the CEL24LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-194 Ports on the CEL24LQFD-GK

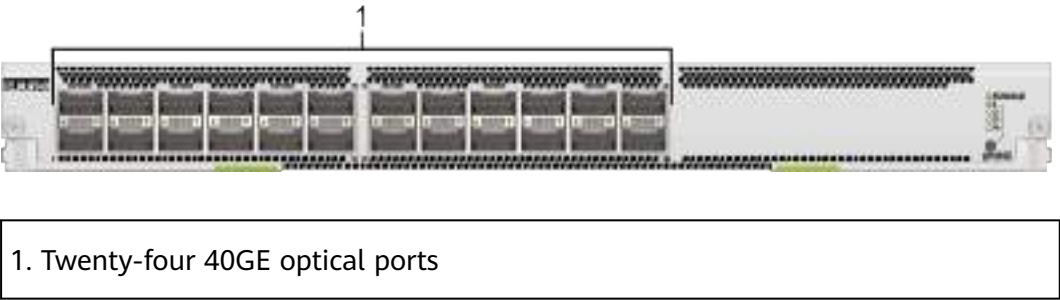


Table 7-611 Ports on the CEL24LQFD-GK

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-612 Functions and features of the CEL24LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-613 Technical specifications of the CEL24LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.3 CEL24LQFD1-G (24-port 40GE interface card (FD1-G, QSFP+))

Overview

Table 7-614 Basic information about the CEL24LQFD1-G

Item	Details
Description	24-port 40GE interface card (FD1-G, QSFP+)
Part Number	03059669
Model	CEL24LQFD1-G
Silkscreen	CEL24LQFD1-G
Card Type	Interface Card

Appearance

Figure 7-195 Appearance of the CEL24LQFD1-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-615 Mappings between CEL24LQFD1-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-616 Mappings between CEL24LQFD1-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R019C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-617 Mappings between CEL24LQFD1-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F1-G	03059671	V200R019C10	-	-
CloudEngine 16808	CE-SFU08F1-G	03059672	V200R019C10	-	-

Indicators and Buttons

Figure 7-196 Indicators and buttons on the CEL24LQFD1-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-618 Indicators on the CEL24LQFD1-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-619 Buttons on the CEL24LQFD1-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-197 Ports on the CEL24LQFD1-G

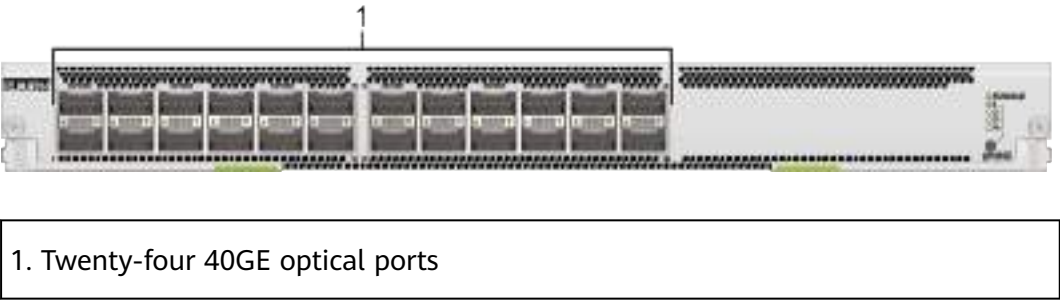


Table 7-620 Ports on the CEL24LQFD1-G

Port	Connector Type	Description	Available Components
24 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 12 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-621 Functions and features of the CEL24LQFD1-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 24 40GE optical ports, which can be split into 96 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 96 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 192 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 384 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 96 x 10GE. All the 24 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 12 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. Their IDs are: 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, and 21. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 24 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 form a group, and ports 12 to 23 form another group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-622 Technical specifications of the CEL24LQFD1-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.3 kg (16.09 lb)
Weight with packaging [kg (lb)]	12.2 kg (26.9 lb)
Typical power consumption [W]	163 W
Typical heat dissipation [BTU/hour]	557 BTU/hour
Static power consumption [W]	120 W
Static heat dissipation [BTU/hour]	410 BTU/hour
Maximum power consumption [W]	307 W
Maximum heat dissipation [BTU/hour]	1048 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.4 CEL36LQFD-G (36-Port 40GE Interface Card (FD-G, QSFP+))

7.5.4.1 CEL36LQFD-G (03058871: 36-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-623 Basic information about the CEL36LQFD-G

Item	Details
Description	36-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058871
Model	CEL36LQFD-G

Item	Details
Silkscreen	CEL36LQFD-G
Card Type	Interface Card

Appearance

Figure 7-198 Appearance of the CEL36LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-624 Mappings between CEL36LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	1-4	-
CloudEngine 16808	V200R005C20	-	1-8	-
CloudEngine 16816	V200R005C20	-	1-16	-

Table 7-625 Mappings between CEL36LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R005C20	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-626 Mappings between CEL36LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-G	03058882	V200R005C 20	-	-
CloudEngin e 16808	CE-SFU08G-G	03058882-001	V200R020C 10	-	-
CloudEngin e 16808	CE-SFU08G-G	03058882-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885	V200R005C 20	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-001	V200R020C 10	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G1	03050KHK	V200R022C 10	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884	V200R005C 20	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-199 Indicators and buttons on the CEL36LQFD-G

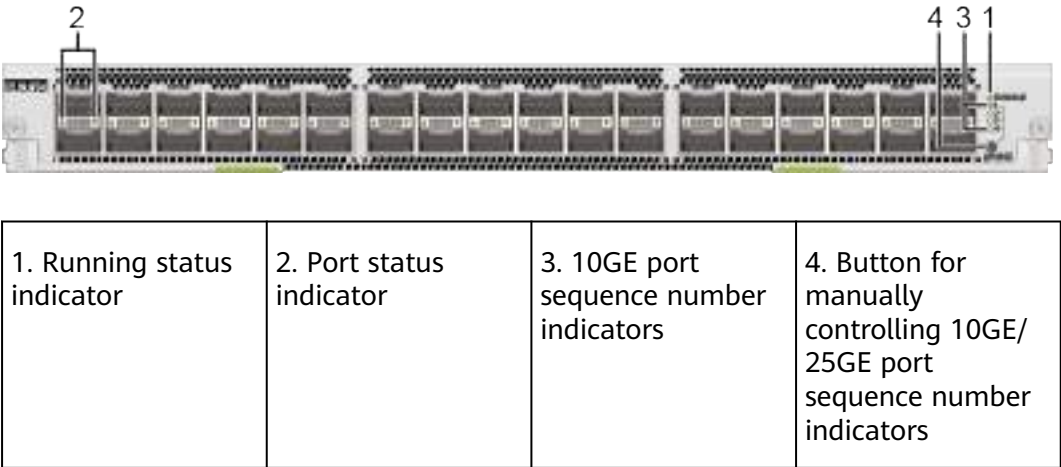


Table 7-627 Indicators on the CEL36LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-628 Buttons on the CEL36LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-200 Ports on the CEL36LQFD-G

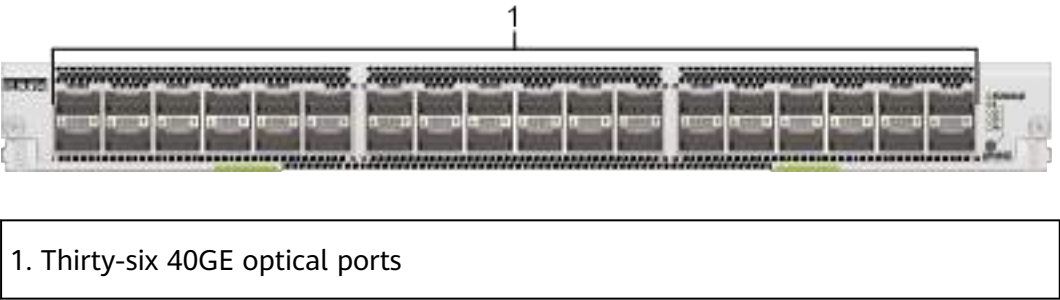


Table 7-629 Ports on the CEL36LQFD-G

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-630 Functions and features of the CEL36LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-631 Technical specifications of the CEL36LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.4.2 CEL36LQFD-G (03058871-001: 36-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-632 Basic information about the CEL36LQFD-G

Item	Details
Description	36-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058871-001
Model	CEL36LQFD-G
Silkscreen	CEL36LQFD-G
Card Type	Interface Card

Appearance

Figure 7-201 Appearance of the CEL36LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-633 Mappings between CEL36LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-634 Mappings between CEL36LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R020C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R020C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-635 Mappings between CEL36LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R020C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-202 Indicators and buttons on the CEL36LQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-636 Indicators on the CEL36LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-637 Buttons on the CEL36LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-203 Ports on the CEL36LQFD-G

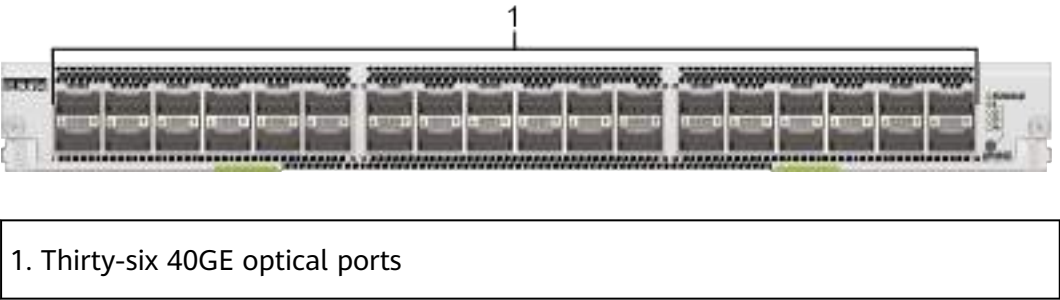


Table 7-638 Ports on the CEL36LQFD-G

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-639 Functions and features of the CEL36LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-640 Technical specifications of the CEL36LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.4.3 CEL36LQFD-G (03058871-002: 36-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-641 Basic information about the CEL36LQFD-G

Item	Details
Description	36-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058871-002
Model	CEL36LQFD-G
Silkscreen	CEL36LQFD-G
Card Type	Interface Card

Appearance

Figure 7-204 Appearance of the CEL36LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-642 Mappings between CEL36LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Table 7-643 Mappings between CEL36LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-644 Mappings between CEL36LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-205 Indicators and buttons on the CEL36LQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-645 Indicators on the CEL36LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-646 Buttons on the CEL36LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-206 Ports on the CEL36LQFD-G

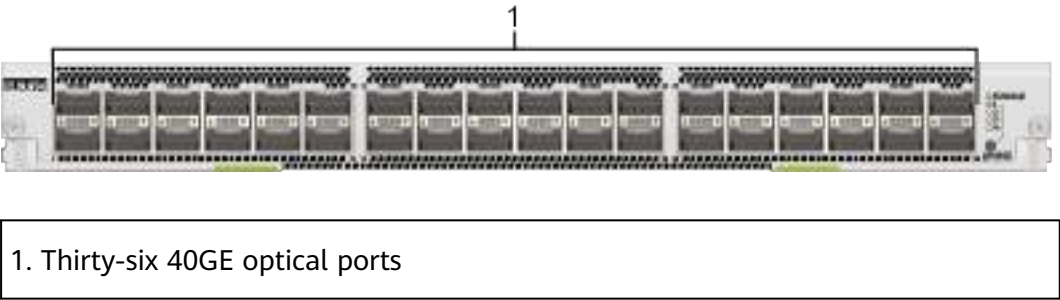


Table 7-647 Ports on the CEL36LQFD-G

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-648 Functions and features of the CEL36LQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-649 Technical specifications of the CEL36LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.4.4 CEL36LQFD-G (03058871-003: 36-Port 40GE Interface Card (FD-G, QSFP+))

Overview

Table 7-650 Basic information about the CEL36LQFD-G

Item	Details
Description	36-Port 40GE Interface Card (FD-G, QSFP+)
Part Number	03058871-003
Model	CEL36LQFD-G
Silkscreen	CEL36LQFD-G
Card Type	Interface Card

Appearance

Figure 7-207 Appearance of the CEL36LQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-651 Mappings between CEL36LQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-652 Mappings between CEL36LQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-653 Mappings between CEL36LQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-G	03058882-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-001	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G	03058885-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16F-G1	03050KHK	V200R022C 10	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-G	03058884-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-208 Indicators and buttons on the CEL36LQFD-G

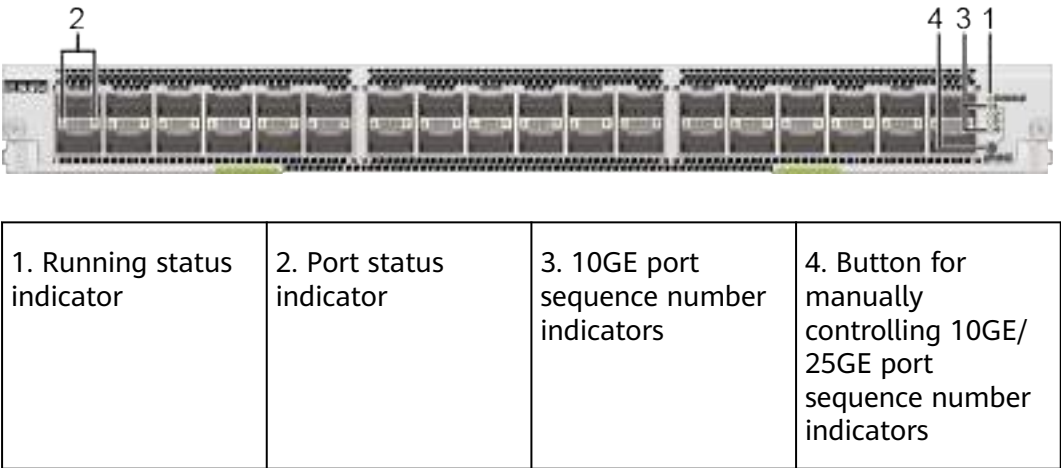


Table 7-654 Indicators on the CEL36LQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of a top-row port, and the indicator with a down arrowhead shows the status of a bottom-row port.	Green	Steady on	A link has been established on the port.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • When the sequence number indicator N (N can be 1, 2, 3, or 4) of a converted 10GE/25GE port is on, the port indicator shows the status of the Nth converted 10GE/25GE port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator still shows the status of the port.

Table 7-655 Buttons on the CEL36LQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The default mode is restored automatically 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-209 Ports on the CEL36LQFD-G

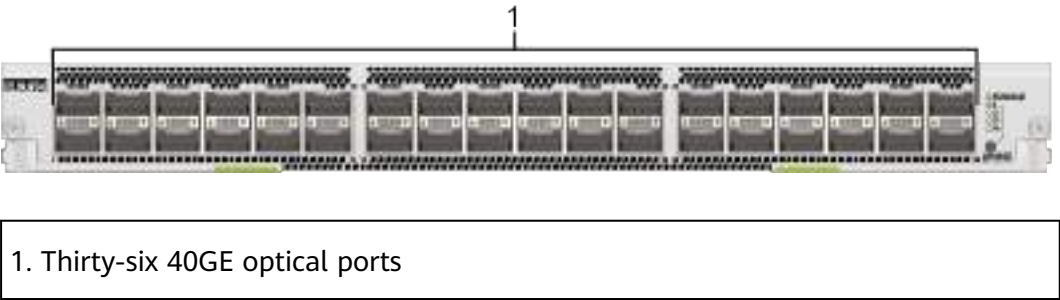


Table 7-656 Ports on the CEL36LQFD-G

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When 18 x 40GE ports (0, 1, 4, 5...) on the card work in 100GE mode, the ports support: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on a port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-657 Functions and features of the CEL36LQFD-G

Functions and Features	Description
Basic function	Provides data packet processing and traffic management on 36 x 40GE optical ports, which can be split into 144 x 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 144 x 40GE ports. • The CloudEngine 16808 chassis provides up to 288 x 40GE ports. • The CloudEngine 16816 chassis provides up to 576 x 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, the card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, 18 x 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by 0, 1, 4, 5... in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200 Gbit/s. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-658 Technical specifications of the CEL36LQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	4-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.5 CEL36LQFD-GK (36-port 40GE interface card (FD-GK, QSFP+))

7.5.5.1 CEL36LQFD-GK (03059592: 36-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-659 Basic information about the CEL36LQFD-GK

Item	Details
Description	36-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059592
Model	CEL36LQFD-GK

Item	Details
Silkscreen	CEL36LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-210 Appearance of the CEL36LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-660 Mappings between CEL36LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-661 Mappings between CEL36LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R019C10	-

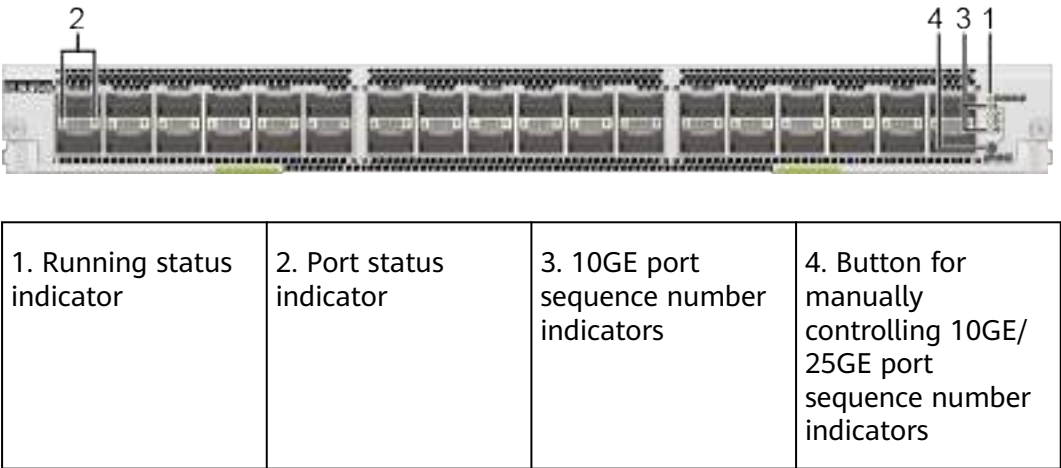
Table 7-662 Mappings between CEL36LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R019C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C00	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R019C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R020C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R019C 10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-211 Indicators and buttons on the CEL36LQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-663 Indicators on the CEL36LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-664 Buttons on the CEL36LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-212 Ports on the CEL36LQFD-GK

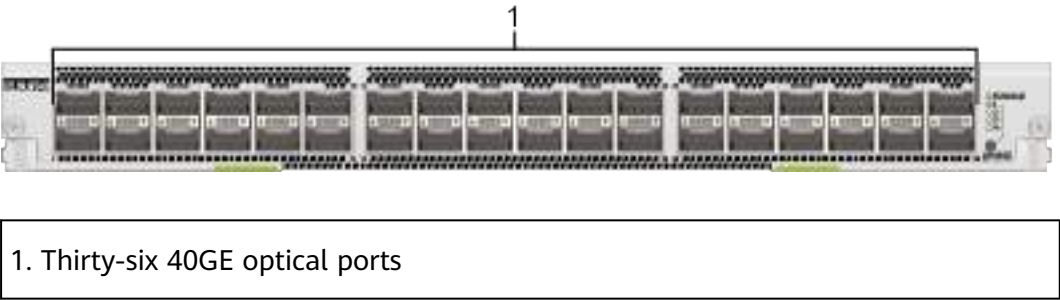


Table 7-665 Ports on the CEL36LQFD-GK

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-666 Functions and features of the CEL36LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-667 Technical specifications of the CEL36LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.5.2 CEL36LQFD-GK (03059592-001: 36-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-668 Basic information about the CEL36LQFD-GK

Item	Details
Description	36-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059592-001
Model	CEL36LQFD-GK
Silkscreen	CEL36LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-213 Appearance of the CEL36LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-669 Mappings between CEL36LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-670 Mappings between CEL36LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R020C10	-

Table 7-671 Mappings between CEL36LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngin e 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU	V200R020C10	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R020C10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R020C10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-214 Indicators and buttons on the CEL36LQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-672 Indicators on the CEL36LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-673 Buttons on the CEL36LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-215 Ports on the CEL36LQFD-GK

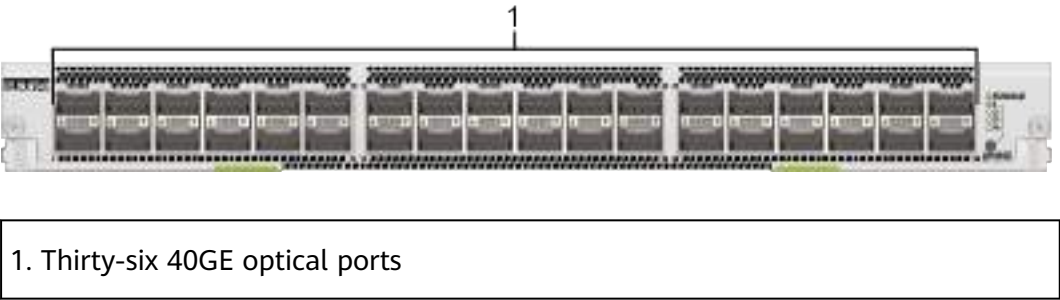


Table 7-674 Ports on the CEL36LQFD-GK

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-675 Functions and features of the CEL36LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-676 Technical specifications of the CEL36LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.5.3 CEL36LQFD-GK (03059592-002: 36-port 40GE interface card (FD-GK, QSFP+))

Overview

Table 7-677 Basic information about the CEL36LQFD-GK

Item	Details
Description	36-port 40GE interface card (FD-GK, QSFP+)
Part Number	03059592-002
Model	CEL36LQFD-GK
Silkscreen	CEL36LQFD-GK
Card Type	Interface Card

Appearance

Figure 7-216 Appearance of the CEL36LQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-678 Mappings between CEL36LQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1-8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-679 Mappings between CEL36LQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R021C10 SPC600	-

Table 7-680 Mappings between CEL36LQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-GK	03059596-001	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-GK	03059596-002	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU-002	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-217 Indicators and buttons on the CEL36LQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-681 Indicators on the CEL36LQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-682 Buttons on the CEL36LQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-218 Ports on the CEL36LQFD-GK

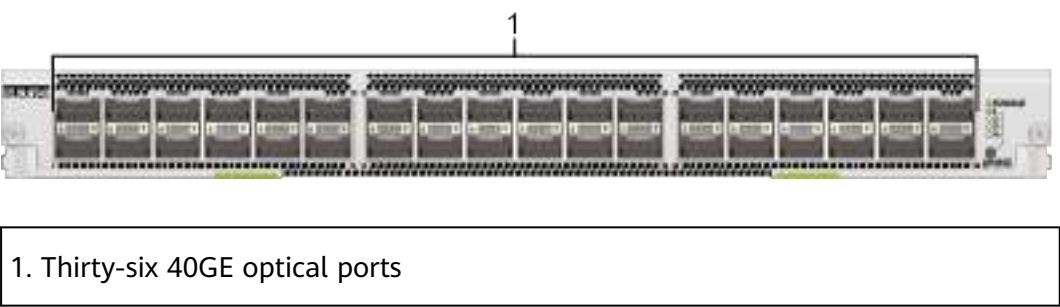


Table 7-683 Ports on the CEL36LQFD-GK

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-684 Functions and features of the CEL36LQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-685 Technical specifications of the CEL36LQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.6 CEL36LQFD1-G (36-port 40GE interface card (FD1-G, QSFP+))

Overview

Table 7-686 Basic information about the CEL36LQFD1-G

Item	Details
Description	36-port 40GE interface card (FD1-G, QSFP+)
Part Number	03059668
Model	CEL36LQFD1-G
Silkscreen	CEL36LQFD1-G
Card Type	Interface Card

Appearance

Figure 7-219 Appearance of the CEL36LQFD1-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-687 Mappings between CEL36LQFD1-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-688 Mappings between CEL36LQFD1-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R019C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R019C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-689 Mappings between CEL36LQFD1-G and SFU

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04F1-G	03059671	V200R019C10	-	-
CloudEngin e 16808	CE-SFU08F1-G	03059672	V200R019C10	-	-

Indicators and Buttons

Figure 7-220 Indicators and buttons on the CEL36LQFD1-G



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-690 Indicators on the CEL36LQFD1-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-691 Buttons on the CEL36LQFD1-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-221 Ports on the CEL36LQFD1-G

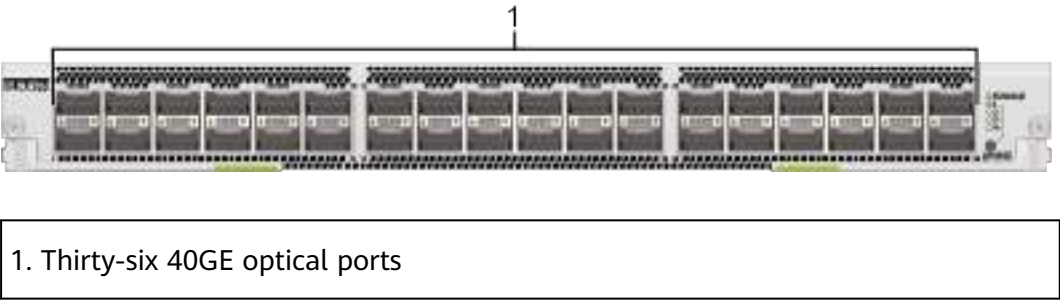


Table 7-692 Ports on the CEL36LQFD1-G

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-693 Functions and features of the CEL36LQFD1-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.

Functions and Features	Description
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions. The 36 ports on the card are divided into groups, each of which contains four ports starting from port 0. The total port rate of a group cannot exceed 200GE. When port n is used as a 100GE port, port n+2 becomes unavailable and is in DOWN(Port unavailable) state. For example: Ports 0 to 3 are in the same group. If ports 0 and 1 are used as 100GE ports, ports 2 and 3 become unavailable and are in DOWN(Port unavailable) state. If only one port (for example, port 0) is used as a 100GE port, port 2 becomes unavailable and is in DOWN(Port unavailable) state. Ports 1 and 3 can work as 40GE ports or each of them can be split into four 10GE ports.
Hot swapping	Supported
Restrictions and remarks	Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in 5+1 backup mode.

Technical Specifications

Table 7-694 Technical specifications of the CEL36LQFD1-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.0 kg (17.64 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)
Typical power consumption [W]	256 W
Typical heat dissipation [BTU/hour]	874 BTU/hour
Static power consumption [W]	217 W
Static heat dissipation [BTU/hour]	741 BTU/hour
Maximum power consumption [W]	443 W
Maximum heat dissipation [BTU/hour]	1513 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.5.7 CEL36LQFD-A (36-port 40GE interface card (FD-A, QSFP+))

7.5.7.1 CEL36LQFD-A (03050ECB: 36-port 40GE interface card (FD-A, QSFP+))

Overview

Table 7-695 Basic information about the CEL36LQFD-A

Item	Details
Description	36-port 40GE interface card (FD-A, QSFP+)
Part Number	03050ECB
Model	CEL36LQFD-A
Silkscreen	CEL36LQFD-A

Item	Details
Card Type	Interface Card

Appearance

Figure 7-222 Appearance of the CEL36LQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-696 Mappings between CEL36LQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	1-4	-
CloudEngine 16808	V200R021C00 SPC100	-	1-8	-
CloudEngine 16816	V200R021C10 SPC600	-	1-16	-

Table 7-697 Mappings between CEL36LQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-698 Mappings between CEL36LQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R021C00SPC100	-	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-A	03050EGY	V200R021C00SPC100	-	-
CloudEngine 16804	CE-SFU04G-A	03050EGY-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-A	03050ECD	V200R021C00SPC100	-	-
CloudEngine 16808	CE-SFU08F-A	03050ECD-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-A	03050EQK	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-A	03050EQJ	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-223 Indicators and buttons on the CEL36LQFD-A



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-699 Indicators on the CEL36LQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-700 Buttons on the CEL36LQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-224 Ports on the CEL36LQFD-A

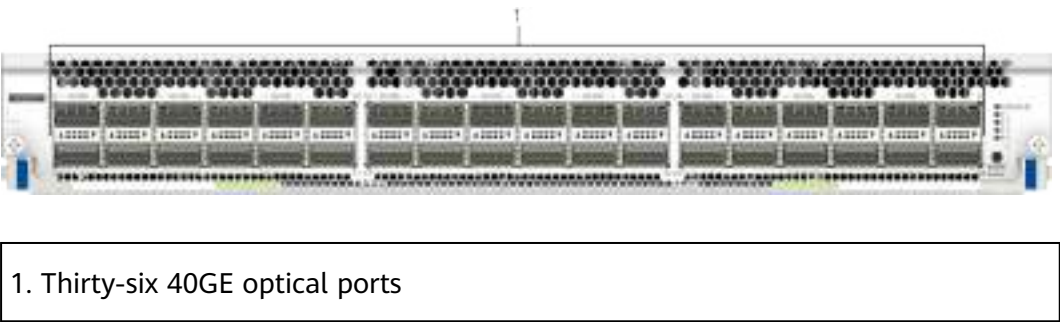


Table 7-701 Ports on the CEL36LQFD-A

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable

Functions and Features

Table 7-702 Functions and features of the CEL36LQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	All LPUs are of this model: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions.

Functions and Features	Description
Restrictions and remarks	When 18 ports on an LPU are configured to work at 100 Gbit/s, the other 18 ports are configured to work at 40 Gbit/s, and all these 36 ports work at the same time, they cannot forward data at the line rate. Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. • When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in the 4+1 backup mode. • When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in the 5+1 backup mode.
Hot swapping	Supported

 NOTE

The port split function is supported in V200R021C00SPC200 and later versions.

Technical Specifications

Table 7-703 Technical specifications of the CEL36LQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.9 kg (17.42 lb)
Weight with packaging [kg (lb)]	11.4 kg (25.13 lb)
Typical power consumption [W]	349 W
Typical heat dissipation [BTU/hour]	1191 BTU/hour
Static power consumption [W]	249 W
Static heat dissipation [BTU/hour]	850 BTU/hour

Item	Specification
Maximum power consumption [W]	596 W
Maximum heat dissipation [BTU/hour]	2035 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.5.7.2 CEL36LQFD-A (03050ECB-001: 36-port 40GE interface card (FD-A, QSFP+))

Overview

Table 7-704 Basic information about the CEL36LQFD-A

Item	Details
Description	36-port 40GE interface card (FD-A, QSFP+)
Part Number	03050ECB-001
Model	CEL36LQFD-A
Silkscreen	CEL36LQFD-A
Card Type	Interface Card

Appearance

Figure 7-225 Appearance of the CEL36LQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-705 Mappings between CEL36LQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1-4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1-8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.

Table 7-706 Mappings between CEL36LQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-

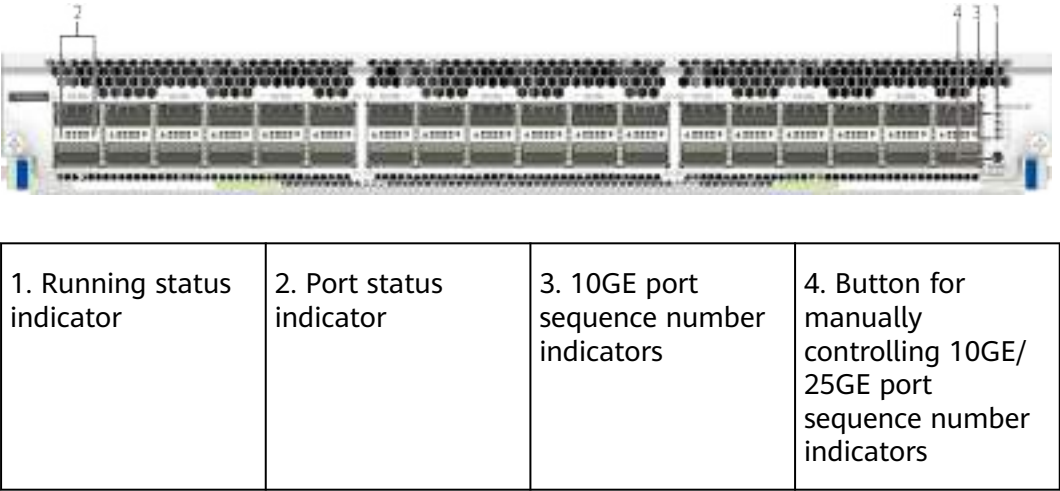
Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-707 Mappings between CEL36LQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04F-A	03050ECE	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04F-A	03050ECE-001	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-226 Indicators and buttons on the CEL36LQFD-A



1. Running status indicator	2. Port status indicator	3. 10GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-708 Indicators on the CEL36LQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator	Green	Steady on	The link on the port is connected.
	One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-709 Buttons on the CEL36LQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-227 Ports on the CEL36LQFD-A

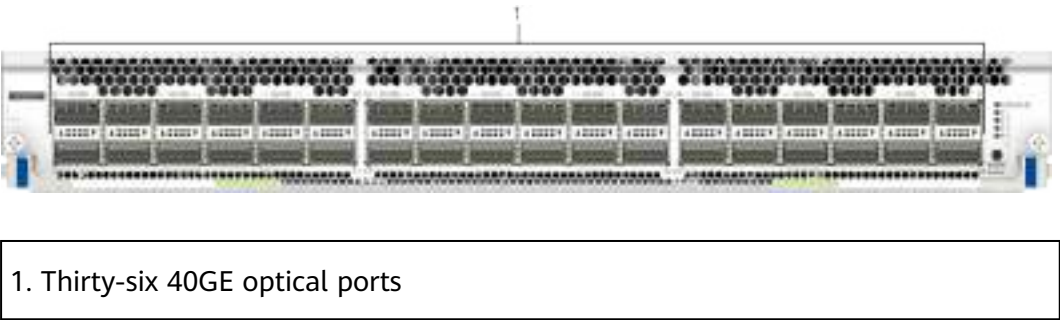


Table 7-710 Ports on the CEL36LQFD-A

Port	Connector Type	Description	Available Components
36 x 40GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE (supported by half of the ports)/40GE optical signals	• 40GE QSFP+ optical modules • QSFP+ to QSFP+ AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable When one of the 18 40GE ports (0, 1, 4, 5...) on the card works in 100GE mode, it can use: • 100GE QSFP28 optical modules • QSFP28 to QSFP28 AOC cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable

Functions and Features

Table 7-711 Functions and features of the CEL36LQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 40GE optical ports, which can be split into 144 10GE optical ports.
Maximum port density	All LPUs are of this model: • The CloudEngine 16804 chassis provides a maximum of 144 40GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 40GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 40GE ports.
40GE port	A 40GE port on the card can be split into four 10GE ports. That is, a card provides a maximum port density of 144 x 10GE. All the 36 40GE ports are independent, and each can be split into four 10GE ports. On the card, the 18 40GE ports that can be used as 100GE ports are marked with silkscreen 40/100G. The port IDs are 0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25, 28, 29, 32, and 33. These port IDs are represented by (0, 1, 4, 5...) in the following descriptions.

Functions and Features	Description
Restrictions and remarks	When 18 ports on an LPU are configured to work at 100 Gbit/s, the other 18 ports are configured to work at 40 Gbit/s, and all these 36 ports work at the same time, they cannot forward data at the line rate. Ports 0 to 11 are in a group, ports 12 to 23 are in a group, and ports 24 to 35 are in a group. - When the total port bandwidth of each group does not exceed 480 Gbit/s, SFUs can forward data at the line rate in the 4+1 backup mode. - When the total port bandwidth of each group does not exceed 600 Gbit/s, SFUs can forward data at the line rate in the 5+1 backup mode.
Hot swapping	Supported

Technical Specifications

Table 7-712 Technical specifications of the CEL36LQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.9 kg (17.42 lb)
Weight with packaging [kg (lb)]	11.4 kg (25.13 lb)
Typical power consumption [W]	349 W
Typical heat dissipation [BTU/hour]	1191 BTU/hour
Static power consumption [W]	249 W
Static heat dissipation [BTU/hour]	850 BTU/hour
Maximum power consumption [W]	596 W
Maximum heat dissipation [BTU/hour]	2035 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed

Item	Specification
Memory	Standard 8 GB

7.6 100GE Interface Card

7.6.1 CEL18CQFD-G (18-Port 100GE Interface Card (FD-G, QSFP28))

7.6.1.1 CEL18CQFD-G (03058870: 18-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-713 Basic information about the CEL18CQFD-G

Item	Details
Description	18-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058870
Model	CEL18CQFD-G
Silkscreen	CEL18CQFD-G
Card Type	Interface Card

Appearance

Figure 7-228 Appearance of the CEL18CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-714 Mappings between CEL18CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	1-4	-
CloudEngine 16808	V200R005C20	-	1-8	-
CloudEngine 16816	V200R005C20	-	1-16	-

Table 7-715 Mappings between CEL18CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R005C20	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-716 Mappings between CEL18CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R005C20	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R005C20	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R005C20	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-229 Indicators and buttons on the CEL18CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-717 Indicators on the CEL18CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-718 Buttons on the CEL18CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-230 Ports on the CEL18CQFD-G

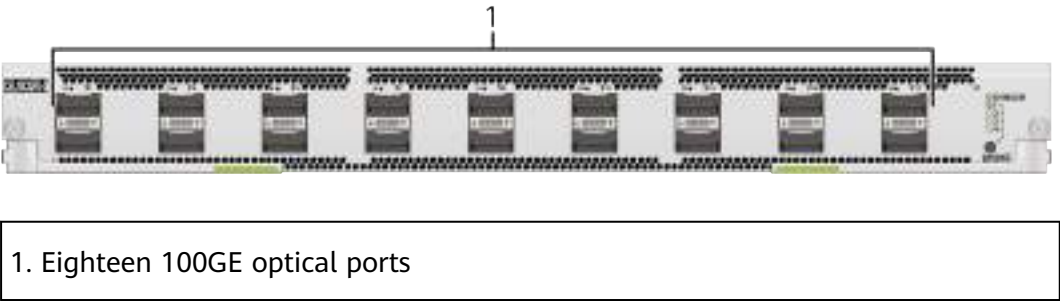


Table 7-719 Ports on the CEL18CQFD-G

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-720 Functions and features of the CEL18CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 72 x 100GE ports. • The CloudEngine 16808 chassis provides up to 144 x 100GE ports. • The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-721 Technical specifications of the CEL18CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.1.2 CEL18CQFD-G (03058870-001: 18-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-722 Basic information about the CEL18CQFD-G

Item	Details
Description	18-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058870-001
Model	CEL18CQFD-G
Silkscreen	CEL18CQFD-G
Card Type	Interface Card

Appearance

Figure 7-231 Appearance of the CEL18CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-723 Mappings between CEL18CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16816	V200R020C10	-	1–16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-724 Mappings between CEL18CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R020C10	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R020C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R020C10	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R020C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-725 Mappings between CEL18CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10 SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10 SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R020C10	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R020C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-232 Indicators and buttons on the CEL18CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-726 Indicators on the CEL18CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.

Silkscreen	Name	Color	Status	Description
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-727 Buttons on the CEL18CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-233 Ports on the CEL18CQFD-G

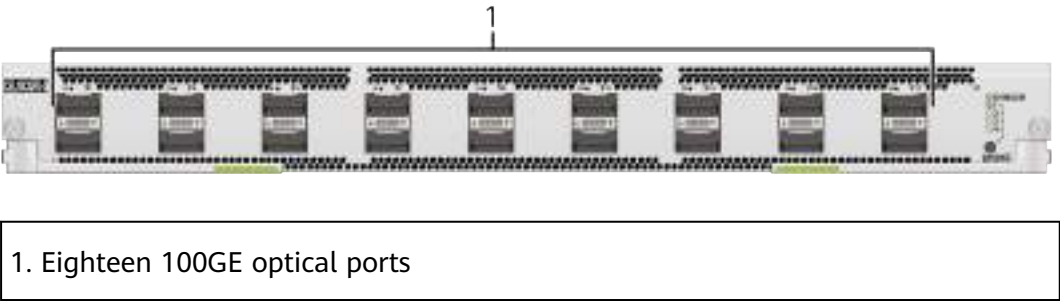


Table 7-728 Ports on the CEL18CQFD-G

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-729 Functions and features of the CEL18CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides up to 72 x 100GE ports. - The CloudEngine 16808 chassis provides up to 144 x 100GE ports. - The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-730 Technical specifications of the CEL18CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.1.3 CEL18CQFD-G (03058870-002: 18-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-731 Basic information about the CEL18CQFD-G

Item	Details
Description	18-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058870-002
Model	CEL18CQFD-G
Silkscreen	CEL18CQFD-G
Card Type	Interface Card

Appearance

Figure 7-234 Appearance of the CEL18CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-732 Mappings between CEL18CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 032 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 010 or a later version.

Table 7-733 Mappings between CEL18CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-734 Mappings between CEL18CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-235 Indicators and buttons on the CEL18CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-735 Indicators on the CEL18CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-736 Buttons on the CEL18CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-236 Ports on the CEL18CQFD-G

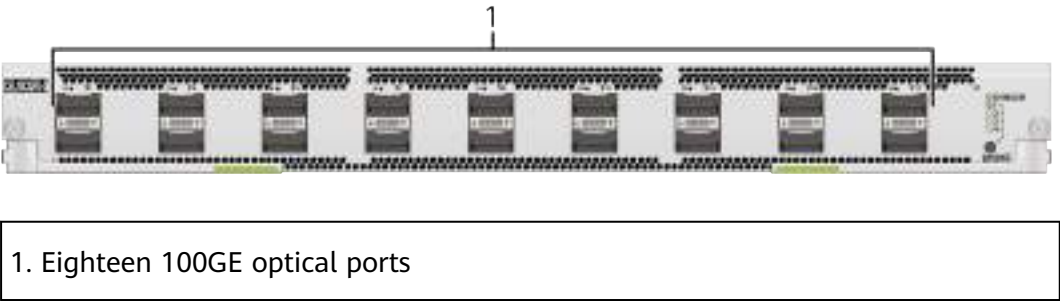


Table 7-737 Ports on the CEL18CQFD-G

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-738 Functions and features of the CEL18CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides up to 72 x 100GE ports. - The CloudEngine 16808 chassis provides up to 144 x 100GE ports. - The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-739 Technical specifications of the CEL18CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.1.4 CEL18CQFD-G (03058870-003: 18-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-740 Basic information about the CEL18CQFD-G

Item	Details
Description	18-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058870-003
Model	CEL18CQFD-G
Silkscreen	CEL18CQFD-G
Card Type	Interface Card

Appearance

Figure 7-237 Appearance of the CEL18CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-741 Mappings between CEL18CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-742 Mappings between CEL18CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-743 Mappings between CEL18CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-238 Indicators and buttons on the CEL18CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-744 Indicators on the CEL18CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of a top-row port, and the indicator with a down arrowhead shows the status of a bottom-row port.	Green	Steady on	A link has been established on the port.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • When the sequence number indicator N (N can be 1, 2, 3, or 4) of a converted 10GE/25GE port is on, the port indicator shows the status of the Nth converted 10GE/25GE port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator still shows the status of the port.

Table 7-745 Buttons on the CEL18CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split, the sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-239 Ports on the CEL18CQFD-G

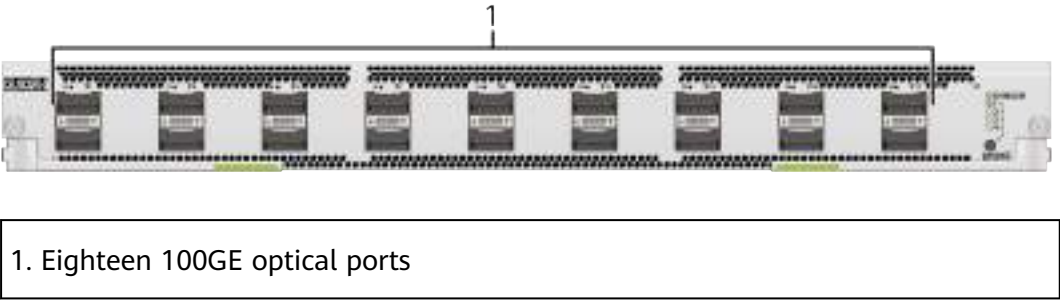


Table 7-746 Ports on the CEL18CQFD-G

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-747 Functions and features of the CEL18CQFD-G

Functions and Features	Description
Basic function	Provides data packet processing and traffic management on 18 x 100GE optical ports, which can be used as 18 x 40GE optical ports or split into 72 x 25GE or 72 x 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 72 x 100GE ports. • The CloudEngine 16808 chassis provides up to 144 x 100GE ports. • The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	QSFP28 and QSFP+ optical modules are supported. A 100GE port on the card can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. The card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 optical ports are independent, and each 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-748 Technical specifications of the CEL18CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	4-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.2 CEL18CQFD-GK (18-port 100GE interface card (FD-GK, QSFP28))

7.6.2.1 CEL18CQFD-GK (03059591: 18-port 100GE interface card (FD-GK, QSFP28))

Overview

Table 7-749 Basic information about the CEL18CQFD-GK

Item	Details
Description	18-port 100GE interface card (FD-GK, QSFP28)
Part Number	03059591
Model	CEL18CQFD-GK
Silkscreen	CEL18CQFD-GK
Card Type	Interface Card

Appearance

Figure 7-240 Appearance of the CEL18CQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-750 Mappings between CEL18CQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-751 Mappings between CEL18CQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R019C10	-

Table 7-752 Mappings between CEL18CQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R019C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C00	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R019C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632	V200R019C10	-	-
CloudEngine 16816	CE-SFU16G-GK	03059632-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-241 Indicators and buttons on the CEL18CQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-753 Indicators on the CEL18CQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-754 Buttons on the CEL18CQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-242 Ports on the CEL18CQFD-GK

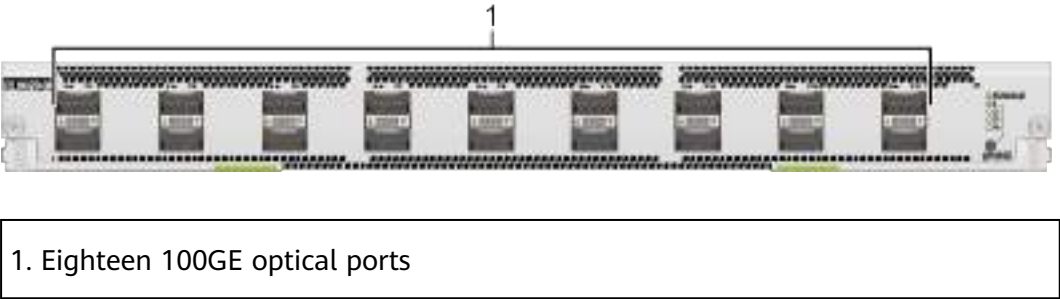


Table 7-755 Ports on the CEL18CQFD-GK

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-756 Functions and features of the CEL18CQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 72 x 100GE ports. • The CloudEngine 16808 chassis provides up to 144 x 100GE ports. • The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-757 Technical specifications of the CEL18CQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.2.2 CEL18CQFD-GK (03059591-001: 18-port 100GE interface card (FD-GK, QSFP28))

Overview

Table 7-758 Basic information about the CEL18CQFD-GK

Item	Details
Description	18-port 100GE interface card (FD-GK, QSFP28)
Part Number	03059591-001
Model	CEL18CQFD-GK
Silkscreen	CEL18CQFD-GK
Card Type	Interface Card

Appearance

Figure 7-243 Appearance of the CEL18CQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-759 Mappings between CEL18CQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R020C10	-	1-4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16808	V200R020C10	-	1-8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.
CloudEngine 16816	V200R020C10	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 007 or a later version.

Table 7-760 Mappings between CEL18CQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R020C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R020C10	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R020C10	-

Table 7-761 Mappings between CEL18CQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R020C10	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10 SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10 SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R020C 10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-244 Indicators and buttons on the CEL18CQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-762 Indicators on the CEL18CQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-763 Buttons on the CEL18CQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-245 Ports on the CEL18CQFD-GK

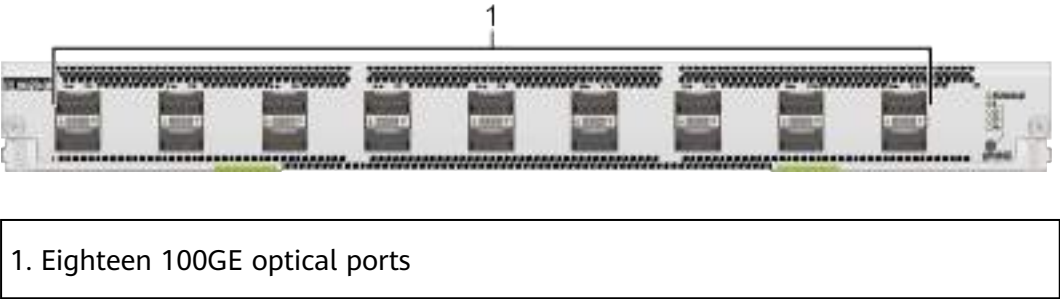


Table 7-764 Ports on the CEL18CQFD-GK

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-765 Functions and features of the CEL18CQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 72 x 100GE ports. • The CloudEngine 16808 chassis provides up to 144 x 100GE ports. • The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-766 Technical specifications of the CEL18CQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.2.3 CEL18CQFD-GK (03059591-002: 18-port 100GE interface card (FD-GK, QSFP28))

Overview

Table 7-767 Basic information about the CEL18CQFD-GK

Item	Details
Description	18-port 100GE interface card (FD-GK, QSFP28)
Part Number	03059591-002
Model	CEL18CQFD-GK
Silkscreen	CEL18CQFD-GK
Card Type	Interface Card

Appearance

Figure 7-246 Appearance of the CEL18CQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-768 Mappings between CEL18CQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-769 Mappings between CEL18CQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R021C10 SPC600	-

Table 7-770 Mappings between CEL18CQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-247 Indicators and buttons on the CEL18CQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-771 Indicators on the CEL18CQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-772 Buttons on the CEL18CQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-248 Ports on the CEL18CQFD-GK

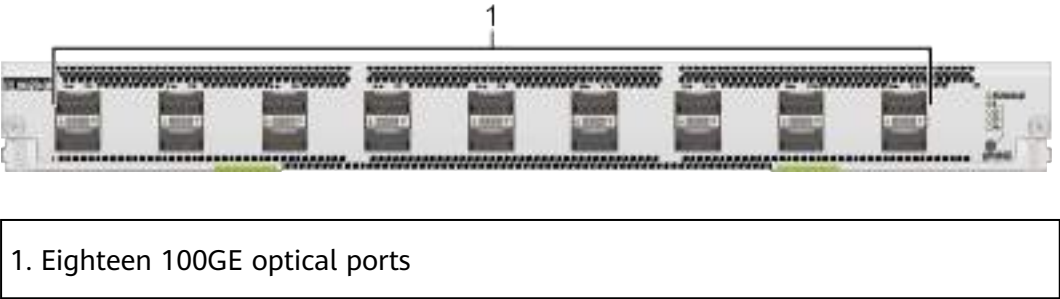


Table 7-773 Ports on the CEL18CQFD-GK

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-774 Functions and features of the CEL18CQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides up to 72 x 100GE ports. - The CloudEngine 16808 chassis provides up to 144 x 100GE ports. - The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-775 Technical specifications of the CEL18CQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	12.5 kg (27.56 lb)
Typical power consumption [W]	251 W
Typical heat dissipation [BTU/hour]	857 BTU/hour
Static power consumption [W]	150 W
Static heat dissipation [BTU/hour]	512 BTU/hour
Maximum power consumption [W]	388 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1325 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 4 GB

7.6.3 CEL18CQFD-A (18-port 100GE interface card (FD-A, QSFP28))

7.6.3.1 CEL18CQFD-A (03050ECC: 18-port 100GE interface card (FD-A, QSFP28))

Overview

Table 7-776 Basic information about the CEL18CQFD-A

Item	Details
Description	18-port 100GE interface card (FD-A, QSFP28)
Part Number	03050ECC
Model	CEL18CQFD-A
Silkscreen	CEL18CQFD-A
Card Type	Interface Card

Appearance

Figure 7-249 Appearance of the CEL18CQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-777 Mappings between CEL18CQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C00 SPC100	-	1–4	-
CloudEngine 16808	V200R021C00 SPC100	-	1–8	-
CloudEngine 16816	V200R021C10 SPC600	-	1–16	-

Table 7-778 Mappings between CEL18CQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C00 SPC100	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-779 Mappings between CEL18CQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R021C00SPC100	-	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R021C 00SPC100	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R021C 00SPC100	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-250 Indicators and buttons on the CEL18CQFD-A



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-780 Indicators on the CEL18CQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-781 Buttons on the CEL18CQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-251 Ports on the CEL18CQFD-A

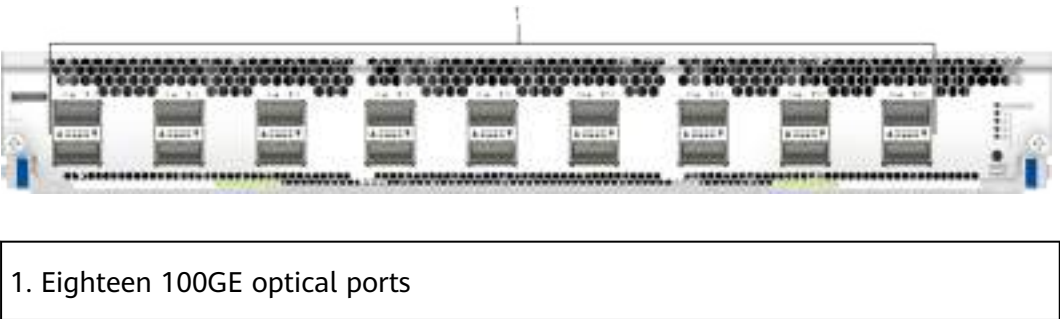


Table 7-782 Ports on the CEL18CQFD-A

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable

Functions and Features

Table 7-783 Functions and features of the CEL18CQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides up to 72 x 100GE ports. - The CloudEngine 16808 chassis provides up to 144 x 100GE ports. - The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-784 Technical specifications of the CEL18CQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	11.1 kg (24.47 lb)
Typical power consumption [W]	362 W
Typical heat dissipation [BTU/hour]	1236 BTU/hour
Static power consumption [W]	279 W
Static heat dissipation [BTU/hour]	953 BTU/hour
Maximum power consumption [W]	506 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1728 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.3.2 CEL18CQFD-A (03050ECC-001: 18-port 100GE interface card (FD-A, QSFP28))

Overview

Table 7-785 Basic information about the CEL18CQFD-A

Item	Details
Description	18-port 100GE interface card (FD-A, QSFP28)
Part Number	03050ECC-001
Model	CEL18CQFD-A
Silkscreen	CEL18CQFD-A
Card Type	Interface Card

Appearance

Figure 7-252 Appearance of the CEL18CQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-786 Mappings between CEL18CQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 022 or a later version.

Table 7-787 Mappings between CEL18CQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-788 Mappings between CEL18CQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04F-A	03050ECE	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04F-A	03050ECE-001	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-253 Indicators and buttons on the CEL18CQFD-A

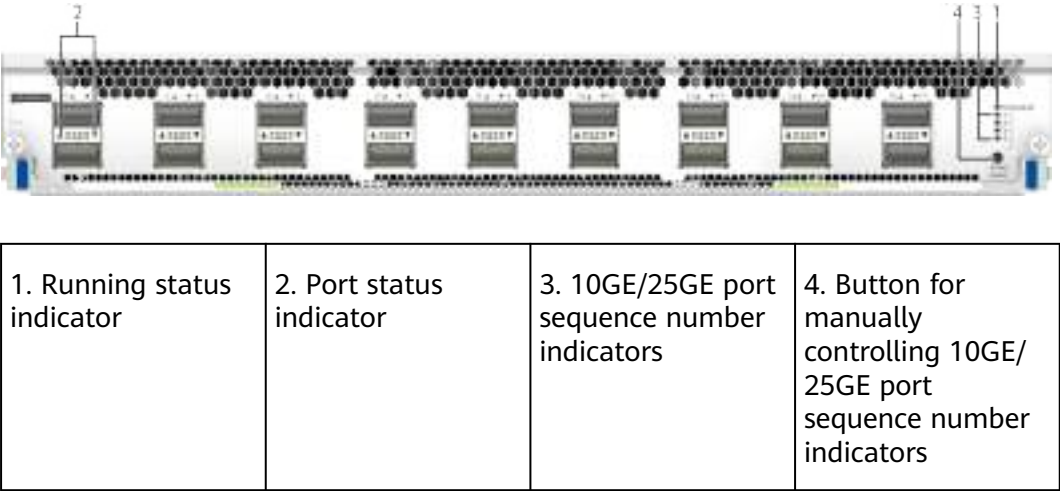


Table 7-789 Indicators on the CEL18CQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-790 Buttons on the CEL18CQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-254 Ports on the CEL18CQFD-A

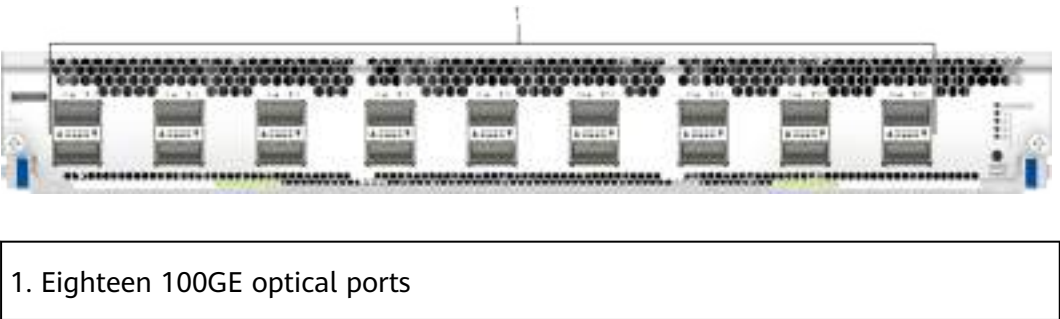


Table 7-791 Ports on the CEL18CQFD-A

Port	Connector Type	Description	Available Components
18 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable

Functions and Features

Table 7-792 Functions and features of the CEL18CQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 18 100GE optical ports, which can be used as 18 40GE optical ports or split into 72 25GE or 72 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides up to 72 x 100GE ports. • The CloudEngine 16808 chassis provides up to 144 x 100GE ports. • The CloudEngine 16816 chassis provides up to 288 x 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 18 x 40GE, 72 x 25GE, or 72 x 10GE. All the 18 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-793 Technical specifications of the CEL18CQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	7.6 kg (16.76 lb)
Weight with packaging [kg (lb)]	11.1 kg (24.47 lb)
Typical power consumption [W]	362 W
Typical heat dissipation [BTU/hour]	1236 BTU/hour
Static power consumption [W]	279 W
Static heat dissipation [BTU/hour]	953 BTU/hour
Maximum power consumption [W]	506 W

Item	Specification
Maximum heat dissipation [BTU/hour]	1728 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.4 CEL36CQFD-G (36-Port 100GE Interface Card (FD-G, QSFP28))

7.6.4.1 CEL36CQFD-G (03058869: 36-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-794 Basic information about the CEL36CQFD-G

Item	Details
Description	36-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058869
Model	CEL36CQFD-G
Silkscreen	CEL36CQFD-G
Card Type	Interface Card

Appearance

Figure 7-255 Appearance of the CEL36CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-795 Mappings between CEL36CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R005C20	-	1-4	-
CloudEngine 16808	V200R005C20	-	1-8	-
CloudEngine 16816	V200R005C20	-	1-16	-

Table 7-796 Mappings between CEL36CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R005C20	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C01	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R019C10	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C00 SPC100	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R005C20	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C01	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R019C10	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C00 SPC100	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-797 Mappings between CEL36CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R005C20	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R005C20	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R020C10	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R005C20	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R005C20	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R020C10	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R005C20	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R020C10	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R005C20	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-256 Indicators and buttons on the CEL36CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-798 Indicators on the CEL36CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-799 Buttons on the CEL36CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-257 Ports on the CEL36CQFD-G

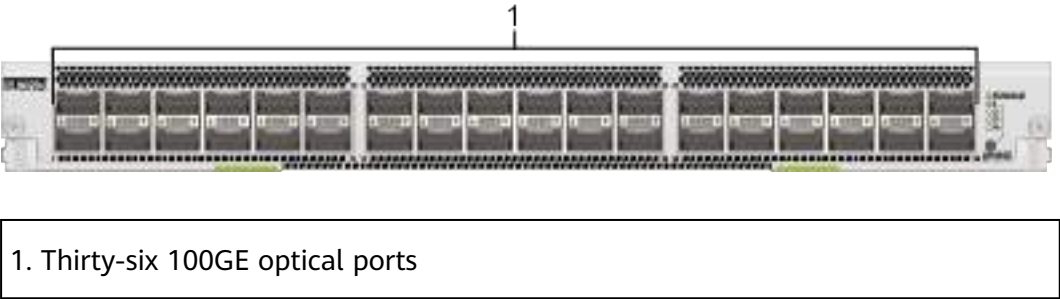


Table 7-800 Ports on the CEL36CQFD-G

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-801 Functions and features of the CEL36CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 36x40GE, 144x25GE, or 144x10GE. All the 36 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-802 Technical specifications of the CEL36CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W

Item	Specification
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.4.2 CEL36CQFD-G (03050EPU: 36-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-803 Basic information about the CEL36CQFD-G

Item	Details
Description	36-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03050EPU
Model	CEL36CQFD-G
Silkscreen	CEL36CQFD-G
Card Type	Interface Card

Appearance

Figure 7-258 Appearance of the CEL36CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-804 Mappings between CEL36CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC500	-	1-4	-
CloudEngine 16808	V200R021C10 SPC500	-	1-8	-
CloudEngine 16816	V200R021C10 SPC500	-	1-16	-

Table 7-805 Mappings between CEL36CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC500	-
CloudEngine 16804	CE-MPUD-HALF	03058876-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC500	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-00 1	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-00 2	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC500	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC500	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-00 1	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC500	-
CloudEngine 16808	CE-MPUD-HALF	03058876-00 1	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC500	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC500	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC500	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC500	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC500	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC500	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC500	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-806 Mappings between CEL36CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10 SPC500	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10 SPC500	-	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC500	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC500	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC500	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC500	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC500	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC500	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC500	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC500	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-259 Indicators and buttons on the CEL36CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-807 Indicators on the CEL36CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-808 Buttons on the CEL36CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-260 Ports on the CEL36CQFD-G

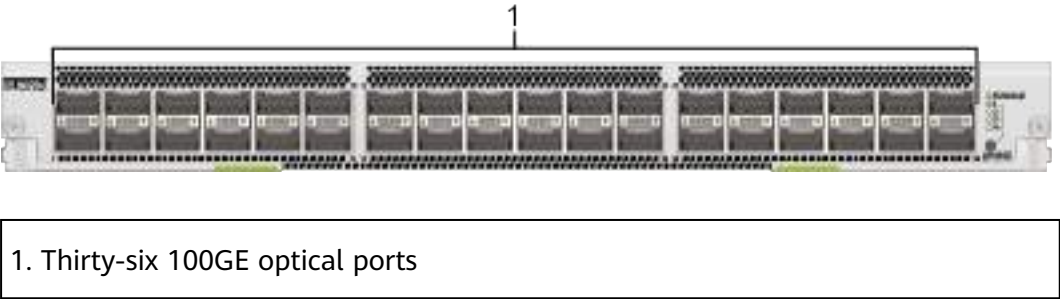


Table 7-809 Ports on the CEL36CQFD-G

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-810 Functions and features of the CEL36CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 36x40GE, 144x25GE, or 144x10GE. All the 36 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-811 Technical specifications of the CEL36CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W

Item	Specification
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.4.3 CEL36CQFD-G (03058869-002: 36-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-812 Basic information about the CEL36CQFD-G

Item	Details
Description	36-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058869-002
Model	CEL36CQFD-G
Silkscreen	CEL36CQFD-G
Card Type	Interface Card

Appearance

Figure 7-261 Appearance of the CEL36CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-813 Mappings between CEL36CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 033 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 011 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 033 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 011 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 033 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 011 or a later version.

Table 7-814 Mappings between CEL36CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-815 Mappings between CEL36CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-262 Indicators and buttons on the CEL36CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-816 Indicators on the CEL36CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-817 Buttons on the CEL36CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-263 Ports on the CEL36CQFD-G

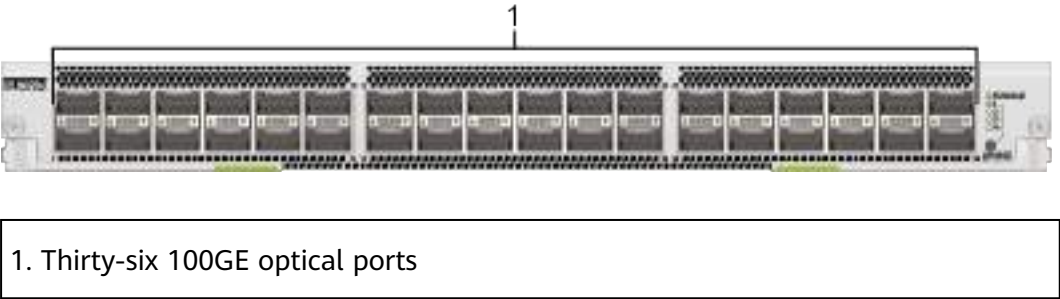


Table 7-818 Ports on the CEL36CQFD-G

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-819 Functions and features of the CEL36CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 36x40GE, 144x25GE, or 144x10GE. All the 36 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-820 Technical specifications of the CEL36CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W

Item	Specification
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.4.4 CEL36CQFD-G (03058869-003: 36-Port 100GE Interface Card (FD-G, QSFP28))

Overview

Table 7-821 Basic information about the CEL36CQFD-G

Item	Details
Description	36-Port 100GE Interface Card (FD-G, QSFP28)
Part Number	03058869-003
Model	CEL36CQFD-G
Silkscreen	CEL36CQFD-G
Card Type	Interface Card

Appearance

Figure 7-264 Appearance of the CEL36CQFD-G



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-822 Mappings between CEL36CQFD-G and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16816	V200R021C10 SPC600	-	1-16	If the card is used in V200R019C10 SPC800, install the patch of V200R019SPH 053 or a later version. If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-823 Mappings between CEL36CQFD-G and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUE-HALF	03059364	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUE-FULL	03059365	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

Table 7-824 Mappings between CEL36CQFD-G and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-G	03058881	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-G	03058881-002	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-G	03058880-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-G	03058883-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-G	03058882-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-001	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G	03058885-002	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16F-G1	03050KHK	V200R022C10	-	-
CloudEngine 16816	CE-SFU16G-G	03058884	V200R021C10SPC600	-	-
CloudEngine 16816	CE-SFU16G-G	03058884-002	V200R021C10SPC600	-	-

Indicators and Buttons

Figure 7-265 Indicators and buttons on the CEL36CQFD-G



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-825 Indicators on the CEL36CQFD-G

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).

Silkscreen	Name	Color	Status	Description
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of a top-row port, and the indicator with a down arrowhead shows the status of a bottom-row port.	Green	Steady on	A link has been established on the port.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • When the sequence number indicator N (N can be 1, 2, 3, or 4) of a converted 10GE/25GE port is on, the port indicator shows the status of the Nth converted 10GE/25GE port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator still shows the status of the port.

Table 7-826 Buttons on the CEL36CQFD-G

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split, the sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-266 Ports on the CEL36CQFD-G

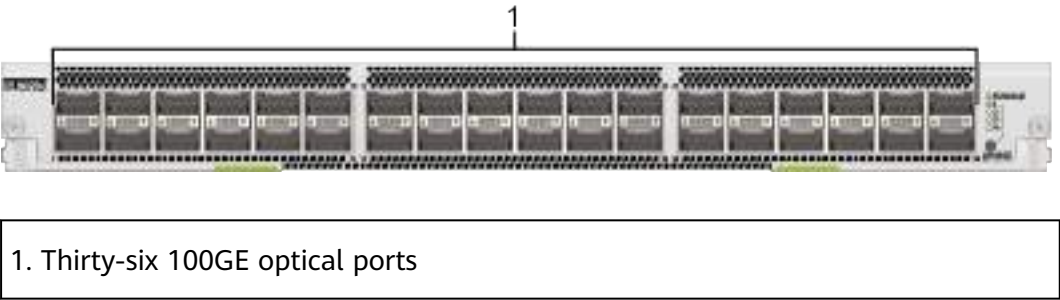


Table 7-827 Ports on the CEL36CQFD-G

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-828 Functions and features of the CEL36CQFD-G

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	All LPUs are of this model: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	Supports QSFP28 and QSFP+ optical modules. 100GE ports can work as 40GE ports through auto-sensing or be split into four 25GE or 10GE ports. Provides up to 36 x 40GE, 144 x 25GE, or 144 x 10GE ports. The 36 optical ports are independent, and each can work as a 40GE port or be split into four 25GE or 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-829 Technical specifications of the CEL36CQFD-G

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour

Item	Specification
CPU	4-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.5 CEL36CQFD-GK (36-port 100GE interface card (FD-GK, QSFP28))

7.6.5.1 CEL36CQFD-GK (03059589: 36-port 100GE interface card (FD-GK, QSFP28))

Overview

Table 7-830 Basic information about the CEL36CQFD-GK

Item	Details
Description	36-port 100GE interface card (FD-GK, QSFP28)
Part Number	03059589
Model	CEL36CQFD-GK
Silkscreen	CEL36CQFD-GK
Card Type	Interface Card

Appearance

Figure 7-267 Appearance of the CEL36CQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-831 Mappings between CEL36CQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R019C10	-	1-4	-
CloudEngine 16808	V200R019C10	-	1-8	-
CloudEngine 16816	V200R019C10	-	1-16	-

Table 7-832 Mappings between CEL36CQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R019C10	-
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C01	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R019C10	-

Table 7-833 Mappings between CEL36CQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-GK	03059596	V200R019C 10	-	-
CloudEngin e 16804	CE-SFU04G-GK	03059596-001	V200R020C 10	-	-
CloudEngin e 16804	CE-SFU04G-GK	03059596-002	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU	V200R020C 00	-	-
CloudEngin e 16808	CE-SFU08F-GK	03050BLU-002	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595	V200R019C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-001	V200R020C 10	-	-
CloudEngin e 16808	CE-SFU08G-GK	03059595-002	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R019C 10	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-268 Indicators and buttons on the CEL36CQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/25GE port sequence number indicators
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Table 7-834 Indicators on the CEL36CQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)

Silkscreen	Name	Color	Status	Description
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-835 Buttons on the CEL36CQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-269 Ports on the CEL36CQFD-GK

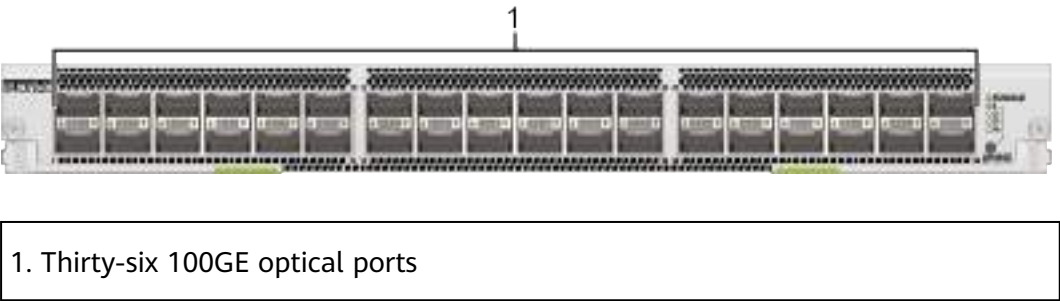


Table 7-836 Ports on the CEL36CQFD-GK

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-837 Functions and features of the CEL36CQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 36x40GE, 144x25GE, or 144x10GE. All the 36 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-838 Technical specifications of the CEL36CQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W

Item	Specification
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.5.2 CEL36CQFD-GK (03059589-001: 36-port 100GE interface card (FD-GK, QSFP28))

Overview

Table 7-839 Basic information about the CEL36CQFD-GK

Item	Details
Description	36-port 100GE interface card (FD-GK, QSFP28)
Part Number	03059589-001
Model	CEL36CQFD-GK
Silkscreen	CEL36CQFD-GK
Card Type	Interface Card

Appearance

Figure 7-270 Appearance of the CEL36CQFD-GK



NOTICE

The device where this card is to be installed cannot house SFUs or LPUs with mixed-card prevention labels. Otherwise, the device will be damaged.

Version Mapping

Table 7-840 Mappings between CEL36CQFD-GK and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16808	V200R021C10 SPC600	-	1–8	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.
CloudEngine 16816	V200R021C10 SPC600	-	1–16	If the card is used in V200R021C01 SPC600, install the patch of V200R021SPH 020 or a later version.

Table 7-841 Mappings between CEL36CQFD-GK and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HK	03059590	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16808	CE-MPUD-HK	03059590	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HK	03059590-001	V200R022C10	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-F2K	03050EQN-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FK	03059636	V200R021C10 SPC600	-

Table 7-842 Mappings between CEL36CQFD-GK and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04G-GK	03059596	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-001	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04G-GK	03059596-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08F-GK	03050BLU-002	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-001	V200R021C10SPC600	-	-
CloudEngine 16808	CE-SFU08G-GK	03059595-002	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16816	CE-SFU16G-GK	03059632	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-GK	03059632-002	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-271 Indicators and buttons on the CEL36CQFD-GK



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-843 Indicators on the CEL36CQFD-GK

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.

Silkscreen	Name	Color	Status	Description
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical alarm about hardware).
		Yellow	Steady on	The card is in power-off state. (For example, the card has been forcibly powered off using the power off command or is about to start.)
-	Port status indicator One single-color indicator for each port The indicator with an up arrowhead shows the status of the port on the top, and the indicator with a down arrowhead shows the status of the port at the bottom.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • If a port is split: When the sequence number indicator N (N can be 1, 2, 3, or 4) is on, the port indicator shows the status of the Nth 10GE or 25GE converted port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator shows the status of the 40GE or 100GE port.

Table 7-844 Buttons on the CEL36CQFD-GK

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 100GE ports are split into 10GE or 25GE ports, these sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the 10GE/25GE port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-272 Ports on the CEL36CQFD-GK

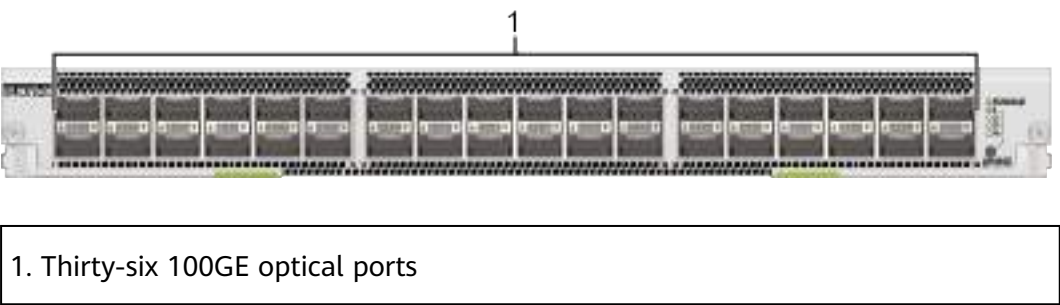


Table 7-845 Ports on the CEL36CQFD-GK

Port	Connector Type	Description	Available Components
36 x 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable NOTE When a QSFP+ to QSFP+ high-speed cable is installed on the port, the cable can only be used as a stack cable or a cable that connects peer-link interfaces in an M-LAG.

Functions and Features

Table 7-846 Functions and features of the CEL36CQFD-GK

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.

Functions and Features	Description
Maximum port density	If all LPUs are these cards: • The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. • The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. • The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	100GE ports on the card support QSFP28 and QSFP+ optical modules. A 100GE port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports. That is, a card can provide a maximum port density of 36x40GE, 144x25GE, or 144x10GE. All the 36 100GE ports are independent, and each port can be used as a 40GE port through auto-sensing, or split into four 25GE or four 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-847 Technical specifications of the CEL36CQFD-GK

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	8.4 kg (18.52 lb)
Weight with packaging [kg (lb)]	13.3 kg (29.32 lb)
Typical power consumption [W]	525 W
Typical heat dissipation [BTU/hour]	1793 BTU/hour
Static power consumption [W]	406 W

Item	Specification
Static heat dissipation [BTU/hour]	1386 BTU/hour
Maximum power consumption [W]	764 W
Maximum heat dissipation [BTU/hour]	2609 BTU/hour
CPU	Quad-core, 1.4 GHz clock speed
Memory	Standard 8 GB

7.6.6 CEL36CQFD-A (36-port 100GE interface card (FD-A, QSFP28))

7.6.6.1 CEL36CQFD-A (03050EPY: 36-port 100GE interface card (FD-A, QSFP28))

Overview

Table 7-848 Basic information about the CEL36CQFD-A

Item	Details
Description	36-port 100GE interface card (FD-A, QSFP28)
Part Number	03050EPY
Model	CEL36CQFD-A
Silkscreen	CEL36CQFD-A
Card Type	Interface Card

Appearance

Figure 7-273 Appearance of the CEL36CQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-849 Mappings between CEL36CQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R021C10 SPC600	-	1–4	-
CloudEngine 16808	V200R021C10 SPC600	-	1–8	-
CloudEngine 16816	V200R021C10 SPC600	-	1–16	-

Table 7-850 Mappings between CEL36CQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R021C10 SPC600	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R021C10 SPC600	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R021C10 SPC600	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R021C10 SPC600	-

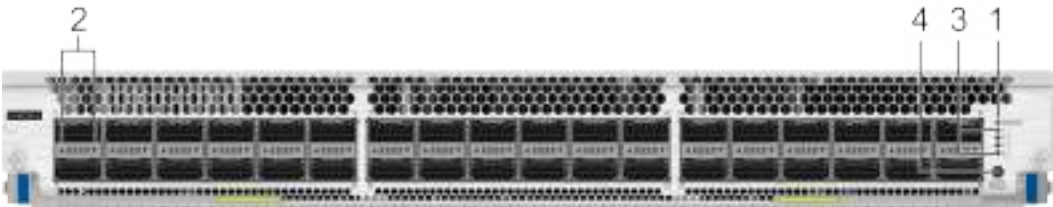
Table 7-851 Mappings between CEL36CQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R021C10SPC600	-	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R021C10SPC600	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R021C 10SPC600	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R021C 10SPC600	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R021C 10SPC600	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R021C 10SPC600	-	-

Indicators and Buttons

Figure 7-274 Indicators and buttons on the CEL36CQFD-A



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-852 Indicators on the CEL36CQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical hardware alarm).
		Yellow	Steady on	The card is not powered on. For example, the card has been forcibly powered off using the power off command or is in the pre-startup state.

Silkscreen	Name	Color	Status	Description
-	Port status indicators Each port has a single-color indicator. The indicator with an upward arrow indicates the status of the top-row port, and the indicator with a downward arrow indicates the status of the bottom-row port.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • When the sequence number indicator N (N can be 1, 2, 3, or 4) of a converted 10GE/25GE port is on, the port indicator shows the status of the Nth converted 10GE/25GE port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator still shows the status of the port.

Table 7-853 Buttons on the CEL36CQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, the sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-275 Ports on the CEL36CQFD-A

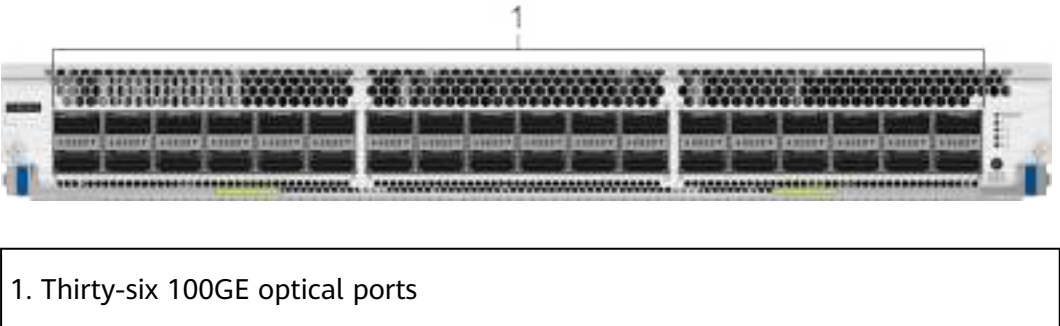


Table 7-854 Ports on the CEL36CQFD-A

Port	Connector Type	Description	Available Components
Thirty-six 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable NOTE When QSFP28 to QSFP28 or QSFP28 to 4*SFP28 high-speed cables are used on a card, only 1 m high-speed cables are supported.

Functions and Features

Table 7-855 Functions and features of the CEL36CQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.
Maximum port density	All LPUs are of this model: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	Supports QSFP28 and QSFP+ optical modules. 100GE ports can work as 40GE ports through auto-sensing or be split into four 25GE or 10GE ports. Provides up to 36 x 40GE, 144 x 25GE, or 144 x 10GE ports. The 36 optical ports are independent, and each can work as a 40GE port or be split into four 25GE or 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-856 Technical specifications of the CEL36CQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	9.2 kg (20.28 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)

Item	Specification
Typical power consumption [W]	584 W
Typical heat dissipation [BTU/hour]	1993 BTU/hour
Static power consumption [W]	494 W
Static heat dissipation [BTU/hour]	1686 BTU/hour
Maximum power consumption [W]	948 W
Maximum heat dissipation [BTU/hour]	3235 BTU/hour
CPU	Eight-core, 1.5 GHz clock speed
Memory	Standard 8 GB

7.6.6.2 CEL36CQFD-A (03050EPY-001: 36-port 100GE interface card (FD-A, QSFP28))

Overview

Table 7-857 Basic information about the CEL36CQFD-A

Item	Details
Description	36-port 100GE interface card (FD-A, QSFP28)
Part Number	03050EPY-001
Model	CEL36CQFD-A
Silkscreen	CEL36CQFD-A
Card Type	Interface Card

Appearance

Figure 7-276 Appearance of the CEL36CQFD-A



NOTICE

The blue labels on the left and right ejector levers are mixed-card prevention labels. The color of mixed-card prevention labels on SFUs and LPUs to be installed on one device must be the same. Otherwise, the device will be damaged.

Version Mapping

Table 7-858 Mappings between CEL36CQFD-A and product

Product	First Supported Version	Unsupported Model	IDs of Available Slots	Limitations
CloudEngine 16804	V200R022C00	-	1-4	-
CloudEngine 16808	V200R022C00	-	1-8	-
CloudEngine 16816	V200R022C00	-	1-16	-

Table 7-859 Mappings between CEL36CQFD-A and MPU

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16804	CE-MPUD-HALF	03058876	V200R022C00	-
CloudEngine 16804	CE-MPUD-HALF	03058876-001	V200R022C00	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL	V200R022C00	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-001	V200R022C00	-
CloudEngine 16804	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA	V200R022C00	-
CloudEngine 16804	CE-MPUF-HALF	03050AVA-001	V200R022C00	-
CloudEngine 16808	CE-MPUD-HALF	03058876	V200R022C00	-

Product	Model	Part Number	First Supported Version	Unsupported Product Model
CloudEngine 16808	CE-MPUD-HALF	03058876-001	V200R022C00	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL	V200R022C00	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-001	V200R022C00	-
CloudEngine 16808	CE-MPUD-HALF2	03050EQL-002	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA	V200R022C00	-
CloudEngine 16808	CE-MPUF-HALF	03050AVA-001	V200R022C00	-
CloudEngine 16816	CE-MPUD-FULL	03058877	V200R022C00	-
CloudEngine 16816	CE-MPUD-FULL	03058877-001	V200R022C00	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP	V200R022C00	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-001	V200R022C00	-
CloudEngine 16816	CE-MPUD-FULL2	03050EQP-002	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY	V200R022C00	-
CloudEngine 16816	CE-MPUF-FULL	03050AUY-001	V200R022C00	-

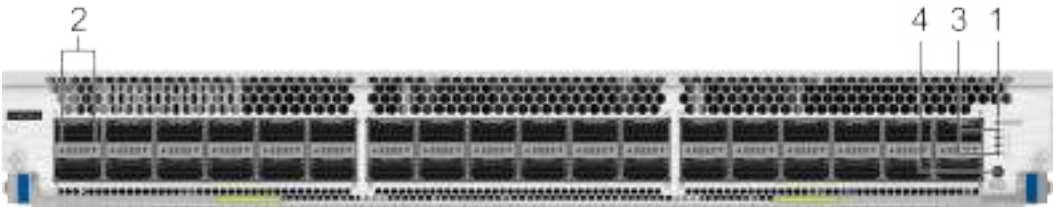
Table 7-860 Mappings between CEL36CQFD-A and SFU

Product	Model	Part Number	First Supported Version	Unsupported Product Model	Limitations
CloudEngine 16804	CE-SFU04F-A	03050ECE	V200R022C00	-	-
CloudEngine 16804	CE-SFU04F-A	03050ECE-001	V200R022C00	-	-

Product	Model	Part Number	First Supported Version	Unsupport ed Product Model	Limitation s
CloudEngin e 16804	CE-SFU04G-A	03050EGY	V200R022C 00	-	-
CloudEngin e 16804	CE-SFU04G-A	03050EGY-001	V200R022C 00	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD	V200R022C 00	-	-
CloudEngin e 16808	CE-SFU08F-A	03050ECD-001	V200R022C 00	-	-
CloudEngin e 16808	CE-SFU08G-A	03050EQK	V200R022C 00	-	-
CloudEngin e 16816	CE-SFU16G-A	03050EQJ	V200R022C 00	-	-

Indicators and Buttons

Figure 7-277 Indicators and buttons on the CEL36CQFD-A



1. Running status indicator	2. Port status indicator	3. 10GE/25GE port sequence number indicators	4. Button for manually controlling 10GE/ 25GE port sequence number indicators
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Table 7-861 Indicators on the CEL36CQFD-A

Silkscreen	Name	Color	Status	Description
RUN/ALM	Running status indicator	Green	Steady on	The card has been powered on but the system software is not running.

Silkscreen	Name	Color	Status	Description
		Green	Slow blinking (0.5 Hz)	The card is working properly.
		Green	Fast blinking (4 Hz)	The card is loading the system software or is resetting.
		Red	Steady on	The card has a fault that affects services and cannot be rectified automatically (critical hardware alarm).
		Yellow	Steady on	The card is not powered on. For example, the card has been forcibly powered off using the power off command or is in the pre-startup state.

Silkscreen	Name	Color	Status	Description
-	Port status indicators Each port has a single-color indicator. The indicator with an upward arrow indicates the status of the top-row port, and the indicator with a downward arrow indicates the status of the bottom-row port.	Green	Steady on	The link on the port is connected.
		Green	Blinking	The port is transmitting or receiving data.

Silkscreen	Name	Color	Status	Description
40/100GE Breakout 1 to 4	10GE/25GE port sequence number indicators (1, 2, 3, 4)	Green	-	When one or more 40GE or 100GE ports are split, these sequence number indicators take effect. The sequence number indicators work with the port indicators to show the status of converted ports. • When the sequence number indicator N (N can be 1, 2, 3, or 4) of a converted 10GE/25GE port is on, the port indicator shows the status of the Nth converted 10GE/25GE port. Sequence number indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s. This is the default

Silkscreen	Name	Color	Status	Description
				automatic mode. You can also press the 40/100GE Breakout button to manually control a specific port sequence number indicator. • If a 40GE or 100GE port is not split, the port indicator still shows the status of the port.

Table 7-862 Buttons on the CEL36CQFD-A

Silkscreen	Name	Description
40/100GE Breakout	Button for manually controlling 10GE/25GE port sequence number indicators	When one or more 40GE or 100GE ports are split, the sequence number indicators take effect. Press the 40/100GE Breakout button to turn on the port sequence number indicators cyclically. For example, if you press the 40/100GE Breakout button when indicator 3 is on, indicator 3 is still on. When you press the button a second time, indicator 4 turns on. Indicators 1 and 2 turn on in sequence when you press the 40/100GE Breakout button a third and fourth time. The manual control mode restores to the default automatic mode 60s after you stop pressing the 40/100GE Breakout button.

Ports

Figure 7-278 Ports on the CEL36CQFD-A

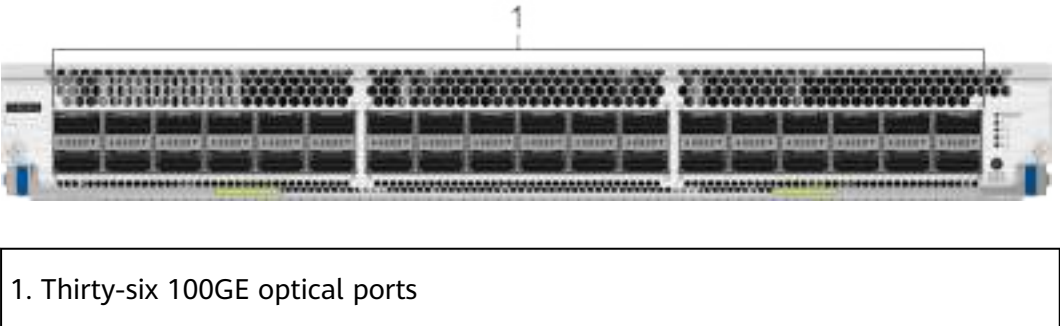


Table 7-863 Ports on the CEL36CQFD-A

Port	Connector Type	Description	Available Components
Thirty-six 100GE optical ports	QSFP28/QSFP+	Port for inputting and outputting 100GE/40GE optical signals	• 40GE QSFP+ optical modules • 100GE QSFP28 optical modules • QSFP+ to QSFP+ AOC cable • QSFP28 to QSFP28 AOC cable • QSFP+ to 4*SFP+ AOC cable • QSFP+ to QSFP+ high-speed cable • QSFP+ to 4*SFP+ high-speed cable • QSFP28 to QSFP28 high-speed cable • QSFP28 to 4*SFP28 high-speed cable NOTE When QSFP28 to QSFP28 or QSFP28 to 4*SFP28 high-speed cables are used on a card, only 1 m high-speed cables are supported.

Functions and Features

Table 7-864 Functions and features of the CEL36CQFD-A

Functions and Features	Description
Basic function	The card provides data packet processing and traffic management on 36 100GE optical ports, which can be used as 36 40GE optical ports or split into 144 25GE or 144 10GE optical ports.
Maximum port density	All LPUs are of this model: - The CloudEngine 16804 chassis provides a maximum of 144 100GE ports. - The CloudEngine 16808 chassis provides a maximum of 288 100GE ports. - The CloudEngine 16816 chassis provides a maximum of 576 100GE ports.
100GE port	Supports QSFP28 and QSFP+ optical modules. 100GE ports can work as 40GE ports through auto-sensing or be split into four 25GE or 10GE ports. Provides up to 36 x 40GE, 144 x 25GE, or 144 x 10GE ports. The 36 optical ports are independent, and each can work as a 40GE port or be split into four 25GE or 10GE ports.
Hot swapping	Supported

Technical Specifications

Table 7-865 Technical specifications of the CEL36CQFD-A

Item	Specification
Dimensions without packaging (H x W x D) [mm (in.)]	55.4 mm x 433.0 mm x 523.2 mm (2.18 in. x 17.05 in. x 20.60 in.)
Dimensions with packaging (H x W x D) [mm (in.)]	225 mm x 590 mm x 750 mm (8.86 in. x 23.23 in. x 29.53 in.)
Weight without packaging [kg (lb)]	9.2 kg (20.28 lb)
Weight with packaging [kg (lb)]	12.9 kg (28.44 lb)

Item	Specification
Typical power consumption [W]	584 W
Typical heat dissipation [BTU/hour]	1993 BTU/hour
Static power consumption [W]	494 W
Static heat dissipation [BTU/hour]	1686 BTU/hour
Maximum power consumption [W]	948 W
Maximum heat dissipation [BTU/hour]	3235 BTU/hour
CPU	Eight-core, 1.5 GHz clock speed
Memory	Standard 8 GB

8 Cables

- [8.1 DC Power Cables](#)
- [8.2 3000 W PM AC&240 V/380 V HVDC Power Cable](#)
- [8.3 Ground Cable](#)
- [8.4 Console Cable](#)
- [8.5 Ethernet Cable](#)
- [8.6 USB-to-Ethernet Cable](#)
- [8.7 Fiber Jumpers](#)
- [8.8 High-Speed Cable](#)
- [8.9 AOC Cable](#)

8.1 DC Power Cables

Overview

DC power cables include –48 V power cables and RTN ground cables. You need to cut the delivered DC power cables into appropriate lengths according to actual situations in your site and make DC power cables onsite. Choose appropriate DC power cables for different scenarios according to the following rules:

- **Optional:**
 - If a device is connected to a power distribution box or frame, use DC power cables with a JG2 90° right angle terminal at one end and an OT terminal at the other end.
- If a device is directly connected to a power distribution frame, use DC power cables with a JG2 90° right angle terminal at one end and determine the terminal type at the other end and the cable length based on the site survey result.

NOTICE

Do not bend OT terminals to 90 degrees onsite.

DC power cables with the following cross-sectional areas are available: 25 mm² and 35 mm². They apply to scenarios that require different power supply distances, as listed in [Table 8-1](#).

Table 8-1 DC power cable usage scenarios

Power Module Number	Max PM Output Power	Rated Input Current Per Channel	25 mm ² Power Cable Length	35 mm ² Power Cable Length
02312GDC	2200 W	46 A	≤27m	≤38m

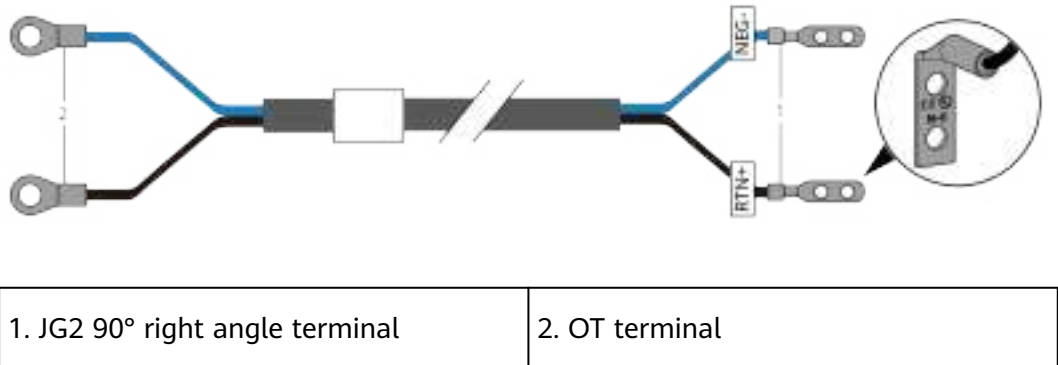
Appearance and Structure

NOTE

The figures provided here are for reference only and may differ from the actually delivered cables.

[Figure 8-1](#) shows a DC power cable used for connection with a power distribution box or frame.

Figure 8-1 Appearance of a DC power cable



[Figure 8-2](#) shows a DC power cable used to connect to a power distribution frame.

Figure 8-2 Appearance of a DC power cable



1. JG2 90° right angle terminal

Connection

A DC power cable transmits DC power to a chassis. Terminals are crimped onsite. A DC power cable is connected in the following way:

- If a device is connected to a power distribution box or frame, the JG2 90° right angle terminal is connected to a terminal block on a DC PEM at the rear of the device chassis, and the OT terminal is connected to the power distribution box or frame.
- If a device is directly connected to a power distribution frame, the JG2 90° right angle terminal is connected to a terminal block on a DC PEM at the rear of the device chassis, and the other end is connected to the power distribution frame.

Technical Specifications

- Device connected to a DC power distribution box

Table 8-2 Specifications of power cables used to connect to a DC power distribution box

Part Number	Model	Specification	Fireproof Level
25030695	P-25mm ² -Blue-LSZH	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Blue,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030638	P-25mm ² -Black-LSZH	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Black,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030738	CR1025GR00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Gray,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1

Part Number	Model	Specification	Fireproof Level
25030719	C1025RE00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Red,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25032791-001	C1025BL05	Electronic Electric Cable,1100V,Y(FR-LSH),25mm ² ,Blue,138A,For India,BIS	IEC60332-1, IEC60332-3-22
25032791	C1025BK03	Electronic Electric Cable,1100V,Y(FR-LSH),25mm ² ,Black,138A,For India,BIS	IEC60332-1, IEC60332-3-22
25034053-001	C60245B01	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Blue,For Saudi Arabia-CB	IEC 60332-1
25034053	C60245B53	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Black,For Saudi Arabia-CB	IEC 60332-1
25034053-002	C6024525RE05	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Red,For Saudi Arabia-CB	IEC 60332-1

Table 8-3 Specifications of power cable terminals used to connect to a DC power distribution box

Part Number	Specification	Terminal Type
02233EQX	MPE Site Materials Kit,GX_48V DC OT Terminal Kit-25mm ²	Naked Crimping Connector,JG2,25mm ² ,M6,80A ,Right angle_Galaxy Naked Crimping Terminal,OT,25mm ² ,M8,Tin Plating,Naked Ring Terminal
14170147	Naked Crimping Terminal,OT,25mm ² ,M6,Tin Plating,Width Less Than 12.4mm,For OEM	-

- Device directly connected to a power distribution frame

Table 8-4 Specifications of power cables used to directly connect to a power distribution frame (25 mm² power cables)

Part Number	Model	Specification	Fireproof Level
25030695	P-25mm ² -Blue-LSZH	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Blue,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030638	P-25mm ² -Black-LSZH	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Black,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030738	CR1025GR00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Gray,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030719	C1025RE00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Red,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030745	C1025BRN0	Electronic Electric Cable,450V/750V,H07Z-K UL3386,25mm ² ,Brown,138A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25032791-001	C1025BL05	Electronic Electric Cable,1100V,Y(FR-LSH),25mm ² ,Blue,138A,For India,BIS	IEC60332-1, IEC60332-3-22
25032791	C1025BK03	Electronic Electric Cable,1100V,Y(FR-LSH),25mm ² ,Black,138A,For India,BIS	IEC60332-1, IEC60332-3-22
25034053	C60245B53	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Black,For Saudi Arabia-CB	IEC 60332-1
25034053-002	C6024525RE05	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Red,For Saudi Arabia-CB	IEC 60332-1
25034053-001	C60245B01	Electronic Electric Cable,450V/750V,60245 IEC 05,25mm ² ,Blue,For Saudi Arabia-CB	IEC 60332-1

Table 8-5 Specifications of power cables used to directly connect to a power distribution frame (35 mm² power cables)

Part Number	Model	Specification	Fireproof Level
25030741	C1035BL01	Electronic Electric Cable,450V/750V,H07Z-K UL3386,35mm ² ,Blue,171A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030742	C1035BK01	Electronic Electric Cable,450V/750V,H07Z-K UL3386,35mm ² ,Black,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030720	EN3M CABRD000	Electronic Electric Cable,450V/750V,H07Z-K UL3386,35mm ² ,Red,171A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030739	C1035GR00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,35mm ² ,Gray,171A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25030740	C1035BR00	Electronic Electric Cable,450V/750V,H07Z-K UL3386,35mm ² ,Brown,171A,LSZH Cable,VDE,UL	IEC 60332-1, VW-1
25032792-001	C1035BL03	Electronic Electric Cable,1100V,Y(FR-LSH),35mm ² ,Blue,171A,For India,BIS	IEC60332-1, IEC60332-3-22
25032792	C1035BK02	Electronic Electric Cable,1100V,Y(FR-LSH),35mm ² ,Black,171A,For India,BIS	IEC60332-1, IEC60332-3-22
25034054-002	C6024535RE05	Electronic Electric Cable,450V/750V,60245 IEC 05,35mm ² ,Red,For Saudi Arabia-CB	IEC 60332-1
25034054-001	C6024535BL05	Electronic Electric Cable,450V/750V,60245 IEC 05,35mm ² ,Blue,For Saudi Arabia-CB	IEC 60332-1

Part Number	Model	Specification	Fireproof Level
25034054	C6024535BK05	Electronic Electric Cable,450V/750V,60245 IEC 05,35mm^2,Black,For Saudi Arabia-CB	IEC 60332-1

Table 8-6 Specifications of power cable terminals used to directly connect to a power distribution frame

Power Cable Specification	Part Number	Model	Specification	Terminal Type
25 mm ² power cable	02233EQX	KITOT025PD01	MPE Site Materials Kit,GX_48V DC OT Terminal Kit-25mm ²	Naked Crimping Connector,JG2,25mm ² ,M6,80A,Right angle_Galaxy Naked Crimping Terminal,OT,25mm ² ,M8,Tin Plating,Naked Ring Terminal
35 mm ² power cable	02233ERA	KITOT035PD01	MPE Site Materials Kit,GX_48V DC OT Terminal Kit-35mm ²	Naked Crimping Connector,35mm ² ,M6,160A,tin,Right angle_Galaxy Naked Crimping Terminal,OT,38mm ² ,M8,Tin Plating,Naked Ring Terminal,for 35mm ² ~38mm ² cable

8.2 3000 W PM AC&240 V/380 V HVDC Power Cable

 NOTE

The AC&240 V/380 V HVDC chassis requires Huawei-customized HVDC connectors. Only AC&240 V/380 V HVDC cables delivered by Huawei can be used.

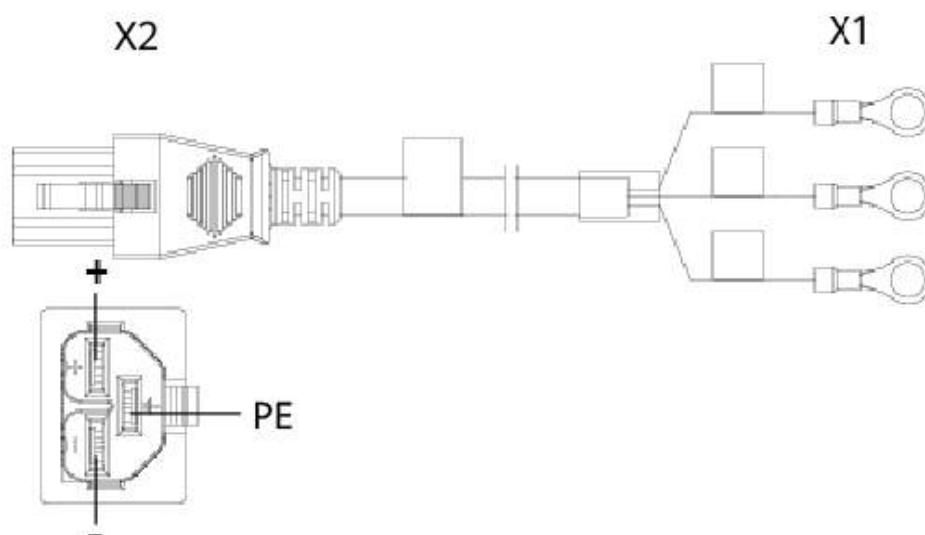
Direct Connection to a PDF

Figure 8-3 and **Figure 8-4** show the appearance and structure of an AC&240 V/380 V HVDC power cable.

Figure 8-3 Appearance of an AC&240 V/380 V HVDC power cable (HVDC 3 straight female – OT M8)



Figure 8-4 Structure of an AC&240 V/380 V HVDC power cable (connector on the device side – connector on the PDF side)



The mappings between terminals and core wires are as follows: (PE) – yellow and green; (+) – brown; (-) – blue.

Cable Connections for Direct Connection to a PDF

An AC&240 V/380 V HVDC power cable is connected in the following way:

- HVDC 3 straight female connector: connects to a socket on a PEM at the rear of the chassis.
- OT M8: connects to a PDF. The brown wire connects to the L terminal, the blue wire connects to the N terminal, and the yellow/green wire connects to the ground terminal. The AC power cables to be delivered may vary according to local regulations or customer requirements.

Select AC&240 V HVDC power cables based on the types of power sockets used in your equipment room. Huawei provides power distribution unit (PDU) power cables and country-specific power cables to suit different types of power sockets.

Connection to a PDU

- C19 PDU power cables: HVDC straight female to C20 straight male AC power cables, delivered with a PDU with C19 straight female sockets. [Figure 8-5](#) shows a PDU power cable.

Figure 8-5 PDU power cable and PDU



- Huawei horizontal PDU power cables: HVDC straight female to HVDC straight male power cables must be delivered for PDUs with HVDC ports.

Figure 8-6 Horizontal PDU power cable and PDU



- Country-specific power cables: delivered in compliance with standards of the destination country or region. For example, AC&240 V HVDC power cables with PI angle male connectors are used in China. [Figure 8-7](#) shows such a power cable.

Figure 8-7 Power cable and power strip used in China



When 380 V HVDC power is supplied, an appropriate Huawei horizontal HVDC PDU is required, and HVDC-straight-male-to-HVDC-straight-female power cables are delivered with the PDU.

NOTE

L indicates the negative terminal, and N indicates the positive terminal.

Figure 8-8 Structure of a C19 PDU power cable

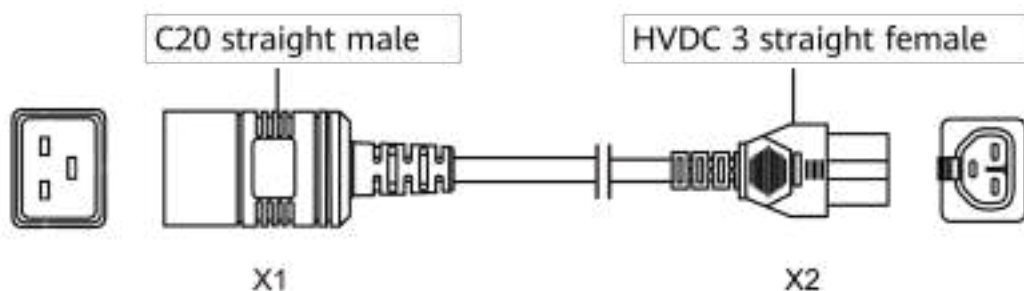


Figure 8-9 Structure of a horizontal PDU power cable

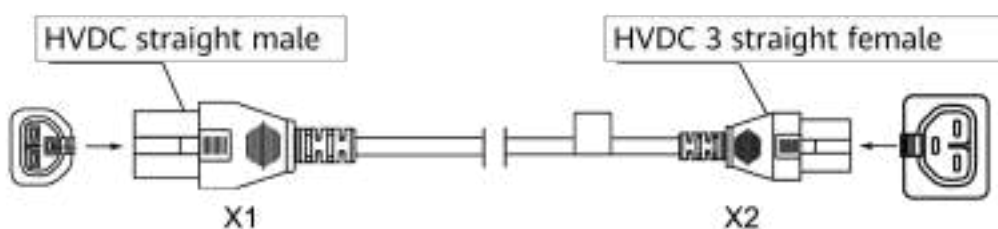


Figure 8-10 Structure of a socket-type power cable used in China

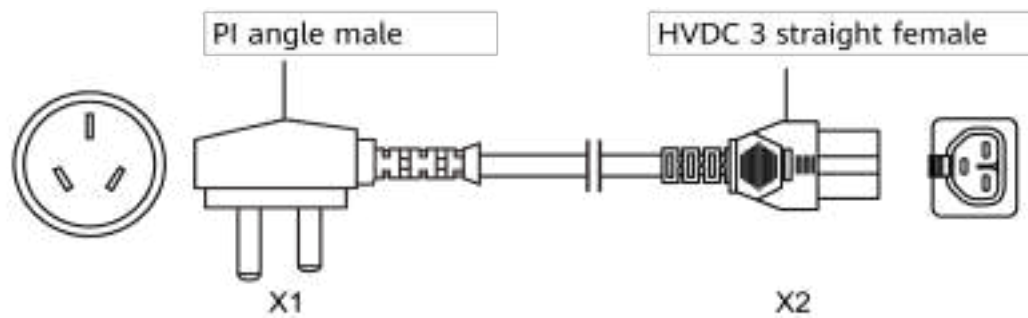
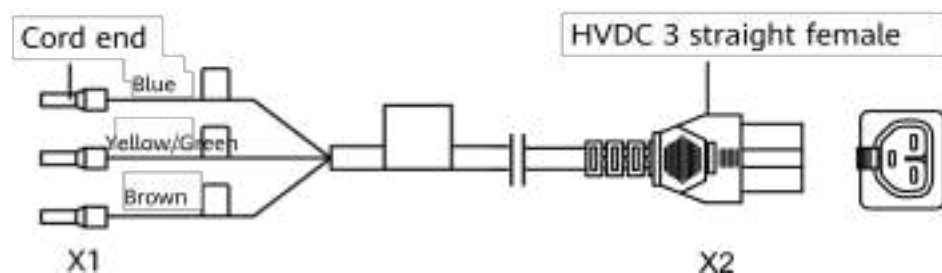


Figure 8-11 Structure of a power cable for the AC PDB



Cable Connections for Connection to a PDU

An AC&240 V/380 V HVDC power cable transmits AC&240 V/380 V HVDC power to a chassis and is connected in the following way:

- The HVDC 3 straight female connector connects to a socket on a PEM at the rear of the chassis.
- The other plug connects to a PDU or an external socket.

Technical Specifications

- Technical specifications of the cable directly connected to the PDF
[Table 8-7](#) lists the technical specifications of a cable directly connected to the PDF. [Figure 8-4](#) shows the cable structure.

Table 8-7 Technical specifications of an AC&240 V/380 V HVDC power cable

Part Number	Country/Region	Description	Connector X1	Connector X2	Length
04151269-002	Global	Power Cable,25m,3*1.5mm ² ,HVDC 3SF,227IEC53-1.5 ² (3C), 3*OT1-8,380V high voltage DC PEM dedicated,NO	3*OT1-8	HVDC 3 straight female	25 m

- Technical specifications of a PDU power cable

Table 8-8 lists the technical specifications of C19 PDU power cables, and **Figure 8-8** shows the cable structure.

Table 8-8 Technical specifications of C19 PDU power cables

Part Number	Model	Country/Region	Description	Connector X1	Connector X2	Length
04151958	HVDC PDU-UPWRCBL	Global	Power Cable,2.5m,3*1.5mm ² ,HVDC 3SF,(H05VV-F-1.5mm ² (3C)+14SJT(3C)),C20SM,PD U	C20 straight male	HVDC 3 straight female	2.5 m
04152510	-	China	Power Cable,5m,3*1.5mm ² ,HVDC 3SF,227IEC53-1.5 ² (3C),C20SM,PDU,Yellow	C20 straight male	HVDC 3 straight female	5 m

Table 8-9 lists the technical specifications of horizontal PDU power cables, and **Figure 8-9** shows the cable structure.

Table 8-9 Technical specifications of horizontal PDU power cables

Part Number	Country/Region	Description	Connector X1	Connector X2	Length
04152989	China	Power Cable,3m,3*1.5mm ² ,Black,HVDC 3SM,227IEC57-1.5 ² (3C),HVDC 3SF,China 380V HVDC PDU Power Cable	HVDC 3 straight male	HVDC 3 straight female	3 m
04152990	United States	Power Cable,3m,3*14AWG,Black ,HVDC 3SM,14SJT3(3C),HVDC 3SF,American 380V HVDC PDU Power Cable	HVDC 3 straight male	HVDC 3 straight female	3 m
04153057	Europe	Power Cable,3m,3*1.5mm ² ,Black,HVDC 3SM,H05V2V2-F-1.5mm ² (3C),HVDC 3SF,Europe 380V HVDC PDU Power Cable	HVDC 3 straight male	HVDC 3 straight female	3 m

Table 8-10 lists the technical specifications of the power cables for the AC PDB, and Figure 8-11 shows the power cable structure.

Table 8-10 Technical specifications of the power cables for the AC PDB

Part Number	Country / Region	Description	Connector X1	Connector X2	Length
04153098	Global	Power Cable,2.5m,3*1.5mm ² ,Black,HVDC 3SF,227IEC57-1.5 ² (3C), 3*T1.5 ² R	Cord end terminal	HVDC 3 straight female	2.5 m

Technical specifications of socket-type power cables

Use power cables complying with local standards in different countries or regions. Select power cables according to [Table 8-11](#). [Figure 8-10](#) shows the structure of a power cable used in China.

Table 8-11 Technical specifications of power cables applicable to different countries or regions

Part Number	Model	Country / Region	Description	Connector X1	Connector X2	Length
04152719	HVDC 3F-PWRC BL-US	United States	Power Cable, 2.5m, HVDC 3SF, 12SJT(3C), NEMA5-20PAM, North America AC Power 125V20A	NEMA5-20P angle male	HVDC 3 straight female	2.5 m
04152721	HVDC - PWRC BL-EU	Europe	Power Cable, 2.5m, HVDC 3SF, H05VVF-1.5 ² (3C), PFAM, Europe AC Power 250V16A	PF angle male	HVDC 3 straight female	2.5 m
04152722	HVDC 3F-PWRC BL-JP	Japan	Power Cable, 2.5m, HVDC 3SF, HVCTF-3.5mm ² (3C), NEMA5-20PAM, Japan AC Power Cable 125V20A	NEMA5-20P angle male	HVDC 3 straight female	2.5 m
04152723	HVDC 3F-PWRC BL-AU	Australia / New Zealand	Power Cable, 2.5m, HVDC 3SF, H05VV-F 2.5mm ² (3C), PISM, Australia AC Power 250V16A	PI straight male	HVDC 3 straight female	2.5 m

Part Number	Model	Country / Region	Description	Connector X1	Connector X2	Length
04152724	HVDC 3F-PWEC BL-IN	India	Power Cable,2.5m,HVDC 3SF,IS 694-1.5mm ² (3C),PM-IIIAM,India AC Power 250V16A	PM-III angle male	HVDC 3 straight female	2.5 m
04152725	HVDC 3F-PWEC BL-CH	Switzerland	Power Cable,2.5m,HVDC 3SF,H05VV-F-1.5mm ² (3C),PJ-II SM,Switzerland AC Power Cable 250V16A	PJ-II straight male	HVDC 3 straight female	2.5 m
04152726	HVDC 3F-PWEC BL-SA	South Africa	Power Cable,2.5m,HVDC 3SF,H05VV-F-1.5mm ² (3C),PMAM ,South Africa AC Power 250V16A	PM angle male	HVDC 3 straight female	2.5 m
04152721	HVDC 3F-PWRC BL-EU	Republic of Korea	Power Cable,2.5m,HVDC 3SF,H05VVF-1.5 ² (3C), PFAM,Europe AC Power 250V16A	PF angle male	HVDC 3 straight female	2.5 m
04152731	HVDC 3F-PWEC BL-BR	Brazil	Power Cable,2.5m,HVDC 3SF,H05VVF-1.5 ² (3C), PN-ISM,Brazil AC Power 250V16A	PN-I straight male	HVDC 3 straight female	2.5 m

Part Number	Model	Country / Region	Description	Connector X1	Connector X2	Length
04151963	HVDC - PWRC BL-CN	China	Power Cable,2.5m,3*1.5mm ² ,HVDC 3SF,227IEC53-1.5 ² (3C),PIAM,China AC Power 250V16A	PI angle male	HVDC 3 straight female	2.5 m

8.3 Ground Cable

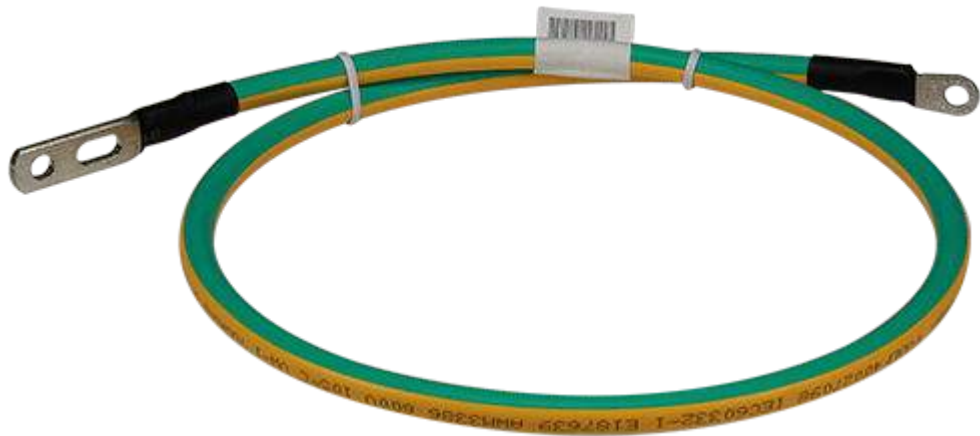
Overview

The ground cable ensures equipotential between the chassis and cabinet. The ground point on the chassis is at the rear of the chassis on the right side, and is identified by a yellow grounding label.

Appearance and Structure

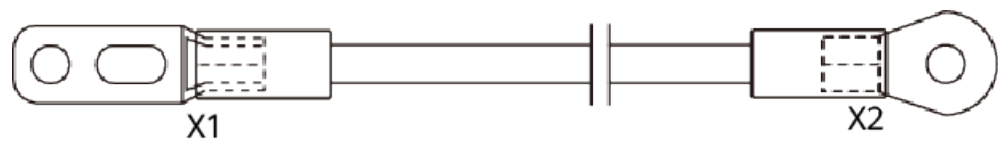
[Figure 8-12](#) shows the appearance of the ground cable.

Figure 8-12 Appearance of the ground cable



[Figure 8-13](#) shows the structure of the ground cable.

Figure 8-13 Structure of the ground cable



Connection

A ground cable grounds a device to protect it from lightning and electromagnetic interference. A ground cable is connected to a chassis in the following way:

- The OT bare crimp terminal X1 connects to the ground point on the chassis.
- The OT bare crimp terminal X2 connects to the ground bar of the cabinet.

Technical Specifications

Table 8-12 Technical specifications of the ground cable

Part Number	Model	Description	X1	X2	Length
04153446	-	Power Cable,0.85m,OT2-25-8,H07Z-K-25^2G&Y,OT25-6	14170121:OT2-25-8	14170147:OT25-6	0.85 m
04153446-001	PE2mOT2-25-8	Power Cable,2m,OT2-25-8,H07Z-K-25^2G&Y,OT25-6	14170121:OT2-25-8	14170147:OT25-6	2 m
04153446-002	PE4mOT2-25-8	Power Cable,4m,OT2-25-8,H07Z-K-25^2G&Y,OT25-6	14170121:OT2-25-8	14170147:OT25-6	4 m

8.4 Console Cable

Appearance and Structure

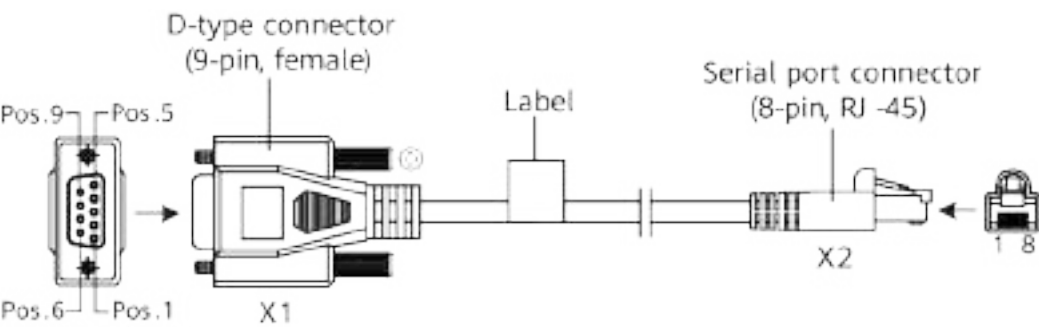
Figure 8-14 shows the appearance of a console cable.

Figure 8-14 Console cable appearance



Figure 8-15 shows the structure of a console cable.

Figure 8-15 Console cable structure



Pin Assignments

Table 8-13 lists the pin assignments of console cable connectors.

Table 8-13 Pin assignments of console cable connectors

Connector	X1 (DB9)	X2 (RJ45)	Description
Pin assignment	2	3	RS232_RX

Connector	X1 (DB9)	X2 (RJ45)	Description
	3	6	RS232_TX
	5	5	GND

Connection

A console cable connects the console port of a switch to the serial port of a console, enabling users to commission or locally maintain the device. A shielded cable or an unshielded cable can be used according to the onsite situation.

A console cable connects a device and a console as follows:

- The 8-pin RJ45 connector is connected to the console port of the switch.
- The DB9 female connector is connected to a maintenance terminal, such as a computer.

8.5 Ethernet Cable

Types of Ethernet Cables

Ethernet cables are classified into straight-through cables and crossover cables:

- Straight-through cable: [Table 8-14](#) lists the pin assignments of the RJ45 connectors on the two ends of a straight-through cable.
- Crossover cable: [Table 8-15](#) lists the pin assignments of the RJ45 connectors on the two ends of a crossover cable.

Appearance and Structure

NOTE

- Straight-through cables and crossover cables are standard unshielded twisted pairs that use RJ45 connectors.
- A straight-through cable and a crossover cable have the same appearance.

[Figure 8-16](#) and [Figure 8-17](#) show the appearance of an Ethernet cable.

Figure 8-16 Ethernet cable appearance (1)

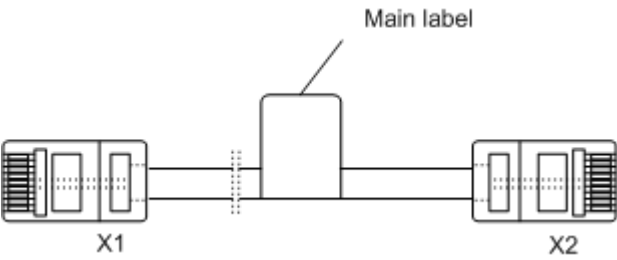


Figure 8-17 Ethernet cable appearance (2)



Figure 8-18 shows the structure of an Ethernet cable.

Figure 8-18 Ethernet cable structure



Pin Assignments

Table 8-14 lists the pin assignments of a straight-through cable.

Table 8-14 Pin assignments of a straight-through cable

X1 Pin	Wire Color	X2 Pin
1	White and orange	1
2	Orange	2
3	White and green	3
4	Blue	4
5	White and blue	5
6	Green	6
7	White and brown	7
8	Brown	8

Table 8-15 lists the pin assignments of a crossover cable.

Table 8-15 Pin assignments of a crossover cable

X1 Pin	Wire Color	X2 Pin
1	White and orange	3
2	Orange	6
3	White and green	1
4	Blue	4
5	White and blue	5
6	Green	2

X1 Pin	Wire Color	X2 Pin
7	White and brown	7
8	Brown	8

 NOTE

To achieve the best electrical transmission performance, ensure that the wires connected to pins 1 and 2 and to pins 3 and 6 are twisted pairs.

Connection

Ethernet cables connect network devices to each other to enable the devices to communicate or to allow local maintenance and remote management.

- A straight-through cable connects a terminal (such as a PC or switch) to a network device.
- A crossover cable connects two terminals (such as PCs and switches).

8.6 USB-to-Ethernet Cable

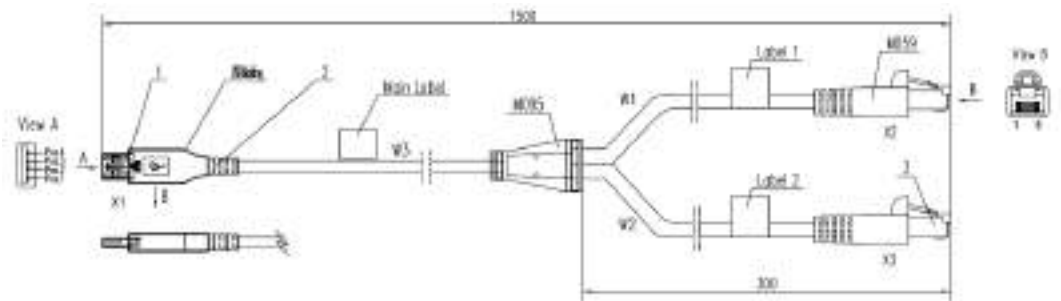
This section describes the structure and technical specifications of the USB-to-Ethernet cable.

Overview

One end of a USB-to-Ethernet cable is a USB port used to connect to a client, and the other end is split into two RJ45 connectors used to connect to switches. The two RJ45 connectors include one standard console port and one non-standard console port. Pin assignment for the two ports is different. For details, see [Table 8-16](#).

Appearance

Figure 8-19 Structure of the USB-to-Ethernet cable



Pin Assignment

Table 8-16 Pin assignment of the USB-to-Ethernet cable

Cable	Relation	Color	Description	Start Pin	Converter	End Pin	Description	Color	Relation	Cable
• Cable: W3 • Cable label: Main Label	Pair	White	+5V DC	X1.1	USB to RS232 conversion	X2.6	RS232_RX	Blue	Pair	• Cable: W1 • Cable label: Label 1 (RS232)
		Blue	Data -	X1.2		X2.3	RS232_TX	White		
	Pair	Orange	Data +	X1.3		X2.5	GND	Orange	-	• Cable: W2 • Cable label: Label 2 (Special RS232)
		White	GND	X1.4		X3.5	RS232_RX	Blue	Pair	
-	-	-	-	-		X3.8	RS232_TX	White		
-	-	-	-	-		X3.4	GND	Orange	-	

Technical Specifications

Table 8-17 Technical specifications of the USB-to-Ethernet cable

Part Number	Model	Description	Connector X1	Connector X2	Length	Number of Cores	Fire Rating
04071851	-	Signal Cable,USB-to-Ethernet cable,1.5m,USB-A(Male),CC2P0.48B(S),2*MP8-II	USB-A(Male)	MP8-II	1.5 m	-	-

NOTICE

- To use the USB-to-Ethernet serial cable, download the driver from the website and configure the driver as required. Download the driver from <http://www.wch-ic.com/search?q=CH340&t=downloads>.
- The device uses the W1 cable number and the cable label is RS232.

8.7 Fiber Jumpers

Overview

A fiber jumper consists of one or more optical fibers of a certain length and the optical connectors at both ends. A fiber jumper connects an optical module to a fiber terminal box.

Comply with the following rules when selecting fiber jumpers:

- Determine the length of fiber jumpers based on the onsite cabling distance.
- Determine the fiber type based on the optical module type.
 - Use a multimode fiber jumper for a multimode optical module.
 - Use a single-mode fiber jumper for a single-mode optical module.
- Determine the optical connector type based on the port type.

Ensure that the optical connector at each end of a fiber jumper is the same type as the port to which it will be connected.

NOTE

The MPO-MPO fibers for CloudEngine series switches use type B connectors (key Up/key Up).

Appearance and Structure

Figure 8-20 shows the appearance of an LC single-mode fiber.

Figure 8-20 Appearance of an LC single-mode fiber



Figure 8-21 shows the appearance of an LC multimode fiber.

Figure 8-21 Appearance of an LC multimode fiber



Figure 8-22 shows the appearance of an MPO-MPO single-mode fiber.

Figure 8-22 Appearance of an MPO-MPO single-mode fiber



Figure 8-23 shows the appearance of an 8-core or 12-core MPO-MPO multi-mode fiber.

Figure 8-23 Appearance of an 8-core or 12-core MPO-MPO multi-mode fiber



Figure 8-24 shows the appearance of an MPO-4*DLC fiber.

Figure 8-24 Appearance of an MPO-4*DLC fiber



Pin Assignments

- Pin assignments of an 8-core MPO-MPO fiber

Figure 8-25 Structure of an 8-core MPO-MPO fiber

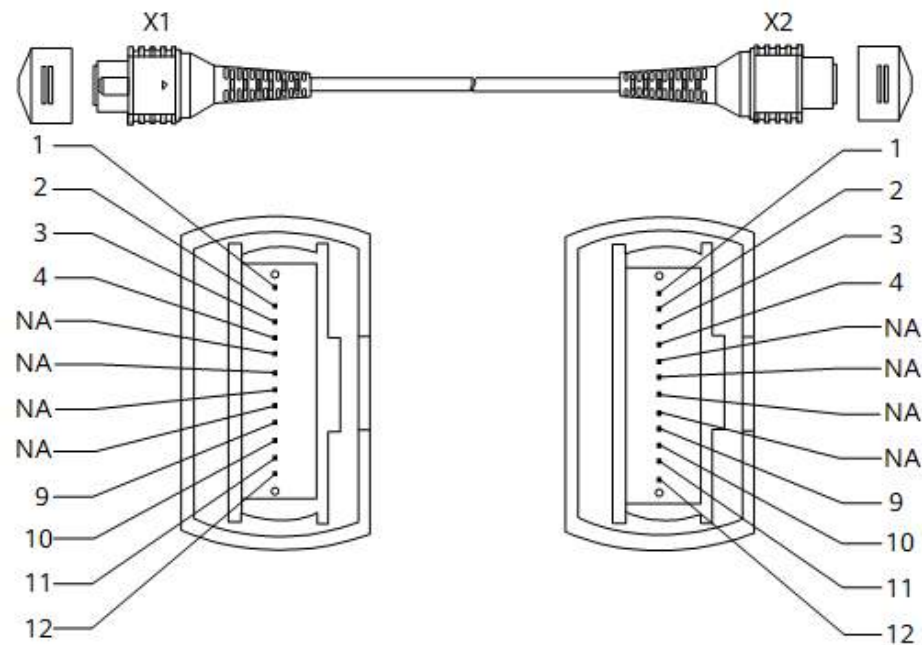


Table 8-18 Pin assignments of an 8-core MPO-MPO fiber

X1 Pin	X2 Pin
1	12
2	11
3	10
4	9
9	4
10	3
11	2
12	1

- Pin assignments of a 12-core MPO-MPO fiber

Figure 8-26 Structure of a 12-core MPO-MPO fiber

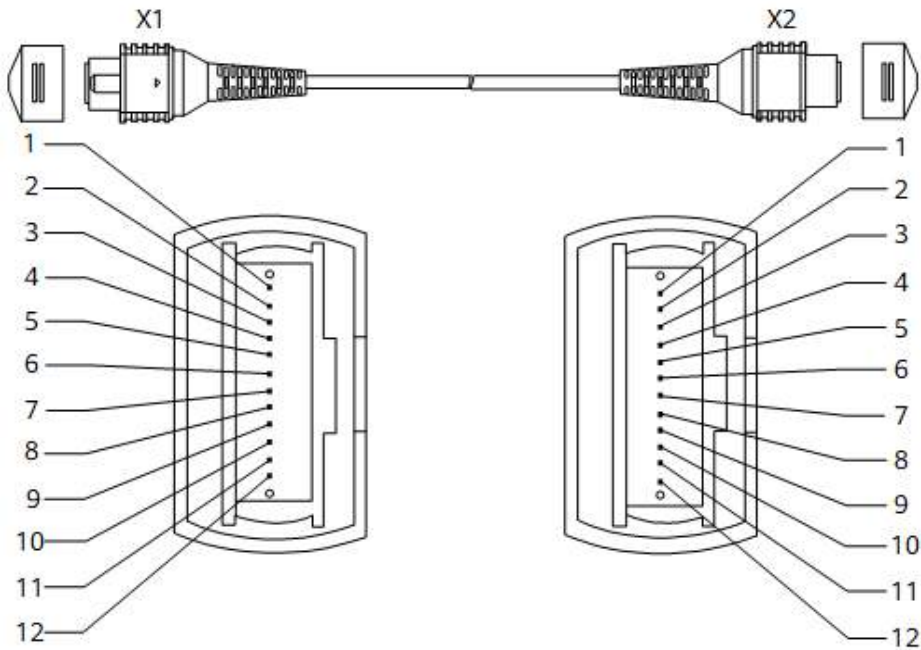


Table 8-19 Pin assignments of a 12-core MPO-MPO fiber

X1 Pin	X2 Pin
1	12
2	11
3	10
4	9
5	8
6	7
7	6
8	5
9	4
10	3
11	2
12	1

- Pin assignments of an MPO-4*DLC fiber

Figure 8-27 Structure of an MPO-4*DLC fiber

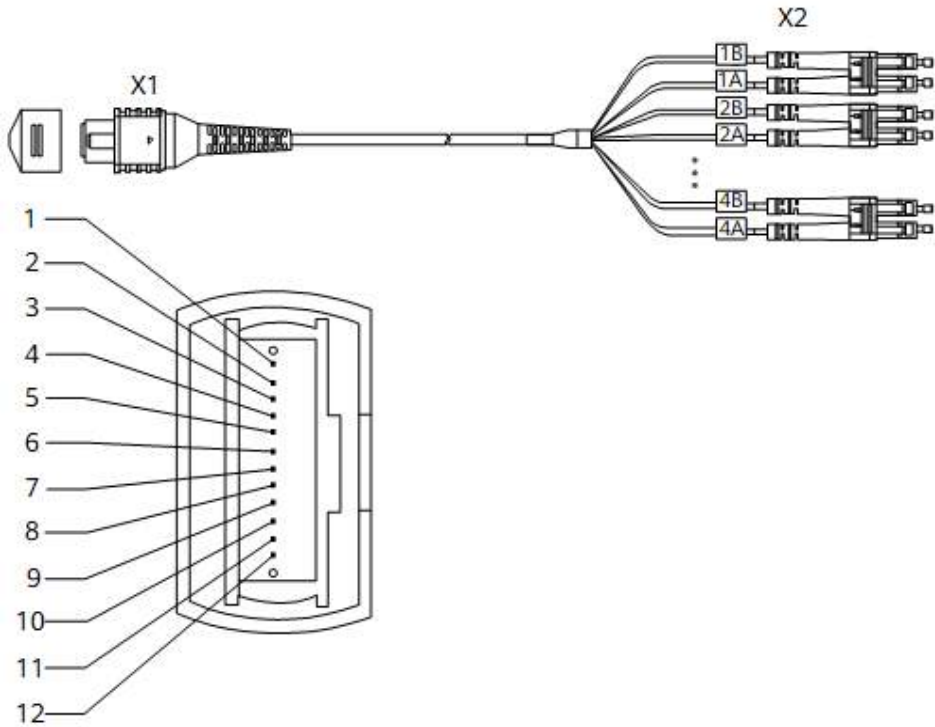


Table 8-20 Pin assignments of an MPO-4*DLC fiber

X1 Pin	X2 Pin
1	1B
2	2B
3	3B
4	4B
9	4A
10	3A
11	2A
12	1A

Optical Fiber, Optical Connector, and Optical Distribution Frame

Optical Fiber

Optical fibers are classified into single-mode fibers and multimode fibers.

- Single-mode fibers have a diameter of 5 μm to 10 μm and transmit laser in a single mode with a specified wavelength. These fibers support a wide frequency band and a large transmission capacity, so they are used for long-

distance transmission. Most single-mode fibers are yellow, as shown in [Figure 8-20](#).

- Multimode fibers have a diameter of 50 μm or 62.5 μm and transmit laser in multiple modes with a specified wavelength. These fibers have a lower transmission capacity than single-mode fibers and are used for short-distance transmission. Modal dispersion occurs during transmission over multimode fibers.

In the latest cabling infrastructure of ISO/IEC 11801, multimode fibers are classified into four categories: OM1, OM2, OM3, and OM4.

- OM1: traditional 62.5 μm /125 μm multimode fibers. OM1 fibers have a large core diameter and numerical aperture, and provide high light gathering ability and bending resistance.
- OM2: traditional 50 μm /125 μm multimode fibers. OM2 fibers have a small core diameter and numerical aperture. Compared with OM1 fibers, OM2 fibers provide higher bandwidth because they significantly reduce the modal dispersion. When transmitting data at 1 Gbit/s with 850 nm wavelength, OM1 and OM2 fibers support maximum link lengths of 220 m and 550 m, respectively. OM1 and OM2 fibers can provide sufficient bandwidth within a distance of 300 m. Generally, OM1 and OM2 fibers are orange, as shown in [Figure 8-21](#).
- OM3: next-generation multimode fibers, with longer transmission distances than OM1 and OM2 fibers.
- OM4: laser optimized multimode fibers with 50 μm core diameter. OM4 is an improvement to OM3 and only increases the modal bandwidth. OM4 fibers provide 4700 MHz*km of modal bandwidth, whereas OM3 fibers provide only 2000 MHz*km of modal bandwidth. Generally, OM3 and OM4 fibers are light green, as shown in [Figure 8-23](#). You can identify OM3 and OM4 fibers by their labels or printed marks.

MPO fibers are used for 40GE and 100GE optical modules. An MPO fiber consists of multiple multimode fiber cores, and each multimode fiber core provides one laser transmission channel. MPO fibers may come in the following specifications: 8-core, 12-core, or 24-core.

- A 40GE optical module uses four channels to transmit laser and four channels to receive laser. That is, a total of eight channels are required for a 40GE optical module. 8-core and 12-core MPO fibers have the same definition of fiber channels. Therefore, they are equivalent in functionality when connecting to 40GE optical modules.
- When 100GE optical modules are used, choose MPO fibers according to the optical module form factor. Choose 8-core or 12-core fibers for QSFP28 optical modules with MPO connectors.

An optical module installed with optical fibers with LC connectors uses one channel to transmit laser and one channel to receive laser. That is, two channels are required for such an optical module. A single-fiber bidirectional optical module uses one channel to transmit and receive laser, and therefore only one channel is required.

Optical Connector

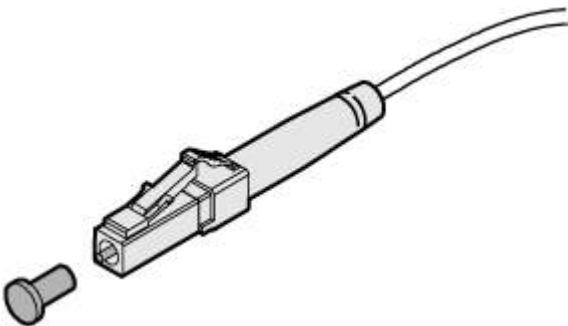
Optical connectors are used to connect optical fibers of the same type. [Table 8-21](#) lists common optical connectors.

Table 8-21 Common optical connectors

Connector Type	Optical Connector			
Square connector	SC/PC connector 	LC/PC connector 	MTRJ/PC connector 	MPO connector 
Round connector	FC/PC connector 	ST/PC connector 	-	-

Figure 8-28 shows an LC/PC optical connector.

Figure 8-28 LC/PC optical connector



NOTICE

- When connecting or removing an LC/PC optical connector, align the connector with the optical port and do not rotate the fiber. Pay attention to the following points:
- To connect a fiber, align the optical connector with the optical port and gently insert the optical fiber into the port.
 - To remove a fiber, press the clip on the connector, push the connector inward slightly, and pull the fiber out.

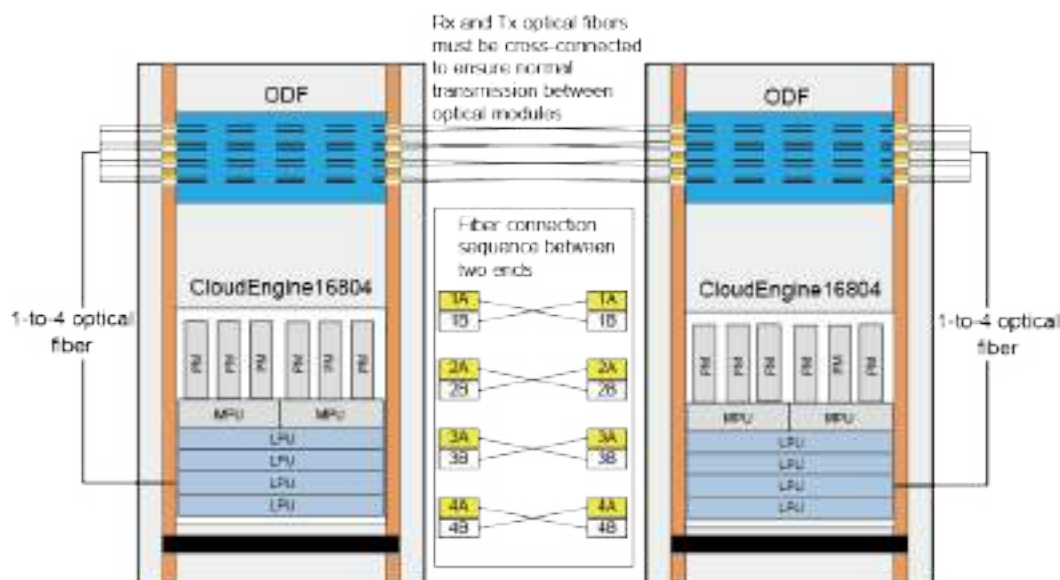
Optical Distribution Frame

Optical ports of two switches are not directly connected by optical fibers. An optical distribution frame (ODF) is usually deployed between the two switches.

Figure 8-29 shows an example of MPO-4*DLC (1-to-4) fiber connection through an ODF. When connecting optical fibers between two devices, pay attention to the following points:

- Determine the fiber connector type (LC/FC) supported by the ODF and select appropriate fiber jumpers for the supported connector type.
- Fiber jumpers from the DC switches at both ends must be connected to the ODF at the same sequence.
- Each pair of Tx and Rx optical fibers must be cross-connected on the ODF to ensure normal transmission between the optical modules at both ends.

Figure 8-29 MPO-4*DL (1-to-4) fiber connection through an ODF



Ceramic Ferrule End Face

Based on the return loss, the end faces of the fiber's ceramic ferrule are classified into three types: PC, UPC, and APC, as shown in [Figure 8-30](#).

Figure 8-30 Polishing types of the fiber's ceramic ferrule end face

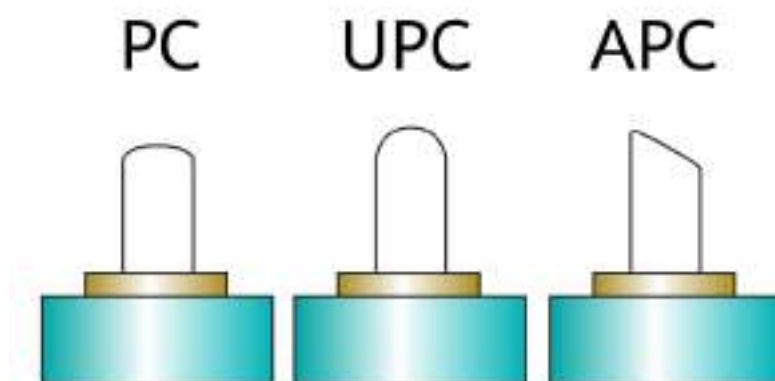


Table 8-22 Polishing types of the fiber's ceramic ferrule end face

Polishing Type	Return Loss	Characteristics	Application Scenario
PC	-35 dB	Polished with a slight curvature	Scenarios with no high requirements on return loss
UPC	-50 dB	Dome-shaped	Scenarios with high requirements on return loss
APC	-60 dB	Polished with an 8-degree angle	

NOTICE

In principle, optical fibers with different ceramic ferrule end faces cannot be directly connected through optical connectors. Interconnection between PC and UPC connectors does not cause permanent physical damage to them. The structure of APC end faces is totally different from that of PC end faces. Therefore, if fibers with APC end faces and fibers with PC end faces are connected through optical connectors, their ceramic ferrule end faces will be damaged. To connect them together, use a fiber jumper. This, however, adversely affects the transmission performance.

Figure 8-31 shows the requirements of different types of ceramic ferrule end face of fibers.

Figure 8-31 Fiber's ceramic ferrule end faces

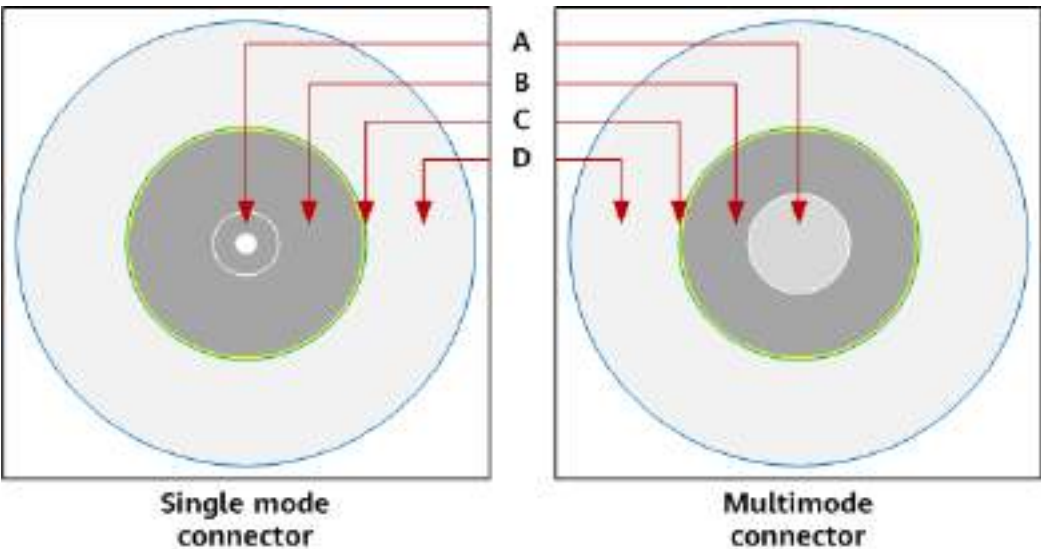


Table 8-23 End face requirements for fiber ceramic ferrules

Type	Zone	Diameter	Defects	Scratches
Single mode connector	A. Core	0-25 μm	None	None
	B. Cladding	25-120 μm	< 2 μm: no limit 2-5 μm: 5 > 5 μm: 0	≤ 3 μm: no limit > 3 μm: 0
	C. Adhesive	120-130 μm	No limit	No limit
	D. Contact	130-250 μm	≥ 10 μm: 0	No limit
Multi mode connector	A. Core	0-65 μm	≤ 5 μm: 4 > 5 μm: 0	≤ 5 μm: no limit > 5 μm: 0
	B. Cladding	65-120 μm	< 2 μm: no limit 2-5 μm: 5 > 5 μm: 0	≤ 5 μm: no limit > 5 μm: 0
	C. Adhesive	120-130 μm	No limit	No limit
	D. Contact	130-250 μm	≥ 10 μm: 0	No limit

8.8 High-Speed Cable

8.8.1 Understanding High-Speed Cables

 NOTE

You are advised to use high-speed cables that are certified for Huawei data center switches. This is because high-speed cables that are not certified for Huawei data center switches cannot ensure transmission reliability and may affect service stability. Huawei is not liable for any problems caused by high-speed cables that are not certified for Huawei data center switches and will not fix such problems.

Appearance and Structure

Figure 8-32 shows the appearance of an SFP+ to SFP+ high-speed cable.

Figure 8-32 Appearance of an SFP+ to SFP+ high-speed cable



Figure 8-33 shows the appearance of a QSFP+ to QSFP+ or QSFP28 to QSFP28 high-speed cable.

Figure 8-33 Appearance of a QSFP+ to QSFP+ or QSFP28 to QSFP28 high-speed cable



Figure 8-34 shows the appearance of a QSFP+ to 4*SFP+ or QSFP28 to 4*SFP28 high-speed cable.

Figure 8-34 Appearance of a QSFP+ to 4*SFP+ or QSFP28 to 4*SFP28 high-speed cable



Figure 8-35 shows the structure of an SFP+ to SFP+ high-speed cable.

Figure 8-35 Structure of an SFP+ to SFP+ high-speed cable

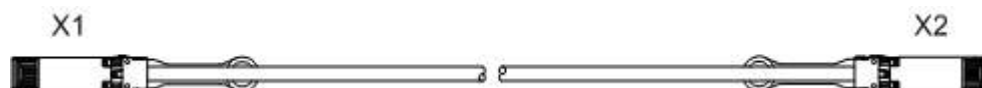


Figure 8-36 shows the structure of a QSFP+ to QSFP+ or QSFP28 to QSFP28 high-speed cable.

Figure 8-36 Structure of a QSFP+ to QSFP+ or QSFP28 to QSFP28 high-speed cable

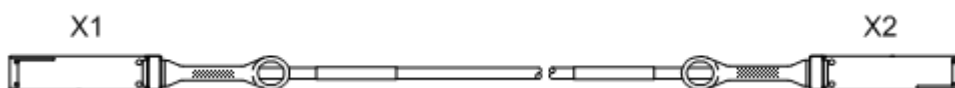
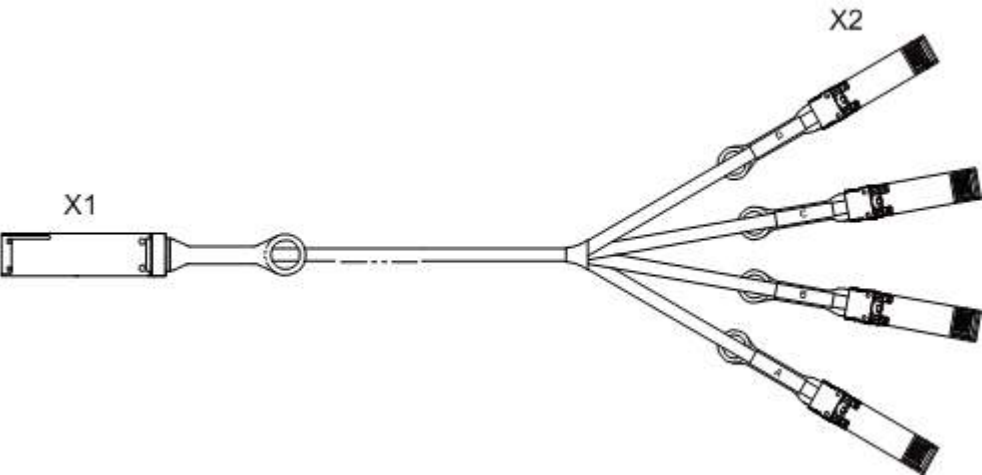


Figure 8-37 shows the structure of a QSFP+ to 4*SFP+ or QSFP28 to 4*SFP28 high-speed cable.

Figure 8-37 Structure of a QSFP+ to 4*SFP+ or QSFP28 to 4*SFP28 high-speed cable



8.8.2 SFP+ to SFP+ High-Speed Cable

8.8.2.1 SFP-10G-CU1M

Technical Specifications

Table 8-24 Technical specifications of the SFP-10G-CU1M

Item	Details
Part Number	02310MUN
Model	SFP-10G-CU1M
Description	SFP+,10G,High Speed Direct-attach Cables,1m,SFP +20M,CC2P0.254B(S),SFP+20M,Used indoor
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)

Item	Details
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.2 SFP-10G-CU1M (02314NVD)

Technical Specifications

Table 8-25 Technical specifications of the SFP-10G-CU1M (02314NVD)

Item	Details
Part Number	02314NVD
Model	SFP-10G-CU1M
Description	10G,SFP+,Passive High Speed Cables,1m,DAC,Used Indoor,LSZH,Compound Bag Package
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.3 SFP-10G-CU3M

Technical Specifications

Table 8-26 Technical specifications of the SFP-10G-CU3M

Item	Details
Part Number	02310MUP
Model	SFP-10G-CU3M

Item	Details
Description	SFP+,10G,High Speed Direct-attach Cables,3m,SFP+20M,CC2P0.254B(S),SFP+20M,Used indoor
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.4 SFP-10G-CU3M (02314NVE)

Technical Specifications

Table 8-27 Technical specifications of the SFP-10G-CU3M (02314NVE)

Item	Details
Part Number	02314NVE
Model	SFP-10G-CU3M
Description	10G,SFP+,Passive High Speed Cables,3m,DAC,Used Indoor,LSZH,Compound Bag Package
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.5 SFP-10G-CU5M

Technical Specifications

Table 8-28 Technical specifications of the SFP-10G-CU5M

Item	Details
Part Number	02310QPR
Model	SFP-10G-CU5M
Description	SFP+,10G,High Speed Cable,5m,SFP +20M,CC2P0.254B(S),SFP +20M,LSFRZH For Indoor
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	5 m (16.40 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	30 mm (1.18 in.)
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.6 SFP-10G-CU5M (02314NVF)

Technical Specifications

Table 8-29 Technical specifications of the SFP-10G-CU5M (02314NVF)

Item	Details
Part Number	02314NVF
Model	SFP-10G-CU5M
Description	10G,SFP+,Passive High Speed Cables,5m,DAC,Used Indoor,LSZH,Compound Bag Package
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	5 m (16.4 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	30 mm (1.18 in.)

Item	Details
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.2.7 SFP-10G-AC10M

Technical Specifications

Table 8-30 Technical specifications of the SFP-10G-AC10M

Item	Details
Part Number	02310MUQ
Model	SFP-10G-AC10M
Description	SFP+,10G,Active High Speed Cables,10m,SFP+20M,CC2P0.32B(S),SFP+20M,Used indoor
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	10 m (32.81 ft.)
Electrical attribute	Active
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	60 mm (2.36 in.)

8.8.3 QSFP+ to QSFP+ High-Speed Cable

8.8.3.1 QSFP-40G-CU1M

Technical Specifications

Table 8-31 Technical specifications of the QSFP-40G-CU1M

Item	Details
Part Number	02310MUG

Item	Details
Model	QSFP-40G-CU1M
Description	QSFP+,40G,High Speed Direct-attach Cables,1m,QSFP +38M,CC8P0.254B(S),QSFP+38M,Used indoor
Connector X1	QSFP+
Connector X2	QSFP+
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	35 mm (1.38 in.)
Minimum Clearance for Cable Routing [mm (in.)]	75 mm (2.95 in.)

8.8.3.2 QSFP-40G-CU3M

Technical Specifications

Table 8-32 Technical specifications of the QSFP-40G-CU3M

Item	Details
Part Number	02310MUH
Model	QSFP-40G-CU3M
Description	QSFP+,40G,High Speed Direct-attach Cables,3m,QSFP +38M,CC8P0.32B(S),QSFP+38M,Used indoor
Connector X1	QSFP+
Connector X2	QSFP+
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	40 mm (1.57 in.)
Minimum Clearance for Cable Routing [mm (in.)]	75 mm (2.95 in.)

8.8.3.3 QSFP-40G-CU5M

Technical Specifications

Table 8-33 Technical specifications of the QSFP-40G-CU5M

Item	Details
Part Number	02310MUJ
Model	QSFP-40G-CU5M
Description	QSFP+,40G,High Speed Direct-attach Cables,5m,QSFP+38M,CC8P0.40B(S),QSFP+38M,Used indoor
Connector X1	QSFP+
Connector X2	QSFP+
Cable length [m(ft.)]	5 m (16.40 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	45 mm (1.77 in.)
Minimum Clearance for Cable Routing [mm (in.)]	75 mm (2.95 in.)

8.8.4 QSFP+ to 4*SFP+ High-Speed Cable

8.8.4.1 QSFP-4SFP10G-CU1M

Technical Specifications

Table 8-34 Technical specifications of the QSFP-4SFP10G-CU1M

Item	Details
Part Number	02310MUK
Model	QSFP-4SFP10G-CU1M
Description	QSFP+,4SFP+10G,High Speed Direct-attach Cables,1m,QSFP+38M,CC8P0.254B(S),4*SFP+20M,Used indoor
Connector X1	QSFP+

Item	Details
Connector X2	4*SFP+
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP+ end: 100 mm (3.94 in.) SFP+ end: 60 mm (2.36 in.)

8.8.4.2 QSFP-4SFP10G-CU3M

Technical Specifications

Table 8-35 Technical specifications of the QSFP-4SFP10G-CU3M

Item	Details
Part Number	02310MUL
Model	QSFP-4SFP10G-CU3M
Description	QSFP+,4SFP+10G,High Speed Direct-attach Cables,3m,QSFP+38M,CC8P0.32B(S),4*SFP+20M,Used indoor
Connector X1	QSFP+
Connector X2	4*SFP+
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	25 mm (0.98 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP+ end: 100 mm (3.94 in.) SFP+ end: 60 mm (2.36 in.)

8.8.4.3 QSFP-4SFP10G-CU5M

Technical Specifications

Table 8-36 Technical specifications of the QSFP-4SFP10G-CU5M

Item	Details
Part Number	02310MUM
Model	QSFP-4SFP10G-CU5M
Description	QSFP+,4SFP+10G,High Speed Direct-attach Cables,5m,QSFP+38M,CC8P0.4B(S),4*SFP+20M,Used indoor
Connector X1	QSFP+
Connector X2	4*SFP+
Cable length [m(ft.)]	5 m (16.40 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	30 mm (1.18 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP+ end: 100 mm (3.94 in.) SFP+ end: 60 mm (2.36 in.)

8.8.5 QSFP28 to QSFP28 High-Speed Cable

8.8.5.1 QSFP28-100G-CU1M

Technical Specifications

Table 8-37 Technical specifications of the QSFP28-100G-CU1M

Item	Details
Part Number	02311KNW
Model	QSFP28-100G-CU1M
Description	QSFP28,100G,High Speed Direct-attach Cables,1m,(QSFP28),CC8P0.254B(S),QSFP28,Used indoor
Connector X1	QSFP28
Connector X2	QSFP28

Item	Details
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.2 QSFP28-100G-CU1M (02314NVH)

Technical Specifications

Table 8-38 Technical specifications of the QSFP28-100G-CU1M (02314NVH)

Item	Details
Part Number	02314NVH
Model	QSFP28-100G-CU1M
Description	100G,QSFP28,Passive High Speed Cables,1m,DAC,Used indoor,Compound Bag Package
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.3 QSFP28-100G-CU3M

Technical Specifications

Table 8-39 Technical specifications of the QSFP28-100G-CU3M

Item	Details
Part Number	02311KNX

Item	Details
Model	QSFP28-100G-CU3M
Description	QSFP28,100G,High Speed Direct-attach Cables,3m,(QSFP28),CC8P0.254B(S),QSFP28,Used indoor
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.4 QSFP28-100G-CU3M (02314NVK)

Technical Specifications

Table 8-40 Technical specifications of the QSFP28-100G-CU3M (02314NVK)

Item	Details
Part Number	02314NVK
Model	QSFP28-100G-CU3M
Description	100G,QSFP28,Passive High Speed Cables,3m,DAC,Used Indoor,Compound Bag Package
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.5 QSFP28-100G-CU3M(02315UCR)

Technical Specifications

Table 8-41 Technical specifications of the QSFP28-100G-CU3M(02315UCR)

Item	Details
Part Number	02315UCR
Model	QSFP28-100G-CU3M
Description	QSFP28,100G,High Speed Direct-attach Cables,3m,(QSFP28),CC8P0.254B(S),QSFP28,Used indoor,SW
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.6 QSFP28-100G-CU5M

Technical Specifications

Table 8-42 Technical specifications of the QSFP28-100G-CU5M

Item	Details
Part Number	02311KNY
Model	QSFP28-100G-CU5M
Description	QSFP28,100G,High Speed Direct-attach Cables,5m,(QSFP28),CC8P0.4B(S),QSFP28,Used indoor
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	5 m (16.40 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)

Item	Details
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.5.7 QSFP28-100G-CU5M(02315UCV)

Technical Specifications

Table 8-43 Technical specifications of the QSFP28-100G-CU5M(02315UCV)

Item	Details
Part Number	02315UCV
Model	QSFP28-100G-CU5M
Description	QSFP28,100G,High Speed Direct-attach Cables,5m,(QSFP28),CC8P0.4B(S),QSFP28,Used indoor,SW
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	5 m (16.4 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	70 mm (2.76 in.)
Minimum Clearance for Cable Routing [mm (in.)]	90 mm (3.54 in.)

8.8.6 QSFP28 to 4*SFP28 High-Speed Cable

8.8.6.1 QSFP-4SFP25G-CU1M

Technical Specifications

Table 8-44 Technical specifications of the QSFP-4SFP25G-CU1M

Item	Details
Part Number	02311MNX

Item	Details
Model	QSFP-4SFP25G-CU1M
Description	100GE QSFP28-4SFP25G,High Speed Direct-attach Cables,1m,(QSFP28), (4*(CC2P0.254B(S))), (4SFP28)
Connector X1	QSFP28
Connector X2	4*SFP28
Cable length [m(ft.)]	1 m (3.28 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	35 mm (1.38 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP28 end: 100 mm (3.94 in.) SFP28 end: 70 mm (2.76 in.)

8.8.6.2 QSFP-4SFP25G-CU3M

Technical Specifications

Table 8-45 Technical specifications of the QSFP-4SFP25G-CU3M

Item	Details
Part Number	02311MNY
Model	QSFP-4SFP25G-CU3M
Description	100GE QSFP28-4SFP25G,High Speed Direct-attach Cables,3m,(QSFP28), (4*(CC2P0.254B(S))), (4SFP28)
Connector X1	QSFP28
Connector X2	4*SFP28
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	35 mm (1.38 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP28 end: 100 mm (3.94 in.) SFP28 end: 70 mm (2.76 in.)

8.8.6.3 QSFP-4SFP25G-CU3M-N

Technical Specifications

Table 8-46 Technical specifications of the QSFP-4SFP25G-CU3M-N

Item	Details
Part Number	02311MPA
Model	QSFP-4SFP25G-CU3M-N
Description	100GE QSFP28-4SFP25G,High Speed Direct-attach Cables,3m,(QSFP28), (4*(CC2P0.4B(S))), (4SFP28)
Connector X1	QSFP28
Connector X2	4*SFP28
Cable length [m(ft.)]	3 m (9.84 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	45 mm (1.77 in.)
Minimum Clearance for Cable Routing [mm (in.)]	QSFP28 end: 100 mm (3.94 in.) SFP28 end: 70 mm (2.76 in.)

8.8.6.4 QSFP-4SFP25G-CU5M

Technical Specifications

Table 8-47 Technical specifications of the QSFP-4SFP25G-CU5M

Item	Details
Part Number	02311MPB
Model	QSFP-4SFP25G-CU5M
Description	100GE QSFP28-4SFP25G,High Speed Direct-attach Cables,5m,(QSFP28), (4*(CC2P0.4B(S))), (4SFP28)
Connector X1	QSFP28
Connector X2	4*SFP28
Cable length [m(ft.)]	5 m (16.40 ft.)
Electrical attribute	Passive
Bend radius [mm (in.)]	45 mm (1.77 in.)

Item	Details
Minimum Clearance for Cable Routing [mm (in.)]	QSFP28 end: 100 mm (3.94 in.) SFP28 end: 70 mm (2.76 in.)

8.9 AOC Cable

8.9.1 Understanding AOC Cables

Overview

An active optical cable (AOC) is an active optical fiber with optical modules at both ends, and therefore is easy to use.

Appearance and Structure

Figure 8-38 Appearance of an SFP+ to SFP+ AOC cable



Figure 8-39 Appearance of a QSFP+ to QSFP+ or QSFP28 to QSFP28 AOC cable



Figure 8-40 Appearance of a QSFP+ to 4*SFP+ AOC cable



8.9.2 SFP+ to SFP+ AOC Cable

8.9.2.1 SFP-10G-AOC-3M

Technical Specifications

Table 8-48 Technical specifications of the SFP-10G-AOC-3M

Item	Details
Part Number	02311BKP
Model	SFP-10G-AOC-3M
Description	Optical transceiver,SFP+,1G~10.5G, (850nm,3m,AOC)
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	3 m (9.84 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

8.9.2.2 SFP-10G-AOC10M

Technical Specifications

Table 8-49 Technical specifications of the SFP-10G-AOC10M

Item	Details
Part Number	02310QWH
Model	SFP-10G-AOC10M
Description	AOC Optical Transceiver,SFP+,850nm,1G~10G,10m
Connector X1	SFP+
Connector X2	SFP+
Cable length [m(ft.)]	10 m (32.81 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

8.9.3 QSFP+ to QSFP+ AOC cable

8.9.3.1 QSFP-H40G-AOC10M

Technical Specifications

Table 8-50 Technical specifications of the QSFP-H40G-AOC10M

Item	Details
Part Number	02310SSH
Model	QSFP-H40G-AOC10M
Description	Optical transceiver,QSFP+,40G,(850nm,10m,AOC)
Connector X1	QSFP+
Connector X2	QSFP+
Cable length [m(ft.)]	10 m (32.81 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

8.9.4 QSFP+ to 4*SFP+ AOC cable

8.9.4.1 QSFP-4SFP10-AOC10M

Technical Specifications

Table 8-51 Technical specifications of the QSFP-4SFP10-AOC10M

Item	Details
Part Number	02310SSJ
Model	QSFP-4SFP10-AOC10M
Description	Optical transceiver,QSFP+,40G, (850nm,10m,AOC)(Connect to four SFP+ Optical Transceiver)
Connector X1	QSFP+
Connector X2	4*SFP+
Cable length [m(ft.)]	10 m (32.81 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

8.9.5 QSFP28 to QSFP28 AOC Cable

8.9.5.1 QSFP-100G-AOC-10M

Technical Specifications

Table 8-52 Technical specifications of the QSFP-100G-AOC-10M

Item	Details
Part Number	02311KNQ
Model	QSFP-100G-AOC-10M
Description	Active Optical Cable ,QSFP28,100G, (850nm,10m,AOC)
Connector X1	QSFP28
Connector X2	QSFP28

Item	Details
Cable length [m(ft.)]	10 m (32.81 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

8.9.5.2 QSFP-100G-AOC-30M

Technical Specifications

Table 8-53 Technical specifications of the QSFP-100G-AOC-30M

Item	Details
Part Number	02311RAH
Model	QSFP-100G-AOC-30M
Description	Active Optical Cable ,QSFP28,100G, (850nm,30m,AOC)
Connector X1	QSFP28
Connector X2	QSFP28
Cable length [m(ft.)]	30 m (98.42 ft.)
Operating Wavelength [nm]	850 nm
Operating Temperature [°C (°F)]	0 °C to 70 °C (32 °F to 158 °F)

9 Pluggable Modules for Interfaces

[9.1 Understanding Optical Modules](#)

[9.2 Understanding Copper Modules](#)

[9.3 GE eSFP Optical Modules](#)

[9.4 GE SFP Copper Modules](#)

[9.5 10GE SFP+ Optical Modules](#)

[9.6 40GE QSFP+ Optical Modules](#)

[9.7 100GE QSFP28 Optical Modules](#)

9.1 Understanding Optical Modules

9.1.1 Optical Module Appearance and Structure

[Figure 9-1](#) shows the structure of an optical module.

Figure 9-1 Optical module structure

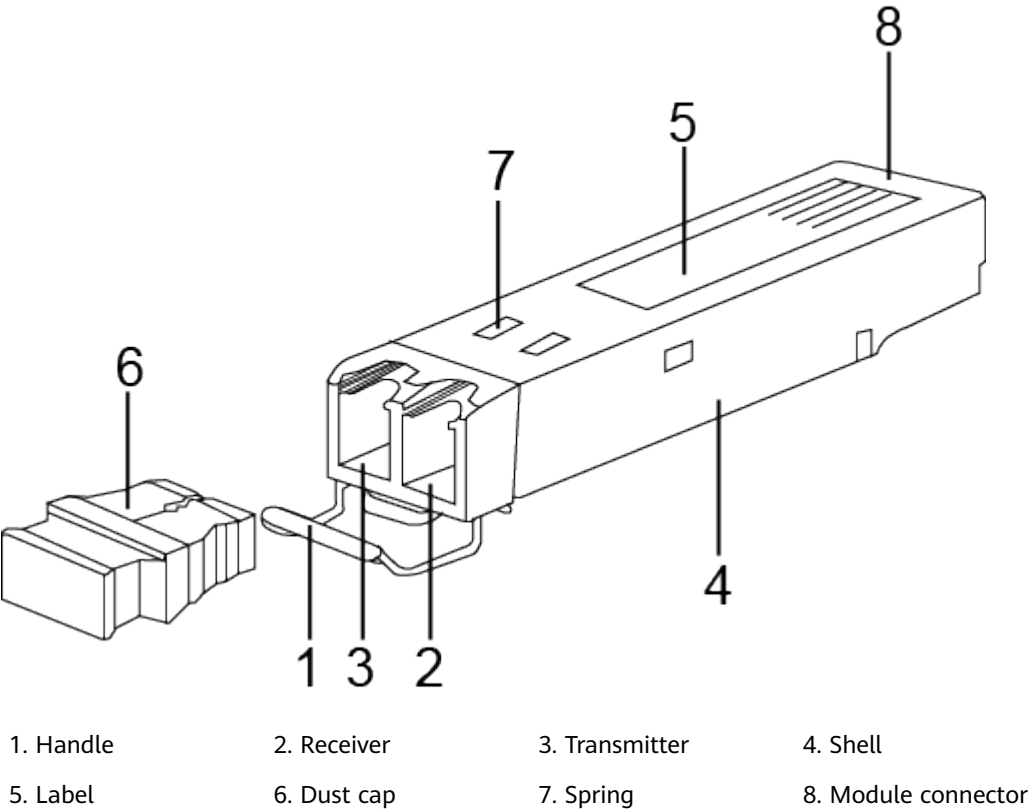


Figure 9-2 shows the appearance of an SFP/SFP+ optical module.

Figure 9-2 SFP/SFP+ optical module



Figure 9-3 shows the appearance of a QSFP+ optical module.

Figure 9-3 QSFP+ optical module

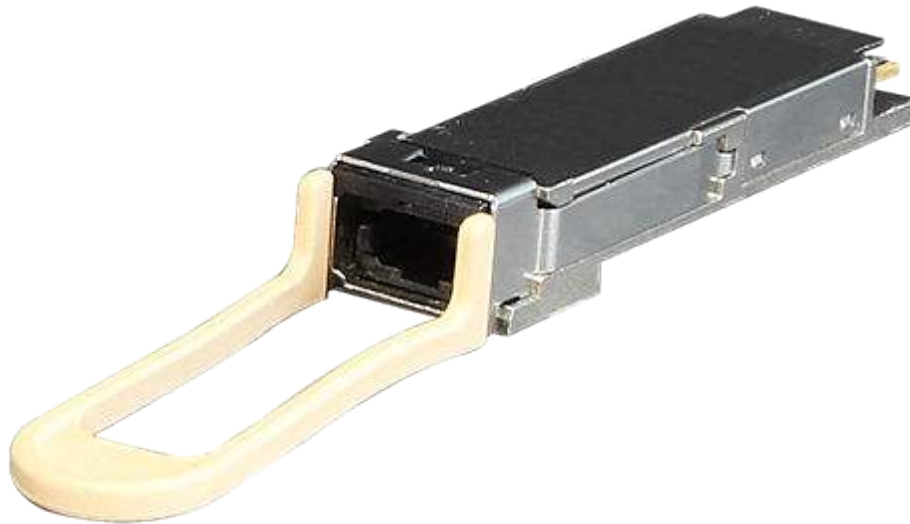


Figure 9-4 and **Figure 9-5** shows the appearance of a QSFP28 optical module.

Figure 9-4 QSFP28 optical module (for LC optical fibers)



Figure 9-5 QSFP28 optical module (for MPO optical fibers)



9.1.2 Types of Optical Modules

Optical modules are available in various types to meet diversified requirements.

- **Classified by transmission rates**

Depending on transmission rates, optical modules are classified into 100GE, 40GE, 25GE, 10GE, FE, and GE optical modules.

- **Classified by encapsulation types**

The higher transmission rate an optical module provides, the more complex structure it has. According to the encapsulation type, optical modules are classified into SFP, eSFP, SFP+, XFP, SFP28, QSFP+, CXP, CFP, and QSFP28. All optical modules are hot swappable.

- SFP: small form-factor pluggable. SFP optical modules support LC fiber connectors.
- eSFP: enhanced small form-factor pluggable. An eSFP module is an SFP module that supports monitoring of voltage, temperature, bias current, transmit optical power, and receive optical power. Because all the SFP optical modules support these monitoring functions, eSFP is also called SFP.
- SFP+: small form-factor pluggable plus, SFP with a higher rate. SFP+ modules are more sensitive to electromagnetic interference (EMI) because they have a higher rate. To reduce EMI, SFP+ modules have more springs than SFP modules and the cages for SFP+ modules on a card are tighter.
- XFP: 10 Gigabit small form-factor pluggable. X is the Roman numeral 10, meaning that all XFP optical modules provide a 10 Gbit/s transmission rate. XFP optical modules support LC fiber connectors. XFP optical modules are wider and longer than SFP+ optical modules.
- SFP28: with the same interface size as an SFP+ module. An SFP28 interface can use a 25GE SFP28 optical module or 10GE SFP+ optical module.

- QSFP+: quad small form-factor pluggable. QSFP+ optical modules support MPO fiber connectors and are larger than SFP+ modules.
- CXP: hot-pluggable high-density parallel optics transceiver form factor, which provides 12 channels of traffic in each of the Tx and Rx directions. It applies only to short multimode links.
- CFP: C form-factor pluggable, a new standard for high-speed, hot-pluggable optical transceivers that support data communication and telecommunication applications. Dimensions of a CFP optical module are 144.75 mm x 82 mm x 13.6 mm (W x D x H).
- QSFP28: with the same interface size as a QSFP+ module. A QSFP28 interface can use a 100GE QSFP28 optical module or a 40GE QSFP+ optical module.

- **Classified by physical layer standards**

Different physical layer standards are defined to allow data transmission in different modes. Therefore, different types of optical modules are produced to comply with these standards. For details, see **Standards compliance** of the specific optical module.

- **Classified by modes**

Optical fibers are classified into single-mode fibers (SMFs) and multi-mode fibers (MMFs). Therefore, optical modules are also classified into single-mode and multi-mode modules to support different optical fibers.

- Single-mode optical modules are used with single-mode fibers. Single-mode fibers support a wide band and large transmission capacity, and are used for long-distance transmission.
- Multi-mode optical modules are used with multi-mode fibers. Multi-mode fibers have lower transmission performance than single-mode fibers because of modal dispersion, but their costs are also lower. They are used for small-capacity, short-distance transmission.

For single-mode optical modules, single-mode optical splitters are not recommended due to considerations on the IEEE standard link budget and the impact of single-mode optical splitters on optical signal attenuation on backbone fiber links. If optical splitters must be used for special scenarios or requirements, strictly follow the link budget and insertion loss specifications of different single-mode optical modules. Ensure that the insertion loss of the single-mode optical splitter plus the inherent attenuation of the backbone link is less than the link budget supported by the corresponding single-mode optical modules, with a certain link budget margin reserved.

Multi-mode optical modules cannot be reliably used in scenarios where optical splitters are used due to considerations on the IEEE standard link budget and the impact of multi-mode optical splitter on multi-mode optical signals.

Wavelength division multiplexing modules differ from other optical modules in center wavelengths. A common optical module has a center wavelength of 850 nm, 1310 nm, or 1550 nm, whereas a wavelength division multiplexing module transmits lights with different center wavelengths. Wavelength division multiplexing modules are classified into two types: coarse wavelength division multiplexing (CWDM) and dense wavelength division multiplexing (DWDM). Within the same band, DWDM modules are available in more types and use wavelength resources more efficiently than CWDM modules. DWDM and CWDM

modules allow lights with different center wavelengths to be transmitted on one fiber without interfering each other. Therefore, a passive multiplexer can be used to combine the lights into one channel, which is then split into multiple channels by a demultiplexer on the remote end. This reduces the optical fibers required. DWDM and CWDM modules are used for long-distance transmission.

The transmit power of a long-distance optical module is often larger than its overload power. Therefore, when using such optical modules, select optical fibers of an appropriate length to ensure that the actual receive power is smaller than the overload power. If the optical fibers connected to a long-distance optical module are too short, use an optical attenuator to reduce the receive power on the remote optical module. Otherwise, the remote optical module may be burnt.

9.1.3 Optical Module Terms

Transmission distance	Maximum distance over which optical signals can transmit. Optical signals sent from different types of sources can transmit over different distances due to negative effects of optical fibers, such as dispersion and attenuation.
Interface rate	Maximum rate of electrical signals that an optical device can transmit without bit errors. The interface rates defined in Ethernet standards include 125 Mbit/s, 1.25 Gbit/s, 10.3125 Gbit/s, and 41.25 Gbit/s.

Encapsulation type	Appearance type of an optical module. Encapsulation types of optical modules include SFP, eSFP, SFP+, XFP, SFP28, QSFP+, CXP, CFP, and QSFP28. • SFP: small form-factor pluggable. • eSFP: enhanced small form-factor pluggable. An eSFP module is an SFP module that supports monitoring of voltage, temperature, bias current, transmit optical power, and receive optical power. Because all the SFP optical modules support these monitoring functions, eSFP is also called SFP. • SFP+: small form-factor pluggable plus, SFP with a higher rate. SFP+ modules are more sensitive to electromagnetic interference (EMI) because they have a higher rate. To reduce EMI, SFP+ modules have more springs than SFP modules and the cages for SFP+ modules on a card are tighter. • XFP: 10GE optical module. X is the Roman numeral 10. • SFP28: with the same interface size as an SFP+ module. An SFP28 interface can use a 25 GE SFP28 optical module or 10GE SFP+ optical module. • QSFP+: Quad SFP+, four-channel SFP+. • CXP: hot-pluggable high-density parallel optics transceiver form factor, which provides 12 channels of traffic in each of the Tx and Rx directions. It applies only to short multimode links. • CFP: C form-factor pluggable, a new standard for high-speed, hot-pluggable optical transceivers that support data communication and telecommunication applications. Dimensions of a CFP optical module are 144.75 mm x 82 mm x 13.6 mm (W x D x H). • QSFP28: with the same interface size as a QSFP+ module. A QSFP28 interface can use a 100GE QSFP28 optical module or a 40GE QSFP+ optical module.
Center wavelength	Wavelength measured at the midpoint of the half-amplitude line in the transmit spectrum.
Fiber mode	Mode of fibers defining based on core diameters and features of optical fibers. Optical fibers are classified into single-mode fibers and multi-mode fibers. Generally, multi-mode fibers have large core diameters and severe dispersion, so they transmit optical signals over short distances when working with multi-mode optical modules. Single-mode fibers have small dispersion and can transmit optical signals over long distances when working with single-mode optical modules.
Modal bandwidth	Bandwidth measured at a point with transmit power several dB lower than that of the point with the peak center wavelength. Modal bandwidth reflects spectrum characteristics of an optical module.
Fiber diameter	Diameter of the core of a fiber. According to international standards for optical fibers, the diameter of a multi-mode fiber is 62.5 μm or 50 μm, and the diameter of a single-mode fiber is 9 μm.

Fiber class	Optical signals with different wavelengths have their best working windows in different optical fibers. To help efficiently adjust wavelengths or dispersion features of optical fibers and change their refractive indexes, the following classes are defined: multi-mode fiber (G.651), common single-mode fiber (G.652), shifted dispersion fiber (G.653), and non-zero shifted dispersion fiber (G.655). G.651 and G.652 are commonly used fiber classes.
Connector type	Type of the interface on an optical module to accommodate a fiber. Commonly used connector types are LC (applicable to all the SFP, SFP+, and XFP modules) and MPO (applicable to 150 m QSFP+, 100 m CFP, and CXP modules).
Transmit optical power	Output optical power of an optical module when it is working properly.
Receiver sensitivity	Minimum average input optical power that the receiver of an optical module can receive within a range of bit error rate ($BER = 10^{-12}$).
Overload optical power	Maximum average input optical power that the receiver of an optical module can receive within a range of bit error rate ($BER = 10^{-12}$).
Extinction ratio	Minimum ratio of the average optical power with signals transmitted against the average optical power without signals transmitted in complete modulation mode. The extinction ratio indicates the capability of an optical module to identify signal 0 and signal 1.

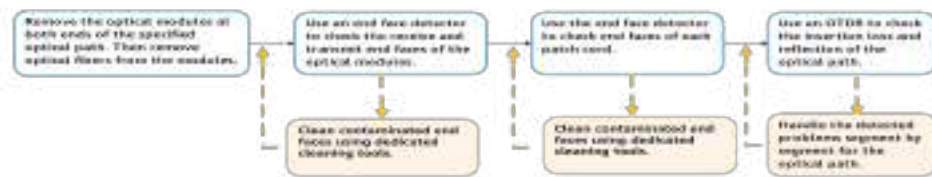
9.1.4 Guide to Using Optical Modules

Precautions for optical module installation:

- When optical path quality is tested using an OTDR, optical fibers must be removed from the associated optical module. This is because the OTDR's transmit optical power is far greater than the optical power damage threshold at the receive end of an optical module.
- For single-mode optical modules, single-mode optical splitters are not recommended due to considerations on the IEEE standard link budget and the impact of single-mode optical splitters on optical signal attenuation on backbone fiber links. If optical splitters must be used for special scenarios or requirements, strictly follow the link budget and insertion loss specifications of different single-mode optical modules. Ensure that the insertion loss of the single-mode optical splitter plus the inherent attenuation of the backbone link is less than the link budget supported by the corresponding single-mode optical modules, with a certain link budget margin reserved.
- Multi-mode optical modules cannot be reliably used in scenarios where optical splitters are used due to considerations on the IEEE standard link budget and the impact of multi-mode optical splitter on multi-mode optical signals.

Method of checking an optical path:

Figure 9-6 Method of checking an optical path



1. Removing and checking the optical module and optical fiber:
Remove the optical module from the device port and remove the optical fiber connector to ensure that the optical path is completely disconnected. Check the appearance of the optical module to ensure that the optical module is not damaged, burnt, or contaminated with foreign objects.
2. Checking the end faces of the optical module:
Use the end face detector to check the receive and transmit ports of the optical module, and check whether the end faces have scratches, contamination, or dust. If an end face is not clean, use the fiber cleaning tool to clean the end face. For details, see *Optical Module Link Cleaning*. Recheck the end faces until they meet the requirements. The end face quality directly affects the insertion loss and reflection characteristics of optical signals.
3. Checking the end faces of the patch cord:
Use the end face detector to check the end faces of the patch cord connector and ensure that they are clean and uncontaminated. End face contamination or damage is a common cause of increased link attenuation and higher bit error rates. If necessary, re-clean the end faces of the patch cord or replace the patch cord. For detailed cleaning methods, see *Optical Module Link Cleaning*.
4. Testing the insertion loss and reflection of the optical path:
Use an OTDR to test the optical path, measuring the link insertion loss and reflection values. Based on the measurement results, determine whether the issue originates from the optical module, patch cord, or the optical cable itself. Then carry out segment-by-segment troubleshooting accordingly.

Inspecting optical fiber connectors:

The Ethernet port rate is increasing and the quality requirement for optical fibers and optical cables is higher. [Table 9-1](#) describes requirements for the loss of optical fiber connectors according to the standard ANSI/TIA-568-C or GBT50312-2016.

Table 9-1 Maximum attenuation of the optical fiber connector

Type	Maximum Attenuation of an Optical Fiber Connector (dB)
Fiber splicing connector	0.3
Optical mechanical connector	0.3
Optical connector	0.75

 NOTE

Fiber cores are connected through connectors, such as the ODF, optical attenuator, and flange, in splicing and mechanical modes.

The loss of optical fiber connectors and reflectance values are tested and processed as follows:

1. After the optical fiber at the peer end is disconnected, use the OTDR meter to test the local end. Check whether the loss and reflection of each link and node are normal. (The loss of a fiber splicing connector should be less than 0.3 dB, the loss of a connector should be less than 0.75 dB, and the reflection of a connector should be less than -30 dB.) If the test result is not within the required range, process the abnormal port.
2. Locate the equipment room where the port resides based on the distance between abnormal points in the OTDR test result. Preliminarily determine the port location, disconnect the port, and perform an OTDR test on the port that reports alarms. Check whether the distance is consistent with that in the previous test. If not, continue to test other ports.
3. After the abnormal port is found, test the port using a fiber end-face inspector. If the port is dirty, clean it.
4. After the port is cleaned, restore the port, and ensure that the connector is tightened. Perform an OTDR test on the port to check whether loss and reflection of each link and node are normal.
5. If the fault persists, replace the flange and perform an OTDR test on the port that reports alarms to check whether loss and reflection of each link and node are normal.
6. If the fault persists, replace the optical fiber and perform an OTDR test on the port that reports alarms to check whether loss and reflection of each link and node are normal.
7. If multiple abnormal points exist on the link, repeat steps 2 to 6.

9.1.5 Optical Module Link Cleaning

Tools and Accessories

The end faces of optical modules and optical fibers are easily affected by contamination such as dust and oil, which may cause abnormal optical power, increased bit errors, or even link interruption. Therefore, when a link exception occurs or during routine maintenance, you need to check and clean the optical module link in a standard manner to ensure the reliability and long-term stable running of the optical link.

You will need to prepare the following tools and accessories:

- ESD wrist strap or ESD gloves
- Dust cap
- Module cleaning pen (with an adapter, used to clean optical fibers)
- Fiber cleaning wipes and lint-free swabs
- End face detector

NOTICE

During the cleaning, strictly comply with ESD control requirements. Wear an ESD wrist strap and ESD gloves, and never touch the module or fiber end faces with bare hands.

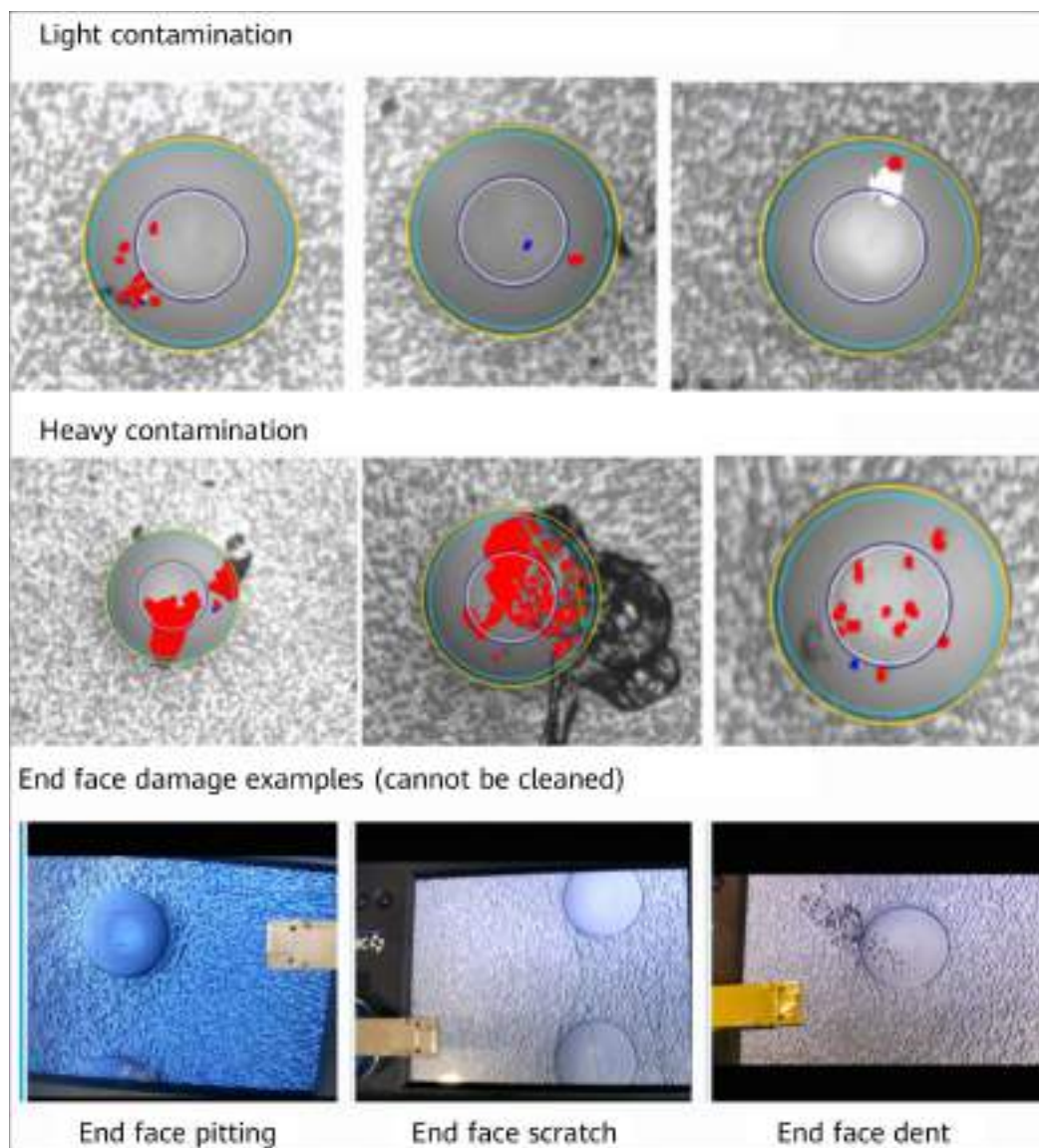
If damage to the optical module or fiber end faces is detected, promptly notify Huawei or the optical fiber supplier.

Procedure

Step 1 Confirm an abnormal link.

Identify the abnormal port and the corresponding optical module link based on the device alarm information or optical module diagnostic information. If necessary, combine optical power readings, interface status, or OTDR test results to make a preliminary judgment on whether the abnormality is related to optical path quality. After the confirmation, clearly mark the port on the device side to avoid misoperation of other normal links. Remove the optical module corresponding to the abnormal port from the device and disconnect the optical fiber. Promptly install dust caps on both the optical module and fiber port to prevent secondary contamination. Before cleaning, use an end face detector to observe the contamination status of the optical module and fiber end faces. If you discover an end face is damaged and cannot be cleaned, replace the patch cord or optical module.

Figure 9-7 Dirt identification using the end face detector



Step 2 Clean the end faces of an optical module.

Method 1: Use a lint-free swab to clean the end faces of the optical module. Remove the dust cap from the optical port of the optical module, and put the dust cap into the dedicated cleaning kit.

1. Select a proper lint-free swab based on the type of the end face to be cleaned. (For SC optical ports, use lint-free swabs with a diameter of 2.5 mm; for LC and MTRJ optical ports, use lint-free swabs with a diameter of 1.25 mm.) Dip the lint-free swab into the cleanser, insert it into the inside of the optical port of the optical module, and clean the optical port by rotating the swab 360 degrees in one direction along the inner wall of the optical port.
2. Insert a dry lint-free swab of the same type into the optical port of the optical module and clean the optical port by rotating the swab 360 degrees in one direction along the inner wall of the optical port.
3. Cover the optical port with a dust cap after you finish cleaning it.

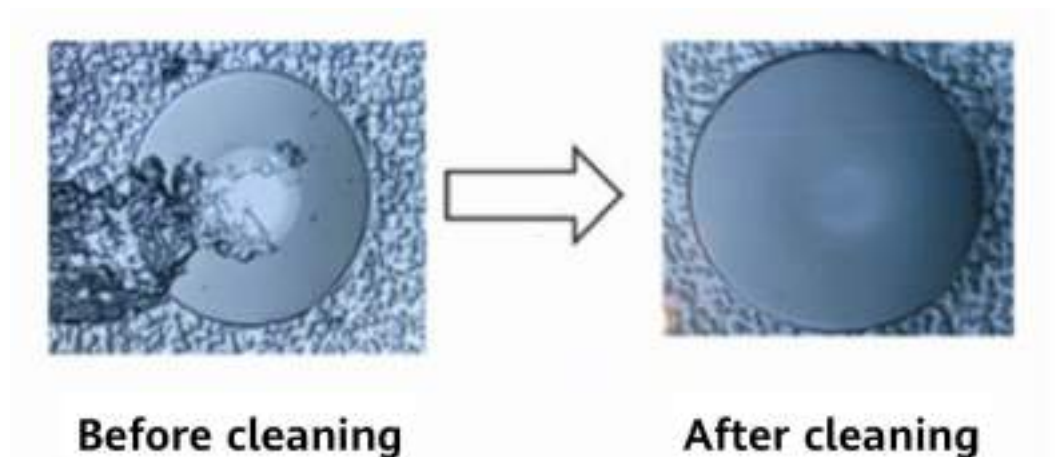
CAUTION

- When cleaning the optical port of an optical module, clean the end faces of associated optical fibers to prevent the optical fibers from contaminating the optical port.
- Typically, each lint-free swab can be used for cleaning only once. If you have checked that a used lint-free swab is clean and can be reused, it can be used for a maximum of three times. For example, a lint-free swab that has been used to dry an optical port can be used for a maximum of three times.

Method 2: Use a module cleaning pen to clean the optical port of an optical module.

1. Hold the cleaning pen (without the adapter), insert it into the optical module, and push the cleaning pen forward to start cleaning. When you hear a click, the cleaning is complete. Repeat this step once more.
2. After cleaning the optical module, use the end face detector to check the cleanliness.

Figure 9-8 Cleaning effect



Step 3 Clean the end faces of an optical fiber.

Before cleaning the end faces of an optical fiber that is in use, ensure that the optical fiber has no optical signals. To achieve this, shut down the ports at both ends of the optical fiber. Then, clean the end faces and insert the optical fiber back into the corresponding port after cleaning.

To clean the end faces of an optical fiber that is not in use, remove the dust cap from the fiber connector (or the patch cord connector of the involved optical component), and put the dust cap into a dedicated cleaning kit. After the cleaning is complete, reinstall the dust cap.

Method 1: Use a module cleaning pen to clean the end faces of an optical fiber.

1. Hold the cleaning pen (with the adapter installed), insert the optical fiber into the adapter, and start cleaning. When you hear a click, the cleaning is complete. Perform this step once more.

2. After the cleaning, use the end face detector to check the cleanliness.

 CAUTION

After cleaning the end faces, clean the adapter. Insert the lint-free swab into the flange and rotate the swab in one direction for four to five rotations to clean the flange.

Method 2: Use fiber cleaning wipes to clean the end faces of an optical fiber.

1. For contamination that is hard to clean, use fiber cleaning wipes. Identify the smooth side of the wipe, and wipe firmly in one direction along the grain of the wipe (avoid wiping back and forth). Spraying a small amount of anhydrous alcohol onto the wipe can further improve the cleaning effect.
2. After the cleaning, use the end face detector to check the cleanliness.

 CAUTION

- Use the smooth surface of the fiber cleaning wipe for cleaning.
 - Ensure that the fiber connector is vertical to the fiber cleaning wipe during cleaning.
 - Wipe the end face along the grain of the wipe.
 - Wipe the end face in one direction only. Do not wipe it back and forth along one route or in multiple directions.
 - Any part of a fiber cleaning wipe can be used only once, and a small fiber cleaning wipe can be used to clean only one connector.
-

Step 4 Restore an abnormal link.

Once you have cleaned the fiber, immediately insert it into the optical module (hold the unlocking position of the fiber and push it forward until you hear a "click" sound, which indicates that the fiber is securely fastened). Do not let the fiber end faces touch any surface.

Check whether the link status of the device port is normal, and whether the optical power, bit errors, and alarms meet the expectation. If the fault persists, replace the patch cord or optical module and use the OTDR to further troubleshoot the fault.

----End

9.1.6 Rules for Optical Module Interoperation

Interoperation Rules

Optical modules with the same standards can interoperate with each other. The standards define the rate, wavelength, and transmission distance of optical modules, but not their encapsulation modes (two interoperated optical modules can have different encapsulation modes).

If you need to achieve interoperability between optical modules with different standards, contact technical support personnel.

When connecting this product to other products, comply with the preceding optical module interoperation rules.

Standards Description

The following describes the standards, using 1000BASE-LX10 as an example:

- 1000 indicates the rate (1000 Mbit/s, in this case). Other rates include 10 Mbit/s, 100 Mbit/s, 10 Gbit/s, 25 Gbit/s, 40 Gbit/s, 100 Gbit/s, 200 Gbit/s .
- BASE indicates baseband transmission.
- L represents a center wavelength of the laser. Currently, the following center wavelengths are available: S (short wavelength: 850 nm), L (long wavelength: 1310 nm), E (extra long wavelength: 1550 nm), and B (single-fiber bidirectional long wavelength).
- X represents the encoding format. The encoding formats include T (twisted pair), X (8B/10B), R (64B/66B), and W (WIS).
- 10 indicates the number of channels. Currently, the value can be 4 or 10. If there is no number, the value is 1.

NOTE

This example provides the definitions in IEEE standards, which are not applicable to all optical modules, for example, non-standard optical modules.

The following organizations or agreements define standards related to optical modules:

- IEEE 802.3, which defines MAC and PHY standards
- Small Form Factor (SFF) committee or Multi-Source Agreements (MSAs), which define optical module hardware, software, and structure standards

9.2 Understanding Copper Modules

Copper modules are also called RJ45 modules. Unlike optical modules, copper modules do not perform electrical-optical conversion. When two optical interfaces have copper modules installed, the interfaces can be connected using a copper cable.

Figure 9-9 shows a copper module.

Figure 9-9 Appearance of a copper module



9.3 GE eSFP Optical Modules

9.3.1 eSFP-GE-SX-MM850 (02315204)

Figure 9-10 Appearance of the eSFP-GE-SX-MM850



 NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-2 eSFP-GE-SX-MM850 specifications

Item	Value
Basic Information	
Module name	eSFP-GE-SX-MM850
Part Number	02315204
Model	eSFP-GE-SX-MM850
Form factor	eSFP
Optical port standard/type	1000BASE-SX
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	-20°C to +85°C (-4°F to +185°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 220 m Multimode fiber (OM1) (with diameter of 62.5 μm): 275 m Multimode fiber (with diameter of 50 μm): 500 m Multimode fiber (OM2) (with diameter of 50 μm): 550 m Multimode fiber (OM3/OM4) (with diameter of 50 μm): 550 m to 1000 m (The distance can exceed 550 m when the fiber quality and the number of fiber connections meet related requirements.)
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-9.5 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-17.0 dBm
Overload power (AVG) [dBm]	0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.3.2 eSFP-GE-SX-MM850 (02313URD)

Figure 9-11 Appearance of the eSFP-GE-SX-MM850



NOTE
The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-3 eSFP-GE-SX-MM850 specifications

Item	Value
Basic Information	
Module name	eSFP-GE-SX-MM850
Part Number	02313URD
Model	eSFP-GE-SX-MM850
Form factor	eSFP
Optical port standard/type	1000BASE-SX
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 220 m Multimode fiber (OM1) (with diameter of 62.5 μm): 275 m Multimode fiber (with diameter of 50 μm): 500 m Multimode fiber (OM2) (with diameter of 50 μm): 550 m Multimode fiber (OM3/OM4) (with diameter of 50 μm): 550 m to 1000 m (The distance can exceed 550 m when the fiber quality and the number of fiber connections meet related requirements.)
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-9.5 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-17.0 dBm
Overload power (AVG) [dBm]	0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-

Item	Value
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.3.3 LE2MGSC40DE0

Figure 9-12 Appearance of the LE2MGSC40DE0



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-4 LE2MGSC40DE0 specifications

Item	Value
Basic Information	
Module name	LE2MGSC40DE0

Item	Value
Part Number	02310KVV
Model	LE2MGSC40DE0
Form factor	eSFP
Connector type	LC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490(RX)1310(TX)
Maximum Tx optical power (AVG) [dBm]	3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-2.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE Single-fiber bidirectional (BIDI) optical modules must be used in pairs. For example, LE2MGSC40DE0 must be used with LE2MGSC40ED0.	

9.3.4 LE2MGSC40ED0

Figure 9-13 Appearance of the LE2MGSC40ED0



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-5 LE2MGSC40ED0 specifications

Item	Value
Basic Information	
Module name	LE2MGSC40ED0
Part Number	02310KVU
Model	LE2MGSC40ED0
Form factor	eSFP
Connector type	LC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC

Item	Value
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310(RX)1490(TX)
Maximum Tx optical power (AVG) [dBm]	3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-2.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE Single-fiber bidirectional (BIDI) optical modules must be used in pairs. For example, LE2MGSC40ED0 must be used with LE2MGSC40DE0.	

9.3.5 SFP-GE-LX-SM1310 (02315200)

Figure 9-14 Appearance of the SFP-GE-LX-SM1310



 NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-6 SFP-GE-LX-SM1310 specifications

Item	Value
Basic Information	
Module name	SFP-GE-LX-SM1310
Part Number	02315200
Model	SFP-GE-LX-SM1310
Form factor	eSFP
Optical port standard/type	1000BASE-LX10
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode (G.652) optical fiber (diameter: 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	-3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-9.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-20.0 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.6 SFP-GE-LX-SM1310 (02313URF)

Figure 9-15 Appearance of the SFP-GE-LX-SM1310



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-7 SFP-GE-LX-SM1310 specifications

Item	Value
Basic Information	
Module name	SFP-GE-LX-SM1310
Part Number	02313URF
Model	SFP-GE-LX-SM1310
Form factor	eSFP
Optical port standard/type	1000BASE-LX10
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode (G.652) optical fiber (diameter: 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	-3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-9.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-20.0 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.7 SFP-GE-LX-SM1310-BIDI

Figure 9-16 Appearance of the SFP-GE-LX-SM1310-BIDI



NOTE
The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-8 SFP-GE-LX-SM1310-BIDI specifications

Item	Value
Basic Information	
Module name	SFP-GE-LX-SM1310-BIDI
Part Number	02315285
Model	SFP-GE-LX-SM1310-BIDI
Form factor	eSFP
Optical port standard/type	1000BASE-BX10
Connector type	LC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode (G.652) optical fiber (diameter: 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1490(RX)1310(TX)
Maximum Tx optical power (AVG) [dBm]	-3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-9.0 dBm
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19.5 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE Single-fiber bidirectional (BIDI) optical modules must be used in pairs. For example, SFP-GE-LX-SM1310-BIDI must be used with SFP-GE-LX-SM1490-BIDI.	

9.3.8 SFP-GE-LX-SM1490-BIDI

Figure 9-17 Appearance of the SFP-GE-LX-SM1490-BIDI



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-9 SFP-GE-LX-SM1490-BIDI specifications

Item	Value
Basic Information	
Module name	SFP-GE-LX-SM1490-BIDI
Part Number	02315286
Model	SFP-GE-LX-SM1490-BIDI
Form factor	eSFP
Optical port standard/type	1000BASE-BX10
Connector type	LC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310(RX)1490(TX)
Maximum Tx optical power (AVG) [dBm]	-3.0 dBm
Minimum Tx optical power (AVG) [dBm]	-9.0 dBm
Minimum extinction ratio [dB]	6 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19.5 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE Single-fiber bidirectional (BIDI) optical modules must be used in pairs. For example, SFP-GE-LX-SM1490-BIDI must be used with SFP-GE-LX-SM1310-BIDI.	

9.3.9 S-SFP-GE-LH40-SM1310

Figure 9-18 Appearance of the S-SFP-GE-LH40-SM1310



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-10 S-SFP-GE-LH40-SM1310 specifications

Item	Value
Basic Information	
Module name	S-SFP-GE-LH40-SM1310
Part Number	02317346
Model	S-SFP-GE-LH40-SM1310
Form factor	eSFP
Optical port standard/type	1000BASE-EX
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0 dBm
Minimum Tx optical power (AVG) [dBm]	-5.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.10 S-SFP-GE-LH80-SM1550

Figure 9-19 Appearance of the S-SFP-GE-LH80-SM1550



NOTE
The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-11 S-SFP-GE-LH80-SM1550 specifications

Item	Value
Basic Information	
Module name	S-SFP-GE-LH80-SM1550
Part Number	02317348
Model	S-SFP-GE-LH80-SM1550
Form factor	eSFP
Optical port standard/type	1000BASE-ZX
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	-2.0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-23 dBm
Overload power (AVG) [dBm]	-3.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.11 CWDM-SFPGE-LH40-1471

Figure 9-20 Appearance of the CWDM-SFPGE-LH40-1471



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-12 CWDM-SFPGE-LH40-1471 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1471
Part Number	02312FWB
Model	CWDM-SFPGE-LH40-1471
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1471 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.12 CWDM-SFPGE-LH40-1491

Figure 9-21 Appearance of the CWDM-SFPGE-LH40-1491



NOTE
The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-13 CWDM-SFPGE-LH40-1491 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1491
Part Number	02312FVX
Model	CWDM-SFPGE-LH40-1491
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1491 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.13 CWDM-SFPGE-LH40-1511

Figure 9-22 Appearance of the CWDM-SFPGE-LH40-1511



 NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-14 CWDM-SFPGE-LH40-1511 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1511
Part Number	02312FWC
Model	CWDM-SFPGE-LH40-1511
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1511 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.14 CWDM-SFPGE-LH40-1531

Figure 9-23 Appearance of the CWDM-SFPGE-LH40-1531



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-15 CWDM-SFPGE-LH40-1531 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1531
Part Number	02312FWQ
Model	CWDM-SFPGE-LH40-1531
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1531 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.15 CWDM-SFPGE-LH40-1551

Figure 9-24 Appearance of the CWDM-SFPGE-LH40-1551



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-16 CWDM-SFPGE-LH40-1551 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1551
Part Number	02312FWR
Model	CWDM-SFPGE-LH40-1551
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1551 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.16 CWDM-SFPGE-LH40-1571

Figure 9-25 Appearance of the CWDM-SFPGE-LH40-1571



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-17 CWDM-SFPGE-LH40-1571 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1571
Part Number	02312FWS
Model	CWDM-SFPGE-LH40-1571
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1571 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.17 CWDM-SFPGE-LH40-1591

Figure 9-26 Appearance of the CWDM-SFPGE-LH40-1591



 NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-18 CWDM-SFPGE-LH40-1591 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1591
Part Number	02312FWT
Model	CWDM-SFPGE-LH40-1591
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1591 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.3.18 CWDM-SFPGE-LH40-1611

Figure 9-27 Appearance of the CWDM-SFPGE-LH40-1611



NOTE

The image of an optical module of the same type (GE eSFP Optical Modules) is used for reference.

Table 9-19 CWDM-SFPGE-LH40-1611 specifications

Item	Value
Basic Information	
Module name	CWDM-SFPGE-LH40-1611
Part Number	02312FWU
Model	CWDM-SFPGE-LH40-1611
Form factor	eSFP
Optical port standard/type	GE-CWDM
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	1Gbit/s
Target transmission distance [km]	Single-mode fiber: 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1611 nm
Maximum Tx optical power (AVG) [dBm]	5.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19 dBm
Overload power (AVG) [dBm]	-3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.4 GE SFP Copper Modules

9.4.1 SFP-1000BaseT (02314171)

Figure 9-28 Appearance of the SFP-1000BaseT



NOTE

The image of an optical module of the same type (GE SFP Copper Modules) is used for reference.

Table 9-20 SFP-1000BaseT specifications

Item	Value
Basic Information	
Module name	SFP-1000BaseT
Part Number	02314171
Model	SFP-1000BaseT
Form factor	SFP
Optical port standard/type	1000BASE-T (SFP-GE-T)
Connector type	RJ45
Target transmission distance [km]	0.1 km
Functions and Features	
DDM options	Not Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	RJ45

9.4.2 SFP-1000BaseT (02313URG)

Figure 9-29 Appearance of the SFP-1000BaseT



 NOTE

The image of an optical module of the same type (GE SFP Copper Modules) is used for reference.

Table 9-21 SFP-1000BaseT specifications

Item	Value
Basic Information	
Module name	SFP-1000BaseT
Part Number	02313URG
Model	SFP-1000BaseT
Form factor	SFP
Optical port standard/type	1000BASE-T (SFP-GE-T)
Connector type	RJ45
Target transmission distance [km]	0.1 km
Functions and Features	
DDM options	Not Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	RJ45

9.4.3 SFP-1000BaseT-MP

Figure 9-30 Appearance of the SFP-1000BaseT-MP



NOTE

The image of an optical module of the same type (GE SFP Copper Modules) is used for reference.

Table 9-22 SFP-1000BaseT-MP specifications

Item	Value
Basic Information	
Module name	SFP-1000BaseT-MP
Part Number	02313GCE
Model	SFP-1000BaseT-MP
Form factor	SFP
Optical port standard/type	1000BASE-T (SFP-GE-T)
Connector type	RJ45
Target transmission distance [km]	0.1 km
Functions and Features	
DDM options	Not Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	RJ45

9.4.4 SFP-1000BaseT-G2 (02314BDD)

Figure 9-31 Appearance of the SFP-1000BaseT-G2



NOTE
The image of an optical module of the same type (GE SFP Copper Modules) is used for reference.

Table 9-23 SFP-1000BaseT-G2 specifications

Item	Value
Basic Information	
Module name	SFP-1000BaseT-G2
Part Number	02314BDD
Model	SFP-1000BaseT-G2
Form factor	SFP
Optical port standard/type	1000BASE-T (SFP-GE-T)
Connector type	RJ45
Target transmission distance [km]	0.1 km
Functions and Features	
DDM options	Not Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	RJ45

 **NOTE**

Version support: V200R022C00 and later versions

9.5 10GE SFP+ Optical Modules

9.5.1 OMXD30000 (02318169)

Figure 9-32 Appearance of the OMXD30000



 **NOTE**

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-24 OMXD30000 specifications

Item	Value
Basic Information	
Module name	OMXD30000
Part Number	02318169
Model	OMXD30000

Item	Value
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm

Item	Value
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.2 OMXD30000 (02313URC)

Figure 9-33 Appearance of the OMXD30000



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-25 OMXD30000 specifications

Item	Value
Basic Information	
Module name	OMXD30000
Part Number	02313URC
Model	OMXD30000
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.3 SFP-10G-SR-MP

Figure 9-34 Appearance of the SFP-10G-SR-MP



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-26 SFP-10G-SR-MP specifications

Item	Value
Basic Information	
Module name	SFP-10G-SR-MP
Part Number	02313AMY
Model	SFP-10G-SR-MP
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	

Item	Value
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.4 SFP-10G-SRV

Figure 9-35 Appearance of the SFP-10G-SRV



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-27 SFP-10G-SRV specifications

Item	Value
Basic Information	
Module name	SFP-10G-SRV
Part Number	02310STB

Item	Value
Model	SFP-10G-SRV
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm

Item	Value
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.5 SFP-10G-SRV-MP

Figure 9-36 Appearance of the SFP-10G-SRV-MP



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-28 SFP-10G-SRV-MP specifications

Item	Value
Basic Information	
Module name	SFP-10G-SRV-MP
Part Number	02313BJJ
Model	SFP-10G-SRV-MP
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.6 OSX010000 (02318170)

Figure 9-37 Appearance of the OSX010000



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-29 OSX010000 specifications

Item	Value
Basic Information	
Module name	OSX010000
Part Number	02318170
Model	OSX010000
Form factor	SFP+
Optical port standard/type	10GBASE-LR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.7 OSX010000 (02313URK)

Figure 9-38 Appearance of the OSX010000



NOTE
The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-30 OSX010000 specifications

Item	Value
Basic Information	
Module name	OSX010000
Part Number	02313URK
Model	OSX010000
Form factor	SFP+
Optical port standard/type	10GBASE-LR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.8 OSX040N01

Figure 9-39 Appearance of the OSX040N01



NOTE
The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-31 OSX040N01 specifications

Item	Value
Basic Information	
Module name	OSX040N01
Part Number	02310CNF
Model	OSX040N01
Form factor	SFP+
Optical port standard/type	10GBASE-ER
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Maximum Tx optical power (AVG) [dBm]	4.0 dBm
Minimum Tx optical power (AVG) [dBm]	-4.7 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-14.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.9 SFP-10G-BXD1

Figure 9-40 Appearance of the SFP-10G-BXD1



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-32 SFP-10G-BXD1 specifications

Item	Value
Basic Information	
Module name	SFP-10G-BXD1
Part Number	02310QDT
Model	SFP-10G-BXD1
Form factor	SFP+
Optical port standard/type	10GBASE-BX
Connector type	LC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	-40°C to +85°C (-40°F to +185°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270(RX)1330(TX)
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-14.4 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.10 SFP-10G-BXU1

Figure 9-41 Appearance of the SFP-10G-BXU1



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-33 SFP-10G-BXU1 specifications

Item	Value
Basic Information	
Module name	SFP-10G-BXU1
Part Number	02310QBJ
Model	SFP-10G-BXU1
Form factor	SFP+
Optical port standard/type	10GBASE-BX
Connector type	LC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	-40°C to +85°C (-40°F to +185°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330(RX)1270(TX)
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-14.4 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.11 SFP-10G-ER-SM1330-BIDI

Table 9-34 SFP-10G-ER-SM1330-BIDI specifications

Item	Value
Basic Information	
Module name	SFP-10G-ER-SM1330-BIDI
Part Number	02311BJB

Item	Value
Model	SFP-10G-ER-SM1330-BIDI
Form factor	SFP+
Application standard/Type	10GBASE-BDER
Connector type	LC
Optical fiber type	SMF
Type of the end face of the fiber ceramic ferrule	PC or UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
DDM options	Supported
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1270(RX)1330(TX)
Maximum Tx optical power (AVG) [dBm]	5 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-18 dBm
Overload power (AVG) [dBm]	-9 dBm

9.5.12 SFP-10G-ER-SM1270-BIDI

Table 9-35 SFP-10G-ER-SM1270-BIDI specifications

Item	Value
Basic Information	
Module name	SFP-10G-ER-SM1270-BIDI
Part Number	02311BJC
Model	SFP-10G-ER-SM1270-BIDI

Item	Value
Form factor	SFP+
Application standard/Type	10GBASE-BDER
Connector type	LC
Optical fiber type	SMF
Type of the end face of the fiber ceramic ferrule	PC or UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
DDM options	Supported
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1330(RX)1270(TX)
Maximum Tx optical power (AVG) [dBm]	5 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-18 dBm
Overload power (AVG) [dBm]	-9 dBm

9.5.13 SFP-10G-ER-1310

Figure 9-42 Appearance of the SFP-10G-ER-1310



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-36 SFP-10G-ER-1310 specifications

Item	Value
Basic Information	
Module name	SFP-10G-ER-1310
Part Number	02311RLX
Model	SFP-10G-ER-1310
Form factor	SFP+
Optical port standard/type	10GBASE-ER
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	4.0 dBm
Minimum Tx optical power (AVG) [dBm]	-2.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-20 dBm
Overload power (AVG) [dBm]	-7.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE When connected to a 10GBASE-ER standard optical module (1550 nm, 10 Gbit/s, 40 km), an SFP-10G-ER-1310 optical module supports only 20 km of maximum transmission distance.	

9.5.14 SFP-10G-iLR

Figure 9-43 Appearance of the SFP-10G-iLR



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-37 SFP-10G-iLR specifications

Item	Value
Basic Information	
Module name	SFP-10G-iLR
Part Number	02311BJJ
Model	SFP-10G-iLR
Form factor	SFP+
Optical port standard/type	10GBASE-iLR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	-40°C to +85°C (-40°F to +185°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 1.4 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-14.4 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.15 SFP-10G-LR (02310QDJ)

Figure 9-44 Appearance of the SFP-10G-LR



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-38 SFP-10G-LR specifications

Item	Value
Basic Information	
Module name	SFP-10G-LR
Part Number	02310QDJ
Model	SFP-10G-LR
Form factor	SFP+
Optical port standard/type	10GBASE-LR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.16 SFP-10G-LR (02313URL)

Figure 9-45 Appearance of the SFP-10G-LR



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-39 SFP-10G-LR specifications

Item	Value
Basic Information	
Module name	SFP-10G-LR
Part Number	02313URL
Model	SFP-10G-LR
Form factor	SFP+
Optical port standard/type	10GBASE-LR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.17 SFP-10G-LR-MP

Figure 9-46 Appearance of the SFP-10G-LR-MP



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-40 SFP-10G-LR-MP specifications

Item	Value
Basic Information	
Module name	SFP-10G-LR-MP
Part Number	02313BJH
Model	SFP-10G-LR-MP
Form factor	SFP+
Optical port standard/type	10GBASE-LR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-12.6 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.18 SFP-10G-USR (02310MNW)

Figure 9-47 Appearance of the SFP-10G-USR



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-41 SFP-10G-USR specifications

Item	Value
Basic Information	
Module name	SFP-10G-USR
Part Number	02310MNW
Model	SFP-10G-USR
Form factor	SFP+
Optical port standard/type	10GBASE-USR
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.7 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.19 SFP-10G-USR (02313URN)

Figure 9-48 Appearance of the SFP-10G-USR



 NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-42 SFP-10G-USR specifications

Item	Value
Basic Information	
Module name	SFP-10G-USR
Part Number	02313URN
Model	SFP-10G-USR
Form factor	SFP+
Optical port standard/type	10GBASE-USR
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.7 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE The optical module complies with Class 1 laser product standards defined in IEC 60825-1 and IEC 60825-2.	

9.5.20 SFP-10G-USR-MP

Figure 9-49 Appearance of the SFP-10G-USR-MP



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-43 SFP-10G-USR-MP specifications

Item	Value
Basic Information	
Module name	SFP-10G-USR-MP
Part Number	02313BJE
Model	SFP-10G-USR-MP
Form factor	SFP+
Optical port standard/type	10GBASE-USR
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.7 dBm
Overload power (AVG) [dBm]	0.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.21 SFP-10G-ZR

Figure 9-50 Appearance of the SFP-10G-ZR



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-44 SFP-10G-ZR specifications

Item	Value
Basic Information	
Module name	SFP-10G-ZR
Part Number	02310SNN
Model	SFP-10G-ZR
Form factor	SFP+
Optical port standard/type	10GBASE-ZR
Connector type	LC/DLC
Optical fiber type	SMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 80 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Maximum Tx optical power (AVG) [dBm]	4.0 dBm
Minimum Tx optical power (AVG) [dBm]	0 dBm
Minimum extinction ratio [dB]	9 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-24.0 dBm
Overload power (AVG) [dBm]	-7.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.22 SFP-10G-ZDWT-L

Table 9-45 SFP-10G-ZDWT-L specifications

Item	Value
Basic Information	
Module name	SFP-10G-ZDWT-L
Part Number	02312DAN

Item	Value
Model	SFP-10G-ZDWT-L
Form factor	SFP+
Application standard/Type	10GBASE-DWDM
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the fiber ceramic ferrule	PC or UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
DDM options	Supported
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 60 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1529.16 nm - 1560.61 nm
Maximum Tx optical power (AVG) [dBm]	3 dBm
Minimum Tx optical power (AVG) [dBm]	-1 dBm
Minimum extinction ratio [dB]	8.2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-24 dBm
Overload power (AVG) [dBm]	-1 dBm

9.5.23 SFP-10G-SR1

Figure 9-51 Appearance of the SFP-10G-SR1



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-46 SFP-10G-SR1 specifications

Item	Value
Basic Information	
Module name	SFP-10G-SR1
Part Number	02314GDS
Model	SFP-10G-SR1
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Random failure rate (60% CL, 50°C case temperature)	300 FIT
Optical power diagnosis accuracy [dB]	2 dB
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	

Item	Value
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.5.24 SFP-10G-ER

Figure 9-52 Appearance of the SFP-10G-ER



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-47 SFP-10G-ER specifications

Item	Value
Basic Information	
Module name	SFP-10G-ER
Part Number	02314GFL
Model	SFP-10G-ER
Form factor	SFP+
Optical port standard/type	10GBASE-ER
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Random failure rate (60% CL, 50°C case temperature)	300 FIT
Optical power diagnosis accuracy [dB]	2 dB
Transmitter Optical Characteristics	
Center wavelength [nm]	1550 nm
Maximum Tx optical power (AVG) [dBm]	4.0 dBm
Minimum Tx optical power (AVG) [dBm]	-4.7 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-14.1 dBm
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	

Item	Value
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.5.25 SFP-10G-SR-E

Figure 9-53 Appearance of the SFP-10G-SR-E



NOTE

The image of an optical module of the same type (10GE SFP+ Optical Modules) is used for reference.

Table 9-48 SFP-10G-SR-E specifications

Item	Value
Basic Information	
Module name	SFP-10G-SR-E
Part Number	02314JUU

Item	Value
Model	SFP-10G-SR-E
Form factor	SFP+
Optical port standard/type	10GBASE-SR
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	10Gbit/s
Target transmission distance [km]	Multimode fiber (with diameter of 62.5 μm): 26 m Multimode fiber (OM1) (with diameter of 62.5 μm): 33 m Multimode fiber (with diameter of 50 μm): 66 m Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber: 160 MHz*km Multimode fiber (OM1): 200 MHz*km Multimode fiber: 400 MHz*km Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	-1.0 dBm
Minimum Tx optical power (AVG) [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm

Item	Value
Overload power (AVG) [dBm]	-1.0 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6 40GE QSFP+ Optical Modules

9.6.1 QSFP-40G-ER4

Figure 9-54 Appearance of the QSFP-40G-ER4



 NOTE

The image of an optical module of the same type (40GE QSFP+ Optical Modules) is used for reference.

Table 9-49 QSFP-40G-ER4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-ER4
Part Number	02311BKT
Model	QSFP-40G-ER4

Item	Value
Form factor	QSFP+
Optical port standard/type	40GBASE-ER4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm
Minimum Tx optical power (AVG) [dBm]	-2.7 dBm
Minimum extinction ratio [dB]	5.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-19.5 dBm
Overload power (AVG) [dBm]	-4.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.6.2 QSFP-40G-eSR4 (02310RMB)

Figure 9-55 Appearance of the QSFP-40G-eSR4



Table 9-50 QSFP-40G-eSR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-eSR4
Part Number	02310RMB
Model	QSFP-40G-eSR4
Form factor	QSFP+
Optical port standard/type	40GBASE-eSR4 10GBASE-SR (four lanes)
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m

Item	Value
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.3 QSFP-40G-eSR4 (02313URU)

Figure 9-56 Appearance of the QSFP-40G-eSR4



Table 9-51 QSFP-40G-eSR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-eSR4
Part Number	02313URU
Model	QSFP-40G-eSR4
Form factor	QSFP+
Optical port standard/type	40GBASE-eSR4 10GBASE-SR (four lanes)
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	

Item	Value
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.4 QSFP-40G-eSR4-MP

Figure 9-57 Appearance of the QSFP-40G-eSR4-MP



Table 9-52 QSFP-40G-eSR4-MP specifications

Item	Value
Basic Information	
Module name	QSFP-40G-eSR4-MP
Part Number	02313GAA
Model	QSFP-40G-eSR4-MP
Form factor	QSFP+
Optical port standard/type	40GBASE-eSR4 10GBASE-SR (four lanes)
Connector type	MPO-12
Optical fiber type	MMF

Item	Value
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 82 m Multimode fiber (OM3) (with diameter of 50 μm): 300 m Multimode fiber (OM4) (with diameter of 50 μm): 400 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.1 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.5 QSFP-40G-iSR4 (02310MHR)

Figure 9-58 Appearance of the QSFP-40G-iSR4



Table 9-53 QSFP-40G-iSR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-iSR4
Part Number	02310MHR
Model	QSFP-40G-iSR4
Form factor	QSFP+
Optical port standard/type	40GBASE-SR4 10GBASE-USR (four lanes)
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m

Item	Value
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.6 QSFP-40G-iSR4 (02313URW)

Figure 9-59 Appearance of the QSFP-40G-iSR4



Table 9-54 QSFP-40G-iSR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-iSR4
Part Number	02313URW
Model	QSFP-40G-iSR4
Form factor	QSFP+
Optical port standard/type	40GBASE-SR4 10GBASE-USR (four lanes)
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	

Item	Value
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE The optical module complies with Class 1 laser product standards defined in IEC 60825-1 and IEC 60825-2.	

9.6.7 QSFP-40G-iSR4-MP

Figure 9-60 Appearance of the QSFP-40G-iSR4-MP



Table 9-55 QSFP-40G-iSR4-MP specifications

Item	Value
Basic Information	
Module name	QSFP-40G-iSR4-MP
Part Number	02313FYY
Model	QSFP-40G-iSR4-MP
Form factor	QSFP+
Optical port standard/type	40GBASE-SR4 10GBASE-USR (four lanes)

Item	Value
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM2) (with diameter of 50 μm): 30 m Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM2): 500 MHz*km Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	0.5 dBm
Minimum Tx optical power (AVG) [dBm]	-7.6 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X10G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.8 QSFP-40G-LR4 (02310MHS)

Figure 9-61 Appearance of the QSFP-40G-LR4



Table 9-56 QSFP-40G-LR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LR4
Part Number	02310MHS
Model	QSFP-40G-LR4
Form factor	QSFP+
Optical port standard/type	40GBASE-LR4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB

Item	Value
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.5 dBm
Overload power (AVG) [dBm]	3.3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.6.9 QSFP-40G-LR4 (02313URY)

Figure 9-62 Appearance of the QSFP-40G-LR4



Table 9-57 QSFP-40G-LR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LR4
Part Number	02313URY
Model	QSFP-40G-LR4
Form factor	QSFP+
Optical port standard/type	40GBASE-LR4

Item	Value
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.5 dBm
Overload power (AVG) [dBm]	3.3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.6.10 QSFP-40G-LR4-Lite (02311YVB)

Figure 9-63 Appearance of the QSFP-40G-LR4-Lite



Table 9-58 QSFP-40G-LR4-Lite specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LR4-Lite
Part Number	02311YVB
Model	QSFP-40G-LR4-Lite
Form factor	QSFP+
Optical port standard/type	40GBASE-LR4 Lite
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum extinction ratio [dB]	3.5 dB

Item	Value
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	2.3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.6.11 QSFP-40G-LR4-Lite (02313URS)

Figure 9-64 Appearance of the QSFP-40G-LR4-Lite



Table 9-59 QSFP-40G-LR4-Lite specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LR4-Lite
Part Number	02313URS
Model	QSFP-40G-LR4-Lite
Form factor	QSFP+
Optical port standard/type	40GBASE-LR4 Lite

Item	Value
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-9 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	2.3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.6.12 QSFP-40G-SR-BD

Figure 9-65 Appearance of the QSFP-40G-SR-BD



NOTE

The image of an optical module of the same type (40GE QSFP+ Optical Modules) is used for reference.

Table 9-60 QSFP-40G-SR-BD specifications

Item	Value
Basic Information	
Module name	QSFP-40G-SR-BD
Part Number	02311FPA
Model	QSFP-40G-SR-BD
Form factor	QSFP+
Optical port standard/type	40GBASE-BIDI
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	10°C to 70°C (50°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm,900 nm
Maximum Tx optical power (AVG) [dBm]	5 dBm
Minimum Tx optical power (AVG) [dBm]	-4 dBm
Minimum extinction ratio [dB]	4.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-4.5 dBm
Overload power (AVG) [dBm]	5 dBm
Functions and Features	
DDM options	Not Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.13 QSFP-40G-eSDLC-PAM

Figure 9-66 Appearance of the QSFP-40G-eSDLC-PAM



NOTE

The image of an optical module of the same type (40GE QSFP+ Optical Modules) is used for reference.

Table 9-61 QSFP-40G-eSDLC-PAM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-eSDLC-PAM
Part Number	02311QTR
Model	QSFP-40G-eSDLC-PAM
Form factor	QSFP+
Optical port standard/type	40GBASE-eSDLC-PAM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 300 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.0 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	

Item	Value
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.14 QSFP-40G-SDLC-PAM

Figure 9-67 Appearance of the QSFP-40G-SDLC-PAM



NOTE

The image of an optical module of the same type (40GE QSFP+ Optical Modules) is used for reference.

Table 9-62 QSFP-40G-SDLC-PAM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-SDLC-PAM
Part Number	02311PUU
Model	QSFP-40G-SDLC-PAM
Form factor	QSFP+
Optical port standard/type	40GBASE-SDLC-PAM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)

Item	Value
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.0 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

9.6.15 QSFP-40G-eSDLC-PAM-G2

Table 9-63 QSFP-40G-eSDLC-PAM-G2 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-eSDLC-PAM-G2
Part Number	02312ELG

Item	Value
Model	QSFP-40G-eSDLC-PAM-G2
Form factor	QSFP+
Application standard/Type	40GBase-eSDLC-PAM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the fiber ceramic ferrule	PC or UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
DDM options	Supported
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 300 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-2 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.0 dBm
Overload power (AVG) [dBm]	2.4 dBm
NOTE The QSFP-40G-eSDLC-PAM optical module cannot be connected to the QSFP-40G-eSDLC-PAM-G2 optical module.	

9.6.16 QSFP-40G-SDLC-PAM-G2

Table 9-64 QSFP-40G-SDLC-PAM-G2 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-SDLC-PAM-G2
Part Number	02312ELH
Model	QSFP-40G-SDLC-PAM-G2
Form factor	QSFP+
Application standard/Type	40GBase-SDLC-PAM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the fiber ceramic ferrule	PC or UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
DDM options	Supported
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 100 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.0 dBm
Overload power (AVG) [dBm]	2.4 dBm

Item	Value
NOTE The QSFP-40G-SDLC-PAM optical module cannot be connected to the QSFP-40G-SDLC-PAM-G2 optical module.	

9.6.17 QSFP-40G-LX4

Figure 9-68 Appearance of the QSFP-40G-LX4

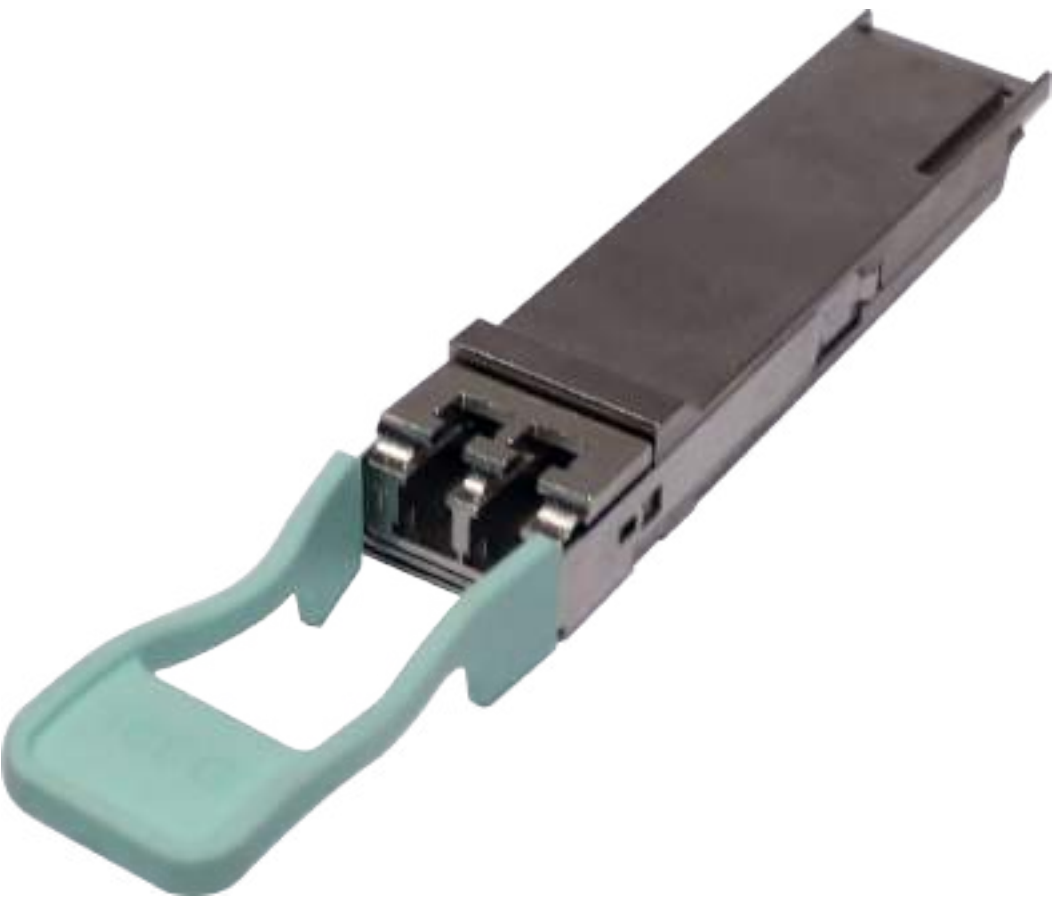


Table 9-65 QSFP-40G-LX4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4
Part Number	02311HNP
Model	QSFP-40G-LX4
Form factor	QSFP+
Optical port standard/type	40GBASE-LX4

Item	Value
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μm): 150 m Multimode fiber (OM4) (with diameter of 50 μm): 150 m Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Overfilled Launch Bandwidth [MHz*km]	Multimode (OM3) optical fiber: 500 MHz*km Multimode (OM4) optical fiber: 500 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.5 dBm
Overload power (AVG) [dBm]	2.3 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

Item	Value
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

9.6.18 QSFP-40G-LX4-MM

Figure 9-69 Appearance of the QSFP-40G-LX4-MM



Table 9-66 QSFP-40G-LX4-MM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4-MM
Part Number	02313NUG

Item	Value
Model	QSFP-40G-LX4-MM
Form factor	QSFP+
Optical port standard/type	40GBASE-LX4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode OM3 optical fiber (diameter: 50 μm): 150 m Multimode OM4 optical fiber (diameter: 50 μm): 150 m
Overfilled Launch Bandwidth [MHz*km]	Multimode (OM3) optical fiber: 500 MHz*km Multimode (OM4) optical fiber: 500 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC

Item	Value
Core Size	50/125μm
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

9.6.19 QSFP-40G-LX4-MM-E

Figure 9-70 Appearance of the QSFP-40G-LX4-MM-E

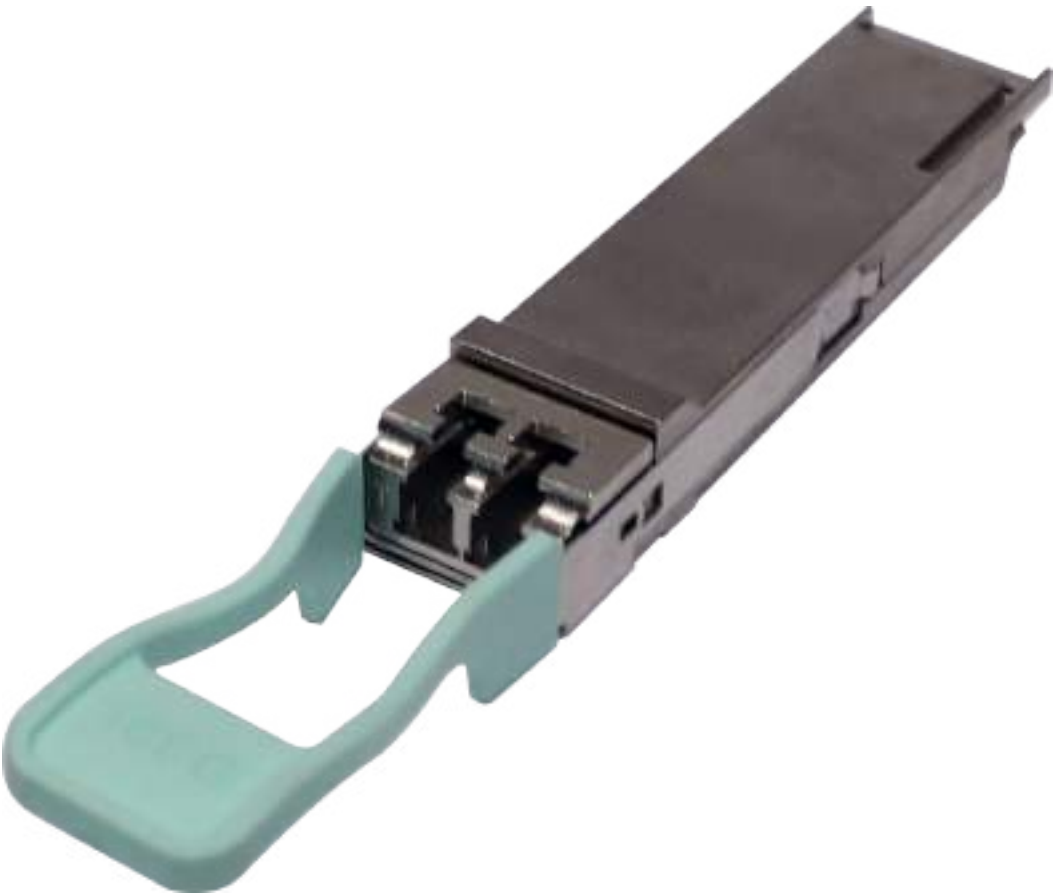


Table 9-67 QSFP-40G-LX4-MM-E specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4-MM-E

Item	Value
Part Number	02314GDU
Model	QSFP-40G-LX4-MM-E
Form factor	QSFP+
Optical port standard/type	40GBASE-LX4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	40Gbit/s
Target transmission distance [km]	Multimode OM3 optical fiber (diameter: 50 μm): 150 m Multimode OM4 optical fiber (diameter: 50 μm): 150 m
Overfilled Launch Bandwidth [MHz*km]	Multimode (OM3) optical fiber: 500 MHz*km Multimode (OM4) optical fiber: 500 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.3 dBm
Minimum Tx optical power (AVG) [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC

Item	Value
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

 **NOTE**

Version support: V200R022C10 and later versions.

9.7 100GE QSFP28 Optical Modules

9.7.1 QSFP28-100G-LR4 (02311KNU)

Figure 9-71 Appearance of the QSFP28-100G-LR4



Table 9-68 QSFP28-100G-LR4 specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-LR4
Part Number	02311KNU
Model	QSFP28-100G-LR4
Form factor	QSFP28

Item	Value
Optical port standard/type	100GBASE-LR4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295 nm,1300 nm,1304 nm,1309 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm
Minimum Tx optical power (AVG) [dBm]	-4.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.6 dBm
Overload power (AVG) [dBm]	4.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.2 QSFP28-100G-LR4 (02313URT)

Figure 9-72 Appearance of the QSFP28-100G-LR4



Table 9-69 QSFP28-100G-LR4 specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-LR4
Part Number	02313URT
Model	QSFP28-100G-LR4
Form factor	QSFP28
Optical port standard/type	100GBASE-LR4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295 nm,1300 nm,1304 nm,1309 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm
Minimum Tx optical power (AVG) [dBm]	-4.3 dBm
Minimum extinction ratio [dB]	4 dB

Item	Value
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.6 dBm
Overload power (AVG) [dBm]	4.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.3 QSFP28-100G-PSM4

Figure 9-73 Appearance of the QSFP28-100G-PSM4



Table 9-70 QSFP28-100G-PSM4 specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-PSM4

Item	Value
Part Number	02311MNM
Model	QSFP28-100G-PSM4
Form factor	QSFP28
Optical port standard/type	100GBASE-PSM4
Connector type	MPO-12
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	APC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 500 m
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	2 dBm
Minimum Tx optical power (AVG) [dBm]	-9.4 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.35 dBm
Overload power (AVG) [dBm]	2.2 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X25G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	APC
Core Size	9/125μm

9.7.4 QSFP28-100G-SR4 (02311GBW)

Figure 9-74 Appearance of the QSFP28-100G-SR4



Table 9-71 QSFP28-100G-SR4 specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-SR4
Part Number	02311GBW
Model	QSFP28-100G-SR4
Form factor	QSFP28
Optical port standard/type	100GBASE-SR4
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 70 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 100 m
Modal bandwidth [MHz*km]	Multimode (OM3) optical fiber: 2000 MHz*km Multimode (OM4) optical fiber: 4700 MHz*km

Item	Value
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X25G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.7.5 QSFP28-100G-SR4 (02313URQ)

Figure 9-75 Appearance of the QSFP28-100G-SR4



Table 9-72 QSFP28-100G-SR4 specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-SR4
Part Number	02313URQ
Model	QSFP28-100G-SR4
Form factor	QSFP28
Optical port standard/type	100GBASE-SR4
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 70 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 100 m
Modal bandwidth [MHz*km]	Multimode optical fiber (OM3): 2000 MHz*km Multimode optical fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X25G

Item	Value
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.7.6 QSFP28-100G-SR4-MP

Figure 9-76 Appearance of the QSFP28-100G-SR4-MP



Table 9-73 QSFP28-100G-SR4-MP specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-SR4-MP
Part Number	02313FYX
Model	QSFP28-100G-SR4-MP
Form factor	QSFP28
Optical port standard/type	100GBASE-SR4
Connector type	MPO-12
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)

Item	Value
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 70 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 100 m
Modal bandwidth [MHz*km]	Multimode (OM3) optical fiber: 2000 MHz*km Multimode (OM4) optical fiber: 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	2.4 dBm
Minimum Tx optical power (AVG) [dBm]	-8.4 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	4X25G
Matching Optical Fiber Specifications	
Fiber connector type	MPO-12
Fiber end face type	PC/UPC
Core Size	50/125μm

9.7.7 QSFP28-100G-BIDI

Figure 9-77 Appearance of the QSFP28-100G-BIDI

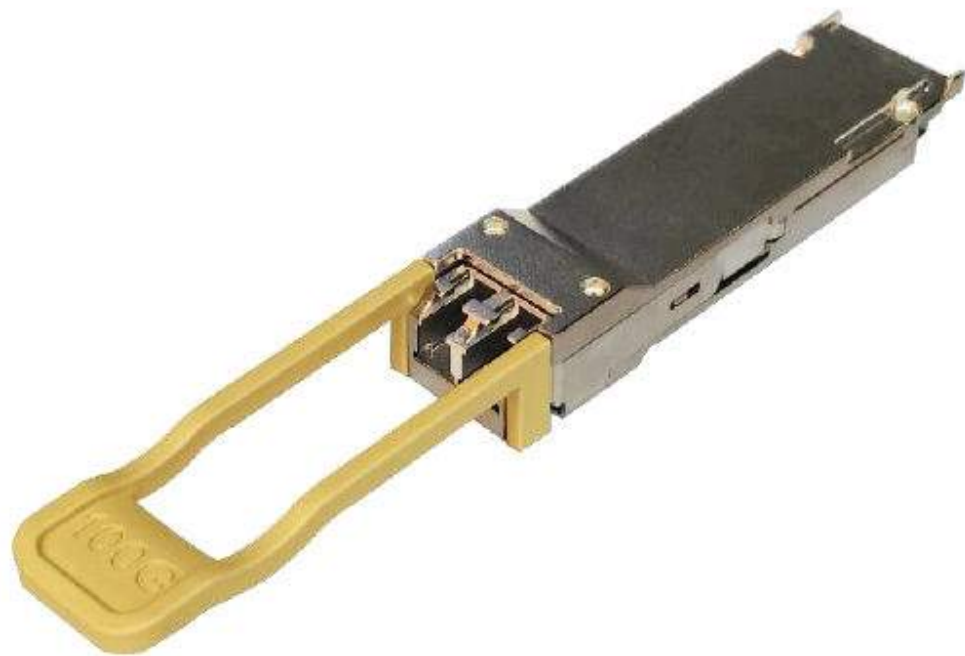


Table 9-74 QSFP28-100G-BIDI specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-BIDI
Part Number	02313EEK
Model	QSFP28-100G-BIDI
Form factor	QSFP28
Optical port standard/type	100G PAM4 BiDi
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	10°C to 70°C (50°F to 158°F)
Transmission rate [bit/s]	100Gbit/s

Item	Value
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 70 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 100 m
Modal bandwidth [MHz*km]	Multimode (OM3) optical fiber: 2000 MHz*km Multimode (OM4) optical fiber: 4700 MHz*km
Bit error ratio (BER)	1e-12
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm/910 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.4 dBm
Maximum Tx optical power (OMA) [dBm]	3 dBm
Minimum Tx optical power (OMA) [dBm]	-2.4 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (AVG) [dBm]	-7.9 dBm
Rx sensitivity (OMA) [dBm]	-5.9 dBm
Overload power (AVG) [dBm]	3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

 NOTE

- Version support:
 - V200R005C20 and V200R019C10 after the corresponding patch is installed
 - V200R020C10 and later versions
- Before installing a QSFP28-100G-BIDI optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP28-100G-BIDI optical module installed, the port status will become Down (Transceiver type mismatch).
- BIDI optical modules must be used in pairs, QSFP-100G-BIDI-G2 cannot be used with QSFP28-100G-BIDI.

9.7.8 QSFP28-100G-DR

Figure 9-78 Appearance of the QSFP28-100G-DR



Table 9-75 QSFP28-100G-DR specifications

Item	Value
Basic Information	
Module name	QSFP28-100G-DR
Part Number	02312VSP

Item	Value
Model	QSFP28-100G-DR
Form factor	QSFP28
Optical port standard/type	100GBASE-DR
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 500 m
Transmitter Optical Characteristics	
Center wavelength [nm]	1311 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-2.9 dBm
Maximum Tx optical power (OMA) [dBm]	4.2 dBm
Minimum Tx optical power (OMA) [dBm]	-0.8 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (AVG) [dBm]	-5.9 dBm
Rx sensitivity (OMA) [dBm]	Max(-3.9,SECQ-5.3)
Overload power (AVG) [dBm]	4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC

Item	Value
Core Size	9/125μm

 NOTE

- Version support: V200R020C10 and later versions
- A port that has a QSFP28-100G-DR optical module installed cannot be used for stack connection.
- Before installing a QSFP28-100G-DR optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP28-100G-DR optical module installed, the port status will become Down (Transceiver type mismatch).

9.7.9 QSFP-100G-4WDM-40

Figure 9-79 Appearance of the QSFP-100G-4WDM-40

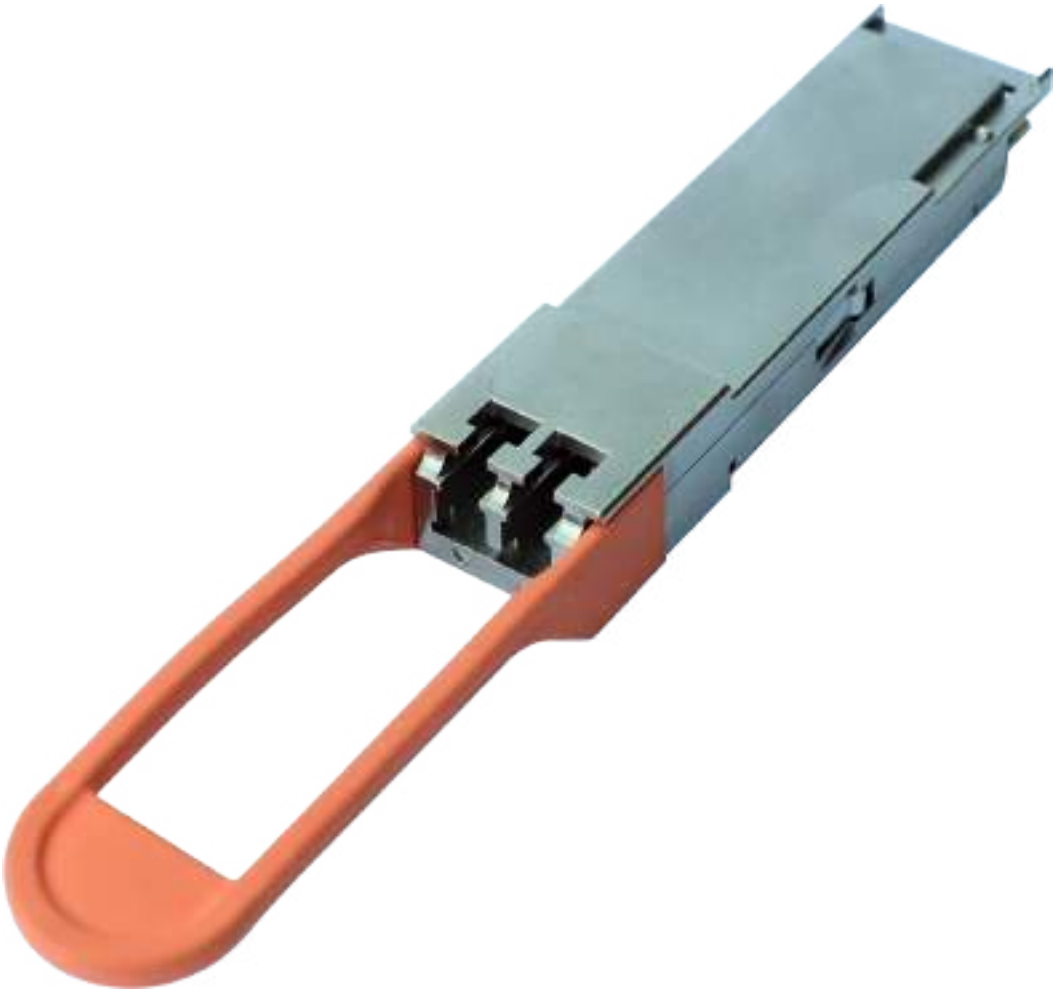


Table 9-76 QSFP-100G-4WDM-40 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-4WDM-40
Part Number	02312QTL
Model	QSFP-100G-4WDM-40
Form factor	QSFP28
Optical port standard/type	100GBASE-4WDM
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	6.5 dBm
Minimum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum extinction ratio [dB]	4.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-18.5 dBm
Overload power (AVG) [dBm]	-3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.10 QSFP-100G-CWDM4 (02311MNN)

Figure 9-80 Appearance of the QSFP-100G-CWDM4



Table 9-77 QSFP-100G-CWDM4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-CWDM4
Part Number	02311MNN
Model	QSFP-100G-CWDM4
Form factor	QSFP28
Optical port standard/type	100GBASE-CWDM4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-6.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.8 dBm
Overload power (AVG) [dBm]	2.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.11 QSFP-100G-CWDM4-Lite

Figure 9-81 Appearance of the QSFP-100G-CWDM4-Lite



Table 9-78 QSFP-100G-CWDM4-Lite specifications

Item	Value
Basic Information	
Module name	QSFP-100G-CWDM4-Lite
Part Number	02312UJN
Model	QSFP-100G-CWDM4-Lite
Form factor	QSFP28
Optical port standard/type	100GBASE-CWDM4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 0.5 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-6.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.8 dBm
Overload power (AVG) [dBm]	2.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.12 QSFP-100G-LR4-Lite

Figure 9-82 Appearance of the QSFP-100G-LR4-Lite



Table 9-79 QSFP-100G-LR4-Lite specifications

Item	Value
Basic Information	
Module name	QSFP-100G-LR4-Lite
Part Number	02311UPS
Model	QSFP-100G-LR4-Lite
Form factor	QSFP28
Optical port standard/type	100GBASE-LR4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295 nm,1300 nm,1304 nm,1309 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	-4.3 dBm
Minimum extinction ratio [dB]	4 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.6 dBm
Overload power (AVG) [dBm]	4.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.13 QSFP-100G-eCWDM4

Figure 9-83 Appearance of the QSFP-100G-eCWDM4



Table 9-80 QSFP-100G-eCWD4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-eCWD4
Part Number	02312DAT
Model	QSFP-100G-eCWD4
Form factor	QSFP28
Optical port standard/type	100GBASE-eCWD4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-6.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-13 dBm
Overload power (AVG) [dBm]	2.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.14 QSFP-100G-ER4-Lite

Figure 9-84 Appearance of the QSFP-100G-ER4-Lite



Table 9-81 QSFP-100G-ER4-Lite specifications

Item	Value
Basic Information	
Module name	QSFP-100G-ER4-Lite
Part Number	02311YXR
Model	QSFP-100G-ER4-Lite
Form factor	QSFP28
Optical port standard/type	100GBASE-ER4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 30 km (FEC OFF)/40 km (FEC ON)
Transmitter Optical Characteristics	

Item	Value
Center wavelength [nm]	1295 nm,1300 nm,1304 nm,1309 nm
Maximum Tx optical power (AVG) [dBm]	2.9 dBm
Minimum Tx optical power (AVG) [dBm]	-2.5 dBm
Minimum extinction ratio [dB]	8 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-18.4 dBm
Overload power (AVG) [dBm]	-3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.15 QSFP-100G-SWDM4 (02311QUK)

Figure 9-85 Appearance of the QSFP-100G-SWDM4



Table 9-82 QSFP-100G-SWDM4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-SWDM4
Part Number	02311QUK
Model	QSFP-100G-SWDM4
Form factor	QSFP28
Optical port standard/type	100G-SWDM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s

Item	Value
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μ m): 75 m Multimode fiber (OM4) (with diameter of 50 μ m): 100 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	3.4 dBm
Minimum Tx optical power (AVG) [dBm]	-7.5 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-10.5 dBm
Overload power (AVG) [dBm]	2.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125 μ m

9.7.16 QSFP-100G-ER4

Figure 9-86 Appearance of the QSFP-100G-ER4



Table 9-83 QSFP-100G-ER4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-ER4
Part Number	02314HES
Model	QSFP-100G-ER4
Form factor	QSFP28
Optical port standard/type	100GBASE-ER4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 40 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1295 nm,1300 nm,1304 nm,1309 nm
Maximum Tx optical power (AVG) [dBm]	2.9 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	-2.9 dBm
Maximum Tx optical power (OMA) [dBm]	4.5 dBm
Minimum Tx optical power (OMA) [dBm]	0.1 dBm
Minimum extinction ratio [dB]	8 dB
Receiver Optical Characteristics	
Rx sensitivity (AVG) [dBm]	-20.9 dBm
Overload power (AVG) [dBm]	-3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

 NOTE

Version support: V200R022C00 and later versions.

9.7.17 QSFP-100G-FR1

Figure 9-87 Appearance of the QSFP-100G-FR1



Table 9-84 QSFP-100G-FR1 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-FR1
Part Number	02314BDE
Model	QSFP-100G-FR1
Form factor	QSFP28
Optical port standard/type	100GBASE-FR1
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode (G.652) optical fiber (diameter: 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1311 nm

Item	Value
Maximum Tx optical power (AVG) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-3.1 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	Max(-4.5,TECQ-5.9)
Overload power (AVG) [dBm]	4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

 NOTE

- Version support: V200R022C00 and later versions.
- A port that has a QSFP-100G-FR1 optical module installed cannot be used for stack connection.
- Before installing a QSFP-100G-FR1 optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP-100G-FR1 optical module installed, the port status will become Down (Transceiver type mismatch).

9.7.18 QSFP-100G-LX4-MM (02314DBX)

Figure 9-88 Appearance of the QSFP-100G-LX4-MM



Table 9-85 QSFP-100G-LX4-MM specifications

Item	Value
Basic Information	
Module name	QSFP-100G-LX4-MM
Part Number	02314DBX
Model	QSFP-100G-LX4-MM
Form factor	QSFP28
Optical port standard/type	100G-LX4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)

Item	Value
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Multimode (OM3) optical fiber (diameter: 50 μm): 100 m Multimode (OM4) optical fiber (diameter: 50 μm): 100 m
Overfilled Launch Bandwidth [MHz*km]	Multimode (OM3) optical fiber: 510 MHz*km Multimode (OM4) optical fiber: 510 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm
Minimum Tx optical power (AVG) [dBm]	-3.5 dBm
Maximum Tx optical power (OMA) [dBm]	4.5 dBm
Minimum Tx optical power (OMA) [dBm]	-3 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.5 dBm
Overload power (AVG) [dBm]	4.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm

Item	Value
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

 **NOTE**

Version support: V200R022C00 and later versions

9.7.19 QSFP-100G-LX4-MM (02314NED)

Figure 9-89 Appearance of the QSFP-100G-LX4-MM

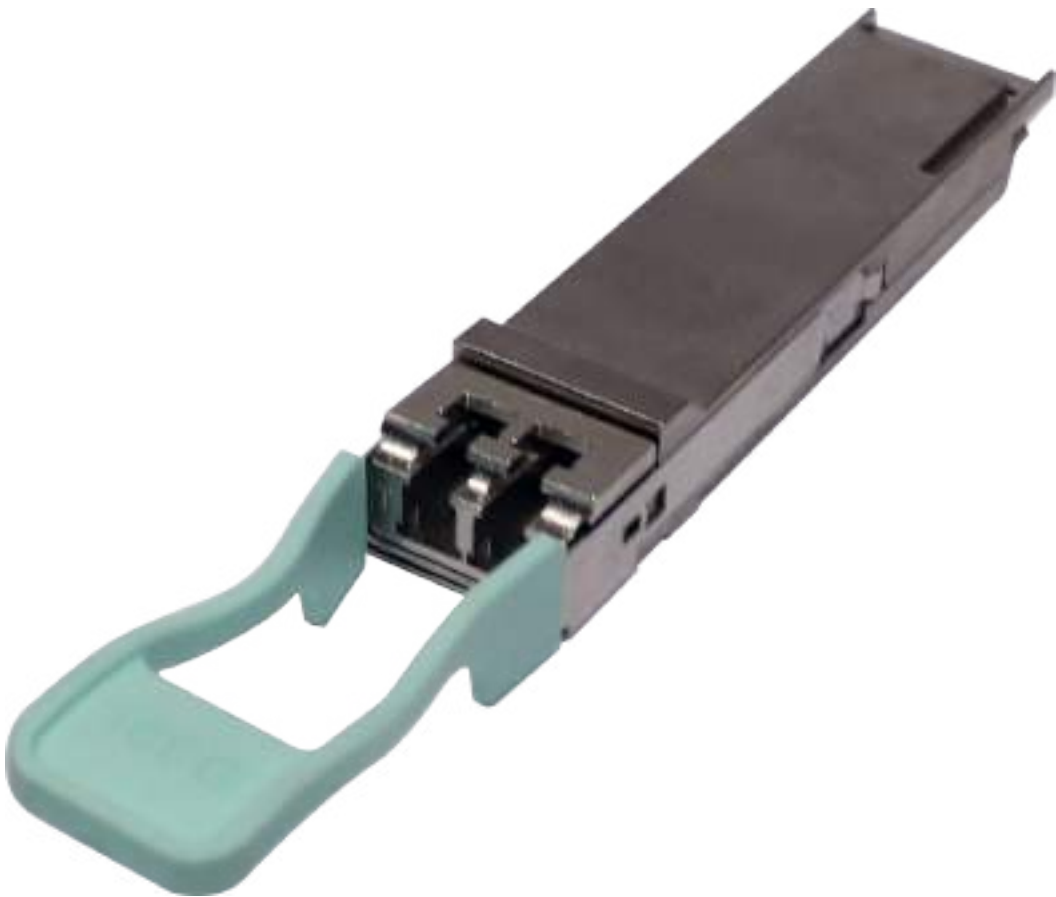


Table 9-86 QSFP-100G-LX4-MM specifications

Item	Value
Basic Information	

Item	Value
Module name	QSFP-100G-LX4-MM
Part Number	02314NED
Model	QSFP-100G-LX4-MM
Form factor	QSFP28
Optical port standard/type	100G-LX4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Multimode (OM3) optical fiber (diameter: 50 μm): 100 m Multimode (OM4) optical fiber (diameter: 50 μm): 100 m
Overfilled Launch Bandwidth [MHz*km]	Multimode (OM3) optical fiber: 510 MHz*km Multimode (OM4) optical fiber: 510 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	4.5 dBm
Minimum Tx optical power (AVG) [dBm]	-3.5 dBm
Maximum Tx optical power (OMA) [dBm]	4.5 dBm
Minimum Tx optical power (OMA) [dBm]	-3 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-11.5 dBm
Overload power (AVG) [dBm]	4.5 dBm
Functions and Features	

Item	Value
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 8. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

 **NOTE**

Version support: V200R022C00 and later versions

9.7.20 QSFP-100G-C4-Lite-E

Figure 9-90 Appearance of the QSFP-100G-C4-Lite-E



Table 9-87 QSFP-100G-C4-Lite-E specifications

Item	Value
Basic Information	
Module name	QSFP-100G-C4-Lite-E

Item	Value
Part Number	02314GDV
Model	QSFP-100G-C4-Lite-E
Form factor	QSFP28
Optical port standard/type	100G-CWDM4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 0.5 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power (AVG) [dBm]	2.5 dBm
Minimum Tx optical power (AVG) [dBm]	-6.5 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.8 dBm
Overload power (AVG) [dBm]	2.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

 NOTE

Version support: V200R022C10 and later versions.

9.7.21 QSFP-100G-LR1

Figure 9-91 Appearance of the QSFP-100G-LR1



Table 9-88 QSFP-100G-LR1 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-LR1
Part Number	02314GYT
Model	QSFP-100G-LR1
Form factor	QSFP28
Optical port standard/type	100GBASE-LR1
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1311 nm
Maximum Tx optical power (AVG) [dBm]	4.8 dBm

Item	Value
Minimum Tx optical power (AVG) [dBm]	-1.9 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	Max(-6.1,TECQ-7.5)
Overload power (AVG) [dBm]	4.8 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm
NOTE The QSFP-100G-LR1 optical module cannot be used for interconnection in stack scenarios.	

 NOTE

- Version support: V200R022C10 and later versions.
- Before installing a QSFP-100G-LR1 optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP-100G-LR1 optical module installed, the port status will become Down (Transceiver type mismatch).

9.7.22 QSFP-100G-BIDI-G2

Figure 9-92 Appearance of the QSFP-100G-BIDI-G2

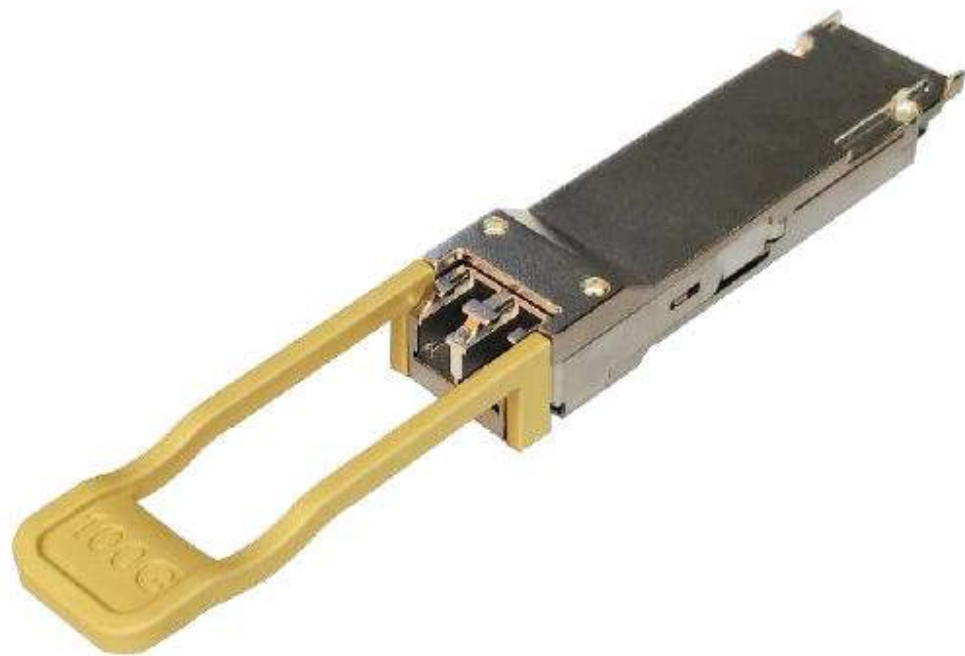


Table 9-89 QSFP-100G-BIDI-G2 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-BIDI-G2
Part Number	02314DBW
Model	QSFP-100G-BIDI-G2
Form factor	QSFP28
Optical port standard/type	100G-BIDI
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s

Item	Value
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 70 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 100 m
Transmitter Optical Characteristics	
Center wavelength [nm]	850,910 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.4 dBm
Maximum Tx optical power (OMA) [dBm]	3 dBm
Minimum Tx optical power (OMA) [dBm]	-3.2 dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	max (-6.6, SECQ - 8)
Overload power (AVG) [dBm]	3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Inbuilt FEC	Yes
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE 1. The optical-layer FEC type of the module is KP4 FEC. Pay attention to the optical-layer interconnection compatibility. 2. The optical-layer FEC of the module introduces extra delay. Pay attention to the impact during application.	

 NOTE

Version support: V200R022C10 and later versions.

BIDI optical modules must be used in pairs, QSFP-100G-BIDI-G2 cannot be used with QSFP28-100G-BIDI.

Before installing a QSFP-100G-BIDI-G2 optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP-100G-BIDI-G2 optical module installed, the port status will become Down (Transceiver type mismatch).

The QSFP-100G-BIDI-G2 optical module cannot be used for interconnection in stack scenarios.

9.7.23 QSFP-100G-CWDM4 (02314LWE)

Figure 9-93 Appearance of the QSFP-100G-CWDM4



Table 9-90 QSFP-100G-CWDM4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-CWDM4
Part Number	02314LWE
Model	QSFP-100G-CWDM4
Form factor	QSFP28
Optical port standard/type	100GBASE-CWDM4
Connector type	LC/DLC
Optical fiber type	SMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)

Item	Value
Transmission rate [bit/s]	100Gbit/s
Target transmission distance [km]	Single-mode fiber (G.652) (with diameter of 9 μm): 2 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power (AVG) [dBm]	per lane:2.5
Minimum Tx optical power (AVG) [dBm]	per lane:-6.5
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-9.8 dBm
Overload power (AVG) [dBm]	2.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	9/125μm

9.7.24 QSFP-100G-SWDM4 (02314LWF)

Figure 9-94 Appearance of the QSFP-100G-SWDM4



Table 9-91 QSFP-100G-SWDM4 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-SWDM4
Part Number	02314LWF
Model	QSFP-100G-SWDM4
Form factor	QSFP28
Optical port standard/type	100G-SWDM4
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s

Item	Value
Target transmission distance [km]	Multimode fiber (OM3) (with diameter of 50 μ m): 75 m Multimode fiber (OM4) (with diameter of 50 μ m): 100 m
Modal bandwidth [MHz*km]	Multimode fiber (OM3): 2000 MHz*km Multimode fiber (OM4): 4700 MHz*km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power (AVG) [dBm]	3.4 dBm
Minimum Tx optical power (AVG) [dBm]	-7.5 dBm
Minimum extinction ratio [dB]	2 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	-8.2 dBm
Overload power (AVG) [dBm]	3.4 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125 μ m

9.7.25 QSFP-100G-eBIDI-G2

Figure 9-95 Appearance of the QSFP-100G-eBIDI-G2

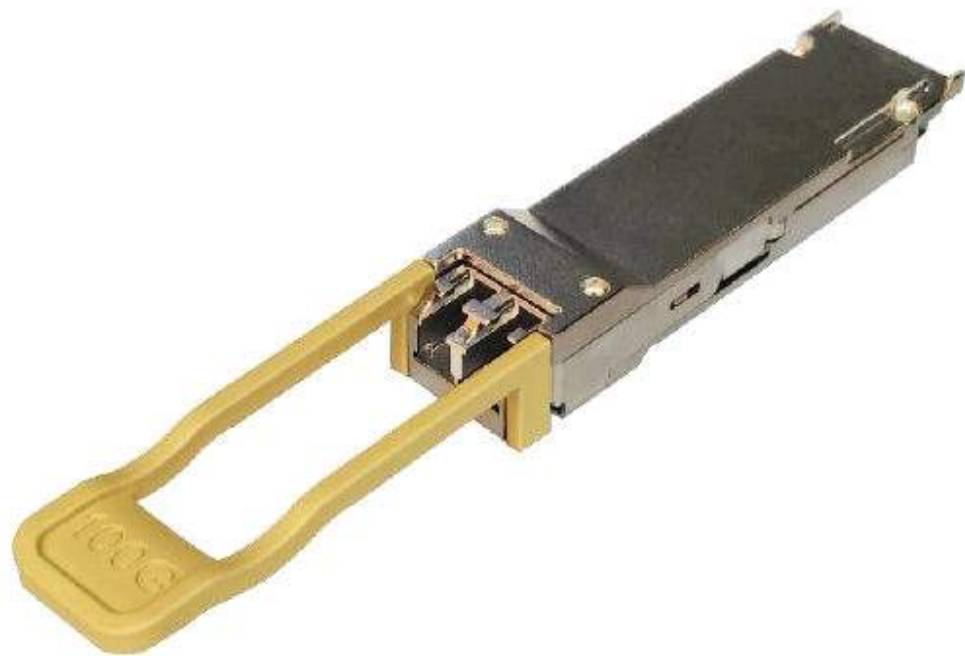


Table 9-92 QSFP-100G-eBIDI-G2 specifications

Item	Value
Basic Information	
Module name	QSFP-100G-eBIDI-G2
Part Number	02314LWG
Model	QSFP-100G-eBIDI-G2
Form factor	QSFP28
Optical port standard/type	100G-BIDI
Connector type	LC/DLC
Optical fiber type	MMF
Type of the end face of the ceramic ferrule on optical ports	PC/UPC
Working case temperature [°C (°F)]	0°C to 70°C (32°F to 158°F)
Transmission rate [bit/s]	100Gbit/s

Item	Value
Target transmission distance [km]	Multi-mode (OM3) optical fiber (diameter: 50 μm): 100 m Multi-mode (OM4) optical fiber (diameter: 50 μm): 150 m
Transmitter Optical Characteristics	
Center wavelength [nm]	850,910 nm
Maximum Tx optical power (AVG) [dBm]	4 dBm
Minimum Tx optical power (AVG) [dBm]	-4.4 dBm
Maximum Tx optical power (OMA) [dBm]	3 dBm
Minimum Tx optical power (OMA) [dBm]	850nm:-2.6dBm; 910nm:-2dBm
Minimum extinction ratio [dB]	3 dB
Receiver Optical Characteristics	
Rx sensitivity (OMA) [dBm]	max (-6.6, SECQ - 8)
Overload power (AVG) [dBm]	3.5 dBm
Functions and Features	
DDM options	Supported
Port splitting	-
Inbuilt FEC	Yes
Matching Optical Fiber Specifications	
Fiber connector type	LC/DLC
Fiber end face type	PC/UPC
Core Size	50/125μm
NOTE 1. The optical-layer FEC type of the module is KP4 FEC. Pay attention to the optical-layer interconnection compatibility. 2. The optical-layer FEC of the module introduces extra delay. Pay attention to the impact during application.	

 NOTE

Version support: V200R023C00 and later versions.

BIDI optical modules must be used in pairs, QSFP-100G-eBIDI-G2 cannot be used with QSFP28-100G-BIDI.

Before installing a QSFP-100G-eBIDI-G2 optical module on a port, you need to disable the FEC function on the port. For example, if the RS-FEC function is enabled on a port that has a QSFP-100G-eBIDI-G2 optical module installed, the port status will become Down (Transceiver type mismatch).

The QSFP-100G-eBIDI-G2 optical module cannot be used for interconnection in stack scenarios.